



VNS Therapy[®] System Physician's Manual

Pulse™ Generator—Model 102

Pulse Duo™ Generator—Model 102R

Demipulse[®] Generator—Model 103

Demipulse Duo[®] Generator—Model 104

AspireHC[®] Generator—Model 105

AspireSR[®] Generator—Model 106

SenTiva™ Generator—Model 1000

Lead—Model 302

PerenniaFLEX[®] Lead—Model 304

PerenniaDURA[®] Lead—Model 303

For Healthcare Professionals

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US Version

Rx Only

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1 Introduction to the VNS Therapy System

For a list of symbols and glossary terms used with the VNS Therapy System, go to www.livanova.com.

1.1 Brief Device Description

1.1.1 The VNS Therapy System

The LivaNova® VNS Therapy® System, used for vagus nerve stimulation (VNS), consists of the implantable VNS Therapy generator, lead, and external programming system used to change stimulation settings. The generator is an implantable, multiprogrammable pulse generator that delivers electrical signals to the vagus nerve. The generator is housed in a hermetically sealed titanium case and is powered by a single battery. Electrical signals are transmitted from the generator to the vagus nerve by the lead. The lead and the generator make up the implantable portion of the VNS Therapy System.

The VNS Therapy Programming System includes a computer pre-installed with VNS Therapy programming software and a programming wand. The physician uses the programming system to read and change generator settings.

1.1.2 Modes of VNS Therapy

VNS Therapy is available both as a scheduled, cycling stimulation between programmable On- and Off- times (Normal Mode) and as on-demand stimulation (Magnet Mode). For generators with tachycardia detection, there is an additional feature, AutoStim Mode, which when turned ON runs in conjunction with the Normal Mode.

1.1.2.1 Normal Mode

Normal Mode stimulation is the primary operating mode of VNS Therapy. Physicians program the device to preset ON and OFF times in which the device cycles throughout the day. This mode is always on, as long as an output current is programmed to a value greater than 0 mA. The patient can inhibit Normal Mode stimulation by holding the VNS magnet over the implanted generator.

1.1.2.2 Magnet Mode

Magnet Mode is limited to patients with epilepsy. Magnet Mode stimulation is an on-demand stimulation that can be initiated by the patient or caregiver passing the VNS magnet over the implanted generator. The patient or caregiver can use the magnet to initiate Magnet Mode stimulation when the patient has an aura, at seizure onset, or during a seizure, as an attempt to abort or de-intensify an oncoming seizure or a seizure in progress.

1.1.2.3 AutoStim Mode

AutoStim Mode (or Automatic Stimulation) is limited to patients with epilepsy. It is an optional feature in generator models capable of tachycardia detection and can be used in conjunction with the Normal Mode. The AutoStim feature monitors and detects rapid, relative heart rate increases ($\geq 20\%$) that may be associated with seizures. Following detection, on-demand stimulation is delivered. AutoStim Mode runs without patient interaction unless the patient

inhibits stimulation via the VNS magnet or the clinician programs the AutoStim Mode OFF. The purpose of Automatic Stimulation is to deliver stimulation associated with specified heart rate increases that may be associated with seizures.

1.2 Intended Use / Indications

1.2.1 Depression

The VNS Therapy System is indicated for the adjunctive long-term treatment of chronic or *recurrent depression* for patients 18 years of age or older who are experiencing a major depressive episode and have not had an adequate response to four or more adequate *antidepressant treatments*.

1.2.2 Epilepsy

The VNS Therapy System is indicated for use as an adjunctive therapy in reducing the frequency of seizures in patients 4 years of age and older with partial onset seizures that are refractory to antiepileptic medications.

1.3 Contraindications

- **Vagotomy**—The VNS Therapy System cannot be used in patients after a bilateral or left cervical vagotomy.
- **Diathermy**—Do not use shortwave diathermy, microwave diathermy, or therapeutic ultrasound diathermy (hereafter referred to as diathermy) on patients implanted with a VNS Therapy System. Diagnostic ultrasound is not included in this contraindication.

Energy delivered by diathermy may be concentrated into or reflected by implanted products such as the VNS Therapy System. This concentration or reflection of energy may cause heating.

Testing indicates that diathermy can cause heating of the VNS Therapy System well above temperatures required for tissue destruction. The heating of the VNS Therapy System resulting from diathermy can cause temporary or permanent nerve, tissue, or vascular damage. This damage may result in pain or discomfort, loss of vocal cord function, or even possibly death if there is damage to blood vessels.

Because diathermy can concentrate or reflect its energy off any size implanted object, the hazard of heating is possible when any portion of the VNS Therapy System remains implanted, including just a small portion of the lead or electrode. Injury or damage can occur during diathermy treatment whether the VNS Therapy System is turned “ON” or “OFF.”

Diathermy is further prohibited because it may also damage the VNS Therapy System components resulting in loss of therapy, requiring additional surgery for system explantation and replacement. All risks associated with surgery or loss of therapy (loss of seizure control) would then be applicable.

Advise your patients to inform all their healthcare professionals that they should not be exposed to diathermy treatment.

1.4 Warnings

Physicians should inform patients about all potential risks and adverse events discussed in the VNS Therapy System physician's manuals.

1.4.1 All Devices

- **Use (depression)**—This device is a permanent implant. It is only to be used in patients with severe depression who are unresponsive to standard psychiatric management. It should only be prescribed and monitored by physicians who have specific training and expertise in the management of treatment-resistant depression and the use of this device. It should only be implanted by physicians who are trained in surgery of the carotid sheath and have received specific training in the implantation of this device.
- **Not curative (depression)** — Physicians should warn patients that VNS Therapy has not been determined to be a cure for depression. Patients should be counseled to understand that individual results will likely vary. Beneficial results might not become evident for months. Most patients will continue to require antidepressant medications and/or electroconvulsive therapy (ECT) in addition to VNS Therapy.
- **Worsening depression/suicidality (depression)** — Patients being treated with adjunctive VNS Therapy should be observed closely for clinical worsening and suicidality, especially at the time of VNS Therapy stimulation parameter changes or drug or drug dose changes, including either increases or decreases in the stimulation parameters or concomitant treatments. Consideration should be given to changing the therapeutic regimen of VNS Therapy or concomitant treatments, including possibly discontinuing VNS Therapy or the concomitant therapy, in patients whose depression is persistently worse or whose emergent suicidality is severe, abrupt in onset, or was not part of the patient's presenting symptoms.
- **Use (epilepsy)** — The VNS Therapy System should only be prescribed and monitored by physicians who have specific training and expertise in the management of seizures and the use of this device. It should only be implanted by physicians who are trained in surgery of the carotid sheath and have received specific training in the implantation of this device.
- **Not curative (epilepsy)** — Physicians should warn patients that VNS Therapy is not a cure for epilepsy and that since seizures may occur unexpectedly, patients should consult with a physician before engaging in unsupervised activities, such as driving, swimming, and bathing, and in strenuous sports that could harm them or others.
- **Unapproved uses** — The safety and efficacy of the VNS Therapy System have not been established for uses outside the "Intended Use / Indications" section, including (but not limited to) patients with:
 - ◆ Acute suicidal thinking or behavior (depression)
 - ◆ Cardiac arrhythmias or other abnormalities
 - ◆ History of schizophrenia, schizoaffective disorder or delusional disorders (depression)
 - ◆ History of rapid cycling bipolar disorder (depression)
 - ◆ History of previous therapeutic brain surgery or CNS injury

- ♦ History of dysautonomias
 - ♦ History of respiratory diseases or disorders, including dyspnea and asthma
 - ♦ History of ulcers (gastric, duodenal, or other)
 - ♦ History of vasovagal syncope
 - ♦ Only one vagus nerve
 - ♦ Other concurrent forms of brain stimulation
 - ♦ Pre-existing hoarseness
 - ♦ Primary generalized seizures
 - ♦ Progressive neurological diseases other than epilepsy or depression
 - ♦ Under 4 years of age (epilepsy)
 - ♦ Under 18 years of age (depression)
- **Dysfunctional cardiac conduction systems** — The safety and effectiveness of the VNS Therapy System in patients with predisposed dysfunction of cardiac conduction systems (re-entry pathway) have not been established. Evaluation by a cardiologist is recommended if the family history, patient history, or electrocardiogram suggests an abnormal cardiac conduction pathway. Serum electrolytes, magnesium, and calcium should be documented before implantation. Additionally, postoperative bradycardia can occur among patients with certain underlying cardiac arrhythmias. Post-implant electrocardiograms and Holter monitoring are recommended if clinically indicated.
 - **Bradycardia or asystole during implantation** — It is important to follow recommended implantation procedures and intraoperative product testing described in the *Implantation Procedure* chapter. During the intraoperative System Diagnostics (Lead Test), infrequent incidents of bradycardia and/or asystole have occurred. If asystole, severe bradycardia (heart rate < 40 bpm), or a clinically significant change in heart rate is encountered during a System Diagnostics (Lead Test) or during initiation of stimulation, physicians should be prepared to follow guidelines consistent with Advanced Cardiac Life Support (ACLS).

Additionally, postoperative bradycardia can occur among patients with certain underlying cardiac arrhythmias. If a patient has experienced asystole, severe bradycardia (heart rate < 40 bpm), or a clinically significant change in heart rate during a System Diagnostics (Lead Test) at the time of initial device implantation, the patient should be placed on a cardiac monitor during initiation of stimulation.

The safety of this therapy has not been systematically established for patients experiencing bradycardia or asystole during VNS Therapy System implantation.

- **External defibrillation or cardioversion (electrical)** — These procedures may damage the generator, and can temporarily or permanently damage the nerve. Attempt to minimize current flowing through the generator and lead system by following these precautions:
 - ♦ Position defibrillation patches or paddles perpendicular to the generator and lead system, and as far from the generator as possible.

- ♦ Use the lowest clinically appropriate energy output (watt-seconds).
 - ♦ Confirm generator function after any internal or external defibrillation, or cardioversion treatment.
 -  **Magnetic resonance imaging (MRI)** — Patients with the VNS Therapy System, or any part of the VNS Therapy System, implanted should have MRI procedures performed **only as described in the *MRI with the VNS Therapy System instructions for use***. In some cases, surgery will be required to remove the VNS Therapy System if a scan using a transmit RF body coil is needed.
 - **Excessive stimulation**—Excessive stimulation is the combination of an excess duty cycle (i.e. one that occurs when ON time is greater than OFF time) and high frequency stimulation (i.e. stimulation at ≥ 50 Hz). Excessive stimulation has resulted in degenerative nerve damage in laboratory animals. Furthermore, excess duty cycle can be produced by continuous or frequent magnet activation (> 8 hours). While LivaNova limits the maximum programmable frequency to 30 Hz, it is recommended that you do not stimulate with excess duty cycle. Further, physicians should warn patients about continuous or frequent magnet use as this could lead to early battery depletion.
-  **Note:** Use of the magnet to activate stimulation is not recommended for patients with depression. The Magnet Mode output current should remain at 0 mA for patients with depression.
- **Device manipulation** — Patients who manipulate the generator and lead through the skin (Twiddler's Syndrome) may damage or disconnect the lead from the generator and/or possibly cause damage to the vagus nerve. For patients with the Model 1000, recalibration of Prone Position detection may be required. Patients should be warned against manipulating the generator and lead.
 - **Swallowing difficulties** — Difficulty swallowing (dysphagia) may occur with active stimulation, and aspiration may result from the increased swallowing difficulties. Patients with pre-existing swallowing difficulties and those with a history of drooling or hypersalivation are at greater risk for aspiration. Appropriate aspiration precautions should be taken for such patients. Use of the magnet to temporarily stop stimulation while eating may mitigate the risk of aspiration.
 - **Dyspnea or shortness of breath** — Dyspnea (shortness of breath) may occur with active VNS Therapy. Any patient with underlying pulmonary disease or insufficiency, such as chronic obstructive pulmonary disease or asthma, may be at increased risk for dyspnea and should have their respiratory status evaluated prior to implantation and monitored following initiation of stimulation.
 - **Obstructive sleep apnea** — Patients with obstructive sleep apnea (OSA) may have an increase in apneic events during stimulation. Lowering stimulus frequency or prolonging "OFF" time may prevent exacerbation of OSA. Vagus nerve stimulation may also cause new onset sleep apnea in patients who have not previously been diagnosed with this disorder. It is recommended that patients being considered for VNS Therapy who demonstrate signs or symptoms of OSA, or who are at increased risk for developing OSA, should undergo the appropriate evaluation(s) prior to implantation.
 - **Device malfunction** — Device malfunction could cause painful stimulation or direct current stimulation. Either event could cause nerve damage and other associated

problems. Patients, parents, and caregivers should be instructed to use the magnet to stop stimulation if they suspect a malfunction, and then to contact their physician immediately for further evaluation. Prompt surgical intervention may be required if a malfunction occurs.

- **Sudden unexpected death in epilepsy (SUDEP)** — Through August 1996, 10 sudden and unexpected deaths (definite, probable, and possible) were recorded among the 1,000 patients implanted and treated with the VNS Therapy device. During this period, these patients had accumulated 2,017 patient-years of exposure.

Some of these deaths could represent seizure-related deaths in which the seizure was not observed, at night, for example. This number represents an incidence of 5.0 definite, probable, and possible SUDEP deaths per 1,000 patient-years.

Although this rate exceeds that expected in a healthy (nonepileptic) population matched for age and sex, it is within the range of estimates for epilepsy patients not receiving vagus nerve stimulation, ranging from 1.3 SUDEP deaths for the general population of patients with epilepsy, to 3.5 (for definite and probable) for a recently studied antiepileptic drug (AED) clinical trial population similar to the VNS Therapy System clinical cohort, to 9.3 for patients with medically intractable epilepsy who were epilepsy surgery candidates.

1.4.2 Generators with the AutoStim Feature

- **Cardiac arrhythmia** — The AutoStim Mode feature should not be used in patients with clinically meaningful arrhythmias currently being managed by devices or treatments that interfere with normal intrinsic heart rate responses (e.g., pacemaker dependency, implantable defibrillator, beta adrenergic blocker medications). Patients also should not have a history of chronotropic incompetence (commonly seen in patients with sustained bradycardia [heart rate < 50 bpm]).



Note: See also “Effects on other medical devices” in the “Precautions” section of this chapter.

- **Pre-surgical Surface Assessment** — For anticipated use of the AutoStim feature, it is important to follow the recommended pre-surgical surface assessment described in the implantation procedure. For the device to detect heart rate, patients must have a peak-to-peak R-wave amplitude ≥ 0.4 mV on ECG measured from the proposed electrode location in the neck to the proposed generator location in the chest via surface ECG electrodes in the body positions described in the “Determine acceptable device implant locations” section of the *Implantation Procedure* chapter of this physician's manual.

1.4.3 Model 106 Only

- **Potential interruption of therapy (Model 106 Serial Numbers < 80000 only)** — Magnet Mode output current should always be set at least 0.125 mA higher than AutoStim Mode output current. When Magnet Mode output current is less than or equal to Autostim Mode output current, repeated magnet applications may trigger a device safety feature that disables stimulation. While stimulation is disabled the generator will not provide therapy and must be programmed by the physician to

resume treatment. If stimulation output becomes disabled (0 mA), stimulation can be reinstated at the next office visit by programming stimulation output current on.

1.5 Precautions

Physicians should inform patients about all potential risks and adverse events discussed in the VNS Therapy System physician's manuals.

1.5.1 All Devices

- Unless otherwise specified, all indications, contraindications, and possible complications and adverse events are applicable to all implantable parts of the VNS Therapy System. Possible adverse events specifically related to the lead include migration, dislodgement, breakage, and corrosion.
- **Physician training** — Appropriate physician training is very important.
 - ◆ **Prescribing physicians** should be experienced in the diagnosis and treatment of depression or epilepsy and should be familiar with the programming and use of the VNS Therapy System.
 - ◆ **Physicians who implant the VNS Therapy System** should be experienced performing surgery in the carotid sheath and should be trained in the surgical technique relating to implantation of the VNS Therapy System.



Note: See "Physician Training/Information" in the *Implantation Procedure* chapter.

- **Use during pregnancy** — The safety and effectiveness of the VNS Therapy System have not been established for use during pregnancy. There are no adequate and well-controlled studies of VNS Therapy in pregnant women. Reproduction studies have been performed using female rabbits stimulated with the commercially available VNS Therapy System at stimulation dose settings similar to those used for humans. These animal studies have revealed no evidence of impaired fertility or harm to the fetus due to VNS Therapy. Because animal reproduction studies are not always predictive of human response and animal studies cannot address developmental abnormalities, VNS Therapy should be used during pregnancy only if clearly needed. Although the operating ranges of the VNS Therapy System and fetal monitors are dissimilar and no interaction would be expected, testing has not been performed. Therefore, the potential may exist for interaction between the VNS Therapy System and fetal monitoring systems.
- **Vagus nerve placement** — The VNS Therapy System is indicated for use only in stimulating the left vagus nerve in the neck area inside the carotid sheath. The VNS Therapy System is indicated for use only in stimulating the **left vagus nerve below where the superior and inferior cervical cardiac branches separate from the vagus nerve**. The safety and efficacy of the VNS Therapy System have not been established for stimulation of the right vagus nerve or of any other nerve, muscle, or tissue.
- **Effects on other medical devices** — The VNS Therapy System may affect the operation of other implanted devices, such as cardiac pacemakers and implanted defibrillators. Possible effects include sensing problems and inappropriate device responses. If the patient requires concurrent implantable pacemaker, defibrillator therapy, or other types of stimulators, careful programming of each system may be necessary to

optimize the patient's benefit from each device. Furthermore, when the VNS Therapy System and another stimulator are implanted in the same patient, the two stimulators should be placed at least 10 centimeters (4 inches) apart to avoid communication interference. Users should refer to the product labeling for the concurrent device to determine if there are additional precautions that should be observed.

- **Infection control** — It is important to follow infection control procedures. Infections related to any implanted device are difficult to treat and may require that the device be explanted. The patient should be given antibiotics preoperatively. The surgeon should ensure that all instruments are sterile prior to the operation. Frequent irrigation of both incision sites with generous amounts of bacitracin or equivalent solution should be performed prior to closure. To minimize scarring, these incisions should be closed with cosmetic closure techniques. Also, antibiotics should be administered postoperatively at the discretion of the physician. Children (4-11 years of age) may have a greater risk for infection when compared to adolescent and adult patients (≥ 12 years). Careful monitoring for site infection as well as the avoidance of manipulation of the surgical site post implant in children should be stressed.
- **Reversal of lead polarity** — Reversal of lead polarity has been associated with an increased chance of bradycardia in animal studies. It is important that the electrodes are attached to the left vagus nerve in the correct orientation. It is also important to make sure that leads with dual connector pins are correctly inserted (white marker band/serial number to + connection) into the generator's lead receptacle(s).
- **Lead stabilization** — The patient can use a neck brace for the first week to help ensure proper lead stabilization.
- **Programming after surgery** — Do not program the VNS Therapy System to an ON or periodic stimulation treatment for at least 14 days after the initial or replacement implantation. Failure to observe this precaution may result in patient discomfort or adverse events.
- **Vagus nerve damage — Some complications** may be associated with damage to the vagus nerve.
 - ◆ Hoarseness may be caused by device malfunction, nerve constriction, or nerve fatigue. Nerve constriction should be apparent within a few days after implantation and may require explantation of the lead. Nerve fatigue usually occurs after intense stimulation parameters have been used, and might not be associated with any other adverse event. If fatigue is suspected, the generator should be turned off for several days until hoarseness subsides.
 - ◆ Persistent hoarseness *not* associated with stimulation suggests possible nerve irritation and should be immediately investigated.
 - ◆ Trauma to the vagus nerve at the implantation site could result in permanent vocal cord dysfunction.
- **Device reset** — A reset of the device will program the device OFF (output current = 0 mA).
- **Laryngeal irritation** — Laryngeal irritation may result from stimulation. Patients who smoke may have an increased risk of laryngeal irritation.

1.5.2 Model 100 - 102R Only

- **Battery depletion** — For Model 100, 101, 102, and 102R generators, do not use frequencies of 5 Hz or below for long-term stimulation, because these frequencies generate an electromagnetic trigger signal, which results in excessive battery depletion of the implanted generator. Therefore, use these low frequencies for short periods of time only.
- **Device history loss** — For Model 100, 101, 102, and 102R generators, a reset of the device causes all device history information to be lost. The device history information (e.g., programmed patient initials, implant date, device serial number) should be documented before resetting.

1.5.3 Model 103 and Higher

- **Device reset** — When a Model 103 or subsequent model generator is reset, its stimulation output is disabled (0 mA); however, all settings and device history are preserved. After a successful reset, the generator stimulation output may be re-enabled to resume operation at the previously programmed settings.

1.5.4 Generators with AutoStim Only

- **Unintended Stimulation** — For devices that sense changes in heart rate, false positive detection may cause unintended stimulation. Examples of instances where heart rate may increase include exercise, physical activity, and normal autonomic changes in heart rate, both awake and asleep, etc. Depending on the amount of unintended stimulation, adjustments to the AutoStim feature's detection threshold should be considered. If necessary, the feature can be disabled.
- **Device Placement** — For the AutoStim feature, the physical location of the device critically affects its ability to properly sense heart beats. Therefore, care must be taken to follow the implant location selection process outlined in the *Implantation Procedure*. Note that this implant location selection procedure may be performed preoperatively as part of the patient's surgical work-up.
- **Battery Drain** — Talk to your patient about the AutoStim feature. Use of this feature will result in reduced battery longevity leading to more frequent replacements. Since the AutoStim feature can significantly affect the generator battery life, patients should return to their physician at appropriate intervals to evaluate whether they are receiving benefit from the current AutoStim settings.

 **Note:** See Table titled “Estimated Model 106 and 1000 Longevity with Sensing and AutoStim” in the *Model 106 and 1000 Technical Information* chapters.

1.5.5 Model 1000 Only

 **Note: Use of Optional Features**—All detection-related features (i.e., AutoStim, low heart rate detection, prone position detection) and Scheduled Programming are intended for use in patients with epilepsy only. Day-Night Programming may be used in patients with epilepsy or depression.

- **Low Heart Rate and Prone Position Detection**—These features are for informational purposes only. Do not use detected events for alarms or medical diagnosis.

- **Scheduled Programming** — Since this feature allows the generator to apply therapy increases at scheduled intervals, it may not be appropriate for use in patients who are nonverbal or are unable to use the patient magnet to stop undesired stimulation. Similarly, exercise caution for use of this feature in patients with a history of obstructive sleep apnea, shortness of breath, coughing, swallowing difficulties, or aspiration.
- **Day-Night Programming** — When using the optional Day-Night programming feature:
 - ◆ Consider risk and benefits of altering a patient's known efficacious settings before this feature is used or when parameter adjustments are made.
 - ◆ Inform your patients about when to expect a setting change (i.e., when Daytime settings transition into Nighttime settings).
 - ◆ Assess patient tolerability of the alternate parameter set before the patient leaves the office visit.
- **Time-based Features** — Day-Night Programming and Scheduled Programming do not automatically adjust for Day Light Savings or time zone changes. Tell the patient to follow-up with the physician for reprogramming, if needed.

1.5.6 Lead Evaluation and Connection

- **Do not use a lead other than** a VNS Therapy lead — Use a VNS Therapy dual-pin lead with the dual-receptacle generator or a VNS Therapy single-pin lead with the single-receptacle generator because use of other leads may damage the generator or injure the patient.
- **Lead size** — The lead is available in multiple sizes. Since it is not possible to predict in patients what size lead will be needed, it is recommended that at least one alternate lead size be available in the operating room. In addition, backups for leads should be available in the event of compromised sterility or damage induced during surgery.

 **Note:** For lead size availability, see “Product Specifications” in the lead-specific Technical Information chapters.

- **Potential effects of lead breaks** — Lead fractures of the VNS Therapy System may prevent patients from receiving therapy. It may also prevent or interfere with tachycardia detection, if the feature is enabled. Children (4-11 years of age) may be more likely to exhibit certain risk factors for lead failures including a higher activity level and a higher likelihood to manipulate the lead. If a lead fracture is suspected, perform diagnostic testing to evaluate continuity within the system. If diagnostics suggest that a fracture is present, consider turning the VNS generator to zero milliamperes (0 mA) of output current. Continuing stimulation with a fractured lead may result in dissolution of the conductor material resulting in adverse events, such as pain, inflammation, and vocal cord dysfunction. The benefits and risks of leaving the generator ON (actively stimulating) when a lead fracture is present should be evaluated and monitored by the medical professional treating the patient.

 **Note:** For more information on diagnostic testing, see “Troubleshooting” in this manual and in the programming system physician’s manuals.

- **Line powered equipment** — Exercise extreme caution if testing the lead using **line-powered equipment** because leakage current can injure the patient.
- **Setscrew** — Do not insert a lead in the generator lead receptacle(s) without first visually **verifying that the setscrew(s) is sufficiently retracted** to allow insertion. Avoid backing the setscrew(s) out further than needed for lead insertion.
- **Hex screwdriver** — Ensure that the hex screwdriver is fully inserted in the setscrew, and then push in on the hex screwdriver and turn it clockwise until it clicks. To avoid damaging (stripping) the setscrew(s) and/or dislodging the setscrew plug(s), insert the hex screwdriver into the center of the setscrew plug, keeping it perpendicular to the generator.

1.5.7 Environmental and Medical Therapy Hazards

Patients should exercise reasonable caution in avoiding devices that generate a strong electric or magnetic field. If a generator ceases operation while in the presence of electromagnetic interference (EMI), moving away from the source may allow it to return to its normal mode of operation.

1.5.7.1 Hospital and medical environments

- **VNS Therapy System operation should always be checked** — Perform device diagnostics after any of the procedures mentioned in this manual. Additional precautions for these procedures are described below.
- **Mammography** — For clear imaging, patients may need to be specially positioned for mammography procedures because of the location of the generator in the chest. (Most routine diagnostic procedures, such as fluoroscopy and radiography, are not expected to affect system operation.)
- **Therapeutic radiation** — This procedure may damage the generator's circuitry. Sources of such radiation include therapeutic radiation, cobalt machines, and linear accelerators. The radiation effect is cumulative, with the total dosage determining the extent of damage. The effects of exposure to such radiation can range from a temporary disturbance to permanent damage, and may not be detectable immediately.
- **Electrosurgery** — Use of electrosurgery (electrocautery or radio frequency [RF] ablation devices) may damage the generator. During the VNS implantation procedure, do not use electrosurgical equipment after the generator has been introduced to the sterile field. When performing other surgical procedures on a patient implanted with a VNS generator, attempt to minimize the current flowing through the generator and lead system by following these precautions:
 - ◆ Position the electrosurgery electrodes as far as possible from the generator and lead.
 - ◆ Avoid electrode placement that puts the generator or lead in the direct path of current flow or within the part of the body being treated.
 - ◆ Confirm that the generator functions as programmed after electrosurgery.

- **Electrostatic Discharge (ESD)** — ESD may damage the generator. Care should be taken when using the hex screwdriver to avoid touching the metal shaft when the screwdriver is engaged with the setscrew of the generator. This shaft can serve as a path to conduct electrostatic discharges into the device circuitry.
- **Extracorporeal shockwave lithotripsy** — This procedure may damage the generator. If therapeutic ultrasound is required, avoid positioning the area of the body where the generator is implanted in the water bath or in any other position that would expose it to ultrasound therapy. If that positioning cannot be avoided, program the generator output to 0 mA for the treatment, and then after therapy, reprogram the generator to the original parameters.
- **Treatment involving electrical currents** — If the patient receives medical treatment for which electric current is passed through the body (such as from a TENS unit), either the generator output should be set to 0 mA or function of the generator should be monitored during initial stages of treatment.
- **Therapeutic ultrasound** — Routine therapeutic ultrasound could damage the generator and may be inadvertently concentrated by the device, causing harm to the patient.
-  **Magnetic resonance imaging (MRI)** — An MRI should not be performed using a transmit RF body coil for certain VNS Therapy device configurations or under certain specific conditions. In some cases, heating of the lead caused by the transmit RF body coil during MRI may result in serious injury. Static, gradient, and radio frequency (RF) electromagnetic fields associated with MRI may change the generator settings (i.e., reset parameters) or activate the VNS device if the Magnet Mode output remains “ON”.



Note: See the *MRI with the VNS Therapy System* instructions for use for details.

-  **Receive RF coils** — Note that certain magnetic resonance (MR) system head coils operate in receive-only mode and require use of the transmit RF body coil. Other MR systems use a transmit/receive RF head coil. Local or surface coils may also be receive-only RF coils that require the transmit RF body coil for MRI. **The use of a receive RF coil does not alter hazards of the transmit RF body coil.**



Note: See the *MRI with the VNS Therapy System* instructions for use for details.

-  **Transmit RF coils** — Exposure of the VNS Therapy System to any transmit RF coil must be avoided. Do not perform MRI scans using any transmit RF coil in the defined exclusion zones.
- **Routine diagnostic procedures** — Most routine diagnostic procedures, such as fluoroscopy and radiography, are not expected to affect system operation.

1.5.7.2 *Home occupational environments*

- **Not expected to affect the generator** — Properly operating microwave ovens, electrical ignition systems, power transmission lines, theft-prevention devices, and metal detectors are not expected to affect the generator. However, because of their higher energy levels, sources such as transmitting antennas may interfere with the VNS Therapy System. It is suggested that the generator be moved away from equipment—typically at least 1.8 meters (6 feet)—that may be causing interference.



Caution: The patient should seek medical advice before entering environments that are protected by a warning notice preventing entry by patients implanted with a cardiac pacemaker or defibrillator.

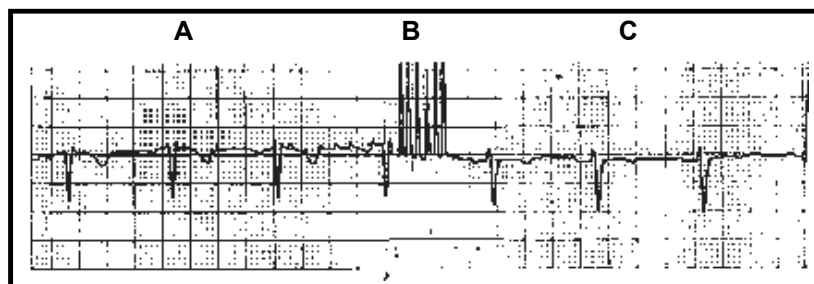
- **Cellular phones** — Based on testing to date, cellular phones have no effect on generator operation.
- **Electronic Article Surveillance (EAS) System tag deactivators** — EAS System tag deactivators can interfere with VNS Therapy when it's operated in proximity of the generator. Potential effects include inhibited pulses and accidental activations (Magnet or AutoStim). Patients should be cautioned to keep at least 60 centimeters (2 feet) away from EAS System tag deactivators to avoid potential interference.
- **Other Electro-mechanical devices** — Strong magnets, tablet computers and their covers, hair clippers, vibrators, loudspeaker magnets, and other similar electrical or electro-mechanical devices, which have a strong static or pulsing magnetic field, can cause accidental magnet activation. Patients should be cautioned to keep such devices at least 20 centimeters (8 inches) away from the generator.

1.5.7.3 *Generator and EMI effects on other devices*

- **Interference during stimulation** — During stimulation, the generator may interfere with devices operating in the 30 kHz to 100 kHz range (e.g., pocket transistor radios and hearing aids). This interference is a theoretical possibility, and no effects on hearing aids have been reported, although the generator can interfere with a transistor radio. No specific testing has been done to date, and no definite information on effects is available. The generator should be moved—typically at least 1.8 meters (6 feet)—away from equipment with which it may be interfering.
- **Interference during programming or interrogation** — Programming or interrogating the generator may momentarily interfere with other sensitive electronic equipment nearby. The generator is not expected to trigger airport metal detectors or theft-protection devices that are further than about 1.8 meters (6 feet).
- **Operation of other implanted devices** — The generator and the VNS Therapy magnet may affect the operation of **other implanted devices**, such as cardiac pacemakers and implantable defibrillators. Possible effects include sensing problems and inappropriate generator responses. If the generator patient requires concurrent implantable pacemaker and/or defibrillator therapy, careful programming of each system is necessary to optimize the patient's benefit from each device.

- **Items affected by strong magnetic fields** — The magnet provided for activation or inhibition of the generator may damage **televisions, computer disks, credit cards, and other items** affected by strong magnetic fields.
- **Effects on ECG monitors** — Generator data communication produces an ECG artifact, an example of which is shown in the ECG tracings in Figure 1:

Figure 1. ECG Artifact Produced by Generator Communication



A Generator Off

B Programming

C Generator On

1.5.8 Sterilization, Storage, and Handling

1.5.8.1 Sterilization

The generator, lead, accessory pack, and tunneler have been sterilized using either hydrogen peroxide (H₂O₂ or HP) gas plasma or ethylene oxide (EO) gas, and are supplied in a sterile pack to permit direct introduction into the operating field. An expiration (or use-before) date and method of sterilization is marked on each package. See Table 1 for sterilization methods.

Table 1. Sterilization Methods

Model	EO	H ₂ O ₂ (HP)
102	x	x
102R	x	x
103	x	x
104	x	x
105	x	x
106	–	x
1000	–	x
302	x	x
303	x	x
304	x	x
402	x	x
502	x	x

- **Sterilization process indicator** — A sterilization process indicator is located on the inner sterile pack. Products labeled as sterile should be used only if the color of the indicator is as follows:

- ♦ H₂O₂ or HP sterilization - gold to bronze range
- ♦ EO sterilization - gray to green range
- **Do not re-sterilize any VNS Therapy System product** — Return any opened devices to LivaNova.

1.5.8.2 Storage

- **Store the VNS Therapy System** between - 20 °C (- 4 °F) and + 55 °C (+ 131 °F). Temperatures outside this range can damage components.
- **Do not store the VNS Therapy System** where it is exposed to water or other liquids. Moisture can damage the seal integrity of the package materials.
- **Nonpyrogenic** — The implantable portions of the VNS Therapy System are nonpyrogenic.

1.5.8.3 Handling

- **Do not implant a device** if any of the following has occurred:
 - ♦ The device has been dropped, because dropping it could damage internal components.
 - ♦ The color of the sterilization process indicator within the inner package is not in the proper range.
 - ♦ The outer or inner storage package has been pierced or altered, because this could have rendered it non-sterile
 - ♦ The expiration (use-before) date has expired, because this can adversely affect the device's longevity and sterility.
- **Do not ultrasonically clean the generator** — Ultrasonically cleaning the generator may damage generator components.
- **Do not reimplant an explanted generator** — The generator and lead are single-use-only devices. **Do not reimplant an explanted generator or lead for any reason**, because sterility, functionality, and reliability cannot be ensured, and infections may occur.
- **Return explanted generators and leads** — Explanted generators and leads should be returned to LivaNova for examination and proper disposal, along with a completed Returned Product Report form. Before returning the generator or lead, disinfect the device components with Betadine[®], Cidex[®] soak, or other similar disinfectant, and double-seal them in a pouch or other container properly labeled with a biohazard warning.
- **Do not incinerate the generator** — The generator contains a sealed chemical battery, and an explosion could result if subjected to incineration or cremation temperatures.

1.6 Education, Training, and Services

LivaNova employs highly trained representatives and engineers located throughout the world to serve you and provide training to prescribers and implanters of LivaNova products. Physicians must contact LivaNova before prescribing or implanting a VNS Therapy System for the first time. In addition to the information provided in this physician's manual, training material includes but is not limited to, surgeon or prescribing physician training slide presentation, surgical video, training block & demo lead, etc. The required training (elements, duration, and frequency) to use LivaNova products may vary depending on the product and physician and can be discussed and arranged with your local LivaNova representative, or you can call or write LivaNova at the appropriate telephone number or address listed in the *Information and Support* chapter of this physician's manual to obtain more information.

2 Technical Information — 102/102R Generators

2.1 Detailed Device Description

2.1.1 Physical Characteristics

The titanium case of the VNS Therapy Pulse™ Model 102 and Pulse Duo™ Model 102R Generators are hermetically sealed and leak rate tested. Specially designed feedthroughs utilizing platinum conductors make the electrical connection from the connector blocks to the circuitry through the hermetically sealed enclosure.

2.1.2 Biological Compatibility

Materials exposed to the subcutaneous environment are biologically compatible. All of these materials have a long history in medical implants and have been found to be tissue compatible in the vast majority of cases.

2.1.3 Power Source

The power source for the Model 102 and Model 102R generators is a Wilson Greatbatch Ltd., Model 2075, lithium carbon monofluoride battery with an open-circuit voltage of 3.3. The battery's maximum available capacity is approximately 1.7 amp-hours. The self-discharge reduces the capacity by less than 1 percent per year. This battery has a gradual drop in voltage near its end of life (EOL).

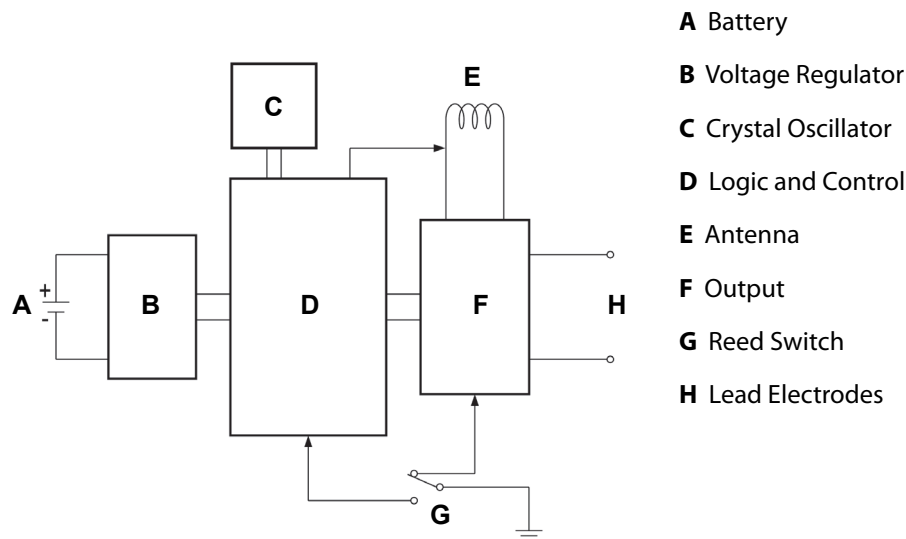
2.1.4 Circuitry

The generator utilizes several complementary metal oxide semiconductor (CMOS) integrated circuits, including a microprocessor. The circuitry is schematically represented in Figure 2.

For descriptive purposes, circuitry of the generator can be divided into the following major functional sections:

Voltage regulator	Regulates the system power supply
Crystal oscillator	Provides a timing reference
Logic and control	Controls overall generator function; receives and implements programming commands; collects and stores telemetry information
Output	Develops and modulates signals delivered to the lead
Antenna	Receives programming signals; transmits telemetry information to the programming wand
Reed switch	Provides a mechanism to place the generator in Magnet Mode or to inhibit its output

Figure 2. Generator Circuitry



2.1.5 Identification

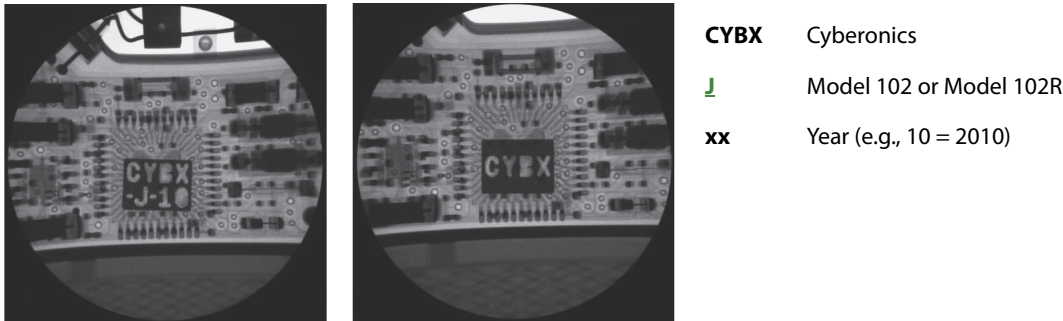
The generator can be identified by x-ray and will appear as shown in Figure 3. The serial number and model number of the generator are marked on its titanium case, but do not appear on the x-ray. The serial number and model number can be identified by interrogating the generator with the software and viewing the Device History screen.

Note: For details on interrogating the generator, see the programming system physician’s manuals.

Note: Serial numbers ≥ 1000000 = Model 102R and serial numbers < 1000000 = Model 102.

The x-rays shown in Figure 3 show possible identification code combinations used for Model 102 and 102R generators. The codes are defined to the right of the images.

Figure 3. X-ray Identification



2.2 VNS Therapy System Compatibility

Models 102 and 102R are compatible with the following system components:

Component	Model	Generator Model	
		102	102R
Lead	dual-pin		x
	single-pin	x	
Programming System	201 wand / 250 software	x	x
	2000 wand / 3000 software	x	x
Tunnelers	402	x	x
Accessory Pack	502	x	x
Magnets	220	x	x

2.3 Directions for Use

2.3.1 Specifications and Product Information

The specifications and product information for the VNS Therapy generators are presented in Table 2.

Table 2. Specifications and Product Information

Stimulation Parameter	Available Parameter Settings
Output current	0-3.5 mA in 0.25-mA steps* $\pm 0.25 \leq 1$ mA, $\pm 10\% > 1$ mA
Signal frequency	1, 2, 5, 10, 15, 20, 25, 30 Hz $\pm 6\%$
Pulse width	130, 250, 500, 750, 1000 μ sec $\pm 10\%$
Signal ON time	7, 14, 21, 30, 60 sec [†] $\pm 15\%$ or + 7 sec, whichever is greater ($\pm 15\%$ or ± 7 sec in Magnet Mode)
Signal OFF time	0.2, 0.3, 0.5, 0.8, 1.1, 1.8, 3 min, and 5 to 180 min (5 to 60 in 5-min steps; 60 to 180 in 30-min steps), + 4.4 / - 8.4 sec
Magnet activation	Provided by magnet application (output current, pulse width, and signal ON time may be independently programmed for this purpose)
Reset parameters	0 mA; 10 Hz; 500 μ sec; ON time, 30 sec; OFF time, 60 min
Nominal parameters	0 mA; 30 Hz; 500 μ sec; ON time, 30 sec; OFF time, 10 min
Telemetry Reports	
Device History Report	Patient code, implant date, model number, and serial number
Device Diagnostic Report	Status messages for programming, telemetry, N EOS, output current, lead impedance, DC-DC converter value, programmed amplitude, and device treatment status
Power Source	
(All Serial Numbers)	
Battery	Wilson Greatbatch Ltd., Model 2075
Chemistry	Lithium carbon monofluoride
Voltage	3.3 V, open circuit
Rated capacity	1.7 amp-hours
Self-discharge rate	< 1% per year

Physical Characteristics		
Materials		
Case	Titanium, hermetically sealed	
Header	Polyurethane—Tecothane™ TT-1075D-M Thermoplastic	
Lead connector blocks	Stainless steel	
Setscrew plug(s)	Silicone [‡]	
Measurements (Typical)		
	Model 102	Model 102R
Lead receptacle(s)	0.126 in (3.2 mm) nominal	0.2 in (5 mm) nominal
Dimensions	2.0 in x 2.0 in x 0.27 in (52 mm x 52 mm x 6.9 mm)	2.0 in x 2.3 in x 0.27 in (52 mm x 58.4 mm x 6.9 mm)
Weight	0.88 oz (25 g)	0.95 oz (27 g)
Connector Retention Strength		
With VNS Therapy lead	> 10N	
Serial Number Range		
Model 102	< 1000000	
Model 102R	≥ 1000000	

* For output currents ≤ 1 mA, the tolerance is ± 0.25mA. Maximum output is 12.5 ± 2.5 V with the exception of 10 Hz, 7 seconds On Time, in which case the maximum output is 4.4 V and 0.25 mA tolerance. This 0.25 mA tolerance also applies to 15 Hz, 7 seconds On Time, 0.5mA output current.

† For Signal ON time > 7 sec, there is no ramp-down at 15 Hz with 0.5 mA and at 10 Hz with 0.5-1.75 or 2.75 mA. For Signal ON time at 30 sec, actual ON time is 40 sec for 10 Hz with 0.25 mA and 38 sec for 15 Hz with 0.25 mA.

‡ No component of the VNS Therapy System is made with natural rubber latex.

2.3.2 Operating Characteristics

2.3.2.1 Communicating with the VNS Therapy System

 **Note:** For proper use of the software, see the programming system physician's manuals.

A programming wand connected to a compatible computer running the programming software is needed to communicate with the generator.

 **Note:** For a list of compatible computers, see the programming system physician's manuals.

 **Note:** For proper placement of the programming wand, connection of the wand to the computer, and use of the wand, see the programming system physician's manuals.

After the program has been initiated, software screens display prompts and messages to aid in communicating with the generator.

The generator “listens” for a communication signal for a 300-msec period every 6.8 seconds. Communication usually takes between 3 and 10 seconds, but may be prolonged in the presence of electromagnetic interference (EMI). The generator listens for and implements interrogations, parameter programming instructions, requests for Device Diagnostics testing,

and Device History inquiries. In response, the generator transmits information on the stimulation parameter settings, changes its parameter settings, responds to requests for Device Diagnostics testing, and provides Device Histories, respectively.

Each time these data are transmitted by the generator, they are saved by the programming software to databases on the storage disk. The generator also transmits a signal for use in evoked potential monitoring.

 **Note:** For details on viewing the database information, see the programming system physician's manuals.

In addition to the programming software and programming wand combination, a magnet can be used for one-way communication to the generator by activating a reed switch in the electronic circuitry. The magnet can be used to initiate stimulation, temporarily inhibit stimulation, perform Magnet Mode diagnostics, and reset the generator.

2.3.2.1.1. Stimulation

After the generator has been programmed, the stimulation will repeat in accordance with the programmed ON and OFF cycle until the generator receives communication from the VNS Therapy programming system or until it is activated (epilepsy only) or inhibited with a magnet. Immediately after successful programming, the generator delivers a programmed stimulation that enables patient response to be evaluated. If programming is performed during stimulation, stimulation will be terminated; after programming, stimulation will begin, using the revised settings.

The generator operation described in the preceding paragraph is done in Normal Mode.

A Magnet Mode stimulation is a single stimulation initiated by applying or passing the magnet over the generator for at least 1 second and then immediately removing it from the area over the generator. Stimulation is delivered after the magnet is removed. The Magnet Mode uses the same frequency as the Normal Mode, but the output current, pulse width, and signal ON time are independently programmable. The Directions for Use are equivalent for epilepsy and depression, with the following exceptions:

- For patients with depression, the Magnet Mode output current should always be programmed at 0 mA, the setting at which the generator is shipped from LivaNova.
- Use of the Magnet Mode is limited to patients with epilepsy. Patients with epilepsy or their caregivers pass the magnet over the implanted generator to activate on-demand delivery of a single train of vagus nerve stimulation and help abort or diminish a seizure.
- Magnet Mode is not used for patients with depression.

An interrogation is made to determine the present settings of the stimulation parameters. If interrogation is made during stimulation, completion of stimulation will be delayed until the interrogation is finished.

A graphic representation of stimulation (Figure 4) depicts the relationship of the stimulation programmable parameters. The programmable parameters are independently variable, offering multiple setting combinations from which the physician may select optimal stimulation for the patient. Figure 4 shows that the output pulse can be varied both by

amplitude (output current) and duration (pulse width). The number of output pulses delivered per second determines the frequency.

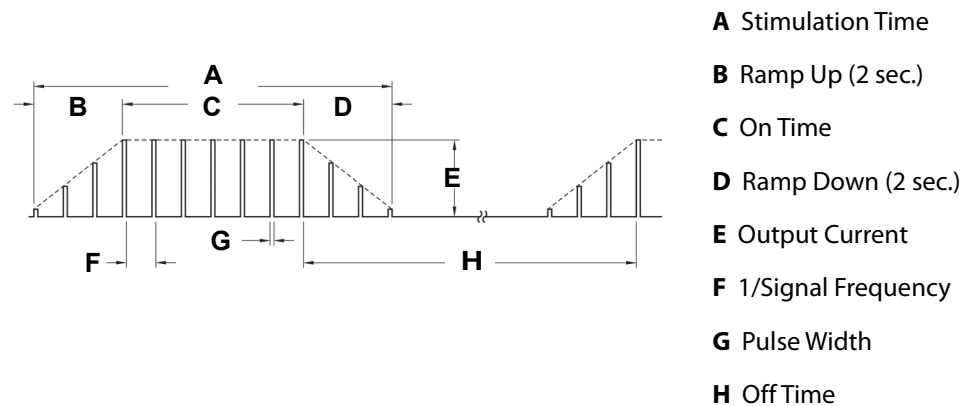
The percentage of time the generator is stimulating is called a “duty cycle.” A duty cycle is calculated by dividing the stimulation time (programmed ON time plus 2 seconds of ramp-up time and 2 seconds of ramp-down time) by the sum of the ON and OFF times.

Note: The various parameter settings for stimulation are listed in “Specifications and Product Information”.

When selecting a combination of parameter settings that will deliver optimal stimulation, the physician should also consider that some combinations will decrease battery life faster than others.

Note: See “Effects of programmed settings on generator projected lifetime”.

Figure 4. Stimulation (Frequencies < 10 Hz do not ramp)



2.3.2.2 Initiating stimulation with the magnet

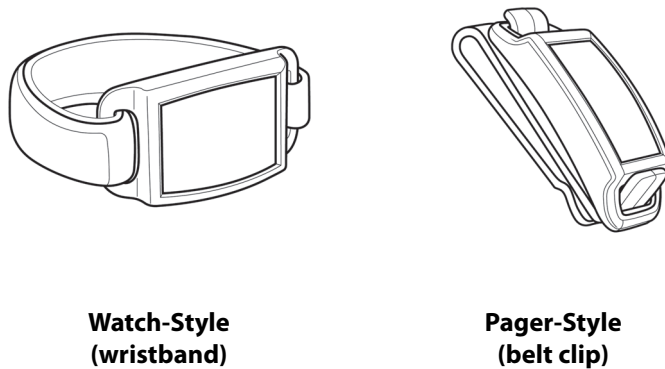
There are four possible uses of the magnet:

- **Epilepsy** —To provide on-demand stimulation as an attempt to abort or de-intensify an oncoming seizure
- To temporarily inhibit stimulation
- To reset the generator (in combination with the programming wand)
- **Epilepsy** — To test daily the functioning of the generator, it is recommended that patients be instructed to use the magnet to activate stimulation. During an aura or at the start of a seizure, magnet activation may be initiated by the patient, a companion, or the physician by applying or passing the magnet over the generator to activate a reed switch in the generator’s electronic circuitry. This action changes the generator from Normal Mode to Magnet Mode.

 **Caution:** Possible nerve damage with ON time > OFF time— Excessive stimulation has resulted in degenerative nerve damage in laboratory animals. One component of excessive stimulation is when ON time exceeds OFF time, which can be produced by continuous or frequent magnet activation (> 8hours). It is recommended that stimulation at these combinations of ranges be avoided.

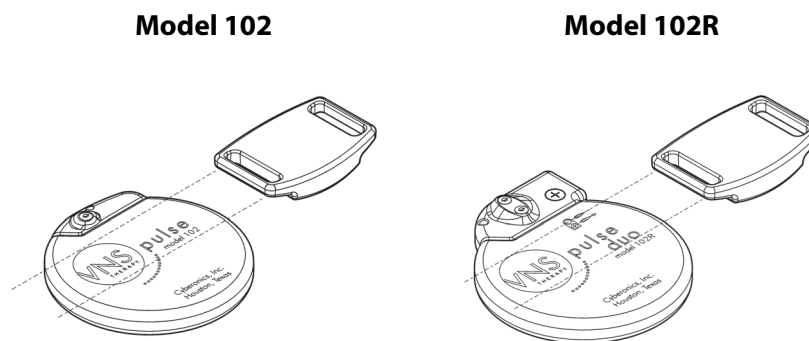
Two identical VNS Therapy magnets (see Figure 5), each providing a minimum of 50 gauss at 1 inch, are supplied by LivaNova. The watch-style accessory attaches the magnet to the wrist with a wrist band and the pager-style accessory holds the magnet in a belt clip with a quick-release mechanism.

Figure 5. Magnet Accessories



Epilepsy —The proper orientation and motion for initiating magnet activation is shown in Figure 6. The magnet is shown without the belt clip or wristband to illustrate the proper orientation of the magnet to the generator.

Figure 6. Initiating Magnet Activation



 **Caution:** Label side of magnet should face the generator while activating or stopping stimulation.

All magnets may lose their effectiveness over time. Avoid dropping the VNS Therapy magnets or storing them near other magnets.

Epilepsy —To initiate stimulation, apply or pass the magnet over the generator for at least 1 second, and then immediately remove it from the area over the generator. Removal of the

magnet causes the generator to operate in Magnet Mode, delivering a single stimulation with the programmed magnet pulse width, magnet current, and magnet signal ON time settings. The frequency is the programmed value for Normal Mode. Any Normal Mode programmed stimulation will always be overridden by a Magnet Mode, even if the Magnet Mode output current is set to 0 mA. If Magnet Mode stimulation is not desired, the Magnet Mode output current may be programmed to 0 mA. Magnet use does not restart the normal OFF time. Therefore, depending on the timing of the magnet use, the patient could receive a second stimulation quickly after magnet activation.

Epilepsy — It is recommended that tests of the magnet output be performed while the patient is still in the physician's office to ensure tolerability of the magnet output.

2.3.2.2.1. Inhibiting generator output with a magnet

Application of the magnet during stimulation will inhibit the output. In addition, holding the magnet in place for at least 65 seconds will prevent the initiation of a Magnet Mode stimulation and will terminate any ongoing Normal Mode stimulation. After the magnet is removed, Normal Mode operation will resume with stimulation when one complete OFF time has elapsed.



Caution: If stimulation becomes painful, the patient should be instructed to stop the stimulation with the magnet.

In the unlikely event of continuous stimulation or other malfunction, the patient must be advised to apply the magnet, secure it in place, and immediately notify his or her physician.

2.3.2.2.2. Resetting the microprocessor using the magnet and programming wand

The VNS Therapy System allows the generator microprocessor to be reset in the event of a malfunction. Resetting will be necessary only in the rare case of microprocessor memory malfunction, which might be caused by conditions described in the "Environmental Hazards" section in the *Introduction to the VNS Therapy System* chapter. Microprocessor reset is indicated when communication with the generator becomes impossible. (For suggestions in solving communication difficulties, see "Troubleshooting" in the programming System physician's manuals.)



Caution: When the generator is reset, all device history information is lost, and the reset parameters (0 mA, 10 Hz; 500 μ sec; ON time, 30 seconds; OFF time, 60 minutes) are internally programmed. Resetting the generator turns the device off (output current = 0 mA). After a successful reset, the generator patient code must be re-entered, and the generator reprogrammed to the desired parameters.

For instructions on resetting the microprocessor, see the programming system physician's manuals. It is recommended, except in cases of a medical emergency, that the physician consults a LivaNova technical representative before resetting is performed.

2.3.2.2.3. Effects of the daily reset of the internal clock

The Model 102 and 102R generators contain an internal clock that rolls over (i.e., restarts) every 24 hours. This daily rollover of the internal clock is a normal device function. Every time the clock restarts, a stimulation cycle beginning with the programmed ON time is delivered. Patients may notice a shorter OFF time between the last stimulation cycle just prior to the clock restart and the first stimulation cycle after the clock restart.

Note: The time that the clock restarts each day corresponds with the time of day the most recent programming event occurred. Holding the magnet over the generator for an extended period of time will put all timekeeping functions on hold and will delay the time that the internal clock rolls over each day.

Some patients may be more sensitive to this shorter OFF time and may exhibit common stimulation related side effects (e.g. coughing, voice changes). These side effects will only occur once a day at the time of the daily clock restart. In the rare reported instances in which side effects occurred with the daily clock restart, it was noted that the most common programmed duty cycle was 30 seconds ON and 3 minutes OFF along with a high output current (> 2 mA).

Note: For a complete list of side effects, see “Adverse Events” in the indication specific information chapters.

As with any normal side effect, adjusting settings for tolerability (i.e., decreasing pulse width, signal frequency, and/or output current) has been shown to be successful in resolving stimulation related side effects associated with the 24-hour rollover event. However, since this 24-hour rollover event is directly related to the programmed ON and OFF times, adjusting the duty cycle may be a better option. Optimizing the patient's benefit from therapy should be considered when making the decision as to which parameter should be adjusted. For example, if the patient is responding well clinically at a particular output current, adjusting a different parameter or duty cycle may be considered. Table 3 shows several ON and OFF time combinations that may be better options when trying to resolve stimulation related side effects associated with the daily clock restart.

Table 3. ON/OFF Time — Options for Optimizing Therapy for Patients Affected by the Internal Clock Cycle

ON Time (sec)	OFF Time (min)
7	0.3
14	0.5
21	0.5
7	0.8
14	1.1
30	1.1
60	1.1
30	1.8
7	3.0
14	3.0
60	5.0
14	10.0

Note: For a comprehensive list of duty cycle settings, see Table 5.

2.3.2.3 *Device history*

The Device History consists of the generator serial number, model number, the patient code (usually three initials), implantation date, and other information pertinent to diagnostic and programming events. Use the software to access and view Device History information.

2.3.2.4 *Device diagnostics*

Information from generator Diagnostics (Pre-Implant Test) aids the physician in determining if the generator is operating properly before it is implanted, if the generator output current is being delivered at the programmed value, if lead impedance is within an acceptable range, and in which mode the generator is operating.



Note: For details on available diagnostic tests, see the programming system physician's manuals.

2.3.2.4.1. *Reasons for high lead impedance readings*

The System Diagnostics (Lead Test) is the most appropriate of the device diagnostic tests to evaluate lead impedance for the VNS Therapy System. The System Diagnostics (Lead Test) is performed at 1 mA, 500 μ sec. Use Table 4 to find the DC-DC Converter Code displayed by the System Diagnostics (Lead Test) screen to determine an estimate of lead impedance in kOhms.

The use of Table 4 with the DC-DC Converter Code from diagnostic screens other than the System Diagnostics (Lead Test) and Generator Diagnostics (Pre-Implant Test) is not appropriate, unless the generator output parameters are the values indicated in the tables. High lead impedance is defined as any DC-DC Converter Code greater than or equal to four with 1 mA of diagnostic current.

Table 4. DC-DC Converter Codes and Lead Impedance (Models 102 and 102R)

DC-DC Converter Code ¹	Estimated Lead Impedance ²
	1 mA, 500 μ sec
0	≤ 1.7 kOhms
1	1.8-2.8 kOhms
2	2.9-4.0 kOhms
3	4.1-5.2 kOhms
4	5.3-6.5 kOhms
5	6.6-7.7 kOhms
6	7.8-8.9 kOhms
7	≥ 9 kOhms

¹ DC-DC Converter Codes are displayed during System Diagnostics (Lead Test).

² Tolerance is ± 10 percent.



Caution: Possible causes of high lead impedance readings are thought to include lead discontinuity, lead disconnection from the generator, fibrosis between the nerve and the electrode, electrode detachment from the nerve, or in rare instances, a defective generator or high battery impedance approaching end of service (EOS).

High lead impedance, in the absence of other device-related complications, is not an indication of a lead or generator malfunction. High lead impedance in combination with the patient's failure to feel even the maximum output stimulus may indicate a lead wire fracture or other type of electrical discontinuity in the lead. Patients experiencing high lead impedance, no sensation of maximum output stimulation, and an increase in seizures/depressive symptoms should be further evaluated for possible lead replacement.

Note: For additional instructions on performing the System Diagnostics (Lead Test), see the programming system physician's manuals.

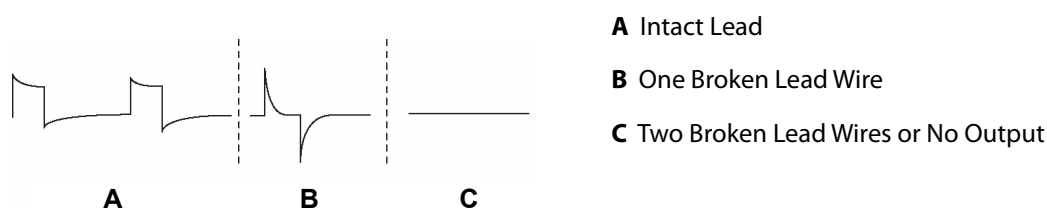
2.3.2.4.2. Short-circuit conditions within the lead

In rare instances, a short-circuit condition within the lead (e.g., direct contact between lead wires) may result in a DC-DC Converter Code of "0" on the System Diagnostics (Lead Test). A significant decrease in DC-DC Converter Code value on the System Diagnostics (e.g., "3" to "1") from prior System Diagnostics may also indicate a lead problem. Patients affected by this event may experience conditions such as an increase in seizures/depressive symptoms, painful stimulation, a feeling of erratic stimulation or no stimulation, or a feeling of stimulation in an atypical anatomical location. In the absence of these adverse events or device-related complications, a DC-DC Converter Code of "0" or decrease in DC-DC Converter code value likely does not indicate a lead malfunction.

2.3.2.4.3. Lead problem troubleshooting

Either evoked potential monitoring equipment or an oscilloscope can be used to analyze the stimulus waveform from the neck for verification of an electrical discontinuity. A differentiated waveform with narrowed pulses or no wave form at all can confirm a discontinuity. Figure 7 shows simulated waveforms expected from skin electrodes for a lead that is intact and for a lead that has a fracture in one or both wires. In addition to these approaches, lead discontinuities can sometimes be identified in x-rays of the implant site.

Figure 7. Typical Waveforms Obtained from Skin Electrodes



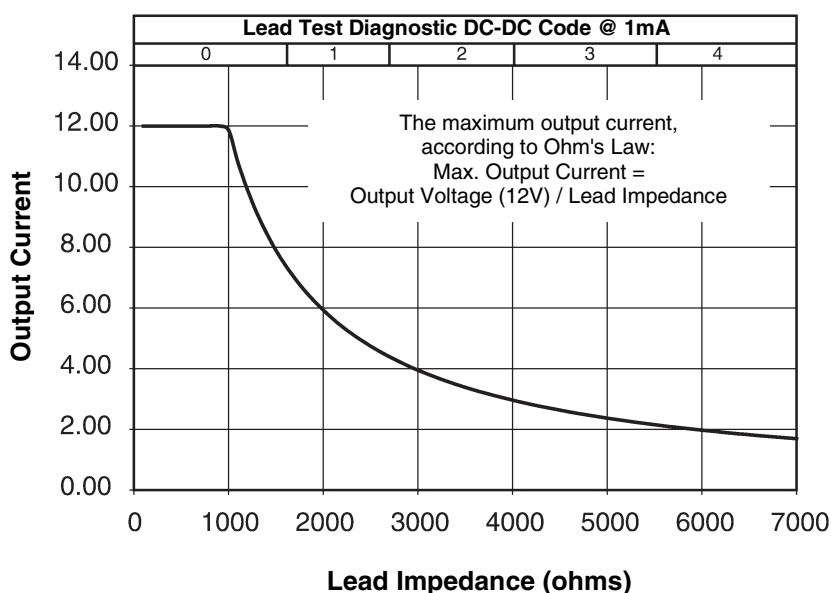
2.3.2.5 Delivery of programmed output current

If the diagnostic tests indicate the output current is LIMIT, the generator may not deliver the programmed output current. Reasons for failure to deliver the programmed output current include a high output current, high lead impedance, and low battery voltage. Figure 8 demonstrates the relationship of lead impedance to maximum deliverable output current.

If the generator fails to deliver the programmed output current, the physician can reprogram to a lower output current and attempt to compensate for a decrease in delivered energy by widening the pulse width. For example, if the output current is at LIMIT for a generator

programmed at 2.5 mA, 30 Hz, 500 μ sec with 30 seconds ON time, then the parameters may be changed by lowering the output current to 2 mA and widening the pulse width to 750 μ sec.

Figure 8. Relationship of Lead Impedance to Maximum Deliverable Output Current

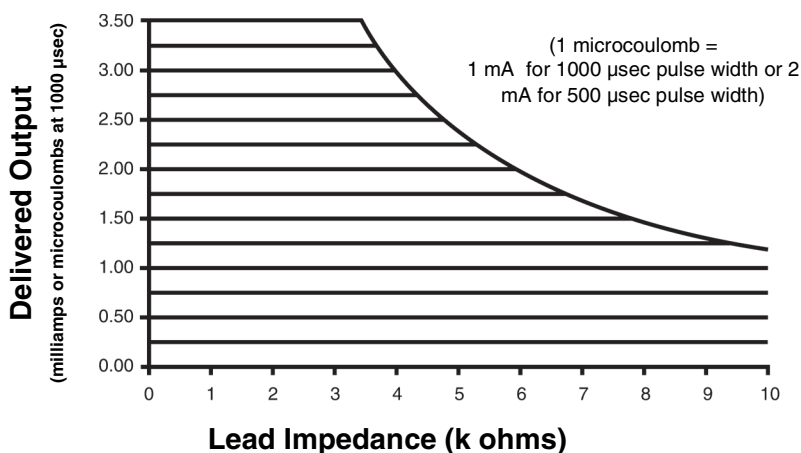


2.3.2.6 Charge delivered per pulse

The charge delivered per pulse is the parameter most important in evaluating stimulation output. It is defined as a microcoulomb (μ C), which is the product of current and time—that is, the output current (mA) multiplied by the pulse width (msec). Figure 9 shows the relationship of programmed output current (μ C) to lead impedance for an output current of 0 to 3.5 mA.

Caution: Do not use frequencies of 5 Hz or below for long-term stimulation. Because these frequencies generate an electromagnetic trigger signal, their use results in excessive battery depletion of the implanted generator and, therefore, should be used for short periods of time only.

Figure 9. Relationship of Programmed Output Current to Lead Impedance



2.3.2.7 Effects of programmed settings on generator projected lifetime

The choice of settings for output parameters affects generator battery life. Generally, a high duty cycle will deplete the battery over a shorter period of time than a low duty cycle will. Table 5 shows duty cycles for typical ON time and OFF time settings (except for frequency settings < 10 Hz). Nominal values of 2 second ramp-up time and 2 second ramp-down time are used for the duty cycle calculations, although actual ramp-up and ramp-down times may vary at lower frequencies or output currents. As depicted in the graphic representation of stimulation in Figure 4, neither of the ramp times is considered part of the programmed ON time; however, they are included in calculations for determining the percentage of stimulation time (duty cycle) and predicting battery life.

 **Note:** More frequent follow-up may be required as the generator approaches the near end of service (N EOS) indicator. The time from N EOS to EOS is highly dependent on the programmed parameters and the lead impedance. See Table 6 and Table 7.

Table 5. Duty Cycles for Various ON and OFF Time Settings

	OFF Time (min)								
	0.2	0.3	0.5	0.8	1.1	1.8	3	5	10
ON Time (sec)	Duty Cycles* (% ON Time)								
7	58	44	30	20	15	10	6	4	2
14	69	56	41	29	23	15	9	6	3
21	76	64	49	36	29	19	12	8	4
30	81	71	57	44	35	25	16	10	5
60	89	82	71	59	51	38	27	18	10

* A duty cycle is calculated by dividing stimulation time (programmed ON time plus 2 seconds of ramp-up time and 2 seconds of ramp-down time) by the sum of the ON time and the OFF time.

* The duty cycles in gray are not recommended as they represent parameter combinations with ON Time > OFF Time.

 **Caution:** Possible nerve damage with ON time > OFF time— Excessive stimulation has resulted in degenerative nerve damage in laboratory animals. One component of excessive stimulation is when ON time exceeds OFF time, which can be produced by continuous or frequent magnet activation (> 8 hours). It is recommended that stimulation at these combinations of ranges be avoided.

For Models 102 and 102R, the approximate battery lifetime predicted at programmed settings of 20 Hz with a 500 μ sec pulse width and 2 mA output current, a lead impedance of 4 kOhms, and a duty cycle of 10 percent is 8.4 years (100.8 months). These generators have an N EOS that provides a warning period prior to EOS.

Table 6 and Table 7 provide estimated battery lifetimes under a variety of stimulation conditions, including lead impedance. Because of the number of possible parameter combinations, it is impractical to provide the projected life for all possible combinations. The tables should not be used to predict battery end of service (EOS), but they give some

indication of the effect of various parameter changes on battery life and can be used to assist in the selection of parameter settings. They also indicate that battery life can be maximized at low duty cycles and low frequencies (10 to 20 Hz) for stimulation. Once N EOS is detected, the generator should be replaced as soon as possible.

Table 6. Estimated Battery Life - Nominal Longevity Estimates from Beginning of Life (BOL) to End of Service (EOS)

Output Current (mA)	Frequency (Hz)	Pulse Width (µsec)	DC-DC Converter Code	Nominal Estimated Battery Life (Years)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
1	10	130	2	15.3	11.3	9.5
1	10	130	3	15.1	11.1	9.2
1	10	130	5	14.8	10.5	8.7
1	10	130	7	14.4	9.8	8.0
1	10	500	2	14.2	9.6	7.7
1	10	500	3	13.8	8.9	7.1
1	10	500	5	13.0	7.9	6.1
1	10	500	7	12.4	7.3	5.6
1	10	1000	2	12.8	7.6	5.9
1	10	1000	3	12.2	6.9	5.3
1	10	1000	5	10.9	5.7	4.2
1	10	1000	7	10.3	5.2	3.8
1	20	130	2	14.2	9.5	7.6
1	20	130	3	13.8	9.0	7.2
1	20	130	5	13.4	8.5	6.7
1	20	130	7	12.7	7.6	5.9
1	20	500	2	12.3	7.1	5.4
1	20	500	3	11.7	6.5	4.9
1	20	500	5	10.6	5.5	4.0
1	20	500	7	10.0	4.9	3.6
1	20	1000	2	10.3	5.2	3.8
1	20	1000	3	9.6	4.6	3.3
1	20	1000	5	8.2	3.6	2.6
1	20	1000	7	7.5	3.2	2.3
1	30	130	2	13.1	8.1	6.3
1	30	130	3	12.7	7.6	5.9
1	30	130	5	12.2	7.0	5.3
1	30	130	7	11.4	6.2	4.6
1	30	500	2	10.9	5.7	4.2
1	30	500	3	10.2	5.1	3.7
1	30	500	5	9.0	4.2	3.0
1	30	500	7	8.3	3.7	2.6
1	30	1000	2	8.7	3.9	2.8
1	30	1000	3	7.9	3.5	2.4
1	30	1000	5	6.6	2.7	1.8
1	30	1000	7	5.9	2.3	1.6
1.5	10	130	2	14.7	10.3	8.4
1.5	10	130	3	14.4	9.8	7.9

Output Current (mA)	Frequency (Hz)	Pulse Width (µsec)	DC-DC Converter Code	Nominal Estimated Battery Life (Years)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
1.5	10	130	5	13.7	8.8	7.0
1.5	10	130	7	13.8	8.9	7.1
1.5	10	500	2	12.4	7.3	5.6
1.5	10	500	3	12.0	6.7	5.1
1.5	10	500	5	10.9	5.7	4.3
1.5	10	500	7	11.2	6.0	4.5
1.5	10	1000	2	10.3	5.2	3.8
1.5	10	1000	3	9.6	4.6	3.3
1.5	10	1000	5	8.4	3.8	2.7
1.5	10	1000	7	8.9	4.1	2.9
1.5	20	130	2	13.1	8.0	6.2
1.5	20	130	3	12.6	7.5	5.8
1.5	20	130	5	11.8	6.5	4.9
1.5	20	130	7	11.8	6.6	5.0
1.5	20	500	2	10.0	5.0	3.6
1.5	20	500	3	9.4	4.5	3.2
1.5	20	500	5	8.2	3.7	2.6
1.5	20	500	7	8.6	3.9	2.8
1.5	20	1000	2	7.5	3.2	2.2
1.5	20	1000	3	6.8	2.8	2.0
1.5	20	1000	5	5.7	2.2	1.5
1.5	20	1000	7	6.2	2.4	1.7
1.5	30	130	2	11.8	6.5	4.9
1.5	30	130	3	11.3	6.1	4.5
1.5	30	130	5	10.3	5.2	3.8
1.5	30	130	7	10.4	5.3	3.9
1.5	30	500	2	8.4	3.8	2.7
1.5	30	500	3	7.7	3.3	2.4
1.5	30	500	5	6.6	2.7	1.9
1.5	30	500	7	7.0	2.9	2.0
1.5	30	1000	2	5.9	2.3	1.6
1.5	30	1000	3	5.3	2.0	1.4
1.5	30	1000	5	4.3	1.6	1.1
1.5	30	1000	7	4.7	1.8	1.2
2	10	130	2	14.1	9.4	7.5
2	10	130	3	13.5	8.5	6.7
2	10	130	5	13.5	8.5	6.7
2	10	130	7	13.7	8.8	7.0
2	10	500	2	11.2	6.0	4.4
2	10	500	3	10.1	5.0	3.6
2	10	500	5	10.5	5.4	3.9

Output Current (mA)	Frequency (Hz)	Pulse Width (µsec)	DC-DC Converter Code	Nominal Estimated Battery Life (Years)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
2	10	500	7	11.1	5.9	4.3
2	10	1000	2	8.4	3.8	2.7
2	10	1000	3	7.4	3.1	2.2
2	10	1000	5	7.9	3.5	2.4
2	10	1000	7	8.6	3.9	2.8
2	20	130	2	12.2	7.0	5.3
2	20	130	3	11.3	6.0	4.5
2	20	130	5	11.4	6.2	4.6
2	20	130	7	11.7	6.5	4.9
2	20	500	2	8.4	3.8	2.7
2	20	500	3	7.3	3.1	2.2
2	20	500	5	7.8	3.4	2.4
2	20	500	7	8.4	3.8	2.7
2	20	1000	2	5.5	2.1	1.5
2	20	1000	3	4.8	1.8	1.2
2	20	1000	5	5.3	2.0	1.4
2	20	1000	7	5.9	2.3	1.6
2	30	130	2	10.8	5.6	4.1
2	30	130	3	9.7	4.7	3.4
2	30	130	5	9.9	4.9	3.5
2	30	130	7	10.2	5.1	3.8
2	30	500	2	6.8	2.8	1.9
2	30	500	3	5.7	2.2	1.5
2	30	500	5	6.2	2.5	1.7
2	30	500	7	6.8	2.8	1.9
2	30	1000	2	4.0	1.4	1.0
2	30	1000	3	3.6	1.3	0.8
2	30	1000	5	4.0	1.4	1.0
2	30	1000	7	4.6	1.7	1.1
3.5	10	130	2	12.6	7.5	5.7
3.5	10	130	3	12.9	7.8	6.0
3.5	10	130	5	13.3	8.3	6.5
3.5	10	130	7	13.5	8.6	6.8
3.5	10	500	2	8.6	3.9	2.8
3.5	10	500	3	9.2	4.4	3.1
3.5	10	500	5	10.1	5.0	3.7
3.5	10	500	7	10.8	5.6	4.1
3.5	10	1000	2	5.8	2.3	1.6
3.5	10	1000	3	6.5	2.6	1.8
3.5	10	1000	5	7.5	3.2	2.3
3.5	10	1000	7	8.3	3.7	2.6

Output Current (mA)	Frequency (Hz)	Pulse Width (μsec)	DC-DC Converter Code	Nominal Estimated Battery Life (Years)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
3.5	20	130	2	10.2	5.1	3.8
3.5	20	130	3	10.6	5.5	4.0
3.5	20	130	5	11.1	5.9	4.4
3.5	20	130	7	11.5	6.3	4.7
3.5	20	500	2	5.9	2.3	1.6
3.5	20	500	3	6.5	2.6	1.8
3.5	20	500	5	7.4	3.1	2.2
3.5	20	500	7	8.1	3.5	2.5
3.5	20	1000	2	3.6	1.3	0.9
3.5	20	1000	3	4.1	1.5	1.0
3.5	20	1000	5	5.0	1.9	1.3
3.5	20	1000	7	5.6	2.2	1.5
3.5	30	130	2	8.6	3.9	2.8
3.5	30	130	3	9.0	4.2	3.0
3.5	30	130	5	9.6	4.6	3.3
3.5	30	130	7	10.0	4.9	3.6
3.5	30	500	2	4.5	1.7	1.1
3.5	30	500	3	5.0	1.9	1.3
3.5	30	500	5	5.8	2.3	1.6
3.5	30	500	7	6.5	2.6	1.8
3.5	30	1000	2	2.7	0.9	0.6
3.5	30	1000	3	3.0	1.0	0.7
3.5	30	1000	5	3.7	1.3	0.9
3.5	30	1000	7	4.3	1.6	1.1

Table 7. Estimated Battery Life - Worst Case Longevity Estimates from Beginning of Life (BOL) to Near End of Service (N EOS)

Output Current (mA)	Frequency (Hz)	Pulse Width (µsec)	DC-DC Converter Code	Worst Case Estimated Battery Life (Years)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
1	10	130	2	9.3	7.1	6.0
1	10	130	3	9.3	7.2	6.1
1	10	130	5	8.8	6.2	5.1
1	10	130	7	8.8	6.2	5.0
1	10	500	2	9.1	6.8	5.7
1	10	500	3	8.9	6.4	5.2
1	10	500	5	8.2	5.3	4.2
1	10	500	7	8.0	5.0	3.9
1	10	1000	2	8.3	5.4	4.3
1	10	1000	3	8.0	5.1	4.0
1	10	1000	5	7.2	4.1	3.1
1	10	1000	7	6.8	3.7	2.8
1	20	130	2	9.1	6.7	5.6
1	20	130	3	8.9	6.4	5.3
1	20	130	5	8.6	5.9	4.8
1	20	130	7	8.2	5.3	4.2
1	20	500	2	8.2	5.2	4.2
1	20	500	3	7.8	4.8	3.7
1	20	500	5	6.9	3.8	2.8
1	20	500	7	6.7	3.6	2.7
1	20	1000	2	6.9	3.7	2.8
1	20	1000	3	6.6	3.5	2.6
1	20	1000	5	5.7	2.8	2.0
1	20	1000	7	5.2	2.4	1.7
1	30	130	2	8.6	5.9	4.7
1	30	130	3	8.4	5.6	4.4
1	30	130	5	8.0	5.0	3.9
1	30	130	7	7.5	4.5	3.4
1	30	500	2	7.4	4.3	3.3
1	30	500	3	7.0	3.9	2.9
1	30	500	5	6.1	3.0	2.2
1	30	500	7	5.7	2.8	2.0
1	30	1000	2	5.8	2.8	2.0
1	30	1000	3	5.6	2.7	1.9
1	30	1000	5	4.7	2.1	1.5
1	30	1000	7	4.1	1.7	1.2
1.5	10	130	2	9.2	6.9	5.9
1.5	10	130	3	8.9	6.5	5.4

Output Current (mA)	Frequency (Hz)	Pulse Width (µsec)	DC-DC Converter Code	Worst Case Estimated Battery Life (Years)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
1.5	10	130	5	8.3	5.4	4.3
1.5	10	130	7	8.3	5.5	4.4
1.5	10	500	2	7.9	4.9	3.8
1.5	10	500	3	7.8	4.8	3.7
1.5	10	500	5	7.1	4.0	3.0
1.5	10	500	7	7.2	4.1	3.1
1.5	10	1000	2	7.0	3.9	2.9
1.5	10	1000	3	6.6	3.5	2.6
1.5	10	1000	5	5.8	2.8	2.0
1.5	10	1000	7	6.0	3.0	2.2
1.5	20	130	2	8.5	5.7	4.6
1.5	20	130	3	8.2	5.3	4.2
1.5	20	130	5	7.6	4.5	3.5
1.5	20	130	7	7.6	4.6	3.5
1.5	20	500	2	6.9	3.8	2.8
1.5	20	500	3	6.5	3.4	2.5
1.5	20	500	5	5.7	2.7	2.0
1.5	20	500	7	5.9	2.9	2.1
1.5	20	1000	2	5.3	2.5	1.8
1.5	20	1000	3	4.9	2.2	1.5
1.5	20	1000	5	4.2	1.7	1.2
1.5	20	1000	7	4.5	1.9	1.3
1.5	30	130	2	7.8	4.8	3.8
1.5	30	130	3	7.5	4.5	3.4
1.5	30	130	5	6.9	3.7	2.8
1.5	30	130	7	6.9	3.8	2.8
1.5	30	500	2	5.9	2.9	2.1
1.5	30	500	3	5.5	2.6	1.9
1.5	30	500	5	4.8	2.1	1.5
1.5	30	500	7	5.0	2.2	1.6
1.5	30	1000	2	4.3	1.8	1.3
1.5	30	1000	3	3.9	1.6	1.1
1.5	30	1000	5	3.3	1.2	0.8
1.5	30	1000	7	3.5	1.4	1.0
2	10	130	2	8.8	6.3	5.2
2	10	130	3	8.0	5.0	4.0
2	10	130	5	8.2	5.3	4.2
2	10	130	7	8.3	5.5	4.4
2	10	500	2	7.4	4.3	3.3
2	10	500	3	6.6	3.5	2.6
2	10	500	5	6.9	3.7	2.8

Output Current (mA)	Frequency (Hz)	Pulse Width (µsec)	DC-DC Converter Code	Worst Case Estimated Battery Life (Years)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
2	10	500	7	7.2	4.0	3.1
2	10	1000	2	5.6	2.6	1.9
2	10	1000	3	5.1	2.3	1.7
2	10	1000	5	5.5	2.6	1.9
2	10	1000	7	5.9	2.9	2.1
2	20	130	2	8.0	5.0	3.9
2	20	130	3	7.3	4.2	3.2
2	20	130	5	7.4	4.3	3.3
2	20	130	7	7.6	4.5	3.4
2	20	500	2	5.8	2.8	2.0
2	20	500	3	5.2	2.3	1.7
2	20	500	5	5.4	2.5	1.8
2	20	500	7	5.8	2.8	2.0
2	20	1000	2	3.7	1.4	1.0
2	20	1000	3	3.6	1.4	1.0
2	20	1000	5	3.9	1.6	1.1
2	20	1000	7	4.3	1.8	1.3
2	30	130	2	7.3	4.1	3.1
2	30	130	3	6.5	3.4	2.5
2	30	130	5	6.7	3.5	2.6
2	30	130	7	6.8	3.7	2.8
2	30	500	2	4.7	2.1	1.5
2	30	500	3	4.2	1.7	1.2
2	30	500	5	4.5	1.9	1.3
2	30	500	7	4.9	2.1	1.5
2	30	1000	2	2.9	1.0	0.7
2	30	1000	3	2.7	1.0	0.7
2	30	1000	5	3.1	1.1	0.8
2	30	1000	7	3.4	1.3	0.9
3.5	10	130	2	7.9	4.9	3.8
3.5	10	130	3	8.0	5.1	4.0
3.5	10	130	5	8.2	5.3	4.2
3.5	10	130	7	8.3	5.5	4.4
3.5	10	500	2	5.9	2.9	2.1
3.5	10	500	3	6.2	3.1	2.3
3.5	10	500	5	6.7	3.6	2.7
3.5	10	500	7	7.0	3.9	2.9
3.5	10	1000	2	4.2	1.8	1.2
3.5	10	1000	3	4.6	2.0	1.4
3.5	10	1000	5	5.2	2.4	1.7
3.5	10	1000	7	5.7	2.7	2.0

Output Current (mA)	Frequency (Hz)	Pulse Width (μsec)	DC-DC Converter Code	Worst Case Estimated Battery Life (Years)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
3.5	20	130	2	6.8	3.7	2.7
3.5	20	130	3	7.0	3.9	2.9
3.5	20	130	5	7.3	4.2	3.2
3.5	20	130	7	7.4	4.4	3.3
3.5	20	500	2	4.3	1.8	1.3
3.5	20	500	3	4.7	2.0	1.4
3.5	20	500	5	5.2	2.4	1.7
3.5	20	500	7	5.6	2.7	1.9
3.5	20	1000	2	2.8	1.0	0.7
3.5	20	1000	3	3.1	1.2	0.8
3.5	20	1000	5	3.7	1.5	1.0
3.5	20	1000	7	4.1	1.7	1.2
3.5	30	130	2	6.0	2.9	2.1
3.5	30	130	3	6.2	3.1	2.3
3.5	30	130	5	6.5	3.4	2.5
3.5	30	130	7	6.7	3.6	2.7
3.5	30	500	2	3.4	1.3	0.9
3.5	30	500	3	3.7	1.5	1.0
3.5	30	500	5	4.3	1.8	1.2
3.5	30	500	7	4.7	2.0	1.4
3.5	30	1000	2	2.1	0.7	0.5
3.5	30	1000	3	2.4	0.8	0.6
3.5	30	1000	5	2.9	1.1	0.7
3.5	30	1000	7	3.2	1.2	0.8

The projected battery life decreases as lead impedance increases. Although 1.5 k to 3 kOhms (DC-DC Converter Code 1-2) appears to be the typical lead impedance at time of implantation, the impedance may increase to 3 k to 5 kOhms (DC-DC Converter Code 2-3) during the life of the implant.

2.3.2.7.1. Generator replacement

All VNS Therapy generators will eventually require surgical replacement due to battery depletion. Generator replacement does not, of itself, require lead replacement unless a lead discontinuity is suspected. Generator replacement or removal requires dissection to the generator's pocket, with care being taken not to damage or cut the lead. Generator removal is the reverse of the placement procedure. The entire surgical procedure is generally short, about one hour.

2.3.2.8 End-of-service and replacement information

It is recommended that patients be instructed to activate the generator manually with the magnet on a daily basis to test for the presence of stimulation. The most common reason for the absence of stimulation is battery depletion, though there may be other reasons. The magnet output current should be programmed to a level that is perceptible to the patient. Patients must be instructed to call their physician when they notice that magnet-activated stimulation is no longer perceptible.



Caution: Generator EOS may result in increased seizure frequency and/or seizure intensity and/or seizure duration, in some cases to levels greater than those reported before stimulation.

The lifetime of the generator battery depends on programmed parameters and lead impedance (see Table 6 and Table 7). Immediately before end of service (EOS), the generator may provide unscheduled stimulation. The stimulation output may be above or below the programmed output. When EOS occurs, the generator will not deliver any output, the patient will not feel the stimulation, and communication with the generator will not be possible.

The time from N EOS to EOS is highly dependent on the programmed parameters and the lead impedance. Table 8 and Table 9 provide estimated times from N EOS to EOS under a variety of stimulation conditions, including lead impedance. Because of the number of possible parameter combinations, it is impractical to provide the estimated times for all possible combinations. The tables should not be used to predict battery EOS, but they give some indication of the time remaining once the N EOS flag is set. Due to the uncertainty of knowing when the N EOS flag is set, once N EOS is detected, the generator should be replaced as soon as possible.



Note: More frequent follow-up may be required as the generator approaches N EOS. See Table 8 and Table 9.

Table 8. Estimated Battery Life - Nominal N EOS to EOS Time Estimates

Output Current (mA)	Frequency (Hz)	Pulse Width (μsec)	DC-DC Converter Code	Nominal Time from N EOS to EOS (Months)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
1	10	130	2	9.4	6.7	5.5
1	10	130	3	9.3	6.5	5.3
1	10	130	5	9.1	6.2	5.0
1	10	130	7	8.8	5.8	4.6
1	10	500	2	8.7	5.6	4.4
1	10	500	3	8.4	5.2	4.1
1	10	500	5	7.9	4.6	3.5
1	10	500	7	7.5	4.2	3.2
1	10	1000	2	7.7	4.4	3.4
1	10	1000	3	7.3	4.0	3.1
1	10	1000	5	6.5	3.3	2.5
1	10	1000	7	6.2	3.0	2.2
1	20	130	2	8.6	5.5	4.4
1	20	130	3	8.4	5.3	4.1
1	20	130	5	8.2	4.9	3.8
1	20	130	7	7.7	4.4	3.4
1	20	500	2	7.4	4.1	3.1
1	20	500	3	7.0	3.8	2.8
1	20	500	5	6.3	3.2	2.3
1	20	500	7	5.9	2.9	2.1
1	20	1000	2	6.2	3.0	2.2
1	20	1000	3	5.7	2.7	2.0
1	20	1000	5	4.8	2.1	1.5
1	20	1000	7	4.4	1.9	1.4
1	30	130	2	8.0	4.7	3.6
1	30	130	3	7.7	4.4	3.4
1	30	130	5	7.4	4.1	3.1
1	30	130	7	6.9	3.6	2.7
1	30	500	2	6.5	3.3	2.4
1	30	500	3	6.1	3.0	2.2
1	30	500	5	5.3	2.4	1.8
1	30	500	7	4.9	2.2	1.6
1	30	1000	2	5.1	2.3	1.7
1	30	1000	3	4.7	2.0	1.5
1	30	1000	5	3.9	1.6	1.1
1	30	1000	7	3.3	1.3	0.9
1.5	10	130	2	9.0	6.0	4.9
1.5	10	130	3	8.8	5.7	4.6
1.5	10	130	5	8.4	5.2	4.0

Output Current (mA)	Frequency (Hz)	Pulse Width (μsec)	DC-DC Converter Code	Nominal Time from N EOS to EOS (Months)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
1.5	10	130	7	8.4	5.2	4.1
1.5	10	500	2	7.5	4.2	3.2
1.5	10	500	3	7.2	3.9	3.0
1.5	10	500	5	6.6	3.3	2.5
1.5	10	500	7	6.7	3.5	2.6
1.5	10	1000	2	6.1	3.0	2.2
1.5	10	1000	3	5.7	2.7	2.0
1.5	10	1000	5	5.0	2.2	1.6
1.5	10	1000	7	5.3	2.4	1.7
1.5	20	130	2	7.9	4.7	3.6
1.5	20	130	3	7.6	4.4	3.3
1.5	20	130	5	7.1	3.8	2.8
1.5	20	130	7	7.1	3.8	2.9
1.5	20	500	2	6.0	2.9	2.1
1.5	20	500	3	5.6	2.6	1.9
1.5	20	500	5	4.9	2.2	1.6
1.5	20	500	7	5.1	2.3	1.7
1.5	20	1000	2	4.4	1.9	1.4
1.5	20	1000	3	4.0	1.7	1.2
1.5	20	1000	5	3.1	1.3	0.9
1.5	20	1000	7	3.6	1.5	1.1
1.5	30	130	2	7.1	3.8	2.9
1.5	30	130	3	6.8	3.5	2.6
1.5	30	130	5	6.1	3.0	2.2
1.5	30	130	7	6.2	3.1	2.3
1.5	30	500	2	5.0	2.2	1.6
1.5	30	500	3	4.6	2.0	1.4
1.5	30	500	5	3.9	1.6	1.2
1.5	30	500	7	4.1	1.7	1.2
1.5	30	1000	2	3.2	1.3	0.9
1.5	30	1000	3	2.9	1.1	0.8
1.5	30	1000	5	2.4	0.9	0.7
1.5	30	1000	7	2.6	1.0	0.7
2	10	130	2	8.6	5.5	4.3
2	10	130	3	8.2	5.0	3.9
2	10	130	5	8.2	5.0	3.9
2	10	130	7	8.3	5.1	4.0
2	10	500	2	6.7	3.5	2.6
2	10	500	3	6.0	2.9	2.1
2	10	500	5	6.3	3.1	2.3
2	10	500	7	6.6	3.4	2.5

Output Current (mA)	Frequency (Hz)	Pulse Width (μsec)	DC-DC Converter Code	Nominal Time from N EOS to EOS (Months)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
2	10	1000	2	5.0	2.2	1.6
2	10	1000	3	4.3	1.8	1.3
2	10	1000	5	4.7	2.0	1.5
2	10	1000	7	5.1	2.3	1.7
2	20	130	2	7.4	4.1	3.1
2	20	130	3	6.8	3.5	2.6
2	20	130	5	6.9	3.6	2.7
2	20	130	7	7.0	3.8	2.8
2	20	500	2	5.0	2.2	1.6
2	20	500	3	4.3	1.8	1.3
2	20	500	5	4.6	2.0	1.4
2	20	500	7	5.0	2.2	1.6
2	20	1000	2	3.0	1.2	0.9
2	20	1000	3	2.6	1.0	0.7
2	20	1000	5	2.9	1.2	0.8
2	20	1000	7	3.3	1.3	0.9
2	30	130	2	6.4	3.2	2.4
2	30	130	3	5.8	2.8	2.0
2	30	130	5	5.9	2.8	2.1
2	30	130	7	6.1	3.0	2.2
2	30	500	2	4.0	1.7	1.2
2	30	500	3	3.2	1.3	0.9
2	30	500	5	3.6	1.5	1.1
2	30	500	7	4.0	1.7	1.2
2	30	1000	2	2.2	0.9	0.6
2	30	1000	3	2.0	0.8	0.6
2	30	1000	5	2.2	0.9	0.6
2	30	1000	7	2.5	1.0	0.7
3.5	10	130	2	7.6	4.3	3.3
3.5	10	130	3	7.8	4.5	3.5
3.5	10	130	5	8.1	4.8	3.7
3.5	10	130	7	8.2	5.0	3.9
3.5	10	500	2	5.1	2.3	1.7
3.5	10	500	3	5.5	2.5	1.8
3.5	10	500	5	6.0	2.9	2.1
3.5	10	500	7	6.4	3.2	2.4
3.5	10	1000	2	3.2	1.3	0.9
3.5	10	1000	3	3.8	1.6	1.1
3.5	10	1000	5	4.4	1.9	1.4
3.5	10	1000	7	4.9	2.2	1.6
3.5	20	130	2	6.1	3.0	2.2

Output Current (mA)	Frequency (Hz)	Pulse Width (μsec)	DC-DC Converter Code	Nominal Time from N EOS to EOS (Months)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
3.5	20	130	3	6.3	3.2	2.3
3.5	20	130	5	6.7	3.4	2.6
3.5	20	130	7	6.9	3.6	2.7
3.5	20	500	2	3.3	1.3	0.9
3.5	20	500	3	3.8	1.6	1.1
3.5	20	500	5	4.3	1.9	1.3
3.5	20	500	7	4.8	2.1	1.5
3.5	20	1000	2	2.0	0.8	0.6
3.5	20	1000	3	2.3	0.9	0.6
3.5	20	1000	5	2.7	1.1	0.8
3.5	20	1000	7	3.1	1.2	0.9
3.5	30	130	2	5.1	2.3	1.7
3.5	30	130	3	5.4	2.5	1.8
3.5	30	130	5	5.7	2.7	2.0
3.5	30	130	7	6.0	2.9	2.1
3.5	30	500	2	2.5	1.0	0.7
3.5	30	500	3	2.8	1.1	0.8
3.5	30	500	5	3.2	1.3	0.9
3.5	30	500	7	3.8	1.6	1.1
3.5	30	1000	2	1.5	0.6	0.4
3.5	30	1000	3	1.7	0.7	0.5
3.5	30	1000	5	2.1	0.8	0.6
3.5	30	1000	7	2.4	0.9	0.7

Table 9. Estimated Battery Life - Worst Case N EOS to EOS Time Estimates

Output Current (mA)	Frequency (Hz)	Pulse Width (µsec)	DC-DC Converter Code	Worst Case Time from N EOS to EOS (Months)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
1	10	130	2	7.7	5.6	4.7
1	10	130	3	7.8	5.7	4.8
1	10	130	5	7.2	4.9	4.0
1	10	130	7	7.2	4.9	3.9
1	10	500	2	7.6	5.4	4.4
1	10	500	3	7.3	5.0	4.1
1	10	500	5	6.7	4.2	3.3
1	10	500	7	6.5	3.9	3.0
1	10	1000	2	6.8	4.2	3.3
1	10	1000	3	6.6	4.0	3.1
1	10	1000	5	5.9	3.2	2.4
1	10	1000	7	5.5	2.9	2.2
1	20	130	2	7.5	5.3	4.4
1	20	130	3	7.4	5.1	4.1
1	20	130	5	7.1	4.7	3.7
1	20	130	7	6.7	4.1	3.2
1	20	500	2	6.7	4.1	3.2
1	20	500	3	6.4	3.8	2.9
1	20	500	5	5.6	3.0	2.2
1	20	500	7	5.4	2.8	2.1
1	20	1000	2	5.6	2.9	2.2
1	20	1000	3	5.3	2.8	2.1
1	20	1000	5	4.6	2.2	1.6
1	20	1000	7	4.1	1.9	1.4
1	30	130	2	7.1	4.6	3.7
1	30	130	3	6.9	4.4	3.5
1	30	130	5	6.5	3.9	3.0
1	30	130	7	6.1	3.5	2.7
1	30	500	2	6.0	3.4	2.6
1	30	500	3	5.7	3.0	2.3
1	30	500	5	4.9	2.4	1.8
1	30	500	7	4.6	2.2	1.6
1	30	1000	2	4.6	2.2	1.6
1	30	1000	3	4.5	2.1	1.6
1	30	1000	5	3.8	1.7	1.2
1	30	1000	7	3.1	1.3	0.9
1.5	10	130	2	7.6	5.5	4.6
1.5	10	130	3	7.4	5.1	4.2
1.5	10	130	5	6.8	4.3	3.4

Output Current (mA)	Frequency (Hz)	Pulse Width (µsec)	DC-DC Converter Code	Worst Case Time from N EOS to EOS (Months)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
1.5	10	130	7	6.8	4.3	3.4
1.5	10	500	2	6.4	3.8	3.0
1.5	10	500	3	6.4	3.8	2.9
1.5	10	500	5	5.7	3.1	2.3
1.5	10	500	7	5.9	3.3	2.5
1.5	10	1000	2	5.6	3.0	2.3
1.5	10	1000	3	5.3	2.7	2.0
1.5	10	1000	5	4.6	2.2	1.6
1.5	10	1000	7	4.8	2.4	1.7
1.5	20	130	2	7.0	4.5	3.6
1.5	20	130	3	6.7	4.2	3.3
1.5	20	130	5	6.2	3.5	2.7
1.5	20	130	7	6.2	3.6	2.7
1.5	20	500	2	5.6	3.0	2.2
1.5	20	500	3	5.2	2.7	2.0
1.5	20	500	5	4.6	2.2	1.6
1.5	20	500	7	4.7	2.3	1.7
1.5	20	1000	2	4.3	2.0	1.4
1.5	20	1000	3	3.9	1.7	1.3
1.5	20	1000	5	3.1	1.3	0.9
1.5	20	1000	7	3.5	1.5	1.1
1.5	30	130	2	6.4	3.8	2.9
1.5	30	130	3	6.1	3.5	2.7
1.5	30	130	5	5.5	2.9	2.2
1.5	30	130	7	5.6	3.0	2.2
1.5	30	500	2	4.8	2.3	1.7
1.5	30	500	3	4.4	2.1	1.5
1.5	30	500	5	3.8	1.7	1.2
1.5	30	500	7	4.0	1.8	1.3
1.5	30	1000	2	3.3	1.4	1.0
1.5	30	1000	3	2.9	1.2	0.9
1.5	30	1000	5	2.4	1.0	0.7
1.5	30	1000	7	2.6	1.1	0.8
2	10	130	2	7.3	4.9	4.0
2	10	130	3	6.5	4.0	3.1
2	10	130	5	6.7	4.2	3.3
2	10	130	7	6.8	4.3	3.4
2	10	500	2	6.0	3.4	2.6
2	10	500	3	5.4	2.8	2.1
2	10	500	5	5.5	2.9	2.2
2	10	500	7	5.8	3.2	2.4

Output Current (mA)	Frequency (Hz)	Pulse Width (µsec)	DC-DC Converter Code	Worst Case Time from N EOS to EOS (Months)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
2	10	1000	2	4.5	2.1	1.5
2	10	1000	3	4.1	1.9	1.3
2	10	1000	5	4.4	2.1	1.5
2	10	1000	7	4.7	2.3	1.7
2	20	130	2	6.5	3.9	3.1
2	20	130	3	5.9	3.3	2.5
2	20	130	5	6.0	3.4	2.6
2	20	130	7	6.1	3.5	2.7
2	20	500	2	4.7	2.2	1.6
2	20	500	3	4.1	1.9	1.4
2	20	500	5	4.3	2.0	1.5
2	20	500	7	4.6	2.2	1.6
2	20	1000	2	2.7	1.1	0.8
2	20	1000	3	2.7	1.1	0.8
2	20	1000	5	2.9	1.2	0.9
2	20	1000	7	3.2	1.4	1.0
2	30	130	2	5.9	3.3	2.5
2	30	130	3	5.3	2.7	2.0
2	30	130	5	5.4	2.8	2.1
2	30	130	7	5.5	2.9	2.2
2	30	500	2	3.8	1.7	1.2
2	30	500	3	3.1	1.3	0.9
2	30	500	5	3.6	1.5	1.1
2	30	500	7	3.9	1.7	1.2
2	30	1000	2	2.1	0.8	0.6
2	30	1000	3	2.0	0.8	0.6
2	30	1000	5	2.3	0.9	0.7
2	30	1000	7	2.6	1.0	0.7
3.5	10	130	2	6.4	3.8	3.0
3.5	10	130	3	6.6	4.0	3.1
3.5	10	130	5	6.7	4.2	3.3
3.5	10	130	7	6.8	4.3	3.4
3.5	10	500	2	4.7	2.3	1.7
3.5	10	500	3	5.0	2.5	1.8
3.5	10	500	5	5.4	2.8	2.1
3.5	10	500	7	5.7	3.1	2.3
3.5	10	1000	2	3.2	1.3	1.0
3.5	10	1000	3	3.7	1.6	1.2
3.5	10	1000	5	4.2	1.9	1.4
3.5	10	1000	7	4.6	2.2	1.6
3.5	20	130	2	5.5	2.9	2.2

Output Current (mA)	Frequency (Hz)	Pulse Width (µsec)	DC-DC Converter Code	Worst Case Time from N EOS to EOS (Months)		
				10% Duty Cycle Life	33% Duty Cycle Life	50% Duty Cycle Life
3.5	20	130	3	5.7	3.0	2.3
3.5	20	130	5	5.9	3.3	2.5
3.5	20	130	7	6.1	3.4	2.6
3.5	20	500	2	3.2	1.4	1.0
3.5	20	500	3	3.7	1.6	1.2
3.5	20	500	5	4.2	1.9	1.4
3.5	20	500	7	4.5	2.1	1.5
3.5	20	1000	2	2.1	0.8	0.6
3.5	20	1000	3	2.3	0.9	0.7
3.5	20	1000	5	2.8	1.1	0.8
3.5	20	1000	7	3.1	1.3	0.9
3.5	30	130	2	4.8	2.3	1.7
3.5	30	130	3	5.0	2.5	1.8
3.5	30	130	5	5.2	2.7	2.0
3.5	30	130	7	5.4	2.8	2.1
3.5	30	500	2	2.5	1.0	0.7
3.5	30	500	3	2.8	1.1	0.8
3.5	30	500	5	3.2	1.3	1.0
3.5	30	500	7	3.7	1.6	1.2
3.5	30	1000	2	1.6	0.6	0.5
3.5	30	1000	3	1.8	0.7	0.5
3.5	30	1000	5	2.1	0.8	0.6
3.5	30	1000	7	2.4	1.0	0.7

Epilepsy —Either the physician or the patient through magnet activation, also can verify operation of generators when output current is programmed to a level sufficient to cause strong tingling and/or hoarseness. Failure to detect stimulation after activation with the magnet may indicate EOS or high lead impedance. If this occurs, the physician should try to reset the generator following instructions in the programming system physician's manuals. If the generator cannot be reset, replacement is indicated.

 **Caution:** LivaNova recommends prompt replacement of the generator at the end of battery life. Prompt replacement may help minimize any possible relapse in seizure control.

LivaNova recommends daily magnet activation by the patient as the primary test for battery depletion due to the wide variation in the time (which can sometimes be very short) from the near end of service (N EOS) indicator to the end of service (EOS).

 **Caution:** A generator explanted for any reason should not be reimplanted. An explanted generator should be returned to LivaNova in a double-sealed pouch or container for examination and disposal, accompanied by a completed Returned Product Report. Before returning the generator, disinfect the device components with Betadine®, Cidex® soak, or other similar disinfectant, and double-seal them in a pouch or other container properly labeled with a biohazard warning.

3 Technical Information — 103/104 Generators

3.1 Detailed Device Description

3.1.1 Physical Characteristics

The titanium cases of the VNS Therapy Demipulse® Model 103 and Demipulse Duo® Model 104 Generators are hermetically sealed and leak-rate tested. Specially designed feedthroughs using platinum conductors form the electrical connection from the connector blocks to the circuitry through the hermetically sealed enclosure. The Model 103 accepts the single-pin lead, and the Model 104 accepts the dual-pin lead.

3.1.2 Biological Compatibility

Materials exposed to the subcutaneous environment are biologically compatible. All of these materials have a long history in medical implants and have been found to be tissue compatible.

3.1.3 Power Source

The power source for the Model 103 and Model 104 generators is a Wilson Greatbatch Ltd, Model 2183, lithium carbon monofluoride battery with an open-circuit voltage of 3.3 V. The battery's maximum available capacity is approximately 1 Amp-hour. The self-discharge reduces the capacity by less than 1 percent per year. The voltage in this battery gradually decreases as the battery nears its end of service (EOS).

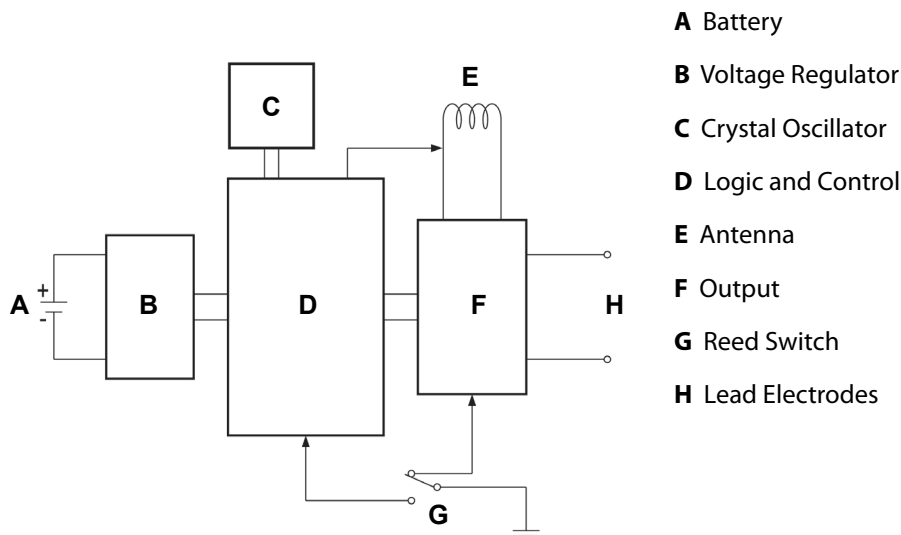
3.1.4 Circuitry

The generator uses complementary metal oxide semiconductor (CMOS) integrated circuits, including a microprocessor. The circuitry is functionally represented in Figure 10.

For descriptive purposes, circuitry of the generator can be divided into the following major functional sections:

Voltage regulator	Regulates the system power supply
Crystal oscillator	Provides a timing reference
Logic and control	Controls overall generator function; receives and implements programming commands; collects and stores telemetry information
Output	Develops and modulates signals delivered to the lead
Antenna	Receives programming signals; transmits telemetry information to the programming wand
Reed switch	Provides a mechanism to place the generator in Magnet Mode or to inhibit its output

Figure 10. Generator Circuitry



3.1.5 Identification

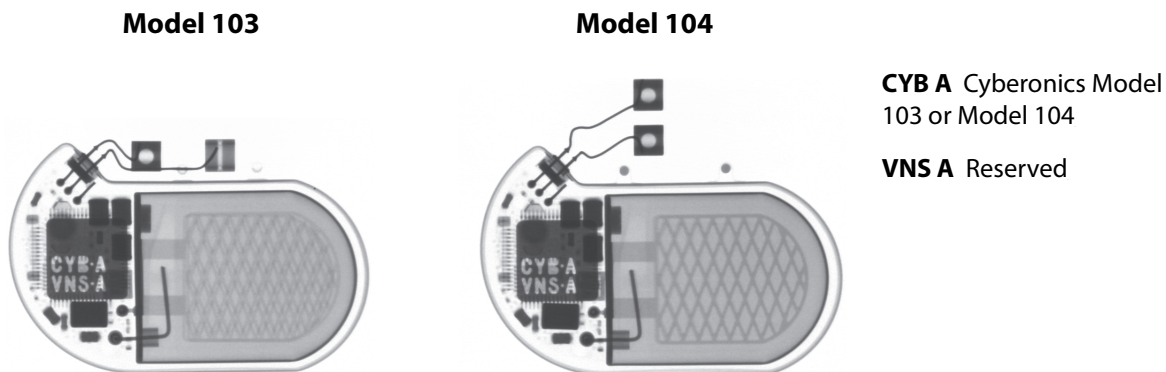
The generator can be identified on an x-ray and will appear as shown in Figure 11. The serial number and model number of the generator are marked on its titanium case, but do not appear on the x-ray.

The serial number, model number and year of manufacture depending upon programming software version can be identified by interrogating the generator with the programming software.

Note: See the programming system physician's manuals for details.

The x-rays in Figure 11 show a Model 103 and a Model 104. The x-ray tag included uses the code CYB A and VNS A. The codes are defined to the right of the images.

Figure 11. X-ray Identification



3.2 VNS Therapy System Compatibility

Models 103 and 104 are compatible with these system components:

		Generator Model	
Component	Model	103	104
Lead	dual-pin		x
	single-pin	x	
Programming System	201 wand / 250 software	x	x
	2000 wand / 3000 software	x	x
Tunnelers	402	x	x
Accessory Pack	502	x	x
Magnets	220	x	x

3.3 Directions for Use

3.3.1 Specifications and Product Information

The specifications and product information for the VNS Therapy generators are presented in Table 10.

Table 10. Specifications and Product Information

Stimulation Parameters	Available Parameter Settings
Output current	0-3.5 mA in 0.25-mA steps $\pm 0.25 \leq 1$ mA, $\pm 10\% > 1$ mA
Signal frequency	1, 2, 5, 10, 15, 20, 25, 30 Hz $\pm 6\%$
Pulse width	130, 250, 500, 750, 1000 μ sec $\pm 10\%$
Signal ON time	7, 14, 21, 30, 60 sec $\pm 15\%$ or + 7 sec, whichever is greater ($\pm 15\%$ or ± 7 sec in Magnet Mode)
Signal OFF time	0.2, 0.3, 0.5, 0.8, 1.1, 1.8, 3 min, and 5 to 180 min (5 to 60 in 5-min steps; 60 to 180 in 30-min steps) + 4.4 / - 8.4 sec or $\pm 1\%$, whichever is greater
Magnet activation	Provided by magnet application (output current, pulse width, and signal ON time may be independently programmed for this purpose)
Reset parameters	Settings are unchanged, but output is disabled (0 mA)
Telemetry Reports	
Device History Report	Patient ID, implant date, model number, serial number, magnet activations, total ON time, total operating time, and manufacturing date
Device Diagnostic Report	Patient ID, model ID, serial number, implant date, communication status, output current status, measured current delivered, lead impedance, and battery status indicators (IFI, N EOS, EOS)
Power Source	
Battery	Wilson Greatbatch Ltd., Model 2183
Chemistry	Lithium carbon monofluoride
Voltage	3.3 V, open circuit
Rated capacity	1 Amp-hour
Self-discharge rate	< 1% per year

Physical Characteristics—Materials		
Case	Titanium, hermetically sealed	
Header	Polyurethane — Tecothane™ TT-1075D-M Thermoplastic	
Lead connector blocks	Stainless steel	
Setscrew plug(s)	Silicone*	
Measurements (Typical)		
	Model 103	Model 104
Lead receptacle(s)	0.126 in (3.2 mm)	0.2 in (5 mm)
Dimensions	1.8 in x 1.3 in x 0.27 in (45 mm x 32 mm x 6.9 mm)	1.8 in x 1.6 in x 0.27 in (45 mm x 39 mm x 6.9 mm)
Weight	0.56 oz (16 g)	0.63 oz (17 g)
Connector Retention Strength		
With VNS Therapy lead	> 10 N	

* No component of the VNS Therapy System is made with natural rubber latex.

3.3.2 Operating Characteristics

3.3.2.1 Communicating with the VNS Therapy System

3.3.2.1.1. Programming software

The generator can be programmed with the VNS Therapy programming software.

 **Note:** For more information, including a list of computers that have been qualified for use with this software, see the programming system physician's manuals.

The programming software is used on a computer, supplied by LivaNova, that is dedicated only to programming the VNS Therapy System.

3.3.2.1.2. Programming wand

A programming wand connected to a compatible computer running the programming software is needed to communicate with the generator.

 **Note:** For proper placement of the programming wand, connection of the wand to the computer, and use of the wand, see the programming system physician's manuals.

3.3.2.1.3. Prompts and messages

After the program has been initiated, the software screens display prompts and messages to aid in communicating with the generator.

3.3.2.1.4. Communication

The generator “listens” for a communication signal from the programming wand. Communication usually takes between 1 and 4 seconds, but may be prolonged or interrupted in the presence of electromagnetic interference (EMI). The generator listens for and implements interrogations, parameter programming instructions, requests for Device Diagnostics testing, and Device History inquiries.

In response, the generator transmits information on the stimulation parameter settings, changes its parameter settings, responds to requests for Device Diagnostics testing, and provides device histories, respectively. Each time these data are transmitted by the generator, they are saved by the programming software to a database.

 **Note:** For details on viewing database information, see the programming system physician's manuals.

In addition to the programming software and programming wand combination, the magnet can be used for one-way communication to the generator by activating a reed switch in the electronic circuitry. The magnet can be used to initiate stimulation, temporarily inhibit stimulation, perform Magnet Mode diagnostics, and reset the generator.

3.3.2.1.5. Normal Mode

After the generator has been programmed, the stimulation will repeat in accordance with the programmed ON and OFF cycle (Normal Mode) until the generator receives communication from the VNS Therapy programming system or until it is inhibited with the magnet. Immediately after successful programming, the generator delivers a programmed stimulation that enables the programmer to evaluate patient response. If programming is performed during stimulation, stimulation will be terminated; after programming, stimulation will begin, using the revised settings.

3.3.2.1.6. Magnet Mode

Magnet Mode produces on-demand stimulation for the programmed magnet ON time. Stimulation is initiated by applying or passing the magnet over the generator for at least 1 second and then immediately removing it from the area over the generator. Stimulation is delivered after the magnet is removed. The Magnet Mode uses the same frequency as the Normal Mode, but the output current, pulse width, and signal ON time are independently programmable. The Directions for Use are equivalent for epilepsy and depression, with the following exceptions:

- For patients with depression, the Magnet Mode output current should always be programmed at 0 mA, the setting at which the generator is shipped from LivaNova.
- Use of the Magnet Mode is limited to patients with epilepsy. Patients with epilepsy or their caregivers pass the magnet over the implanted generator to activate on-demand delivery of a single train of vagus nerve stimulation and help abort or diminish a seizure.
- Magnet Mode is not used for patients with depression.

3.3.2.1.7. Generator interrogation

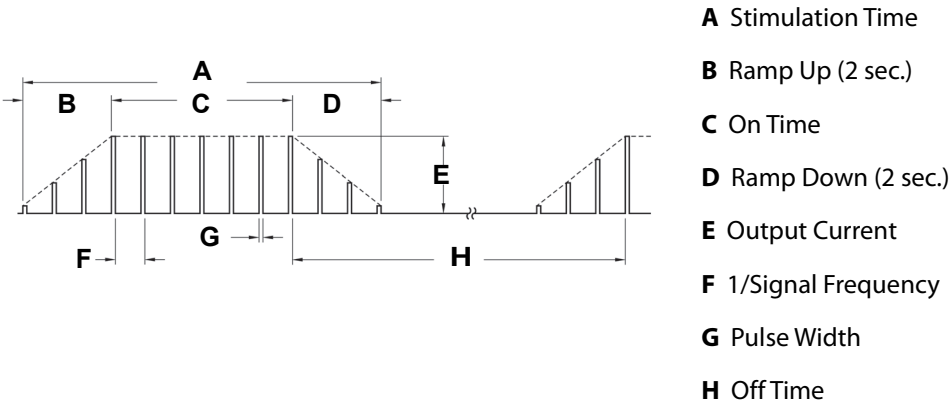
The generator can be interrogated to determine the present settings of the stimulation parameters.

3.3.2.1.8. Programmable parameters

A graphic representation of stimulation (Figure 12) depicts the relationship of the programmable parameters. Each parameter can be independently programmed, thereby offering multiple setting combinations from which the physician may select optimal stimulation for the patient.

Figure 12 shows that the output pulse can be varied both by amplitude (output current) and duration (pulse width). The number of output pulses delivered per second determines the frequency.

Figure 12. Stimulation (Frequencies < 10 Hz do not ramp)



3.3.2.1.9. Duty cycle

The percentage of time the generator is stimulating is called a “duty cycle.” A duty cycle is calculated by dividing the stimulation time (programmed ON time plus, if frequency is ≥ 10 Hz, 2 seconds of ramp-up time and 2 seconds of ramp-down time) by the sum of the ON and OFF times. The various parameter settings for stimulation are listed in “Specifications and Product Information”.

⚠ Caution: *Possible nerve damage with ON time > OFF time*— Excessive stimulation has resulted in degenerative nerve damage in laboratory animals. One component of excessive stimulation is when ON time exceeds OFF time, which can be produced by continuous or frequent magnet activation (> 8 hours). It is recommended that stimulation at these combinations of ranges be avoided.

Table 11 shows duty cycles for typical ON time and OFF time settings.

Table 11. Duty Cycles for Various ON and OFF Time Settings

	OFF Time (min)					
	0.2	0.3	0.5	0.8	1.1	1.8
ON Time (sec)	Duty Cycles* (% ON Time)					
7	58	44	30	20	15	10
14	69	56	41	29	23	15
21	76	64	49	36	29	19
30	81	71	57	44	35	25
60	89	82	71	59	51	38

* A duty cycle is calculated by dividing stimulation time (programmed ON time plus 2 seconds of ramp-up time and 2 seconds of ramp-down time) by the sum of the ON time and the OFF time.

* The duty cycles in gray are not recommended as they represent parameter combinations with ON Time > OFF Time.

3.3.2.1.10. Parameter settings and battery life

When selecting a combination of parameter settings for stimulation, the physician should also consider that some combinations would decrease battery life faster than others.



Note: See “Generator battery longevity”.

3.3.2.2 VNS Therapy magnets

There are four possible uses of the magnet:

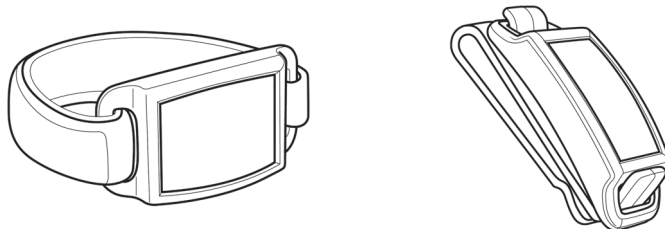
- **Epilepsy Only** — To provide on-demand stimulation as an attempt to abort or de-intensify an oncoming seizure
- To temporarily inhibit stimulation
- To reset the generator (in combination with the programming wand)
- **Epilepsy Only** — To test daily the functioning of the generator, it is recommended that patients be instructed to use the magnet to activate stimulation. During an aura or at the start of a seizure, magnet activation may be initiated by the patient, a companion, or the physician by applying or passing the magnet over the generator to activate a reed switch in the generator’s electronic circuitry. This action changes the generator from Normal Mode to Magnet Mode.



Caution: Possible nerve damage with ON time > OFF time— Excessive stimulation has resulted in degenerative nerve damage in laboratory animals. One component of excessive stimulation is when ON time exceeds OFF time, which can be produced by continuous or frequent magnet activation (> 8 hours). It is recommended that stimulation at these combinations of ranges be avoided.

Two identical VNS Therapy magnets (see Figure 13), each providing a minimum of 50 gauss at 1 inch, are supplied by LivaNova. The watch-style accessory attaches the magnet to the wrist with a wristband and the pager-style accessory holds the magnet in a belt clip with a quick-release mechanism like a pager.

Figure 13. Magnet Accessories



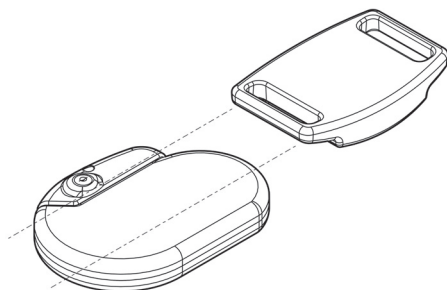
**Watch-Style
(wristband)**

**Pager-Style
(belt clip)**

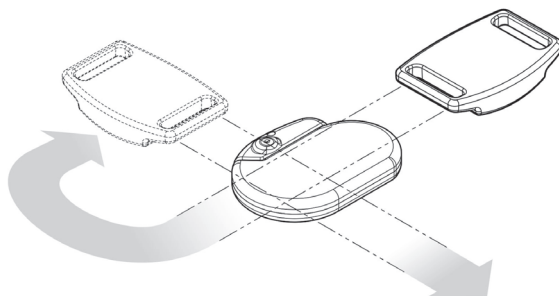
Epilepsy Only — The proper orientation and motion for initiating magnet activation is shown in Figure 14. The magnet is shown without the belt clip or wristband to illustrate the proper orientation of the magnet to the generator.

Figure 14. Initiate Magnet Activation (*Epilepsy Only*)

Standard Magnet Activation



Optional Cross-Pattern Magnet Activation



Note: The magnet swipe patterns shown above can be used with all generators.



Caution: To activate or stop stimulation, the label side of the magnet should face the generator.

3.3.2.2.1. Service life of magnet

All magnets may lose their effectiveness over time. VNS Therapy magnets contain a high-power magnet that is surrounded by a plastic casing in the shape of a watch. These magnets should be operated and stored at temperatures ranging from - 20 °C (- 4 °F) to + 55 °C (+ 131 °F). With normal use, they should remain powerful for approximately 3 years. Avoid dropping the magnets or storing them near other magnets.

3.3.2.2.2. Magnet activation technique (*epilepsy only*)

To initiate stimulation, apply or pass the magnet over the generator for at least 1 second, and then immediately remove it from the area over the generator. Removal of the magnet causes the generator to operate in Magnet Mode, delivering a single stimulation at the programmed magnet pulse width, magnet current, and magnet signal ON time settings. The frequency is the programmed value for Normal Mode. A Magnet Mode stimulation will always override any Normal Mode programmed stimulation, even if the Magnet Mode output current is set to 0 mA. If Magnet Mode stimulation is not desired, the Magnet Mode output current may be programmed to 0 mA.

It is recommended that tests of the magnet output be performed while the patient is still in the physician's office to ensure tolerability of the magnet output.

3.3.2.2.3. Inhibit generator output with the magnet

Application of the magnet during stimulation will inhibit the output. In addition, holding the magnet in place for at least 65 seconds will terminate any ongoing Normal Mode stimulation. After the magnet is removed, Normal Mode operation will resume with stimulation when one complete OFF time has elapsed.

 **Caution:** If stimulation becomes painful, the patient should be instructed to stop the stimulation with the magnet.

In the unlikely event of continuous stimulation or other malfunction, the patient must be advised to apply the magnet, secure it in place, and immediately notify their physician.

 **Note:** See "Adverse Events" in the indication-specific information chapters.

3.3.2.2.4. Reset the microprocessor with the magnet and the programming wand

The VNS Therapy System allows the generator microprocessor to be reset in the event of a malfunction. Resetting is necessary only in the rare case of microprocessor memory malfunction, which might be caused by conditions described in the *Introduction to the VNS Therapy System* chapter. Resetting the microprocessor may be appropriate when the generator and the programming wand are unable to communicate.

 **Caution:** *Generator reset*— When the generator is reset, its stimulation output is disabled (0 mA); however, all settings and device history are preserved. After a successful reset, the generator stimulation output may be re-enabled to resume operation at the previously programmed settings.

For suggestions in solving communication difficulties, see "Troubleshooting" in the programming system physician's manuals.

For instructions on how to reset the microprocessor, see the programming system physician's manuals. After a microprocessor reset has been attempted, wait at least 30 seconds before establishing communication with the programming software. It is recommended, except in cases of a medical emergency, that the physician consult a LivaNova technical representative before a reset is performed.

3.3.2.3 *Device history*

The Device History consists of the generator serial number, model number, the patient code (usually three initials), implantation date, and other information pertinent to diagnostic and programming events. Use the programming software to access and view Device History information.

3.3.2.4 *Device diagnostics*

Information from device diagnostic tests can help the physician determine whether the:

- Generator is operating properly before it is implanted
- Generator output current is being delivered at the programmed value
- Generator is operating in Normal or Magnet Mode
- Lead impedance is within an acceptable range

 **Note:** For details on available diagnostic tests, see the programming system physician's manuals.

3.3.2.4.1. System Diagnostics test

The System Diagnostics evaluates the lead impedance of the VNS Therapy System, as well as the generator's ability to deliver the programmed stimulation. When the output current is programmed to any value greater than 0 mA, the generator will deliver a pulse at 0.25 mA, 130 µsec to assess lead impedance. The generator will then deliver the programmed output stimulus. The programming software will report the lead impedance and whether the programmed stimulus was delivered.

If the output current is programmed to 0 mA, as would be expected during the implantation procedure, the generator will deliver a pulse at 0.25 mA, 130 µsec followed by delivery of stimulation at 1 mA, 20 Hz, 500 µsec for approximately 14 seconds. Likewise, the programming software will report the lead impedance and whether the parameters listed here are deliverable.

3.3.2.4.2. Reasons for high or low lead impedance readings

The System Diagnostics evaluates lead impedance of the VNS Therapy System. The lead impedance measurement is performed with a single pulse at 0.25 mA, 130 µsec. Additionally, the generator automatically performs a lead impedance measurement once every 24 hours. If a HIGH or LOW impedance is detected, the user is notified after an interrogation with the programming software.

High lead impedance is defined as any value ≥ 5300 Ohms. Low lead impedance is defined as any value ≤ 600 Ohms.



Caution: Possible causes of high lead impedance readings are thought to include: lead discontinuity, lead disconnection from the generator, fibrosis between the nerve and the electrode, electrode detachment from the nerve, or a defective generator.



Caution: Possible causes of low lead impedance readings are thought to include: short-circuit condition within the lead or a defective generator.

3.3.2.4.3. High lead impedance: possible implications

High lead impedance (≥ 5300 Ohms), in the absence of other device-related complications, is not an indication of a lead or generator malfunction. High lead impedance in combination with the patient's failure to feel even the maximum output stimulus may indicate a lead wire fracture or other type of electrical discontinuity in the lead. Patients experiencing high lead impedance, no sensation of maximum output stimulation, and an increase in seizures/depressive symptoms should be further evaluated for possible lead replacement.



Note: For additional instructions on how to perform the System Diagnostics, see the programming system physician's manuals.



Note: To troubleshoot high or low impedance, see "Troubleshooting" in the programming system physician's manuals.

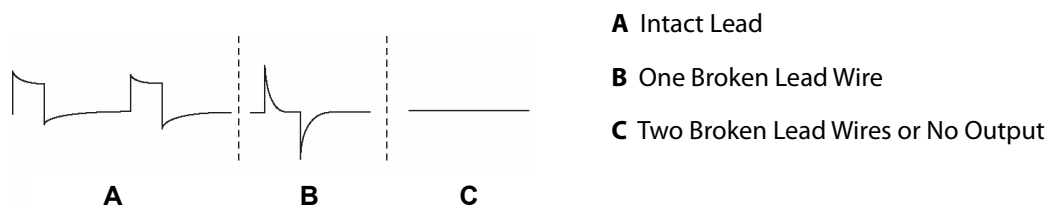
3.3.2.4.4. Low lead Impedance: possible implications

Low lead impedance (≤ 600 Ohms) likely indicates the existence of a short-circuit condition, although an impedance value of greater than 600 Ohms does not exclude the possibility. A sudden decrease in impedance value in combination with device-related complications (e.g., increase in seizures/depressive symptoms or painful stimulation; patient perception of feeling erratic, limited, or no stimulation) may also indicate a short-circuit condition in the lead.

3.3.2.4.5. Stimulus waveform analysis

Either evoked potential monitoring equipment or an oscilloscope can be used to analyze the stimulus waveform from the neck for verification of an electrical discontinuity. A differentiated waveform with narrowed pulses or no waveform at all can confirm a discontinuity. Figure 15 shows characteristic waveforms obtained from skin electrodes for a lead that is intact and for a lead that has a fracture in one or both wires. In addition, lead discontinuities can sometimes be identified on x-ray of the implant site.

Figure 15. Typical Waveforms Obtained from Skin Electrodes



3.3.2.5 *Delivery of programmed output current*

3.3.2.5.1. **LOW as output current**

If the diagnostic tests indicate LOW output current, the generator may not be delivering the programmed output current. Reasons for failure to deliver the programmed output current include high programmed output current and high lead impedance. The maximum deliverable output current, according to Ohm's Law, equals the maximum output voltage (approximately 12 V) divided by the lead impedance.

3.3.2.5.2. **Reprogram to a lower current**

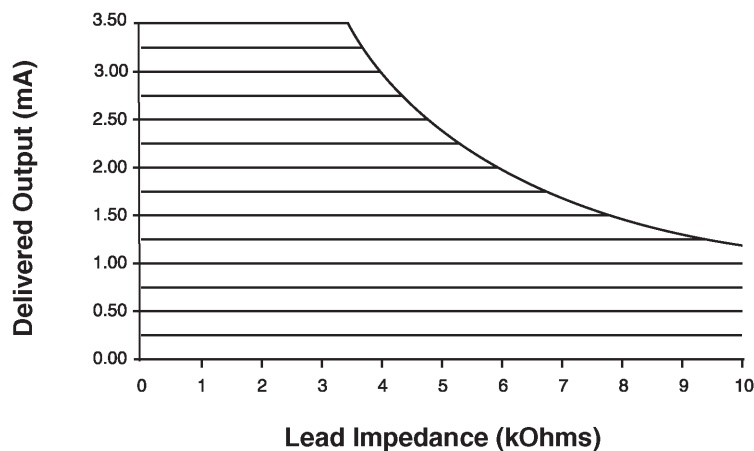
If the generator is failing to deliver the programmed output current, the physician can reprogram the device to a lower output current and attempt to compensate for the decrease in delivered energy by widening the pulse width. For example, if the diagnostics read LOW for a generator programmed at 2.5 mA, 30 Hz, 500 μ sec with 30 seconds ON time, then the parameters may be changed by lowering the output current to 2 mA and widening the pulse width to 750 μ sec.

3.3.2.6 *Charge delivered per pulse*

3.3.2.6.1. **Output current x pulse width = charge delivered per pulse**

The charge delivered per pulse is the most important parameter in evaluating stimulation output. It is defined as a microcoulomb (μ C), which is the product of current and time—that is, the output current (mA) multiplied by the pulse width (msec). Figure 16 shows the relationship of delivered output current (mA) to lead impedance for a 1000 μ sec pulse with output currents from 0 to 3.5 mA.

Figure 16. Relationship of Delivered Output Current to Lead Impedance



3.3.2.7 *Generator battery longevity*

3.3.2.7.1. **Battery longevity and programmed setting choices**

The anticipated longevity of the generator battery varies, depending on the choice of programmed settings. Higher output currents, frequencies, pulse widths, and duty cycles generally deplete the battery over a shorter period of time than lower settings. Generally, the increase in battery depletion rate is proportional to the increase in the programmed setting.



Caution: *Undeliverable output currents*— Programming the generator to a high output current that cannot be delivered due to a high lead impedance may disproportionately increase the battery depletion rate and should be avoided.

Other factors, such as the lead impedance and, if applicable, magnet usage, also affect the anticipated longevity of the generator battery. The anticipated battery longevity decreases as lead impedance increases. Although 1.5 k to 3 kOhms may be a typical lead impedance at implantation, the impedance may increase to 3 k to 5 kOhms during the life of the implant.

For Models 103 and 104, the approximate battery longevity predicted is greater than 6 years at programmed settings of 20 Hz with a 500-μsec pulse width and 2-mA output current, a lead impedance of 4 kOhms, and a duty cycle of 10 percent. Appendix A — Model 103/104 Battery Longevity and Programmed Setting Choices provides estimated battery lifetimes under a variety of stimulation conditions, including lead impedance. Because of the number of possible parameter combinations, it is impractical to provide the projected life for all possible combinations. The tables should not be used to predict battery End of Service (EOS), but they give some indication of the effect of various parameter changes on battery life and can be used to assist in the selection of parameter settings. They also indicate that battery life can be maximized at low duty cycles and low frequencies (10 to 20 Hz) for stimulation.

3.3.2.7.2. Battery status indicators

The programming software displays a battery indicator for the generator similar to an indicator that may be found in cell phones. The visual indicator illustrates the approximate remaining battery capacity.



Caution: *Battery evaluation at cold temperatures*—Low storage temperatures may affect the battery status indicators. In such cases, the battery status indicators should be re-evaluated using the System Diagnostics or Generator Diagnostics after the generator has been at room or body temperature for 30 minutes.

The programming software will display warning messages after an interrogation or programming of the generator if the battery has been depleted to a level where action is recommended due to approaching or reaching End of Service (EOS). See the programming system physician's manuals for additional information in these indicators.

3.3.3 Generator Replacement

All VNS Therapy generators eventually require surgical replacement as a result of battery depletion. Generator replacement does not, of itself, require lead replacement unless a lead discontinuity is suspected. Generator replacement or removal requires dissection to the generator's pocket, with care being taken not to damage or cut the lead. The entire surgical procedure generally requires about 1 hour.

3.3.4 Signs of End of Service

The most common reason for the absence of stimulation is battery depletion, although there may be other reasons. When end of service (EOS) occurs, the generator will disable stimulation and no output will be delivered. If the generator is not explanted or replaced at EOS, the battery voltage will continue to gradually decrease and communication with the generator may not be possible.



Caution: Generator EOS may result in increased frequency, intensity, or duration of signs and symptoms of the patient's disorder, in some cases to levels greater than those reported before stimulation.

3.3.5 Replacement Based on Battery Status Indicators

The generators and programming software have battery status indicators. These indicators provide warnings that a generator battery should be monitored more frequently, is near EOS, or has reached EOS. Once these warning messages appear, see recommendations in the programming system physician's manuals.



Note: See "Battery status indicators".



Caution: *Prompt generator replacement*—Prompt replacement of the generator at or before EOS is recommended. Prompt replacement may help minimize any possible relapse.



Caution: *Explanted generator*—A generator explanted for any reason should not be reimplanted. An explanted generator should be returned to LivaNova. (For instructions on returning an explanted generator, see the *Introduction to the VNS Therapy System* chapter.)

4 Technical Information — 105 Generator

4.1 Detailed Device Description

4.1.1 Physical Characteristics

The titanium case of the VNS Therapy AspireHC[®] Model 105 Generator is hermetically sealed and leak-rate tested. Specially designed feedthroughs using platinum conductors form the electrical connection from the connector blocks to the circuitry through the hermetically sealed enclosure. The Model 105 accepts the single-pin lead.

4.1.2 Biological Compatibility

Materials exposed to the subcutaneous environment are biologically compatible. All of these materials have a long history in medical implants and have been found to be tissue compatible.

4.1.3 Power Source

The power source for the Model 105 generators is a Wilson Greatbatch Ltd, Model 2075, lithium carbon monofluoride battery with an open-circuit voltage of 3.3 V. The battery's maximum available capacity is approximately 1.7 Amp-hours. The self-discharge reduces the capacity by less than 1 percent per year. The voltage in this battery gradually decreases as the battery nears its end of service (EOS).

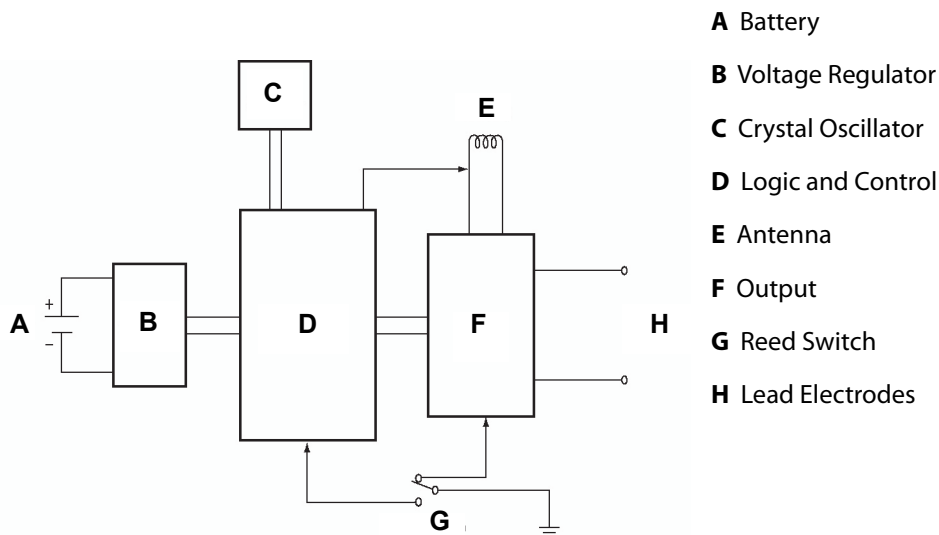
4.1.4 Circuitry

The generator uses complementary metal oxide semiconductor (CMOS) integrated circuits, including a microprocessor. The circuitry is functionally represented in Figure 17.

For descriptive purposes, circuitry of the generator can be divided into the following major functional sections:

Voltage regulator	Regulates the system power supply
Crystal oscillator	Provides a timing reference
Logic and control	Controls overall generator function; receives and implements programming commands; collects and stores telemetry information
Output	Develops and modulates signals delivered to the lead
Antenna	Receives programming signals; transmits telemetry information to the programming wand
Reed switch	Provides a mechanism to place the generator in Magnet Mode or to inhibit its output

Figure 17. Generator Circuitry



4.1.5

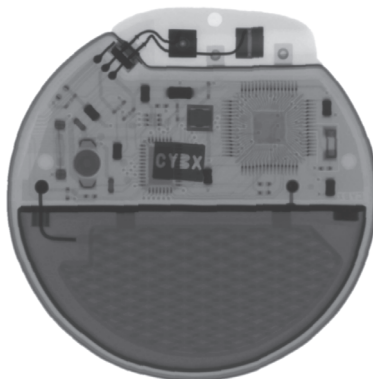
Identification

The generator can be identified on an x-ray and will appear as shown in Figure 18. The serial number and model number of the generator are marked on its titanium case, but do not appear on the x-ray.

The serial number, model number, and year of manufacture depending upon programming software version can be identified by interrogating the generator with the programming software.

Note: See the programming system physician's manuals for details.

Figure 18. X-ray Identification



Note: CYBX is the identification code for Cyberonics.

4.2 VNS Therapy System Compatibility

Model 105 is compatible with these system components:

Component	Model
Lead	single-pin
Programming System	201 wand /250 software
	2000 Wand / 3000 software
Tunnelers	402
Accessory Pack	502
Magnets	220

4.3 Directions for Use

4.3.1 Specifications and Product Information

The specifications and product information for the VNS Therapy Model 105 generator is presented in Table 12.

Table 12. Specifications and Product Information

Stimulation Parameters	Available Parameter Settings
Output current	0-3.5 mA in 0.25-mA steps (± 0.1 mA or $\pm 10\%$; whichever is greater)
Signal frequency	1, 2, 5, 10, 15, 20, 25, 30 Hz $\pm 6\%$
Pulse width	130, 250, 500, 750, 1000 μ sec $\pm 10\%$
Signal ON time	7, 14, 21, 30, 60 sec Normal Mode (+ 7sec / - 15%) Magnet Mode (+ 15% / - 7sec)
Signal OFF time	0.2, 0.3, 0.5, 0.8, 1.1, 1.8, 3 min, and 5 to 180 min (5 to 60 in 5-min steps; 60 to 180 in 30-min steps) + 4.4 / - 8.4 sec or $\pm 1\%$, whichever is greater
Magnet activation	Provided by magnet application (output current, pulse width, and signal ON time may be independently programmed for this purpose)
Reset parameters	Settings are unchanged, but output is disabled (0 mA)
Telemetry Reports	
Device History Report	Patient ID, implant date, model number, serial number, magnet activations, total ON time, total operating time, and manufacturing date
Device Diagnostic Report	Patient ID, model ID, serial number, implant date, communication status, output current status, measured current delivered, lead impedance, and battery status indicators (IFI, N EOS, EOS)
Power Source	
Battery	Wilson Greatbatch Ltd., Model 2075
Chemistry	Lithium carbon monofluoride
Voltage	3.3 V, open circuit
Rated capacity	1.7 Amp-hour
Self-discharge rate	< 1% per year

Physical Characteristics—Materials	
Case	Titanium, hermetically sealed
Header	Polyurethane — Tecothane™ TT-1075D-M Thermoplastic
Lead connector blocks	Stainless steel
Setscrew plug(s)	Silicone*
Measurements (Typical)	
Lead receptacle(s)	0.126 in (3.2 mm) nominal
Dimensions	2.0 in x 2.0 in x 0.27 in (52 mm x 52 mm x 6.9 mm)
Weight	0.88 oz (25 g)
Connector Retention Strength	
With VNS Therapy lead	> 10 N

* No component of the VNS Therapy System is made with natural rubber latex.

4.3.2 Operating Characteristics

4.3.2.1 Communicating with the VNS Therapy System

4.3.2.1.1. Programming software

The generator can be programmed with the VNS Therapy programming software.

 **Note:** For more information, including a list of computers that have been qualified for use with this software, see the programming system physician's manual.

The programming software is used on a computer, supplied by LivaNova, that is dedicated only to programming the VNS Therapy System.

4.3.2.1.2. Programming wand

A programming wand connected to a compatible computer running the programming software is needed to communicate with the generator.

 **Note:** For proper placement of the programming wand, connection of the wand to the computer, and use of the wand, see the programming system physician's manual.

4.3.2.1.3. Prompts and messages

After the program has been initiated, the software screens display prompts and messages to aid in communicating with the generator.

4.3.2.1.4. Communication

The generator “listens” for a communication signal from the programming wand. Communication usually takes between 1 and 4 seconds, but may be prolonged or interrupted in the presence of electromagnetic interference (EMI). The generator listens for and implements interrogations, parameter programming instructions, requests for Device Diagnostics testing, and Device History inquiries.

In response, the generator transmits information on the stimulation parameter settings, changes its parameter settings, responds to requests for Device Diagnostics testing, and provides device histories, respectively. Each time these data are transmitted by the generator, they are saved by the programming software to a database.



Note: For details on viewing database information, see the programming system physician's manuals.

In addition to the programming software and programming wand combination, the magnet can be used for one-way communication to the generator by activating a reed switch in the electronic circuitry. The magnet can be used to initiate stimulation, temporarily inhibit stimulation, perform Magnet Mode diagnostics, and reset the generator.

4.3.2.1.5. Normal Mode

After the generator has been programmed, the stimulation will repeat in accordance with the programmed ON and OFF cycle (Normal Mode) until the generator receives communication from the VNS Therapy programming system or until it is inhibited with the magnet. Immediately after successful programming, the generator delivers a programmed stimulation that enables the programmer to evaluate patient response. If programming is performed during stimulation, stimulation will be terminated; after programming, stimulation will begin, using the revised settings.

4.3.2.1.6. Magnet Mode

Magnet Mode produces on-demand stimulation for the programmed magnet ON time. Stimulation is initiated by applying or passing the magnet over the generator for at least 1 second and then immediately removing it from the area over the generator. Stimulation is delivered after the magnet is removed. The Magnet Mode uses the same frequency as the Normal Mode, but the output current, pulse width, and signal ON time are independently programmable. The Directions for Use are equivalent for epilepsy and depression, with the following exceptions:

- For patients with depression, the Magnet Mode output current should always be programmed at 0 mA, the setting at which the generator is shipped from LivaNova.
- Use of the Magnet Mode is limited to patients with epilepsy. Patients with epilepsy or their caregivers pass the magnet over the implanted generator to activate on-demand delivery of a single train of vagus nerve stimulation and help abort or diminish a seizure.
- Magnet Mode is not used for patients with depression.

4.3.2.1.7. Generator interrogation

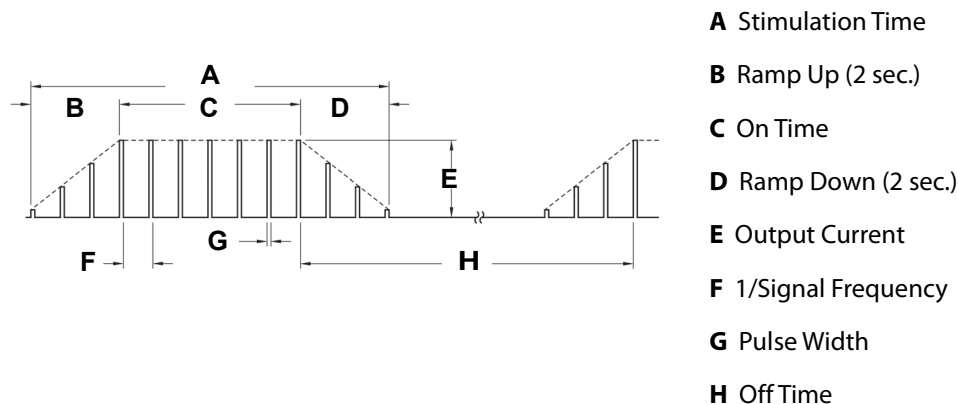
The generator can be interrogated to determine the present settings of the stimulation parameters.

4.3.2.1.8. Programmable parameters

A graphic representation of stimulation (Figure 19) depicts the relationship of the programmable parameters. Each parameter can be independently programmed, thereby offering multiple setting combinations from which the physician may select optimal stimulation for the patient.

Figure 19 shows that the output pulse can be varied both by amplitude (output current) and duration (pulse width). The number of output pulses delivered per second determines the frequency.

Figure 19. Stimulation (Frequencies < 10 Hz do not ramp)



4.3.2.1.9. Duty cycle

The percentage of time the generator is stimulating is called a “duty cycle.” A duty cycle is calculated by dividing the stimulation time (programmed ON time plus, if frequency is ≥ 10 Hz, 2 seconds of ramp-up time and 2 seconds of ramp-down time) by the sum of the ON and OFF times. The various parameter settings for stimulation are listed in “Specifications and Product Information”.

⚠ Caution: Possible nerve damage with ON time > OFF time— Excessive stimulation has resulted in degenerative nerve damage in laboratory animals. One component of excessive stimulation is when ON time exceeds OFF time, which can be produced by continuous or frequent magnet activation (> 8 hours). It is recommended that stimulation at these combinations of ranges be avoided.

Table 13 shows duty cycles for typical ON time and OFF time settings.

Table 13. Duty Cycles for Various ON and OFF Time Settings

	OFF Time (min)					
	0.2	0.3	0.5	0.8	1.1	1.8
ON Time (sec)	Duty Cycles* (% ON Time)					
7	58	44	30	20	15	10
14	69	56	41	29	23	15
21	76	64	49	36	29	19
30	81	71	57	44	35	25
60	89	82	71	59	51	38

* A duty cycle is calculated by dividing stimulation time (programmed ON time plus 2 seconds of ramp-up time and 2 seconds of ramp-down time) by the sum of the ON time and the OFF time.

* The duty cycles in gray are not recommended as they represent parameter combinations with ON Time > OFF Time.

4.3.2.1.10. Parameter settings and battery life

When selecting a combination of parameter settings for stimulation, the physician should also consider that some combinations would decrease battery life faster than others.

 **Note:** See “Generator battery longevity”.

4.3.2.2 VNS Therapy magnets

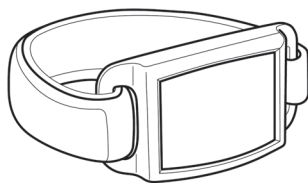
There are four possible uses of the magnet:

- **Epilepsy Only** — To provide on-demand stimulation as an attempt to abort or de-intensify an oncoming seizure
- To temporarily inhibit stimulation
- To reset the generator (in combination with the programming wand)
- **Epilepsy Only** — To test daily the functioning of the generator, it is recommended that patients be instructed to use the magnet to activate stimulation. During an aura or at the start of a seizure, magnet activation may be initiated by the patient, a companion, or the physician by applying or passing the magnet over the generator to activate a reed switch in the generator’s electronic circuitry. This action changes the generator from Normal Mode to Magnet Mode.

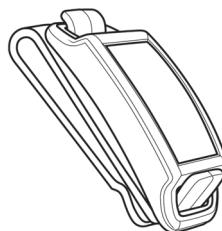
 **Caution:** Possible nerve damage with ON time > OFF time— Excessive stimulation has resulted in degenerative nerve damage in laboratory animals. One component of excessive stimulation is when ON time exceeds OFF time, which can be produced by continuous or frequent magnet activation (> 8 hours). It is recommended that stimulation at these combinations of ranges be avoided.

Two identical VNS Therapy magnets (see Figure 20), each providing a minimum of 50 gauss at 1 inch, are supplied by LivaNova. The watch-style accessory attaches the magnet to the wrist with a wristband and the pager-style accessory holds the magnet in a belt clip with a quick-release mechanism like a pager.

Figure 20. Magnet Accessories



**Watch-Style
(wristband)**

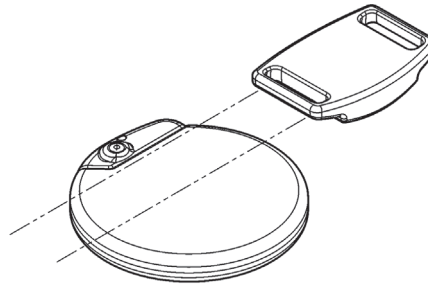


**Pager-Style
(belt clip)**

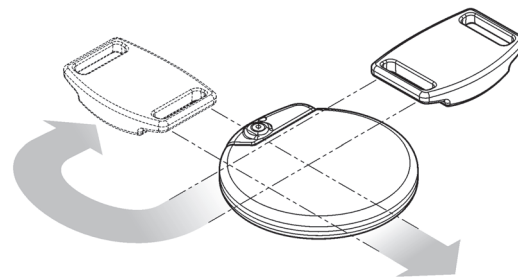
Epilepsy Only — The proper orientation and motion for initiating magnet activation is shown in Figure 21. The magnet is shown without the belt clip or wristband to illustrate the proper orientation of the magnet to the generator.

Figure 21. Initiate Magnet Activation (*Epilepsy Only*)

Standard Magnet Activation



Optional Cross-Pattern Magnet Activation



Caution: To activate or stop stimulation, the label side of the magnet should face the generator.

4.3.2.2.1. Service life of magnet

All magnets may lose their effectiveness over time. VNS Therapy magnets contain a high-power magnet that is surrounded by a plastic casing in the shape of a watch. These magnets should be operated and stored at temperatures ranging from - 20 °C (- 4 °F) to + 55 °C (+ 131 °F). With normal use, they should remain powerful for approximately 3 years. Avoid dropping the magnets or storing them near other magnets.

4.3.2.2.2. Magnet activation technique (*epilepsy only*)

To initiate stimulation, apply or pass the magnet over the generator for at least 1 second and then immediately remove it from the area over the generator. A cross-pattern swipe, as shown in Figure 21, may also be used by the patient or caregiver to activate the Magnet Mode if difficulty is encountered with a single pass of the magnet. Removal of the magnet causes the generator to operate in Magnet Mode, delivering a single stimulation at the programmed magnet pulse width, magnet current, and magnet signal ON time settings. The frequency is the programmed value for Normal Mode. A Magnet Mode stimulation will always override any Normal Mode programmed stimulation, even if the Magnet Mode output current is set to 0 mA. If Magnet Mode stimulation is not desired, the Magnet Mode output current may be programmed to 0 mA.



Caution: The cross-pattern swipe technique may cause duplicate magnet activation entries to be shown in the programming software database. This is an expected occurrence due to device design and is not considered a device malfunction. See the programming system physician's manuals for more details.

It is recommended that tests of the magnet output be performed while the patient is still in the physician's office to ensure tolerability of the magnet output.

4.3.2.2.3. Inhibit generator output with the magnet

Application of the magnet during stimulation will inhibit the output. In addition, holding the magnet in place for at least 65 seconds will terminate any ongoing Normal Mode stimulation. After the magnet is removed, Normal Mode operation will resume with stimulation when one complete OFF time has elapsed.



Caution: If stimulation becomes painful, the patient should be instructed to stop the stimulation with the magnet.

In the unlikely event of continuous stimulation or other malfunction, the patient must be advised to apply the magnet, secure it in place, and immediately notify their physician.



Note: See "Adverse Events" in the indication-specific information chapters.

4.3.2.2.4. Reset the microprocessor with the magnet and the programming wand

The VNS Therapy System allows the generator microprocessor to be reset in the event of a malfunction. Resetting is necessary only in the rare case of microprocessor memory malfunction, which might be caused by conditions described in the *Introduction to the VNS Therapy System* chapter. Resetting the microprocessor may be appropriate when the generator and the programming wand are unable to communicate.



Caution: *Generator reset*— When the generator is reset, its stimulation output is disabled (0 mA); however, all settings and device history are preserved. After a successful reset, the generator stimulation output may be re-enabled to resume operation at the previously programmed settings.

For suggestions in solving communication difficulties, see "Troubleshooting" in the programming system physician's manuals.

For instructions on how to reset the microprocessor, see the programming system physician's manuals. After a microprocessor reset has been attempted, wait at least 30 seconds before establishing communication with the programming software. It is recommended, except in cases of a medical emergency, that the physician consult a LivaNova technical representative before a reset is performed.

4.3.2.3 Device history

The Device History consists of the generator serial number, model number, the patient code (usually three initials), implantation date, year of manufacture, and other information pertinent to diagnostic and programming events. Use the programming software to access and view Device History information.

4.3.2.4 **Device diagnostics**

Information from device diagnostic tests can help the physician determine whether the:

- Generator is operating properly before it is implanted
- Generator output current is being delivered at the programmed value
- Generator is operating in Normal or Magnet Mode
- Lead impedance is within an acceptable range



Note: For details on available diagnostic tests, see the programming system physician's manuals.

4.3.2.4.1. **System Diagnostics test**

The System Diagnostics evaluates the lead impedance of the VNS Therapy System, as well as the generator's ability to deliver the programmed stimulation. When the output current is programmed to any value greater than 0 mA, the generator will deliver a pulse at 0.25 mA, 130 μ sec to assess lead impedance. The generator will then deliver the programmed output stimulus. The programming software will report the lead impedance and whether the programmed stimulus was delivered.

If the output current is programmed to 0 mA, as would be expected during the implantation procedure, the generator will deliver a pulse at 0.25 mA, 130 μ sec followed by delivery of stimulation at 1 mA, 20 Hz, 500 μ sec for approximately 14 seconds. Likewise, the programming software will report the lead impedance and whether the parameters listed here are deliverable.

4.3.2.4.2. **Reasons for high or low lead impedance readings**

The System Diagnostics evaluates lead impedance of the VNS Therapy System. The lead impedance measurement is performed with a single pulse at 0.25 mA, 130 μ sec. Additionally, the generator automatically performs a lead impedance measurement once every 24 hours. If a HIGH or LOW impedance is detected, the user is notified after an interrogation with the programming software.

High lead impedance is defined as any value ≥ 5300 Ohms. Low lead impedance is defined as any value ≤ 600 Ohms.



Caution: Possible causes of high lead impedance readings are thought to include: lead discontinuity, lead disconnection from the generator, fibrosis between the nerve and the electrode, electrode detachment from the nerve, or a defective generator.



Caution: Possible causes of low lead impedance readings are thought to include: short-circuit condition within the lead or a defective generator.

4.3.2.4.3. **High lead impedance: possible implications**

High lead impedance (≥ 5300 Ohms), in the absence of other device-related complications, is not an indication of a lead or generator malfunction. High lead impedance in combination with the patient's failure to feel even the maximum output stimulus may indicate a lead wire fracture or other type of electrical discontinuity in the lead. Patients experiencing high lead

impedance, no sensation of maximum output stimulation, and an increase in seizures/depressive symptoms should be further evaluated for possible lead replacement.

Note: For additional instructions on how to perform the System Diagnostics, see the programming system physician's manuals.

Note: To troubleshoot high or low impedance see "Troubleshooting" in the programming system physician's manuals.

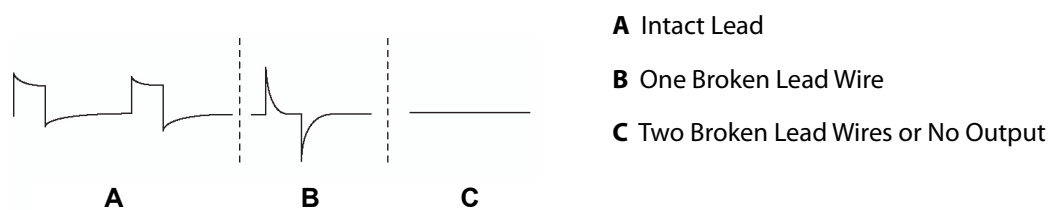
4.3.2.4.4. Low lead Impedance: possible implications

Low lead impedance (≤ 600 Ohms) likely indicates the existence of a short-circuit condition, although an impedance value of greater than 600 Ohms does not exclude the possibility. A sudden decrease in impedance value in combination with device-related complications (e.g., increase in seizures/depressive symptoms or painful stimulation; patient perception of feeling erratic, limited, or no stimulation) may also indicate a short-circuit condition in the lead.

4.3.2.4.5. Stimulus waveform analysis

Either evoked potential monitoring equipment or an oscilloscope can be used to analyze the stimulus waveform from the neck for verification of an electrical discontinuity. A differentiated waveform with narrowed pulses or no waveform at all can confirm a discontinuity. Figure 22 shows characteristic waveforms obtained from skin electrodes for a lead that is intact and for a lead that has a fracture in one or both wires. In addition, lead discontinuities can sometimes be identified on x-ray of the implant site.

Figure 22. Typical Waveforms Obtained from Skin Electrodes



4.3.2.5 Delivery of programmed output current

4.3.2.5.1. LOW as output current

If the diagnostic tests indicate LOW output current, the generator may not be delivering the programmed output current. Reasons for failure to deliver the programmed output current include high programmed output current and high lead impedance. The maximum deliverable output current, according to Ohm's Law, equals the maximum output voltage (approximately 12 V) divided by the lead impedance.

4.3.2.5.2. Reprogram to a lower current

If the generator is failing to deliver the programmed output current, the physician can reprogram the device to a lower output current and attempt to compensate for the decrease in delivered energy by widening the pulse width. For example, if the diagnostics read LOW for

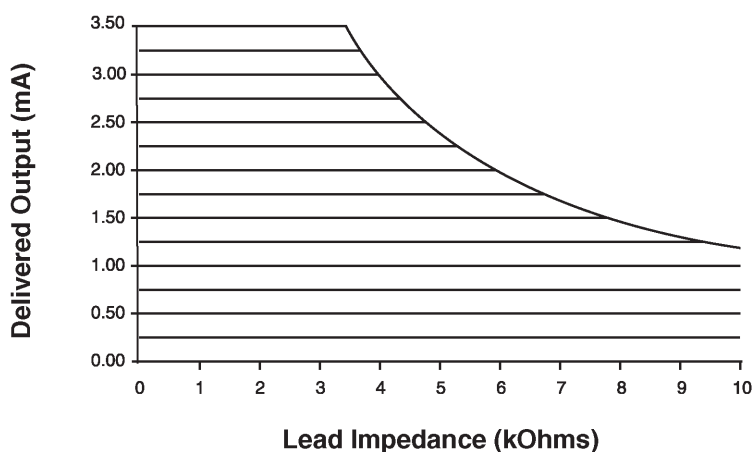
a generator programmed at 2.5 mA, 30 Hz, 500 μ sec with 30 seconds ON time, then the parameters may be changed by lowering the output current to 2 mA and widening the pulse width to 750 μ sec.

4.3.2.6 **Charge delivered per pulse**

4.3.2.6.1. **Output current x pulse width = charge delivered per pulse**

The charge delivered per pulse is the most important parameter in evaluating stimulation output. It is defined as a microcoulomb (μ C), which is the product of current and time—that is, the output current (mA) multiplied by the pulse width (msec). Figure 23 shows the relationship of delivered output current (mA) to lead impedance for a 1000 μ sec pulse with output currents from 0 to 3.5 mA.

Figure 23. Relationship of Delivered Output Current to Lead Impedance



4.3.2.7 **Generator battery longevity**

4.3.2.7.1. **Battery longevity and programmed setting choices**

The anticipated longevity of the generator battery varies, depending on the choice of programmed settings. Higher output currents, frequencies, pulse widths, and duty cycles generally deplete the battery over a shorter period of time than lower settings. Generally, the increase in battery depletion rate is proportional to the increase in the programmed setting.



Caution: *Undeliverable output currents*— Programming the generator to a high output current that cannot be delivered due to a high lead impedance may disproportionately increase the battery depletion rate and should be avoided.

Other factors, such as the lead impedance and, if applicable, magnet usage, also affect the anticipated longevity of the generator battery. The anticipated battery longevity decreases as lead impedance increases. Although 1.5 k to 3 kOhms may be a typical lead impedance at implantation, the impedance may increase to 3 k to 5 kOhms during the life of the implant.

Appendix B — Model 105 Battery Longevity and Programmed Setting Choices provides estimated battery lifetimes under a variety of stimulation conditions, including lead impedance. Because of the number of possible parameter combinations, it is impractical to provide the projected life for all possible combinations. The tables should not be used to predict battery end of service (EOS), but they give some indication of the effect of various parameter changes on battery life and can be used to assist in the selection of parameter

settings. They also indicate that battery life can be maximized at low duty cycles and low frequencies (10 to 20 Hz) for stimulation.

 **Note:** For more information, see the programming system physician's manuals.

4.3.2.7.2. Battery status indicators

The programming software displays a battery indicator similar to an indicator that may be found in cell phones. The visual indicator illustrates the approximate remaining battery capacity.

 **Caution:** *Battery evaluation at cold temperatures*—Low storage temperatures may affect the battery status indicators. In such cases, the battery status indicators should be re-evaluated using the System Diagnostics or Generator Diagnostics after the generator has been at room or body temperature for 30 minutes.

The programming software will display warning messages after an interrogation or programming of the generator if the battery has been depleted to a level where action is recommended due to approaching or reaching End of Service (EOS). See the programming system physician's manuals for additional information on these indicators.

4.3.3 Generator Replacement

All VNS Therapy generators eventually require surgical replacement as a result of battery depletion. Generator replacement does not, of itself, require lead replacement unless a lead discontinuity is suspected. Generator replacement or removal requires dissection to the generator's pocket, with care being taken not to damage or cut the lead. The entire surgical procedure generally requires about 1 hour.

4.3.4 Signs of End of Service

The most common reason for the absence of stimulation is battery depletion, although there may be other reasons. When end of service (EOS) occurs, the generator will disable stimulation and no output will be delivered. If the generator is not explanted or replaced at EOS, the battery voltage will continue to gradually decrease and communication with the generator may not be possible.

 **Caution:** Generator EOS may result in increased frequency, intensity, or duration of signs and symptoms of the patient's disorder, in some cases to levels greater than those reported before stimulation.

4.3.5 Replacement Based on Battery Status Indicators

The generators and programming software have battery status indicators. These indicators provide warnings that a generator battery should be monitored more frequently, is nearing EOS, or has reached EOS. Once these warning messages appear, see recommendations in the programming system physician's manuals.

 **Note:** See "Battery status indicators".

 **Caution:** *Prompt generator replacement*—Prompt replacement of the generator at or before EOS is recommended. Prompt replacement may help minimize any possible relapse.



Caution: *Explanted generator*—A generator explanted for any reason should not be reimplanted. An explanted generator should be returned to LivaNova. (For instructions on returning an explanted generator, see the *Introduction to the VNS Therapy System* chapter.)

5 Technical Information — 106 Generator

5.1 Detailed Device Description

5.1.1 Physical Characteristics

The titanium case of the VNS Therapy® AspireSR® Model 106 generator is hermetically sealed and leak-rate tested. Specially designed feedthroughs using platinum conductors form the electrical connection from the connector blocks to the circuitry through the hermetically sealed enclosure. The Model 106 accepts the single-pin lead.

5.1.2 Biological Compatibility

Materials exposed to the subcutaneous environment are biologically compatible. All of these materials have a long history in medical implants and have been found to be tissue compatible.

5.1.3 Power Source

The power source for the Model 106 generators is a Wilson Greatbatch Ltd, Model 2075, lithium carbon monofluoride battery with an open-circuit voltage of 3.3 V. The battery’s maximum available capacity is approximately 1.7 Amp-hours. The self-discharge reduces the capacity by less than 1 percent per year. The voltage in this battery gradually decreases as the battery nears its end of service (EOS).

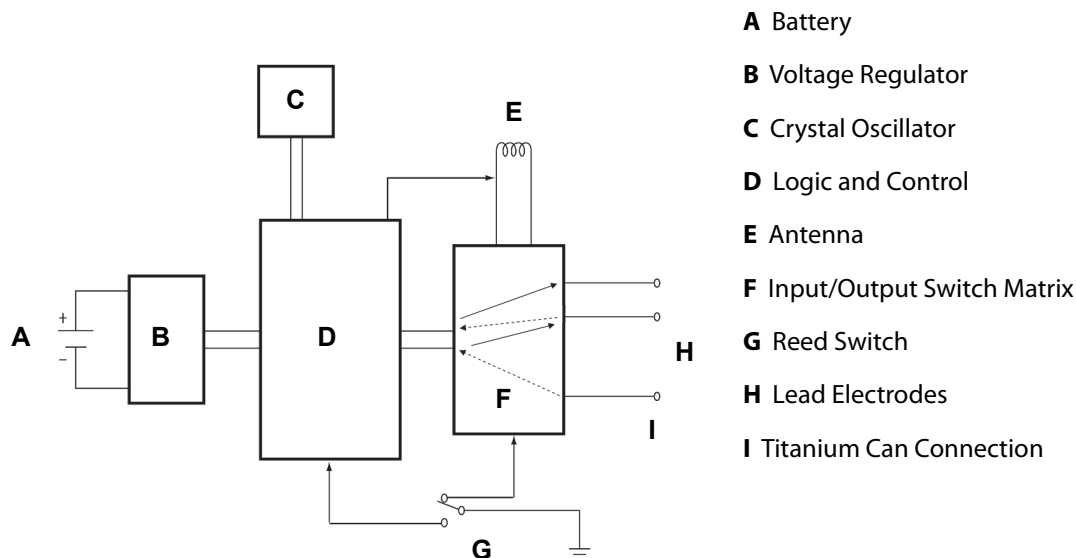
5.1.4 Circuitry

The generator uses complementary metal oxide semiconductor (CMOS) integrated circuits, including a microprocessor. The circuitry is functionally represented in Figure 24.

For descriptive purposes, circuitry of the generator can be divided into the following major functional sections:

Voltage regulators	Regulates the system power supplies
Crystal oscillator	Provides a timing reference
Logic and control	Controls overall generator function; receives and implements programming commands; collects and stores telemetry information, processes sensory input, and controls scheduled and sensory-based therapy outputs
Output	Develops and modulates signals delivered to the lead
Antenna	Receives programming signals; transmits telemetry information to the programming wand
Reed switch	Provides a mechanism to place the generator in Magnet Mode or to inhibit its output
Input/Output Switch Matrix	Allows the traditional VNS electrodes to serve as both therapy outputs and sensing input connections

Figure 24. Generator Circuitry



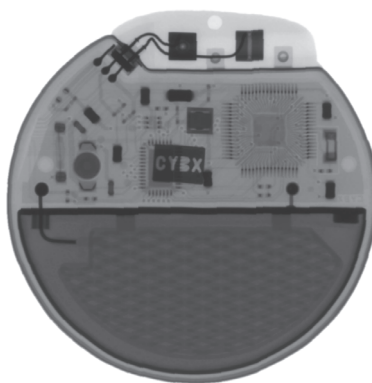
5.1.5 Identification

The generator can be identified on an x-ray and will appear as shown in Figure 25. The serial number and model number of the generator are marked on its titanium case, but do not appear on the x-ray.

The serial number, model number, and year of manufacture depending upon programming software version can be identified by interrogating the generator with the programming software.

Note: See the programming system physician's manuals for details.

Figure 25. X-ray Identification



Note: CYBX is the identification code for Cyberonics.

5.1.6 Heartbeat Detection Performance (Limited to Patients with Epilepsy)

The heartbeat detection performance of the Model 106 device was evaluated against ECG records from standard cardiac databases (i.e. 124 patient files [114 patients] from the

Massachusetts Institute of Technology-Beth Israel Hospital [MIT-BIH] Arrhythmia Database, American Heart Association [AHA] Database and MIT-BIH Noise Stress Test [NST] Database) using a beat-by-beat comparison of the reference beat labels and the device beat labels (in accordance with the ANSI/AAMI EC57 standard). The Model 106 measures heartbeats from the VNS Lead electrode to the generator, an ECG vector unique to VNS Therapy and beat detection. Therefore, each signal in the database was rescaled to amplitudes within the Model 106 heartbeat detection range. The sensitivity of beat detection was 98.8% and Positive Predictive Value (PPV) was 98.3%.

Validation of the Model 106 beat detection was performed in the E-36 clinical study, where cardiac R-wave detection was evaluated against concurrent surface ECG data. Beat detection performance of the device was evaluated through 12 months of follow up. The sensitivity of beat detection exceeded 95.5% and Positive Predictive Value (PPV) exceeded 99.4% at all assessments.

Improper implant location and/or heartbeat detection sensitivity setting could negatively impact R-wave detection performance results. See the *Implantation Procedure* for instruction on how to determine implant location and heartbeat detection setting.

5.2 VNS Therapy System Compatibility

Model 106 is compatible with these system components:

Component	Model
Lead	single-pin
Programming System	201 wand / 250 v11.x software
	2000 wand / 3000 software
Tunnelers	402
Accessory Pack	502
Magnets	220

5.3 Directions for Use

5.3.1 Specifications and Product Information

The specifications and product information for the VNS Therapy Model 106 generator is presented in Table 14.

Table 14. Specifications and Product Information

Stimulation Parameters	Available Parameter Settings
Output current	0-2.0 mA in 0.125-mA steps (± 0.1 mA or $\pm 10\%$; whichever is greater) 2-3.5 mA in 0.25-mA steps (± 0.1 mA or $\pm 10\%$; whichever is greater)
Signal frequency	1, 2, 5, 10, 15, 20, 25, 30 Hz $\pm 6\%$
Pulse width	130, 250, 500, 750, 1000 μ sec $\pm 10\%$
Signal ON time	Normal Mode—7, 14, 21, 30, 60 sec (+ 7 sec/ - 15%) Magnet Mode—7, 14, 21, 30, 60 sec (+ 15%/ - 7 sec) AutoStim Mode—30, 60 sec (+ 15%/ - 7 sec)
Signal OFF time	0.2, 0.3, 0.5, 0.8, 1.1, 1.8, 3 min, and 5 to 180 min (5 to 60 in 5-min steps; 60 to 180 in 30-min steps) ± 4.4 sec or $\pm 1\%$, whichever is greater
Magnet activation	Provided by magnet application (output current, pulse width, and signal ON time may be independently programmed for this purpose)
Reset parameters	Settings are unchanged, but output is disabled (0 mA)
Detection Configuration Parameters	
Tachycardia Detection	ON or OFF; allows VNS to operate in standard mode (i.e. no Tachycardia Detection - "OFF" setting) or a combination of standard VNS with detection (i.e. Tachycardia Detection - "ON" setting)
Threshold for AutoStim	Threshold for heart rate increase which triggers Automatic Stimulation (AutoStim). Setting range is from 20% to 70%. 20% is most sensitive. 70% is least sensitive.
Heartbeat Detection (Sensitivity)	Adjustable sensitivity parameter for heart beat detection, ranging from 1 to 5, with "1" being the least sensitive and "5" being the most sensitive setting. Note that the Model 106 is only capable of detecting heart beats in the range between 32 and 240 bpm ($\pm 10\%$ or 5 bpm, whichever is greater)
Verify Heartbeat Detection	Feature on the programming software that when activated, configures the Model 106 to emit a pulse signal when a heartbeat is detected (for 2 minutes). May be used to check heartbeat detection performance at the currently programmed Heartbeat Detection setting.
Telemetry Reports	
Device History Report	Patient ID, implant date, model number, serial number, magnet activations, total ON time, total operating time, manufacturing date and device settings and stimulation statistics for last 3 office visits
Device Diagnostic Report	Patient ID, model ID, serial number, firmware build number, implant date, communication status, output current status, measured current delivered, lead impedance, and battery status indicators (IFI, N EOS, EOS)
Power Source	
Battery	Wilson Greatbatch Ltd., Model 2075
Chemistry	Lithium carbon monofluoride
Voltage	3.3 V, open circuit
Rated capacity	1.7 Amp-hour
Self-discharge rate	< 1% per year
Physical Characteristics—Materials	
Case	Titanium, hermetically sealed
Header	Polyurethane — Tecothane™ TT-1075D-M Thermoplastic
Lead connector blocks	Stainless steel
Setscrew plug(s)	Silicone*

Measurements (Typical)	
Lead receptacle(s)	0.126 in (3.2 mm) nominal
Dimensions	2.0 in x 2.0 in x 0.27 in (52 mm x 52 mm x 6.9 mm)
Weight	0.88 oz (25 g)
Connector Retention Strength	
With VNS Therapy lead	> 10 N

* No component of the VNS Therapy System is made with natural rubber latex.

5.3.2 Operating Characteristics

5.3.2.1 *Communicating with the VNS Therapy System*

5.3.2.1.1. Programming software

The Model 106 generator can be programmed with the VNS Therapy programming software version 11.0.

 **Note:** See the programming system physician's manuals for details.

5.3.2.1.2. Programming wand

A programming wand connected to a compatible computer running the programming software is needed to communicate with the generator.

 **Note:** For proper placement of the programming wand, connection of the wand to the computer, and use of the wand, see the programming system physician's manuals.

5.3.2.1.3. Prompts and messages

After the program has been initiated, the software screens display prompts and messages to aid in communicating with the generator.

5.3.2.1.4. Communication

The generator “listens” for a communication signal from the programming wand. Communication usually initiates between 1 and 4 seconds, but may be prolonged or interrupted in the presence of electromagnetic interference (EMI). Depending on the type and amount of information being transferred between the generator and the programming wand, complete communication may take up to one minute. Downloading additional information may take more time. The generator listens for and implements interrogations, parameter programming instructions, requests for Device Diagnostics testing, and Device History inquiries.

In response, the generator transmits information on the stimulation parameter settings, changes its parameter settings, responds to requests for Device Diagnostics testing, and provides device histories, respectively. Each time these data are transmitted by the generator, they are saved by the programming software to a database.

 **Note:** For details on viewing database information, see the programming system physician's manuals.

In addition to the programming software and programming wand combination, a magnet can be used for one-way communication to the generator by activating a reed switch in the

electronic circuitry. The magnet can be used to initiate stimulation, temporarily inhibit stimulation, perform Magnet Mode diagnostics, or reset the generator.

5.3.2.1.5. Normal Mode

After the generator has been programmed, the stimulation will repeat in accordance with the programmed ON and OFF cycle (Normal Mode) until the generator receives communication from the VNS Therapy programming system, is inhibited with the magnet, or is activated by the magnet or AutoStim Mode. Immediately after successful programming, the generator delivers a programmed stimulation that enables the programmer to evaluate patient response. If programming is performed during stimulation, stimulation will be terminated; after programming, stimulation will begin using the revised settings.

5.3.2.1.6. Magnet Mode

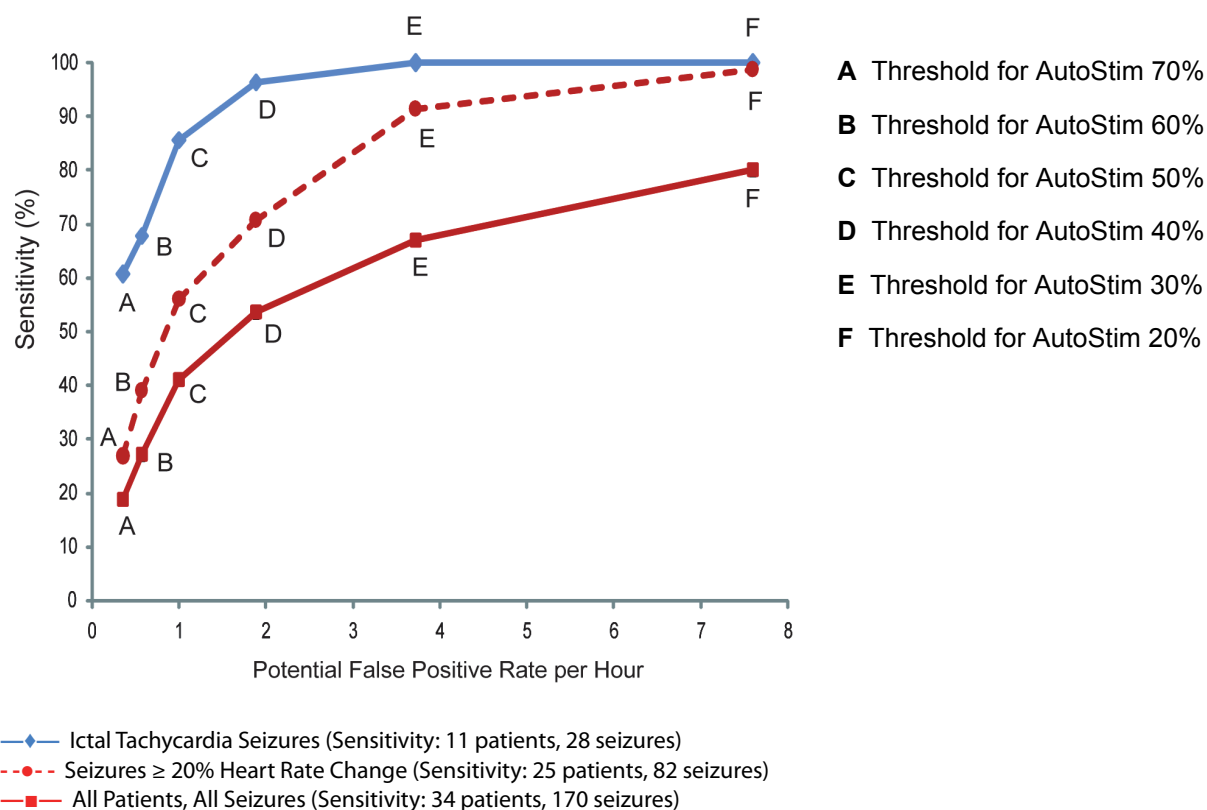
Use of Magnet Mode is limited to patients with epilepsy. Magnet Mode produces on-demand stimulation for the programmed magnet ON time. Stimulation is initiated by applying or passing the magnet over the generator for 1-2 seconds and then immediately removing it from the area over the generator. Magnet Mode stimulation is delivered after the magnet is removed. The Magnet Mode uses the same frequency as the Normal Mode, but the output current, pulse width, and signal ON time are independently programmable. The magnet may also be used to inhibit stimulation. To do so, simply place the magnet over the generator and keep in place for at least 5 seconds. The generator will not stimulate until the magnet is removed.

5.3.2.1.7. AutoStim Mode (Model 106 only - Limited to patients with epilepsy)

AutoStim Mode is an optional feature of the Model 106 that monitors and detects rapid, relative heart rate increases ($\geq 20\%$) that may be associated with seizures. Following detection, on-demand stimulation is delivered.

If AutoStim is enabled, stimulation is initiated automatically upon detection of heart rate rises that exceed the selected threshold for AutoStim. Due to varying physiological conditions among patients, the AutoStim feature has been designed such that the sensitivity of the detection can be adjusted for relative heart rate changes of 20% to 70%. See Figure 26 for Receiver Operating Characteristic (ROC) Curves illustrating the effect of adjustments of the Threshold for AutoStim setting (and Appendix D for 95% confidence intervals). The curves illustrate a trade-off between sensitivity and specificity (potential false positive rate per hour). As one decreases the AutoStim threshold, the sensitivity increases, but at the expense of specificity.

Figure 26. Receiver Operating Characteristic (ROC) Curve for Tachycardia Detection Associated with Seizures



Note: Potential False Positive Rate based on 50 patients, 4516 hours.

Note: Tachycardia detection used for the AutoStim feature requires that the generator is accurately measuring heart rate. Therefore, the heartbeat detection setting should be verified by the physician at each office visit, and changed when needed. See “Troubleshooting” in the programming system physician's manuals.

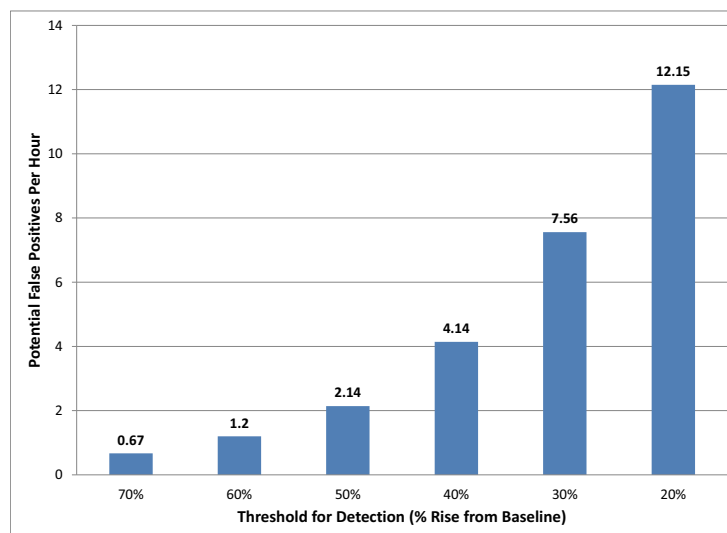
Performance of the AutoStim Mode was demonstrated in patients with Ictal Tachycardia (defined as an increase in heart rate during a seizure to a rate that was greater than 100 bpm and was at least a 55% increase or 35 bpm increase from baseline [resting] heart rate) over the age of 18 in two separate studies (E-36 and E-37).

Figure 26 was generated using prospective clinical study data (E-36 and E-37) acquired from 50 patients with refractory epilepsy during an epilepsy monitoring unit (EMU) stay. EEG data were recorded along with heart rate (ECG) data. During the EMU, anti-epileptic medications were maintained. Of the 50 patients evaluated, 34 experienced seizures while the remaining 16 did not. In addition to annotations from the clinical sites' principal investigator, the EEG data were reviewed by at least three neurologists to identify and confirm, by majority rule, seizure activity. The resulting data from the investigators and external reviewers were used to analyze both the sensitivity and false positive rate of the Tachycardia Detection Algorithm by correlating algorithm detections with seizure onset times determined from patient EEGs. Figure 26 shows three different curves. Curve 1 (blue) includes only seizures that met the criteria for Ictal Tachycardia as defined above. Curve 2 (dashed red) includes all seizures with greater than 20% increase in heart rate during an ictal event, and represents the performance

for all seizures which may be detected based on the design of the Tachycardia Detection Algorithm. Curve 3 (solid red) includes all seizures from all study patients, irrespective of the heart rate change.

In addition, ECG data were collected in a separate clinical study of healthy normal volunteers (E-34) during sub-maximal exercise testing and sleep. The ECG data were analyzed using the Tachycardia Detection Algorithm to assess the impact of exercise (i.e. stair stepping and moderate treadmill) and other activities (i.e. Valsalva maneuvers and sleep) on the AutoStim Potential False Positive rate (see Figure 27).

Figure 27. Non-Seizure Heart Rate Challenges



Note: In Figure 27, N=49 patients

5.3.2.1.8. Limitations of the clinical studies

The data used to assess the sensitivity and false positive rates of the AutoStim feature were collected in the E-36 and E-37 studies from 50 patients during a five day stay in an epilepsy monitoring unit (EMU). During the EMU stay, 170 seizures and 4516 hours of EEG/ECG data were used to establish the performance as depicted in Figure 26. The following limitations should be considered when interpreting the data as an EMU may not be reflective of real world use:

- The EMU represents a controlled setting and thus, the types of seizures and number of seizures associated with tachycardia may vary from those in real world use.
- During the EMU only the AutoStim feature was enabled and the Normal Mode was disabled. Use of AutoStim in conjunction with Normal Mode stimulation and Magnet Mode stimulation may affect tachycardia detections.
- In real world use, physical activities that increase heart rate, including but not limited to exercise, could result in detection of tachycardia that may not be associated with a seizure.

5.3.2.1.9. Generator interrogation

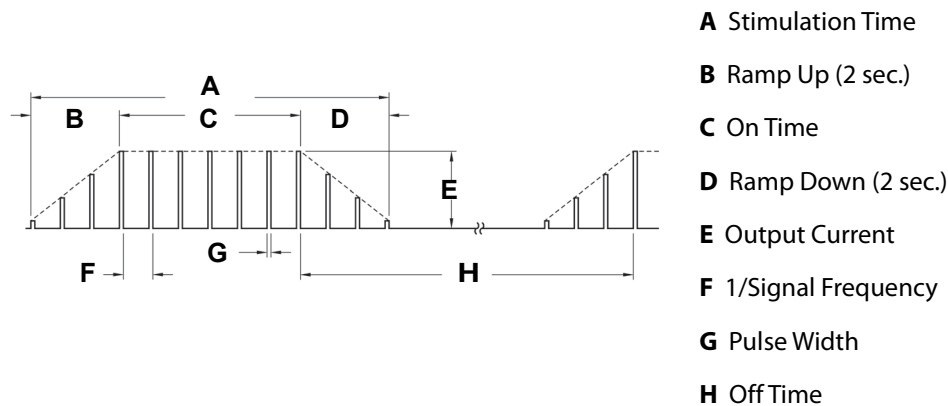
The generator can be interrogated to determine the present settings of the stimulation parameters. A longer interrogation (Extended Generator Memory Download) may be performed under the direction of Clinical Technical Support to obtain additional data relating to tachycardia detection for troubleshooting purposes.

5.3.2.1.10. Programmable parameters

A graphic representation of stimulation (Figure 28) depicts the relationship of the programmable parameters. Each parameter can be independently programmed, thereby offering multiple setting combinations from which the physician may select optimal stimulation for the patient.

Figure 28 shows that the output pulse can be varied both by amplitude (output current) and duration (pulse width). The number of output pulses delivered per second determines the frequency.

Figure 28. Stimulation (Frequencies < 10 Hz do not ramp)



5.3.2.1.11. Duty cycle

The percentage of time the generator is stimulating is called a “duty cycle.” A duty cycle is calculated by dividing the stimulation time (programmed Normal Mode ON time plus, if frequency is ≥ 10 Hz, 2 seconds of ramp-up time and 2 seconds of ramp-down time) by the sum of the ON and OFF times. The various parameter settings for stimulation are listed in “Specifications and Product Information”.

Caution: Possible nerve damage with ON time > OFF time— Excessive stimulation has resulted in degenerative nerve damage in laboratory animals. One component of excessive stimulation is when ON time exceeds OFF time, which can be produced by continuous or frequent magnet activation (> 8 hours). It is recommended that stimulation at these combinations of ranges be avoided.

Table 15 shows duty cycles for typical ON time and OFF time settings.

Table 15. Duty Cycles for Various ON and OFF Time Settings

	OFF Time (min)								
	0.2	0.3	0.5	0.8	1.1	1.8	3	5	10
ON Time (sec)	Duty Cycles* (% ON Time)								
7	58	44	30	20	15	10	6	4	2
14	69	56	41	29	23	15	9	6	3
21	76	64	49	36	29	19	12	8	4
30	81	71	57	44	35	25	16	10	5
60	89	82	71	59	51	38	27	18	10

* A duty cycle is calculated by dividing stimulation time (programmed ON time plus 2 seconds of ramp-up time and 2 seconds of ramp-down time) by the sum of the ON time and the OFF time.

* The duty cycles in grey are not recommended as they represent parameter combinations with ON Time > OFF Time.

5.3.2.1.12. Parameter settings and battery life

When selecting a combination of parameter settings for stimulation, the physician should also consider that some combinations would decrease battery life faster than others. In addition, operating the Model106 with Tachycardia Detection turned ON will also decrease the battery life.

Table 16 shows the longevity impact that the AutoStim Mode feature has when ON time is set to 30 sec and 60 sec with common Normal Mode settings (e.g., 2 mA output current, 20 Hz signal frequency, 500 μ sec pulse width, 10% duty cycle), common lead impedance (3 kOhms) and various detections per hour (e.g., 1, 7, 15).

 **Note:** See “Generator battery longevity”.

Table 16. Estimated Model 106 Longevity with Sensing and AutoStim

AutoStim Feature	Expected Life (yrs)	
Tachycardia Detection / AutoStim “OFF”	12.8	

Tachycardia Detection / AutoStim “ON”	AutoStim ON-Time	
	30 sec	60 sec
AutoStims Per Hour	Expected Life (yrs)	
1	7.1	6.9
7	6.5	5.5
15	5.9	4.4

5.3.2.2 VNS Therapy magnets

There are four possible uses of the magnet:

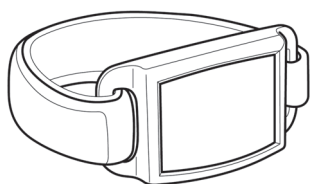
- To provide on-demand stimulation as an attempt to abort or de-intensify an oncoming seizure or a seizure in progress
- To temporarily inhibit stimulation
- To reset the generator (in combination with the programming wand)
- To test daily the functioning of the generator, it is recommended that patients be instructed to use the magnet to activate stimulation. During an aura or at the start of a seizure, magnet activation may be initiated by the patient, a companion, or the physician by applying or passing a magnet over the generator to activate a reed switch in the generator's electronic circuitry. This action changes the generator from Normal Mode or AutoStim to Magnet Mode. AutoStim Mode is only overridden by a request for Magnet Mode Stimulation if the Magnet Mode output is greater than the AutoStim Mode.



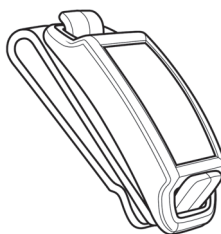
Caution: *Possible nerve damage with ON time > OFF time*— Excessive stimulation has resulted in degenerative nerve damage in laboratory animals. One component of excessive stimulation is when ON time exceeds OFF time, which can be produced by continuous or frequent magnet activation (> 8 hours). It is recommended that stimulation at these combinations of ranges be avoided.

Two identical magnets (see Figure 29), each providing a minimum of 50 gauss at 1 inch, are supplied by LivaNova. The watch-style accessory attaches the magnet to the wrist with a wristband, and the pager-style accessory holds the magnet in a belt clip with a quick-release mechanism like a pager.

Figure 29. Magnet Accessories



**Watch-Style
(wristband)**

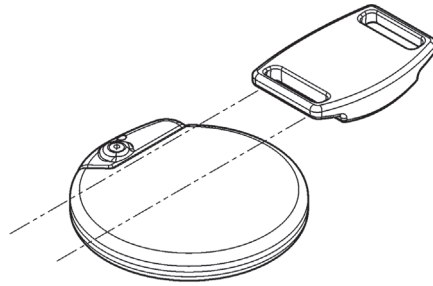


**Pager-Style
(belt clip)**

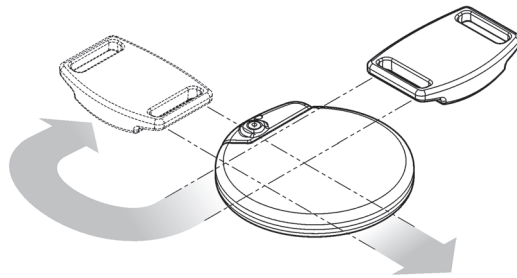
The proper orientation and motion for initiating magnet activation is shown in Figure 30. The magnet is shown without the belt clip or wristband to illustrate the proper orientation of the magnet to the generator.

Figure 30. Initiate Magnet Activation

Standard Magnet Activation



Optional Cross-Pattern Magnet Activation



Caution: To activate or stop stimulation, the label side of the magnet should face the generator.

5.3.2.2.1. Service life of magnet

All magnets may lose their effectiveness over time. VNS Therapy magnets contain a high-power magnet that is surrounded by a plastic casing in the shape of a watch. These magnets should be operated and stored at temperatures ranging from - 20 °C (- 4 °F) to + 55 °C (+ 131 °F). With normal use, they should remain powerful for approximately 3 years. Avoid dropping the magnets or storing them near other magnets.

5.3.2.2.2. Magnet activation technique

To initiate stimulation, apply or pass the magnet over the generator for 1-2 seconds and then immediately remove it from the area over the generator. A cross-pattern swipe, as shown in Figure 30, may also be used by the patient or caregiver to activate the Magnet Mode if difficulty is encountered with a single pass of the magnet. Removal of the magnet causes the generator to operate in Magnet Mode, delivering a single stimulation at the programmed magnet pulse width, magnet current, and magnet signal ON time settings. The frequency is the programmed value for Normal Mode. A Magnet Mode stimulation will always override any Normal Mode programmed stimulation, even if the Magnet Mode output current is set to 0 mA. If Magnet Mode stimulation is not desired, the Magnet Mode output current may be programmed to 0 mA. A Magnet Mode stimulation will only override an AutoStim Mode stimulation if the Magnet Mode output exceeds the AutoStim output.



Caution: The cross-pattern swipe technique may cause duplicate magnet activation entries to be shown in the programming software database. This is an expected occurrence due to device design and is not considered a device malfunction. See the programming system physician's manuals for details.

It is recommended that tests of the magnet output be performed while the patient is still in the physician's office to ensure tolerability of the magnet output.

5.3.2.2.3. Inhibit generator output with the magnet

Application of the magnet during stimulation will inhibit the output. In addition, holding the magnet in place for at least 5 seconds will terminate any ongoing Normal or AutoStim Mode stimulation. After the magnet is removed, Normal Mode operation will resume with stimulation when one complete OFF time has elapsed.



Caution: If stimulation becomes painful, the patient should be instructed to stop the stimulation with the magnet.

In the unlikely event of continuous stimulation or other malfunction, the patient must be advised to apply the magnet, secure it in place, and immediately notify their physician.



Note: See "Adverse Events" in the indication-specific information chapters.

5.3.2.2.4. Reset the microprocessor with the magnet and the programming wand

The VNS Therapy System allows the generator microprocessor to be reset in the event of a malfunction. Resetting is necessary only in the rare case of microprocessor memory malfunction, which might be caused by conditions described in the *Introduction to the VNS Therapy System* chapter. Resetting the microprocessor may be appropriate when the generator and the programming system are unable to communicate.



Caution: *Generator reset*— When the generator is reset, its stimulation output is disabled (0 mA); however, all settings and device history are preserved. After a successful reset, the generator stimulation output may be re-enabled to resume operation at the previously programmed settings.



Caution: (Model 106 Serial Numbers < 80000 only)—If Magnet Mode output current is less than or equal to Autostim Mode output current, repeated magnet applications may trigger a device safety feature that disables stimulation. Stimulation can be re-initiated at the next office visit by programming stimulation output current on.

For suggestions in solving communication difficulties, see "Troubleshooting" in the programming system physician's manuals.

For instructions on how to reset the microprocessor, see the programming system physician's manuals. After a microprocessor reset has been attempted, wait at least 30 seconds before establishing communication with the programming software. It is recommended, except in cases of a medical emergency, that the physician consult a LivaNova technical representative before a reset is performed.

5.3.2.3 Device history

The Device History consists of the generator serial number, model number, the patient code, implantation date, year of manufacture, and other information pertinent to diagnostic and programming events. Use the programming software to access and view Device History information.

5.3.2.4 *Device diagnostics*

Information from device diagnostic tests can help the physician determine whether the:

- Generator is operating properly before it is implanted
- Generator output current is being delivered at the programmed value
- Generator is operating in Normal, Magnet or AutoStim Mode
- Lead impedance is within an acceptable range



Note: For details on available diagnostic tests, see the programming system physician's manuals.

5.3.2.4.1. **System Diagnostics test**

The System Diagnostics evaluates the lead impedance of the VNS Therapy System, as well as the generator's ability to deliver the programmed stimulation. When the output current is programmed to any value greater than 0 mA, the generator will deliver a pulse at 0.25 mA, 130 μ sec to assess lead impedance. The generator will then deliver the programmed output stimulus. The programming software will report the lead impedance and whether the programmed stimulus was delivered.

If the output current is programmed to 0 mA, as would be expected during the implantation procedure, the generator will deliver a pulse at 0.25 mA, 130 μ sec followed by delivery of stimulation at 1 mA, 20 Hz, 500 μ sec for approximately 14 seconds. Likewise, the programming software will report the lead impedance and whether the parameters listed here are deliverable.

5.3.2.4.2. **Reasons for high or low lead impedance readings**

The System Diagnostics evaluates lead impedance of the VNS Therapy System. The lead impedance measurement is performed with a single pulse at 0.25 mA, 130 μ sec. Additionally, the generator automatically performs a lead impedance measurement once every 24 hours. If a HIGH or LOW impedance is detected, the user is notified after an interrogation with the programming software.

High lead impedance is defined as any value ≥ 5300 Ohms. Low lead impedance is defined as any value ≤ 600 Ohms.



Caution: Possible causes of high lead impedance readings are thought to include: lead discontinuity, lead disconnection from the generator, fibrosis between the nerve and the electrode, electrode detachment from the nerve, or a defective generator.



Caution: Possible causes of low lead impedance readings are thought to include: short-circuit condition within the lead or a defective generator.

5.3.2.4.3. **High lead impedance: possible implications**

High lead impedance (≥ 5300 Ohms), in the absence of other device-related complications, is not an indication of a lead or generator malfunction. High lead impedance in combination with the patient's failure to feel even the maximum output stimulus may indicate a lead wire fracture or other type of electrical discontinuity in the lead. Complications with heartbeat sensing may also be indicative of a lead discontinuity. Patients experiencing high lead

impedance, no sensation of maximum output stimulation, and an increase in seizures/depressive symptoms should be further evaluated for possible lead replacement.

Note: For additional instructions on how to perform the System Diagnostics, see the programming system physician's manuals.

Note: To troubleshoot high or low impedance see "Troubleshooting" in the programming System physician's manuals.

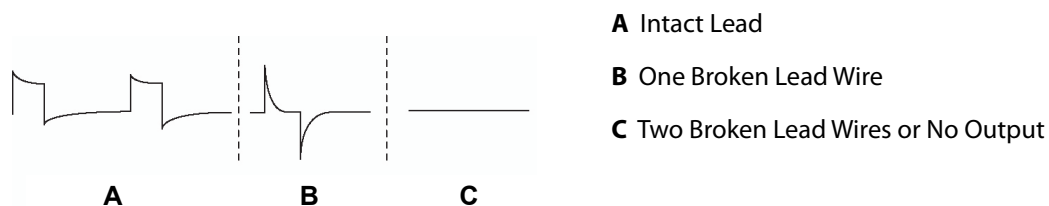
5.3.2.4.4. Low lead Impedance: possible implications

Low lead impedance (≤ 600 Ohms) likely indicates the existence of a short-circuit condition, although an impedance value of greater than 600 Ohms does not exclude the possibility. A sudden decrease in impedance value in combination with device-related complications (e.g., increase in seizures/depressive symptoms or painful stimulation; patient perception of feeling erratic, limited, no stimulation, or complications detecting heartbeats) may also indicate a short-circuit condition in the lead.

5.3.2.4.5. Stimulus waveform analysis

Either evoked potential monitoring equipment or an oscilloscope can be used to analyze the stimulus waveform from the neck for verification of an electrical discontinuity. A differentiated waveform with narrowed pulses or no waveform at all can confirm a discontinuity. Figure 31 shows characteristic waveforms obtained from skin electrodes for a lead that is intact and for a lead that has a fracture in one or both wires. In addition, lead discontinuities can sometimes be identified on an x-ray of the implant site.

Figure 31. Typical Waveforms Obtained from Skin Electrodes



5.3.2.5 Delivery of programmed output current

5.3.2.5.1. LOW as output current

If the diagnostic tests indicate LOW output current, the generator may not be delivering the programmed output current. Reasons for failure to deliver the programmed output current include high programmed output current and high lead impedance. The maximum deliverable output current, according to Ohm's Law, equals the maximum output voltage (approximately 12 V) divided by the lead impedance.

5.3.2.5.2. Reprogram to a lower current

If the generator is failing to deliver the programmed output current, the physician can reprogram the device to a lower output current and attempt to compensate for the decrease

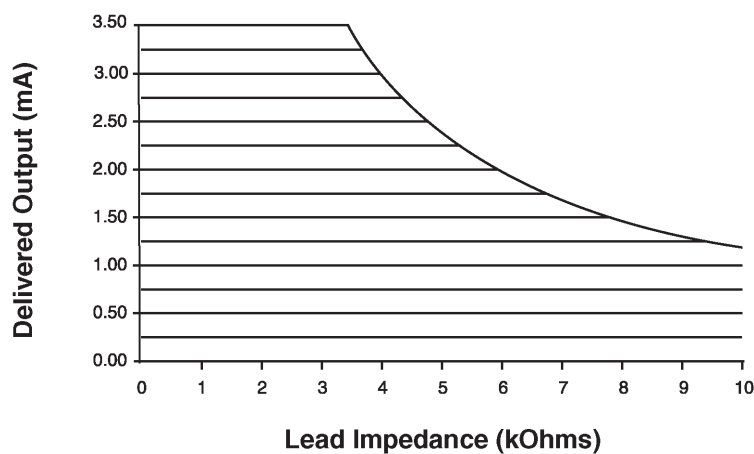
in delivered energy by widening the pulse width. For example, if the diagnostics read LOW for a generator programmed at 2.5 mA, 30 Hz, 500 μ sec with 30 seconds ON time, then the parameters may be changed by lowering the output current to 2 mA and widening the pulse width to 750 μ sec.

5.3.2.6 *Charge delivered per pulse*

5.3.2.6.1. **Output current x pulse width = charge delivered per pulse**

The charge delivered per pulse is the most important parameter in evaluating stimulation output. It is defined as a microcoulomb (μ C), which is the product of current and time—that is, the output current (mA) multiplied by the pulse width (msec). Figure 32 shows the relationship of delivered output current (mA) to lead impedance for a 1000 μ sec pulse with output currents from 0 to 3.5 mA.

Figure 32. Relationship of Delivered Output Current to Lead Impedance



5.3.2.7 *Generator battery longevity*

5.3.2.7.1. **Battery longevity and programmed setting choices**

The anticipated longevity of the generator battery varies, depending on the choice of programmed settings. Higher output currents, frequencies, pulse widths, and duty cycles generally deplete the battery over a shorter period of time than lower settings. Generally, the increase in battery depletion rate is proportional to the increase in the programmed setting in the absence of detection.

Caution: *Undeliverable output currents*— Programming the generator to a high output current that cannot be delivered due to a high lead impedance may disproportionately increase the battery depletion rate and should be avoided.

Other factors, such as the lead impedance and, if applicable, magnet usage or Threshold for AutoStim settings and AutoStim, also affect the anticipated longevity of the generator battery. The anticipated battery longevity decreases as lead impedance increases. Although 1.5 k to 3 kOhms may be a typical lead impedance at implantation, the impedance may increase to 3 k to 5 kOhms during the life of the implant.

The *Appendices* chapter provides estimated battery lifetimes for Model 106 without tachycardia detection under a variety of stimulation conditions, including lead impedance. Because of the number of possible parameter combinations, it is impractical to provide the

projected life for all possible combinations. The tables should not be used to predict battery end of service (EOS), but they give some indication of the effect of various parameter changes on battery life and can be used to assist in the selection of parameter settings. They also indicate that battery life can be maximized at low duty cycles and low frequencies (10 to 20 Hz) for stimulation.

 **Note:** For more information, see the programming system physician's manuals.

5.3.2.7.2. Battery status indicators

The programming software displays a battery indicator similar to an indicator that may be found in cell phones. The visual indicator illustrates the approximate remaining battery capacity.

 **Caution:** *Battery evaluation at cold temperatures*—Low storage temperatures may affect the battery status indicators. In such cases, the battery status indicators should be re-evaluated using the System Diagnostics or Generator Diagnostics after the generator has been at room or body temperature for 30 minutes.

The programming software will display warning messages after an interrogation or programming of the generator if the battery has been depleted to a level where action is recommended due to approaching or reaching End of Service (EOS). See the VNS Therapy programming system physician's manuals for additional information on these indicators.

5.3.3 Generator Replacement

All VNS Therapy generators eventually require surgical replacement as a result of battery depletion. Generator replacement does not, of itself, require lead replacement unless a lead discontinuity is suspected. Generator replacement or removal requires dissection to the generator's pocket, with care being taken not to damage or cut the lead. The entire surgical procedure generally requires about 1 hour.

5.3.4 Signs of End of Service

The most common reason for the absence of stimulation is battery depletion, although there may be other reasons. When end of service (EOS) occurs, the generator will disable stimulation and no output will be delivered. If the generator is not explanted or replaced at EOS, the battery voltage will continue to gradually decrease and communication with the generator may not be possible.

 **Caution:** Generator EOS may result in increased frequency, intensity, or duration of signs and symptoms of the patient's disorder, in some cases to levels greater than those reported before stimulation.

5.3.5 Replacement Based on Battery Status Indicators

The generators and programming software have battery status indicators. These indicators provide warnings that a generator battery should be monitored more frequently, is nearing EOS, or has reached EOS. Once these warning messages appear, see recommendations in the programming system physician's manuals.

 **Note:** See "Battery status indicators".

 **Caution:** *Prompt generator replacement*—Prompt replacement of the generator at or before EOS is recommended. Prompt replacement may help minimize any possible relapse.



Caution: *Explanted generator*—A generator explanted for any reason should not be reimplanted. An explanted generator should be returned to the company. (For instructions on returning an explanted generator, see the *Introduction to the VNS Therapy System* chapter.)

6 Technical Information — 1000 Generator

6.1 Detailed Device Description

6.1.1 Physical Characteristics

The titanium case of the VNS Therapy® SenTiva™ Model 1000 generator is hermetically sealed and leak-rate tested. Specially designed feedthrus using platinum conductors form the electrical connection from the connector blocks to the circuitry through the hermetically sealed enclosure. The Model 1000 accepts the single-pin lead.

6.1.2 Biological Compatibility

Materials exposed to the subcutaneous environment are biologically compatible. All of these materials have a long history in medical implants and have been found to be tissue compatible.

6.1.3 Power Source

The power source for the Model 1000 generator is a Wilson Greatbatch Ltd, Model 2183, lithium carbon monofluoride battery with an open-circuit voltage of 3.3 V.

The battery's maximum available capacity is approximately 1 Amp-hour. The self-discharge reduces the capacity by less than 1 percent per year. The voltage in this battery gradually decreases as the battery nears its end of service (EOS).

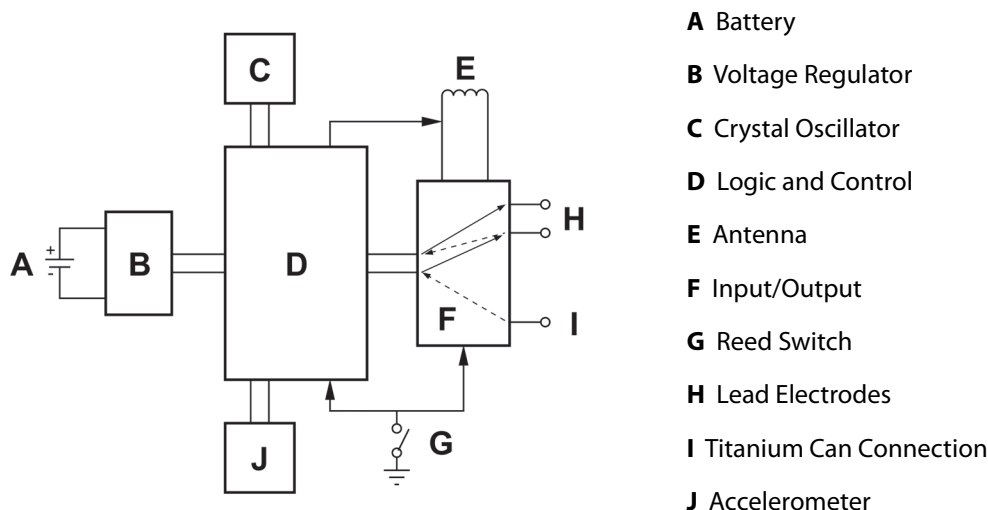
6.1.4 Circuitry

The generator uses complementary metal oxide semiconductor (CMOS) integrated circuits, including a microprocessor. The circuitry is functionally represented in Figure 33.

For descriptive purposes, circuitry of the generator can be divided into the following major functional sections:

Voltage regulators	Regulates the system power supplies
Crystal oscillator	Provides a timing reference
Logic and control	Controls overall generator function; receives and implements programming commands; collects and stores telemetry information, processes sensory input, and controls scheduled and sensory-based therapy outputs
Antenna	Receives programming signals; transmits telemetry information to the programming wand
Reed switch	Provides a mechanism to place the generator in Magnet Mode or to inhibit its output
Input/Output	Develops and modulates signals delivered to the lead; provides amplification of cardiac signals; allows the traditional VNS electrodes to serve as both therapy outputs and sensing input connections
Accelerometer	Provides information related to patient posture

Figure 33. Generator Circuitry



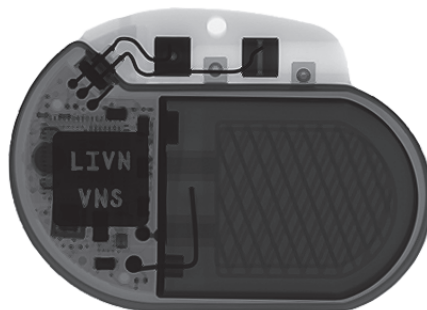
6.1.5 Identification

The generator can be identified on an x-ray (see Figure 34). The serial number and model number of the generator are marked on its titanium case, but do not appear on the x-ray.

The serial number and model number are identified by interrogating the generator with the programming system.

Note: See the programming system physician's manuals for details.

Figure 34. X-ray Identification



Note: LIVN is the identification code for LivaNova.

6.1.6 Heartbeat Detection Performance (Limited to Patients with Epilepsy)

The Model1000 has a heartbeat detection sensitivity of 98% and Positive Predictive Value (PPV) of 98%. Improper implant location and/or inadequate heartbeat detection configuration could negatively impact R-wave detection performance results. See the *Implantation Procedure* for instruction on how to determine implant location and configure heartbeat detection.

6.2 VNS Therapy System Compatibility

Model 1000 is compatible with these system components:

Component	Model
Lead	single-pin
Wand	2000
Software	3000 v1.0+
Tunnelers	402
Accessory Pack	502
Magnets	220

6.3 Directions for Use

6.3.1 Specifications and Product Information

The specifications and product information for the Model 1000 generator is presented in Table 17.

Table 17. Specifications and Product Information

Stimulation Parameters	Available Parameter Settings
Output current	0-2.0 mA in 0.125-mA steps (± 0.1 mA or $\pm 10\%$; whichever is greater) 2-3.5 mA in 0.25-mA steps (± 0.1 mA or $\pm 10\%$; whichever is greater)
Signal frequency	1, 2, 5, 10, 15, 20, 25, 30 Hz $\pm 6\%$
Pulse width	130, 250, 500, 750, 1000 μ sec $\pm 10\%$
Signal ON time	Normal Mode—7, 14, 21, 30, 60 sec AutoStim Mode—30, 60 sec Magnet Mode—7, 14, 21, 30, 60 sec
Signal OFF time	0.2, 0.3, 0.5, 0.8, 1.1, 1.8, 3 min, and 5 to 180 min (5 to 60 in 5-min steps; 60 to 180 in 30-min steps) ± 4.4 sec or $\pm 1\%$, whichever is greater
Magnet activation	Provided by magnet application (output current, pulse width, and signal ON time may be independently programmed for this purpose)
Reset parameters	Settings are unchanged, but output is disabled (no stimulation)
Detection Configuration Parameters	
Tachycardia Detection	Enabled or Disabled; When enabled, allows the device to perform heartbeat sensing and tachycardia detections.
AutoStim Threshold	Threshold for heart rate increase which triggers Automatic Stimulation (AutoStim). Setting range is from 20% to 70%. 20% is most sensitive. 70% is least sensitive.
Heartbeat Detection Setting Sensitivity	Sensitivity parameter for heartbeat detection, ranging from 1 to 5, with “1” being the least sensitive and “5” being the most sensitive setting. Note: Model 1000 is capable of detecting heart beats in the range between 28 and 180 bpm ($\pm 10\%$ or 5 bpm, whichever is greater)
Verify Heartbeat	Feature on the programming software that when activated, configures the Model 1000 to emit a pulse signal when a heartbeat is detected (for 2 minutes). May be used to check heartbeat detection performance at the currently programmed Heartbeat Detection setting.
Low Heart Rate Threshold	Threshold for low heart rate that triggers logging of the event if it occurs after AutoStim or Magnet Mode stimulation. Available selections include OFF, 30, 40, 50, and 60 bpm. Note: “OFF” turns off detection for low heart rate events.

Prone Position Detection	ON or OFF; When ON, configures the Model 1000 to perform detection for prone posture events after an AutoStim or Magnet Mode stimulation
Day-Night Programming	
Day-Night Programming	Enabled or Disabled; When enabled, allows user to program the generator to deliver 2 independent sets of stimulation parameters at different times during a 24-hour period.
Nighttime Period	Time period for which Nighttime Values are active; 1-23 hours in 30 minute increments
Nighttime Values	Programmable parameters for Nighttime stimulation. Includes: ■ Normal Mode, AutoStim Mode, and Magnet Mode output current ■ Normal Mode frequency ■ Normal Mode, AutoStim Mode, and Magnet Mode pulse width ■ Normal Mode, AutoStim Mode and Magnet Mode ON time ■ Normal Mode OFF time ■ Night Time AutoStim threshold AutoStim and Magnet frequency in the nighttime will default to the same value as the Nighttime Normal Mode frequency.
Scheduled Programming Parameters	
Scheduled Programming	Enabled or Disabled When enabled, allows user to schedule automated increases in output current using a protocol of up to 7 steps
Interval Between Steps	Default value: 14 days; range is from 7 days to 28 days
Step Values	Programmable parameters for each step of a protocol: First step: All stimulation parameters Subsequent steps: output currents only
Power Source	
Battery	Wilson Greatbatch Ltd., Model 2183
Chemistry	Lithium carbon monofluoride
Voltage	3.3 V, open circuit
Rated capacity	1 Amp-hour
Self-discharge rate	< 1% per year
Physical Characteristics—Materials	
Case	Titanium, hermetically sealed
Header	Polyurethane — Tecothane™ TT-1075D-M Thermoplastic
Lead connector blocks	Stainless steel
Setscrew plug	Silicone*
Measurements (Typical)	
Lead receptacle(s)	0.126 in (3.2 mm) nominal
Dimensions	1.8 in x 1.3 in x 0.27 in (45 mm x 32 mm x 6.9 mm)
Weight	0.56 oz (16 g)
Connector Retention Strength	
With VNS Therapy lead	> 10 N

* No component of the VNS Therapy System is made with natural rubber latex.

6.3.2 Communicating with the VNS Therapy System

6.3.2.1 *Programming system*

A compatible VNS Therapy programming system is required to communicate with and program the Model 1000 generator. A programming system consists of a programming wand, and a compatible computer running the programming software.

 **Note:** For more information such as the proper placement of the wand, connection of the wand to the computer, and use of the programming system, see the programming system physician's manuals.

6.3.2.2 *Communication*

The generator “listens” for a communication signal from the programming wand. Communication usually initiates between 1 and 4 seconds, but may be prolonged or interrupted in the presence of electromagnetic interference (EMI). Depending on the type and amount of information being transferred between the generator and the programming wand, complete communication may take up to one minute. Downloading additional information may take more time.

The generator listens for and implements interrogations, parameter programming instructions, requests for diagnostics testing, and device history inquiries. In response, the generator transmits information on the stimulation parameter settings, changes its parameter settings, responds to requests for diagnostics testing, and provides device histories, respectively. Each time these data are transmitted by the generator, they are saved by the programming software to a database.

 **Note:** For details on viewing generator information on a programming computer, see the programming system physician's manuals.

In addition to the programming system, a magnet can be used for one-way communication to the generator by activating a reed switch in the electronic circuitry. The magnet can be used to initiate stimulation, temporarily inhibit stimulation, perform Magnet Mode diagnostics, or reset the generator.

6.3.3 Features and Modes

6.3.3.1 *Normal Mode*

After the generator has been programmed, stimulation will repeat in accordance with the programmed ON and OFF cycle (Normal Mode) until the generator receives communication from the programming system, is inhibited with the magnet, or is activated by the magnet, or an AutoStim occurs. Immediately after successful programming, the generator delivers a programmed stimulation that enables the programmer to evaluate patient response. If programming is performed during stimulation, stimulation will be terminated; after programming, stimulation will begin using the revised settings.

6.3.3.2 *Magnet Mode*

Use of Magnet Mode is limited to patients with epilepsy. Magnet Mode produces on-demand stimulation for the programmed magnet ON time. Stimulation is initiated by applying or passing the magnet over the generator for 1-2 seconds and then immediately removing it

from the area over the generator. Magnet Mode stimulation is delivered after the magnet is removed. The Magnet Mode uses the same frequency as the Normal Mode, but the output current, pulse width, and signal ON time are independently programmable.

The magnet may also be used to inhibit stimulation. To do so, simply place the magnet over the generator and keep in place. The generator will not stimulate until the magnet is removed.

6.3.3.3 **AutoStim Mode**

Use of AutoStim Mode is limited to patients with epilepsy. AutoStim Mode is an optional feature that monitors heart rates during stimulation OFF times and detects rapid, relative heart rate increases ($\geq 20\%$) that may be associated with seizures. Following detection, on-demand stimulation is delivered.

If AutoStim is enabled, stimulation is initiated automatically upon detection of heart rate rises that exceed the selected threshold for AutoStim. Due to varying physiological conditions among patients, the AutoStim feature has been designed such that the sensitivity of the detection is adjustable for relative heart rate changes of 20% to 70%.

Tachycardia Detection used for the AutoStim feature requires that the generator is accurately measuring heart rate. Therefore, its heartbeat detection accuracy should be verified by the physician at implant and at each office visit. If heartbeat detection is inaccurate, adjustments of the Heartbeat Detection setting may be needed. See “Troubleshooting” in the programming system physician's manuals for more information.

6.3.3.3.1. **Tachycardia Detection Algorithm Performance —ROC Curves**

 **Note:** The information in this section is applicable for Model 106. The Tachycardia Detection Algorithm in Model 1000 was tested *in silico* and determined to be functionally equivalent to Model 106. Therefore, no new clinical evaluation for algorithm performance was conducted for Model 1000 and the Model 106 results are applicable.

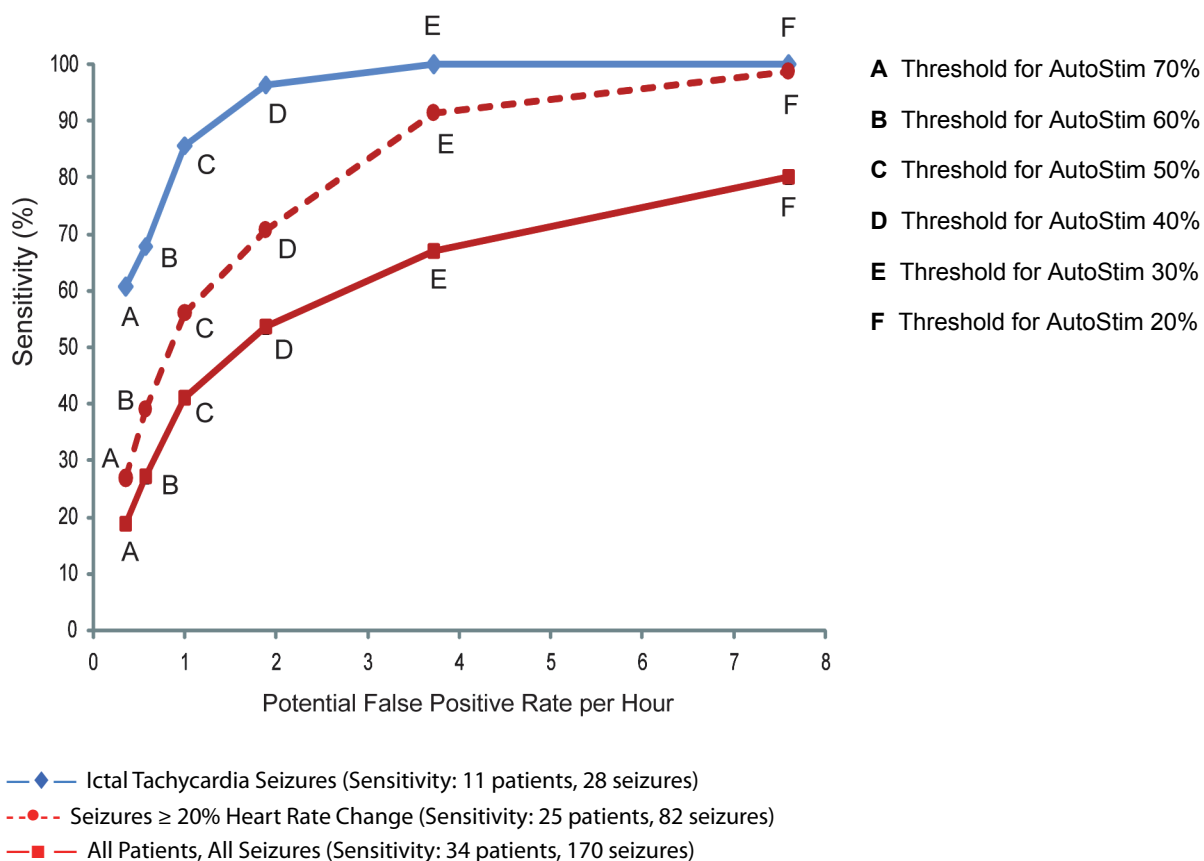
See Figure 35 for Receiver Operating Characteristic (ROC) Curves using data collected in an EMU setting from two previously conducted clinical studies (E36 and E37). Curve 1 (blue) includes only seizures that met the criteria for Ictal Tachycardia (defined as an increase in heart rate during a seizure to a rate that was greater than 100 bpm and was at least a 55% increase or 35 bpm increase from baseline [resting] heart rate). Curve 2 (dashed red) includes all seizures with greater than 20% increase in heart rate during an ictal event, and represents the performance for all seizures which may be detected based on the design of the Tachycardia Detection Algorithm. Curve 3 (solid red) includes all seizures from all study patients, irrespective of the heart rate change. The curves illustrate a trade-off between sensitivity and specificity (potential false positive rate per hour) as the AutoStim Threshold setting is adjusted. As one decreases the AutoStim threshold, the sensitivity increases, but at the expense of specificity.

The following limitations should be considered when interpreting the data in Figure 35, as an EMU may not be reflective of real world use:

- The EMU represents a controlled setting and thus, the types of seizures and number of seizures associated with tachycardia may vary from those in real world use.

- During the EMU only the AutoStim feature was enabled and the Normal Mode was disabled. Use of AutoStim in conjunction with Normal Mode stimulation and Magnet Mode stimulation may affect tachycardia detections.
- In real world use, physical activities that increase heart rate, including but not limited to exercise, could result in detection of tachycardia that may not be associated with a seizure.

Figure 35. Receiver Operating Characteristic (ROC) Curve for Tachycardia Detection Associated with Seizures



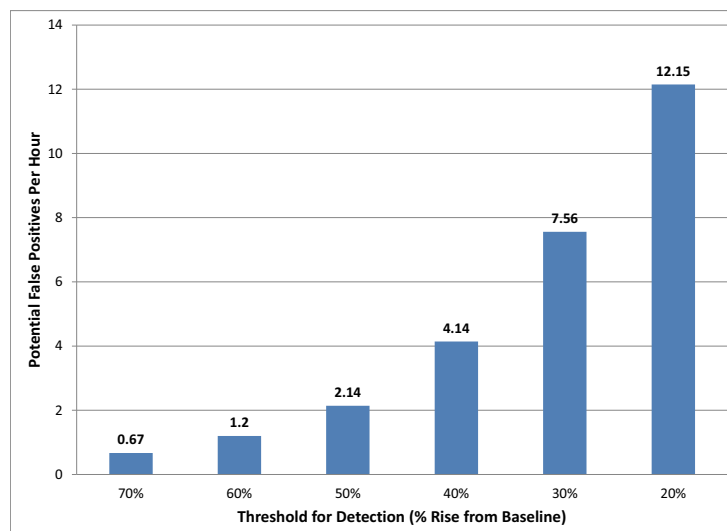
Note: Potential False Positive Rate based on 50 patients, 4516 hours.

6.3.3.3.2. Tachycardia Detection Algorithm Performance —AutoStim Potential False Positives

Note: The information in this section is applicable for Model 106. The Tachycardia Detection Algorithm in Model 1000 was tested *in silico* and determined to be functionally equivalent to Model 106. Therefore, no new clinical evaluation for algorithm performance was conducted for Model 1000 and the Model 106 results are applicable.

ECG data were collected in a previously conducted clinical study of healthy normal volunteers (E-34) during sub-maximal exercise testing and sleep. Figure 36 shows the impact of exercise (i.e. stair stepping and moderate treadmill) and other activities (i.e. Valsalva maneuvers and sleep) on the AutoStim Potential False Positive rate.

Figure 36. Non-Seizure Heart Rate Challenges



Note: In Figure 36, N=49 patients

6.3.3.4 Low heart rate and prone position detection

Caution: Low heart rate and prone position events are for informational purposes only. Detected events are not to be used for alarms or medical diagnosis.

Clinical data suggest that events of cardiac arrest and/or respiratory arrest, possibly aggravated by the prone position, are precursors to instances of Sudden Unexpected Death in Epilepsy (SUDEP)¹. For patients with epilepsy, the M1000 generator can detect and log low heart rate and prone position events if they are of interest to the physician. These events are detected after AutoStim or Magnet Mode stimulations, and Tachycardia Detection must be enabled in order to log low heart rate and prone position events.

Detection for low heart rate and prone position events are independently configurable. For use of low heart rate detection, the physician must define a detection threshold, specific for the patient, from 30 to 60 bpm in 10 bpm increments. For prone position detection, a calibration with the patient in supine and upright positions is required prior to the feature activation. Detected events are stored within the generator's memory and viewable during patient follow-up visits through the programmer.

Note: For additional information on use of the feature, refer to the programming system physician's manuals.

6.3.3.5 Scheduled Programming

Caution: This feature may not be appropriate for use in patients who are nonverbal or are unable to use the patient magnet to stop undesired stimulation. Similarly, exercise caution for use of this feature in patients with a history of obstructive sleep apnea, shortness of breath, coughing, swallowing difficulties, or aspiration.

¹ Ryvlin, Philippe et al. Incidence and mechanisms of cardiorespiratory arrests in epilepsy monitoring units (MORTEMUS): a retrospective study. The Lancet Neurology, Volume 12, Issue 10, 966 - 977

Scheduled Programming is an optional feature intended for use in patients with epilepsy. The feature allows the physician to program the generator so that it automatically increases stimulation therapy parameters while the patient is in the comfort of his or her home. This feature is intended to be used during the titration phase and could potentially lessen the number of office visits the patient will need to travel to and from the clinic for programming increases only. Physicians have the option of creating a custom programming schedule, or selecting and confirming the use of a standard schedule. The programming schedule is limited to a maximum of 7 steps and the physician specifies the parameter settings for each step as well as the time between steps. Once programmed into the generator, the generator will deliver the stimulation increases for each step at the time(s) and date(s) set by the physician.

If using this feature, it is highly recommended that physicians communicate the date(s) and time(s) of the programming schedule to the patient and/or caregiver so the patient is aware of upcoming parameter increases. If a patient is unable to tolerate a scheduled therapy increase, instruct the patient to disable VNS stimulation with the magnet (i.e. place magnet over the generator) and follow-up with the physician for programming adjustment.

 **Note:** For additional information on use of the feature, refer to the programming system physician's manuals.

6.3.3.6 *Day-Night Programming*

 **Caution:** Time-based features (e.g., Scheduled Programming, Day-Night Programming) do not automatically adjust for Day Light Savings or time zone changes. Tell the patient to follow-up with the physician for reprogramming if needed.

Day-Night Programming is an optional feature that allows the generator to deliver 2 independent sets of therapy parameters at different times during a 24-hour period. The physician specifies what parameters will change, and a time period during the 24-hours when the alternate parameter set should be active. After the Day-Night program has been defined, the generator will alternate between the 2 independent parameter sets on a daily basis. This feature provides the physician the ability to further customize the delivery of VNS Therapy to accommodate to each individual patient's needs after a target level has been established for the patient.

As with any therapy setting change, the risk and benefits of altering a patient's known efficacious settings should be considered when making adjustments. Inform your patients about when to expect a setting change (i.e. when Daytime settings transition into Nighttime settings). In addition, patient tolerability of the alternate parameter set should be assessed prior to the patient leaving the office visit.

 **Note:** For additional information on use of the feature, refer to the programming system physician's manuals.

6.3.4 *Stimulation Parameters, Duty Cycle, and Impacts on Battery Life*

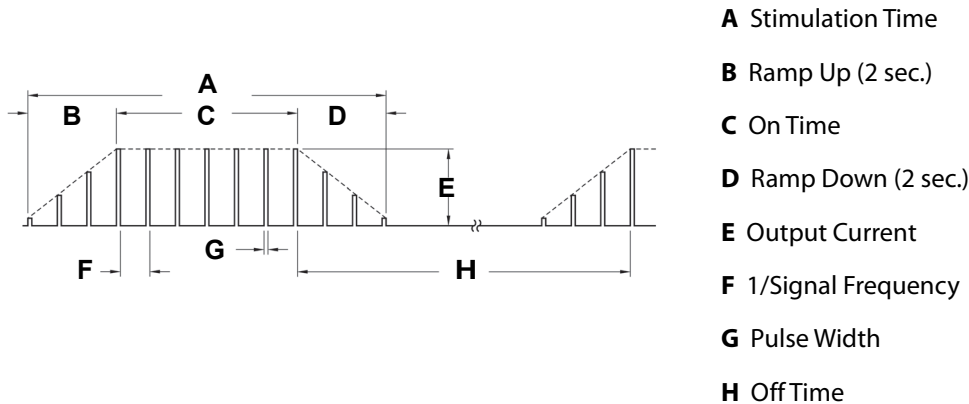
6.3.4.1 *Programmable parameters*

A graphic representation of stimulation (Figure 37) depicts the relationship of the programmable parameters. Each parameter can be independently programmed, thereby

offering multiple setting combinations from which the physician may select optimal stimulation for the patient.

Figure 37 shows that the output pulse can be varied both by amplitude (output current) and duration (pulse width). The number of output pulses delivered per second determines the frequency.

Figure 37. Stimulation (Frequencies < 10 Hz do not ramp)



6.3.4.2 Duty cycle

The percentage of time the generator is stimulating is called a “duty cycle.” A duty cycle is calculated by dividing the stimulation time (programmed Normal Mode ON time plus, if frequency is ≥ 10 Hz, 2 seconds of ramp-up time and 2 seconds of ramp-down time) by the sum of the ON and OFF times. The various parameter settings for stimulation are listed in “Specifications and Product Information”.

⚠ Caution: *Possible nerve damage with ON time > OFF time*— Excessive stimulation has resulted in degenerative nerve damage in laboratory animals. One component of excessive stimulation is when ON time exceeds OFF time, which can be produced by continuous or frequent magnet activation (> 8 hours). LivaNova recommends that stimulation at these combinations of ranges be avoided.

Table 18 shows duty cycles for typical ON time and OFF time settings.

Table 18. Duty Cycles for Various ON and OFF Time Settings

	OFF Time (min)								
	0.2	0.3	0.5	0.8	1.1	1.8	3	5	10
ON Time (sec)	Duty Cycles* (% ON Time)								
7	58	44	30	20	15	10	6	4	2
14	69	56	41	29	23	15	9	6	3
21	76	64	49	36	29	19	12	8	4
30	81	71	57	44	35	25	16	10	5
60	89	82	71	59	51	38	27	18	10

* A duty cycle is calculated by dividing stimulation time (programmed ON time plus 2 seconds of ramp-up time and 2 seconds of ramp-down time) by the sum of the ON time and the OFF time.

* The duty cycles in gray are not recommended as they represent parameter combinations with ON Time > OFF Time.

Note: If Tachycardia Detection is enabled and AutoStim output current is > 0 mA, Normal Mode OFF Times < 1.1 minutes are not available for programming.

6.3.4.3 Parameter settings and battery life

When selecting a combination of parameter settings for stimulation, the physician should also consider that some combinations would decrease battery life faster than others. In addition, operating the M1000 with Tachycardia Detection and/or additional features such as prone position detection turned “ON” will also decrease the battery life.

Note: See “Generator battery longevity”.

Table 19 shows the longevity impact that the AutoStim Mode feature has with a typical lead impedance (3 kOhms) and the following Normal mode settings: 2 mA output current, 20 Hz signal frequency, 250 µsec pulse width, and 10% duty cycle (30 sec ON, 5 min OFF).

Table 19. Estimated Model 1000 Longevity with Sensing and AutoStim

AutoStim Feature	Expected Life (yrs)
Tachycardia Detection / AutoStim “OFF”	11.5

	AutoStim ON-Time	
Tachycardia Detection / AutoStim “ON”	30 sec	60 sec
AutoStims Per Hour	Expected Life (yrs)	
1	7.6	7.5
7	7.0	6.1
15	6.4	4.9

6.3.5 VNS Therapy magnets

There are four possible uses of the magnet:

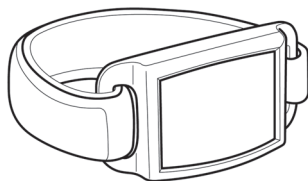
- To provide on-demand stimulation as an attempt to abort or de-intensify an oncoming seizure or a seizure in progress. During an aura or at the start of a seizure, magnet activation may be initiated by the patient, a companion, or the physician by applying or passing a magnet over the generator to activate a reed switch in the generator's electronic circuitry. This action changes the generator from Normal Mode or AutoStim to Magnet Mode.
- To temporarily inhibit stimulation
- To reset the generator (in combination with the programming system)
- To test daily the functioning of the generator, LivaNova recommends that patients be instructed to use the magnet to activate stimulation. Note that this indirectly tests the generator through the ability of the patient to perceive Magnet Mode stimulation. Since patients may become accustomed to their stimulation settings over time, it is recommended that physicians always use the Diagnostics testing available in the programming software in order to formally test the implanted system.



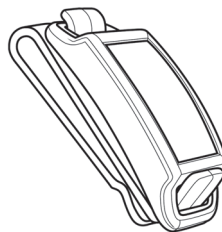
Caution: *Possible nerve damage with ON time > OFF time*— Excessive stimulation has resulted in degenerative nerve damage in laboratory animals. One component of excessive stimulation is when ON time exceeds OFF time, which can be produced by continuous or frequent magnet activation (> 8 hours). LivaNova recommends that stimulation at these combinations of ranges be avoided.

Two identical VNS Therapy magnets (see Figure 38), each providing a minimum of 50 gauss at 1 inch, are supplied inside the patient kit by LivaNova. The watch-style accessory attaches the magnet to the wrist with a wristband, and the pager-style accessory holds the magnet in a belt clip with a quick-release mechanism like a pager.

Figure 38. Magnet Accessories



**Watch-Style
(wristband)**

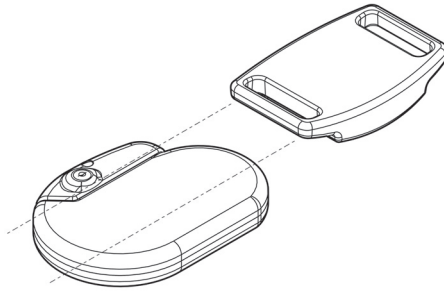


**Pager-Style
(belt clip)**

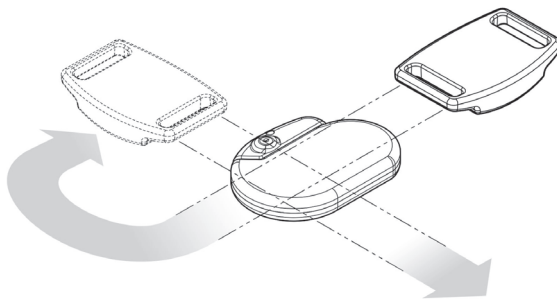
The proper orientation and motion for initiating magnet activation is shown in Figure 39. The magnet is shown without the belt clip or wristband to illustrate the proper orientation of the magnet to the generator.

Figure 39. Initiate Magnet Activation

Standard Magnet Activation



Optional Cross-Pattern Magnet Activation



Caution: To activate or stop stimulation, the label side of the magnet should face the generator.

6.3.5.1 Service life of magnet

All magnets may lose their effectiveness over time. Both styles of VNS Therapy magnets contain a high-power magnet that is surrounded by a plastic casing in the shape of a watch. These magnets should be operated and stored at temperatures ranging from - 20 °C (- 4 °F) to + 55 °C (+ 131 °F). With normal use, they should remain powerful for approximately 3 years. Avoid dropping the magnets or storing them near other magnets.

6.3.5.2 Magnet activation technique

To initiate stimulation, apply or pass the magnet over the generator for 1-2 seconds and then immediately remove it from the area over the generator. A cross-pattern swipe, as shown in Figure 39, may also be used by the patient or caregiver to activate the Magnet Mode if difficulty is encountered with a single pass of the magnet. Removal of the magnet causes the generator to operate in Magnet Mode, delivering a single stimulation at the programmed magnet pulse width, magnet current, and magnet signal ON time settings. The frequency is the programmed value for Normal Mode. A Magnet Mode stimulation will always override any Normal Mode programmed stimulation. If Magnet Mode stimulation is not desired, the Magnet Mode output current may be programmed to 0 mA.



Caution: The cross-pattern swipe technique may cause duplicate magnet activation entries to be shown in the programming software database. This is an expected occurrence due to device design and is not considered a device malfunction.

LivaNova recommends that tests of the magnet output be performed while the patient is still in the physician's office to ensure tolerability of the magnet output.

6.3.5.3 *Inhibit generator output with the magnet*

Application of the magnet during stimulation will inhibit the output. In addition, holding the magnet in place for at least 10 seconds will terminate any ongoing Normal or AutoStim Mode stimulation. After the magnet is removed, Normal Mode operation will resume with stimulation when one complete OFF time has elapsed.



Caution: If stimulation becomes painful, the patient should be instructed to stop the stimulation with the magnet.

In the unlikely event of continuous stimulation or other malfunction, the patient must be advised to apply the magnet, secure it in place, and immediately notify their physician.



Note: See "Adverse Events" in the indication-specific information chapters.

6.3.5.4 *Reset the microprocessor with the magnet and the programming system*

The VNS Therapy System allows the generator microprocessor to be reset in the event of a malfunction. Resetting is necessary only in the rare case of microprocessor memory malfunction, which might be caused by conditions described in the *Introduction to the VNS Therapy System* chapter. Resetting the microprocessor may be appropriate when the generator and the programming system are unable to communicate.



Caution: *Generator reset*— When the generator is reset, its optional features (such as Day-Night Programming) and stimulation output are disabled (0 mA); however, all settings and device history are preserved. After a successful reset, the generator stimulation output may be re-enabled to resume operation at the previously programmed settings and optional features reactivated.

For suggestions in solving communication difficulties, see "Troubleshooting" in the programming system physician's manuals.

For instructions on how to reset the microprocessor, see the programming system physician's manuals. It is recommended, except in cases of a medical emergency, that the physician consult a LivaNova technical representative before a reset is performed.

6.3.6 **Device History**

The generator device history consists of generator serial number, model number, patient code, implantation date, and other information pertinent to diagnostic and programming events. Use the programming system to access and view generator device history information.

6.3.7 **Device Diagnostics**

Information from device diagnostic tests can help the physician determine whether the:

- Generator output current is being delivered at the programmed value
- Lead impedance is within an acceptable range
- Generator battery is at a sufficient level



Note: For details on available diagnostic tests, see the programming system physician's manuals.

6.3.7.1 ***System Diagnostics test***

The System Diagnostics evaluates the lead impedance of the VNS Therapy System, as well as the generator's ability to deliver the programmed Normal Mode stimulation. During this test, the generator delivers a pulse at 0.25 mA, 130 μ sec to assess lead impedance. The generator will then deliver the programmed output stimulus. The programming software will report the lead impedance and whether the programmed stimulus was delivered.

6.3.7.2 ***Reasons for high or low lead impedance readings***

The System Diagnostics evaluates lead impedance of the VNS Therapy System. The lead impedance measurement is performed with a single pulse at 0.25 mA, 130 μ sec. Additionally, once programmed on, the generator automatically performs a lead impedance measurement once every 24 hours. If a HIGH or LOW impedance is detected, the user is notified after an interrogation with the programming system.

High lead impedance is defined as any value ≥ 5300 Ohms. Low lead impedance is defined as any value ≤ 600 Ohms.



Caution: Possible causes of high lead impedance readings are thought to include: lead discontinuity, lead disconnection from the generator, fibrosis between the nerve and the electrode, electrode detachment from the nerve, or a defective generator.



Caution: Possible causes of low lead impedance readings are thought to include: short-circuit condition within the lead or a defective generator.

6.3.7.3 ***High lead impedance: possible implications***

High lead impedance (≥ 5300 Ohms), in the absence of other device-related complications, is not an indication of a lead or generator malfunction. High lead impedance in combination with the patient's failure to feel even the maximum output stimulus may indicate a lead wire fracture or other type of electrical discontinuity in the lead. Complications with heartbeat sensing may also be indicative of a lead discontinuity. Patients experiencing high lead impedance, no sensation of maximum output stimulation, and an increase in seizures/depressive symptoms should be further evaluated for possible lead replacement.



Note: For additional instructions on how to perform the System Diagnostics, see the programming system physician's manuals.



Note: To troubleshoot high or low impedance see "Troubleshooting" in the programming system physician's manuals.

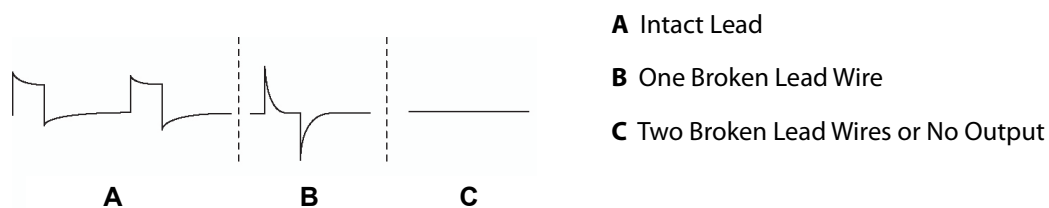
6.3.7.4 **Low lead Impedance: possible implications**

Low lead impedance (≤ 600 Ohms) likely indicates the existence of a short-circuit condition, although an impedance value of greater than 600 Ohms does not exclude the possibility. A sudden decrease in impedance value in combination with device-related complications (e.g., increase in seizures/depressive symptoms or painful stimulation; patient perception of feeling erratic, limited, no stimulation, or complications detecting heartbeats) may also indicate a short-circuit condition in the lead.

6.3.7.5 **Stimulus waveform analysis**

Either evoked potential monitoring equipment or an oscilloscope can be used to analyze the stimulus waveform from the neck for verification of an electrical discontinuity. A differentiated waveform with narrowed pulses or no waveform at all can confirm a discontinuity. Figure 40 shows characteristic waveforms obtained from skin electrodes for a lead that is intact and for a lead that has a fracture in one or both wires. In addition, lead discontinuities can sometimes be identified on an x-ray of the implant site.

Figure 40. Typical Waveforms Obtained from Skin Electrodes



6.3.8 **Delivery of programmed output current**

6.3.8.1 **LOW as output current**

If the diagnostic tests indicate LOW output current, the generator may not be delivering the programmed output current. Reasons for failure to deliver the programmed output current include high programmed output current and high lead impedance. The maximum deliverable output current, according to Ohm's Law, equals the maximum output voltage (approximately 12 V) divided by the lead impedance.

6.3.8.2 **Reprogram to a lower current**

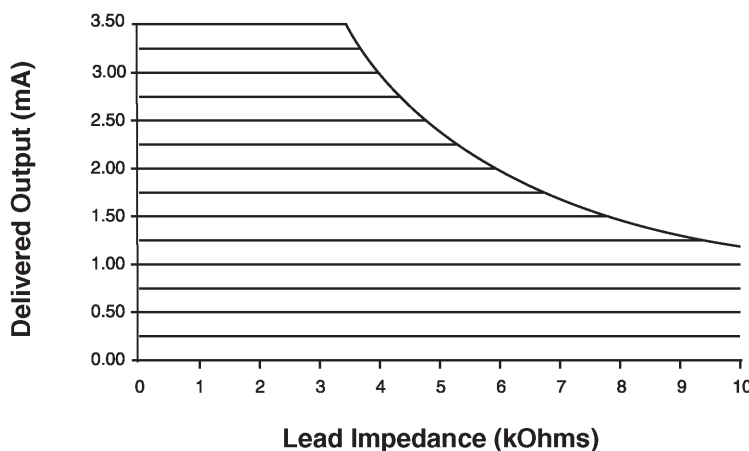
If the generator is failing to deliver the programmed output current, the physician can reprogram the device to a lower output current and attempt to compensate for the decrease in delivered energy by widening the pulse width. For example, if the diagnostics read LOW for a generator programmed at 2.5 mA, 30 Hz, 500 μ sec with 30 seconds ON time, then the parameters may be changed by lowering the output current to 2 mA and widening the pulse width to 750 μ sec.

6.3.9 **Charge delivered per pulse**

Output current x pulse width = charge delivered per pulse

The charge delivered per pulse is the most important parameter in evaluating stimulation output. It is defined as a microcoulomb (μC), which is the product of current and time—that is, the output current (mA) multiplied by the pulse width (msec). Figure 41 shows the relationship of delivered output current (mA) to lead impedance for a 1000 μsec pulse with output currents from 0 to 3.5 mA.

Figure 41. Relationship of Delivered Output Current to Lead Impedance



6.3.10 Generator battery longevity

6.3.10.1 Battery longevity and programmed setting choices

The anticipated longevity of the generator battery varies, depending on the choice of programmed settings. Higher output currents, frequencies, pulse widths, and duty cycles generally deplete the battery over a shorter period of time than lower settings. Generally, the increase in battery depletion rate is proportional to the increase in the programmed setting in the absence of detection.

Caution: *Undeliverable output currents*— Programming the generator to a high output current that cannot be delivered due to a high lead impedance may disproportionately increase the battery depletion rate and should be avoided.

Other factors, such as lead impedance, magnet usage or use of optional features (e.g., AutoStim Threshold settings, AutoStim) also affect the anticipated battery longevity. The anticipated battery longevity decreases as lead impedance increases. Although 1.5 k to 3 kOhms may be a typical lead impedance at implantation, the impedance may increase to 3 k to 5 kOhms during the life of the implant.

The *Appendices* chapter provides Model 1000 estimated battery lifetimes under a variety of stimulation conditions, with and without the AutoStim feature enabled. Because of the number of possible parameter combinations, it is impractical to provide the projected life for all possible combinations. The tables should not be used to predict battery end of service (EOS), but they give some indication of the effect of various parameter changes on battery life and can be used to assist in the selection of parameter settings. They also indicate that battery life can be maximized at low duty cycles and low frequencies (e.g., 20 Hz) for stimulation.

Note: For more information, see the programming system physician's manuals.

6.3.10.2 **Battery status indicators**

The programming software displays a battery indicator similar to an indicator that may be found in cell phones. The visual indicator illustrates the approximate remaining battery capacity.

The programming software will display warning messages after an interrogation or programming of the generator if the battery has been depleted to a level where action is recommended due to approaching or reaching End of Service (EOS). Please refer to the programming system physician's manuals for additional information on these indicators.

 **Caution:** *Battery evaluation at cold temperatures*—Low storage temperatures may affect the battery status indicators. In such cases, the battery status indicators should be re-evaluated using the System Diagnostics or Generator Diagnostics after the generator has been at room or body temperature for 30 minutes.

6.3.11 **Generator Replacement**

All VNS Therapy generators eventually require surgical replacement as a result of battery depletion. Generator replacement does not, of itself, require lead replacement unless a lead discontinuity is suspected. Generator replacement or removal requires dissection to the generator's pocket, with care being taken not to damage or cut the lead. The entire surgical procedure generally requires about 1 hour.

6.3.11.1 **Signs of End of Service**

The most common reason for the absence of stimulation is battery depletion, although there may be other reasons. When end of service (EOS) occurs, the generator will disable stimulation and no output will be delivered. If the generator is not explanted or replaced at EOS, the battery voltage will continue to gradually decrease and communication with the generator may not be possible.

 **Caution:** Generator EOS may result in increased frequency, intensity, or duration of signs and symptoms of the patient's disorder, in some cases to levels greater than those reported before stimulation.

6.3.11.2 **Replacement Based on Battery Status Indicators**

The generators and the programming system have battery status indicators. These indicators provide warnings that a generator battery should be monitored more frequently, is nearing EOS, or has reached EOS. Once these warning messages appear, see recommendations in the programming system physician's manuals.

 **Note:** See "Battery status indicators".

 **Caution:** *Prompt generator replacement*—LivaNova recommends prompt replacement of the generator at or before EOS. Prompt replacement may help minimize any possible relapse.

 **Caution:** *Explanted generator*—A generator explanted for any reason should not be reimplanted. An explanted generator should be returned to LivaNova. (For instructions on returning an explanted generator, see the *Introduction to the VNS Therapy System* chapter.)

7 Technical Information — Leads

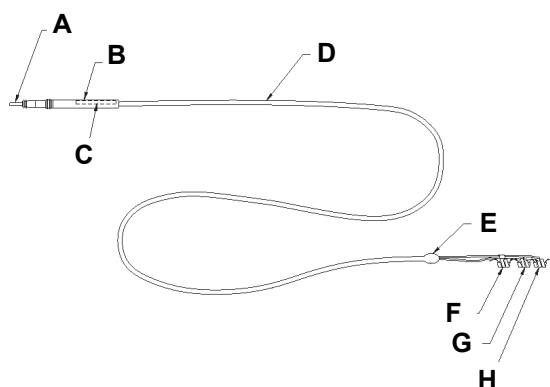
7.1 Brief Device Description

7.1.1 Description

The VNS Therapy Model 302, PerenniaDURA[®] Model 303, and PerenniaFLEX[®] Model 304 (*where available*) Leads are bifurcated at one end and have a single connector pin at the other end, as shown in Figure 42 and Figure 43.

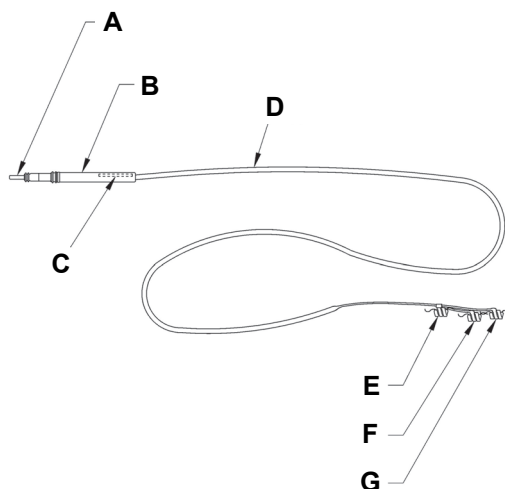
The lead, which delivers the electrical signal from the generator to the vagus nerve, is insulated with silicone. It is available in two sizes (2.0 and 3.0 mm electrode inner diameter) to ensure optimal electrode fit on different size nerves. The lead has two helical electrodes and an anchor tether, which are coiled around the left vagus nerve. The connector end of the lead is tunneled subcutaneously to the generator pocket.

Figure 42. Model 302 and 304 Lead



- A** Connector Pin
- B** Lead Connector
- C** Model/Serial Number Tag
- D** Lead Body
- E** Electrode Bifurcation
- F** Anchor Tether
- G** Electrode (+) (White Suture)
- H** Electrode (-) (Green Suture)

Figure 43. Model 303 Lead



- A** Connector Pin
- B** Lead Connector
- C** Model/Serial Number Tag
- D** Lead Body
- E** Anchor Tether
- F** Electrode (+) (White Suture)
- G** Electrode (-) (Green Suture)

7.1.2 VNS Therapy System Lead Compatibility

The VNS Therapy leads are compatible with the VNS Therapy single-receptacle generators and the VNS Therapy System.

7.2 Product Specifications

Table 20. Product Specifications

	Model 302	Model 303	Model 304
Lead Connector			
Diameter	3.2 mm (0.127 in)		
Material	Silicone*		
Connector Pin			
Diameter	1.27 mm (0.05 in)		
Material	300 series Stainless Steel		
Connector Ring			
Diameter	2.67 mm (0.105 in)		
Material	300 series Stainless Steel		
Lead Body			
Diameter	2 mm (0.08 in)		
Insulation	Silicone*		
Conductor coil construction	Helical, quadfilar	Helical, trifilar	Helical, quadfilar
Conductor material	MP-35N alloy		
Overall length	43 cm (17 in)		
Lead resistance (connector pin/ ring to electrode)	120 to 180 Ohms	180 to 250 Ohms	120 to 180 Ohms
Electrodes and Anchor Tether			
Helical material	Silicone elastomer*		
Conductor material	Platinum/Iridium alloy		
Separation	8 mm (0.31 in) center to center		
Suture material	Polyester		
Inner Diameter of Helix			
-20	2 mm (0.08 in) inner diameter		
-30	3 mm (0.12 in) inner diameter		
Tie-downs			
Dimensions	5.7 mm x 7.7 mm (0.22 in x 0.30 in)		
Material	Radiopaque silicone*		
Connector Assembly			
Lead connector	One (1)		
Connector Retention Strength			
With VNS Therapy generator	> 10 N		

* No component of the VNS Therapy System is made with natural rubber latex.

Integrity information about the lead can be obtained using the Programming System. The software includes a System Diagnostics (Lead Test) feature that can be used to assess lead impedance.

7.3 Lead Lifespan and Replacement

The lead's lifespan is undetermined at this time. A lead would require replacement if a lead fracture were suspected, accompanied by increased symptoms (e.g., seizure frequency). Events that can shorten the life expectancy of the lead are as follows:

- Blunt trauma to the neck and/or any area of the body beneath which the lead is implanted
- Patient's twisting or picking at either the implanted lead or pulse generator
- Improper surgical implantation of the VNS Therapy System, including (but not limited to) providing an inadequate strain relief loop, placing sutures directly on the lead body, not using the tie-downs, and suturing to muscle



Caution: Lead replacement or removal due to lack of efficacy is a medical judgment based on the patient's desires and health status, and must be carefully weighed against the known and unknown risks of surgery. At present, there are no known long-term hazards or risks associated with leaving the lead implanted, beyond those already mentioned in this physician's manual.

8 Troubleshooting

8.1 Model 102 and 102R

8.1.1 “Patient Cannot Feel Stimulation” at Follow-up Visit (Models 102-102R)

A patient may not feel stimulation if any of the following situations exist:

- Patient has become accustomed to the programmed setting
- Device is approaching its end of service (EOS)
- “High” lead impedance
- Short-circuit condition within the lead
- Generator issue

To determine the cause of the situation, perform the following steps:

1. Swipe the magnet. Ask the patient if they feel the magnet activation, experience any voice alteration, or experience any other common side effect to indicate the presence of stimulation.



Note: Ensure that the technique for swiping the magnet over the device is correct according to the section “Initiating stimulation with a magnet” in the 102/102R Technical Information chapter. Also see “Potential Adverse Events” in the indication-specific information chapters for a complete list of possible adverse events.

2. Interrogate the generator.
3. Perform a System Diagnostics (Lead Test) and record the results.

IF...	THEN...
Model 250 version 11.0 software and below: If the DC-DC Converter Code is “0” or there has been a significant decrease in DC-DC Converter Code value (e.g., “3” to “1”) in respect to prior System Diagnostics Model 3000 version 1.0 software and above: If the impedance is ≤ 1700 Ohms or if there has been a sudden change in impedance range (e.g., 4100-5200 Ohms to 1800-2800) in respect to prior System Diagnostics	A short-circuit condition may be present within the lead and the patient may not be receiving the intended therapy. For more information, see “Short-circuit conditions within the lead” in the <i>102/102R Technical Information</i> chapter.
Model 250 version 11.0 software and below: If the DC-DC Converter Code is not “0”, there has been no significant decrease in DC-DC Converter Code value (e.g., “3” to “1”) in respect to prior System Diagnostics, and the System Diagnostics test indicates the lead impedance is “OK” Model 3000 version 1.0 software and above: If System Diagnostics test indicates the lead impedance is OK	The system is functioning properly and the patient could have become accustomed to the settings, as do many patients.
If the System Diagnostics test indicates the lead impedance is “High”	See “Troubleshooting” in the programming system physician’s manuals.



Caution: For the System Diagnostics (Lead Test), the software automatically programs the generator to 1 mA, 500 μ sec, and 20 Hz. Patients whose generator output current is normally *less* than these values may experience increased sensation, coughing, a flushed face, or other effects. For a complete list of possible adverse events, see “Potential Adverse Events” in the indication-specific information chapters.

4. Perform a Normal Mode Diagnostics test and record the results.

IF...	THEN...
The Normal Mode Diagnostics test indicates the Output Current is "LIMIT"	The generator cannot deliver programmed output. Consider reducing output current or frequency and widening the pulse width.
The Normal Mode Diagnostics test indicates the Output Current is "OK"	The generator can deliver the programmed output current.  Note: To obtain accurate information from the device diagnostics, the generator must be programmed to a minimum of 0.75 mA, 15 Hz, and at least 30 seconds ON time.
The Normal Mode Diagnostics test indicates "HIGH" lead impedance	See "Troubleshooting" in the programming system physician's manuals.

5. If further assistance is needed, call Clinical Technical Support (CTS) at 1 (866) 882-8804 (U.S. and Canada), +1 (281) 228-7330 (Worldwide), or +32 2 790 27 73 (Europe/EMMEA).

8.1.2 Magnet Activation Not Working at Follow-up Visit (Models 102-102R)

A patient's magnet activation may not be working if any of the following situations exist:

- Patient may have become accustomed to the programmed setting.
- An incorrect technique is used for swiping the magnet.
- Magnet output current is not programmed to ON.
- Device is approaching its end of service (EOS).
- Device was implanted too deep.
- There is an issue with the generator.
- "High" lead impedance
- Short-circuit condition within the lead

To determine the cause of the situation, perform the following steps:

1. Interrogate the device.
2. Confirm that the Magnet Output Current is ≥ 0.25 mA and Magnet ON time is > 7 seconds.
3. Record the number of magnet activations listed under Device History or Events screen of the programming software.
4. Swipe the magnet over the device and watch for a clinical response to the stimulation. Wait 3 to 4 minutes and re-interrogate the device.

 **Note:** Ensure that the technique for swiping the magnet over the device is correct according to the section "Initiating stimulation with a magnet" 102/102R *Technical Information* chapter.

 **Note:** Follow the listed instructions and swipe the magnet just before starting the test. To obtain accurate information from the device diagnostics, the generator must be programmed to a **minimum** of 0.75 mA (Magnet Output Current), 15 Hz (Normal Mode Frequency), and 30 seconds (Magnet ON Time).

5. Record the number of magnet activations again. The number of activations should have increased by 1.

6. If the number of magnet activations increased but the patient does not feel magnet-induced stimulation, increase the magnet output current until the magnet-induced stimulation is felt.
7. If the number of magnet activations did not increase, perform a Magnet Mode Diagnostics test and record all results.

<i>IF...</i>	<i>THEN...</i>
The Magnet Mode Diagnostics test indicates OK results	The magnet is functioning properly and the patient could have become accustomed to the settings, as do many patients.
The Magnet Mode Diagnostics test indicates device status "STANDBY" and output current "****", or gives a message that the magnet swipe was not detected	Perform steps 1 through 7 with an alternate VNS Therapy magnet.
The Magnet Mode Diagnostics test indicates "HIGH" lead impedance	See "Troubleshooting" in the programming system physician's manuals.

8. If further assistance is needed, call CTS at 1 (866) 882-8804 (U.S. and Canada), +1 (281) 228-7330 (Worldwide), or +32 2 790 27 73 (Europe/EMMEA).

8.2 Model 103, 104, 105 106, and 1000

8.2.1 “Patient Cannot Feel Stimulation” at Follow-up Visit (Models 103-106 and 1000)

A patient may not feel stimulation under any of these conditions:

- Patient has become accustomed to the programmed setting
- Generator battery at end of service (EOS)
- “High” lead impedance
- Defective generator
- Disabled generator
- Short-circuit condition within the lead

To determine the cause of the condition, perform these steps (see Figure 44):

1. Interrogate the generator.

- If the following message appears, call CTS: “The generator is currently disabled due to [CAUSE]. The generator is NOT supplying stimulation.”



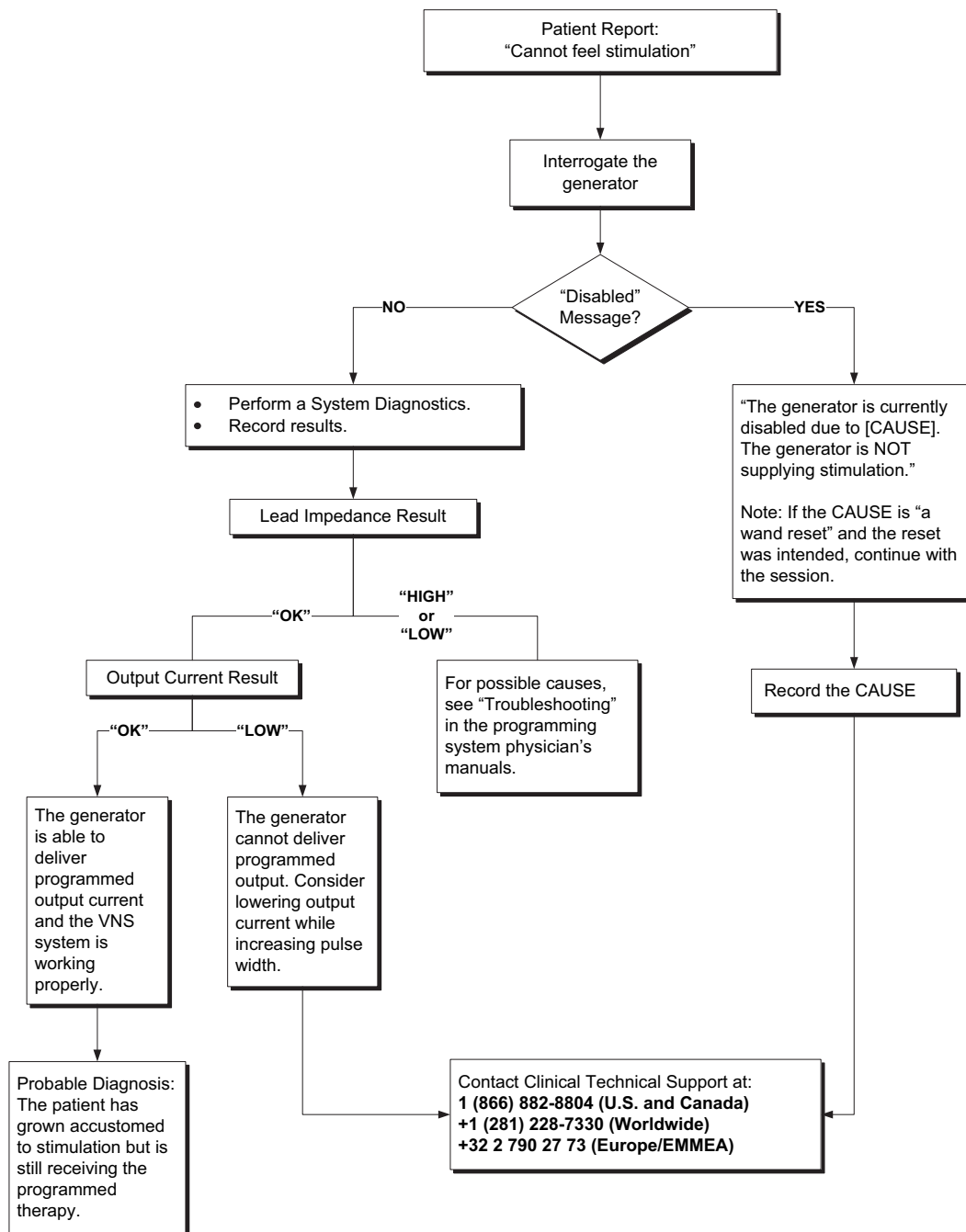
Note: If the CAUSE is “a wand reset” and the reset was intended, continue with the session.

2. Perform a System Diagnostics and record the results.

- If the output current reports “OK” and lead impedance reports “OK,” then the generator is able to deliver the programmed therapy and the patient may have become accustomed to the stimulation, as do many patients.
- If the output current reports “OK” and lead impedance reports “LOW” (≤ 600 Ohms), then there is a possibility of a short-circuit condition within the lead. See “Troubleshooting” in the programming system physician’s manuals.
- If the output current reports “LOW” and lead impedance reports “OK,” then the generator cannot deliver programmed output due to increased impedance. Consider lowering the output current while increasing pulse width.
- If the output current reports “LOW” and lead impedance reports “HIGH” (≥ 5300 Ohms), see “Troubleshooting” in the programming system physician’s manuals.

3. For further assistance, call CTS at 1 (866) 882-8804 (U.S. and Canada), +1 (281) 228-7330 (Worldwide), or +32 2 790 27 73 (Europe/EMMEA).

Figure 44. “Patient Cannot Feel Stimulation” at Follow-Up Visit (Models 103-106 and 1000)



8.2.2 “Patient Cannot Feel Magnet Activation” at Follow-up Visit (Models 103-106 and 1000)

Magnet activation may not be perceived under any of these conditions:

- Patient has become accustomed to the programmed setting
- Incorrect technique for swiping magnet
- Magnet output current is programmed to 0 mA
- Generator battery at end of service (EOS)
- Device implanted too deep in the chest
- Defective generator
- Disabled generator
- “High” lead impedance
- Short-circuit condition within the lead

To determine the cause of the condition, perform these steps (see Figure 45):

1. Interrogate the device.
 - If the following message appears, call CTS: “The generator is currently disabled due to [CAUSE]. The generator is NOT supplying stimulation.”



Note: If the CAUSE is “a wand reset” and the reset was intended, continue with the session.

2. Confirm that the magnet output current is $\geq$ the Normal output current and magnet ON time is >7 seconds.
3. Record the number of magnet activations listed on the Device History or Events screen of the programming software.
4. Swipe the magnet over the device.



Note: Ensure that the technique for swiping the magnet over the device is correct. (See “VNS Therapy magnets” in the 103/104, 105, 106 and 1000 Technical Information chapters.)



The number of magnet activations may increase by 1 or 2 with the cross-pattern swipe technique.

5. Reinterrogate the device.
6. Check the number of magnet activations again. The number of magnet activations should have increased.
 - If the number of magnet activations increased, the generator is delivering the magnet stimulation. Consider raising the magnet current if perceived stimulation is desired.
 - If the number of magnet activations did not increase with the test swipe, perform a Magnet Mode Diagnostics. Record all results.
 - ◆ If the generator did not sense a magnet swipe, then you will see an indication that the generator did not sense a magnet. Swipe the magnet again and repeat Magnet Mode Diagnostics. If the warning appears again, call CTS at

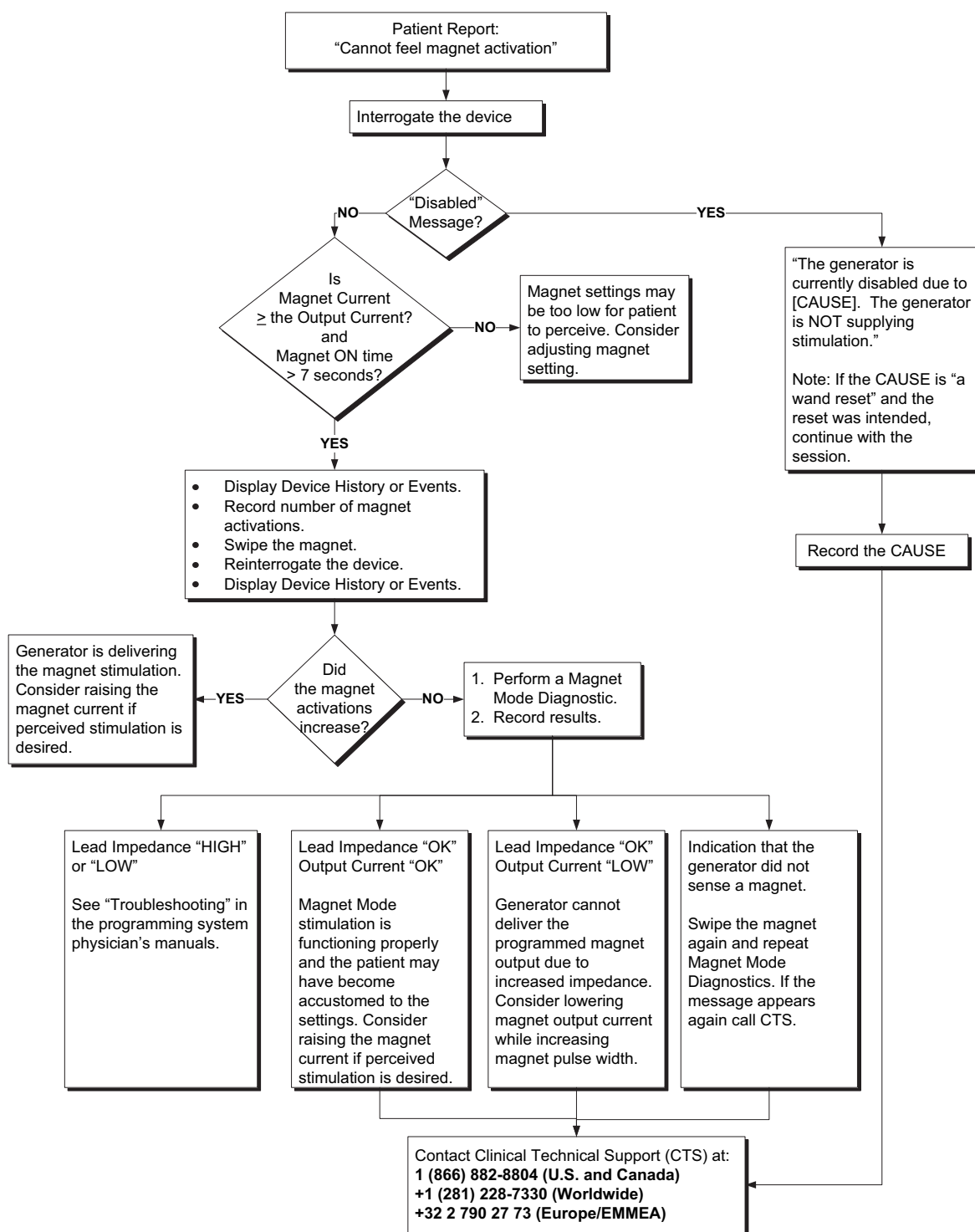
1 (866) 882-8804 (U.S. and Canada), +1 (281) 228-7330 (Worldwide), or +32 2 790 27 73 (Europe/EMMEA).



Note: Follow the listed instructions and swipe the magnet just before starting the test.

- ♦ If the output current reports “OK” and lead impedance reports “OK,” then the Magnet Mode stimulation is functioning properly and the patient may have become accustomed to the settings.
 - ♦ If the output current reports “LOW” and lead impedance reports “OK,” then the generator cannot deliver the programmed magnet output due to increased impedance. Consider lowering the magnet output current while increasing the magnet pulse width.
 - ♦ If the output current reports “LOW” and lead impedance reports “HIGH,” see “Troubleshooting” in the programming system physician’s manuals.
7. For further assistance, call CTS at 1 (866) 882-8804 (U.S. and Canada), +1 (281) 228-7330 (Worldwide), or +32 2 790 27 73 (Europe/EMMEA).

Figure 45. “Patient Cannot Feel Magnet Activation” at Follow-Up Visit (Models 103-106 and 1000)



8.2.3 “Patient Does Not Perceive AutoStim Activation” at Follow-up (Generators with AutoStim only)

AutoStim activation may not be perceived under any of these conditions:

- AutoStim Threshold is too high (e.g., 70% threshold versus 50%)
- Patient has become accustomed to the programmed setting
- AutoStim output current is programmed to 0 mA
- Generator battery at end of service (EOS)
- Defective generator or lead
- Disabled generator

If AutoStim activation is not perceived, perform these steps (see Figure 46):

1. Ensure the programming computer is unplugged and Interrogate the device.
 - If the following message appears, call CTS: “The generator is currently disabled due to [CAUSE]. The generator is NOT supplying stimulation.”



Note: If the CAUSE is “a wand reset” and the reset was intended, continue with the session.

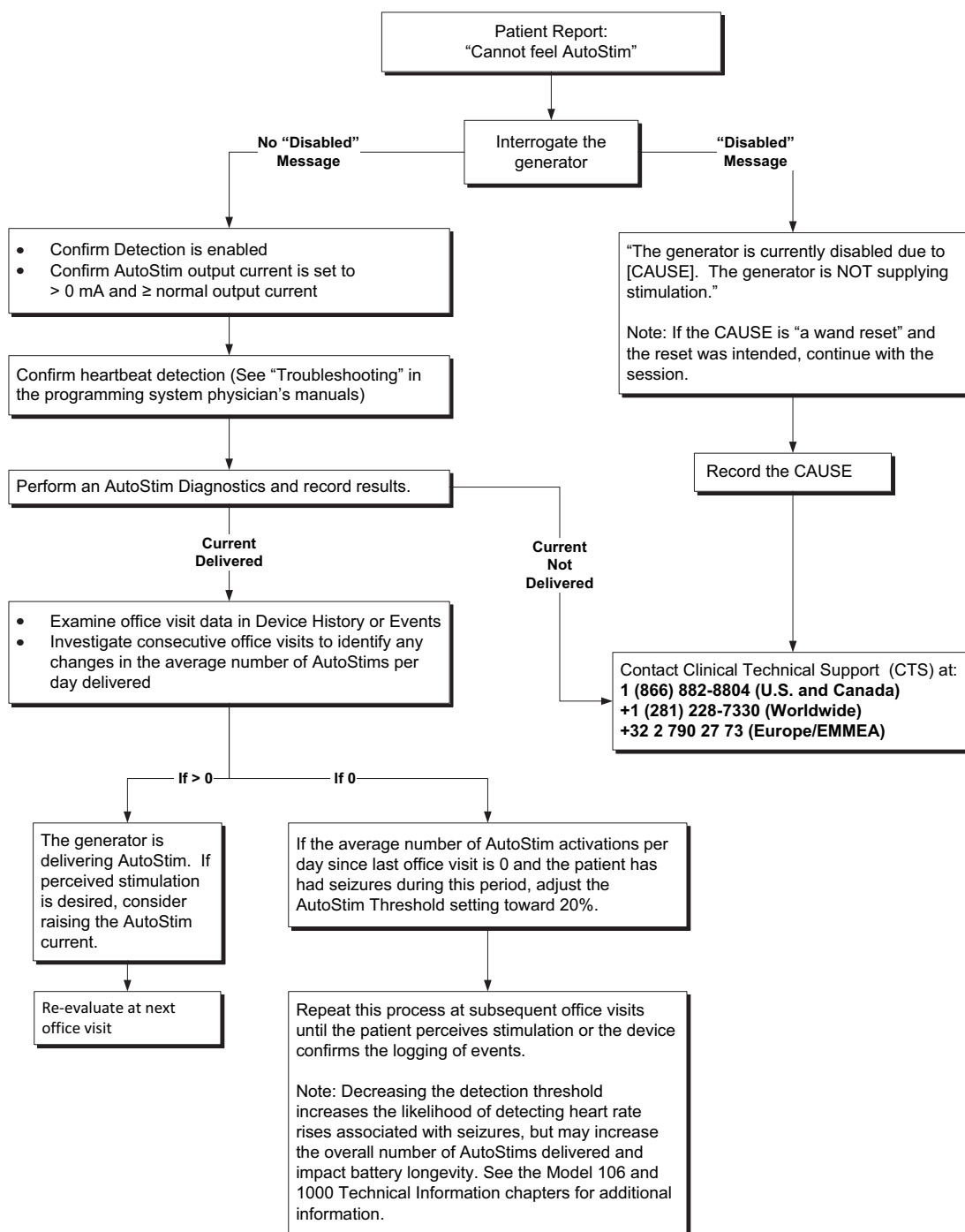
2. Confirm that Detection is enabled and the AutoStim output current is set to a value > 0 mA and $\geq$ the Normal output current.
3. Confirm heartbeat detection (See “Troubleshooting” in the programming system physician’s manuals)
4. Perform AutoStim Diagnostics from the Device Diagnostics Menu.
 - If the diagnostics indicates that the AutoStim output current was delivered, re-evaluate at the next office visit.
5. Look at the Office Visit data under Device History or Events and investigate consecutive office visits to identify any changes in the average number of AutoStims per day delivered by the device.
6. Make note of the average number of AutoStims per day since the last visit.
 - If the average number of AutoStim per day is > 0 , the generator is delivering the AutoStim stimulation as detected by the algorithm. Consider raising the AutoStim current if perceived stimulation is desired.
 - If the average number of AutoStim activations per day since last office visit is 0 and the patient has had seizures during this period, adjust the AutoStim Threshold toward 20%. Repeat this process at subsequent office visits until the patient perceives stimulation or the device confirms the logging of events.



Note: Decreasing the detection threshold increases the likelihood of detecting heart rate rises associated with seizures, but may increase the overall number of AutoStims delivered and impact battery longevity. See Model 106 and 1000 *Technical Information* chapters for additional information.

7. For further assistance, call CTS at 1 (866) 882-8804 (U.S. and Canada), +1 (281) 228-7330 (Worldwide), or +32 2 790 27 73 (Europe/EMMEA).

Figure 46. “Patient Does Not Perceive AutoStim Activation” at Follow-up (Generators with AutoStim Only)



9 Depression Information

9.1 Pivotal and Pilot Clinical Studies

9.1.1 Pivotal and Pilot Studies—Safety

Except where noted otherwise, the safety information presented in this section derives from the pivotal (D-02) study. The D-02 study of VNS Therapy[®] consisted of both an acute and a long-term phase to collect data regarding the safety and efficacy of VNS Therapy as an adjunctive treatment for persons with chronic or recurrent treatment-resistant depression.

 **Note:** For intended use/indications, see the *Introduction to the VNS Therapy System* chapter.

9.1.1.1 Device performance

The VNS Therapy System performed according to its specifications. Most device issues were communication difficulties resolved by repositioning the programming wand or replacing the programming wand batteries. One high lead impedance occurred requiring replacement; a lead break due to fatigue at the electrode bifurcation was noted. Most device complaints were resolved on the day of initial complaint.

9.1.1.2 Adverse events

9.1.1.2.1 Events reported

The number (and percentage) of subjects reporting an adverse event during the 0-3 month period and during the 9-12 month period of the pivotal (D-02) study is depicted in Table 21 for the most commonly reported adverse events. Adverse events were coded using the COSTART 5 dictionary. Note that some subjects may have reported multiple events.

Table 21. Adverse Events Reported During VNS Therapy at 0-3 Months and 9-12 Months (D-02)

Adverse Event	0-3 Months (N=232)	9-12 Months (N=209)
Voice Alteration	135 (58.2%)	113 (54.1%)
Increased Cough	55 (23.7%)	13 (6.2%)
Neck Pain	38 (16.4%)	27 (12.9%)
Dyspnea	33 (14.2%)	34 (16.3%)
Dysphagia	31 (13.4%)	9 (4.3%)
Paresthesia	26 (11.2%)	9 (4.3%)
Laryngismus	23 (9.9%)	10 (4.8%)
Pharyngitis	14 (6.0%)	11 (5.3%)
Nausea	13 (5.6%)	4 (1.9%)
Pain	13 (5.6%)	13 (6.2%)
Headache	12 (5.2%)	8 (3.8%)
Insomnia	10 (4.3%)	2 (1.0%)
Palpitation	9 (3.9%)	6 (2.9%)
Chest Pain	9 (3.9%)	4 (1.9%)

Adverse Event	0-3 Months (N=232)	9-12 Months (N=209)
Dyspepsia	8 (3.4%)	4 (1.9%)
Hypertonia	6 (2.6%)	10 (4.8%)
Hypesthesia	6 (2.6%)	2 (1.0%)
Anxiety	5 (2.2%)	6 (2.9%)
Ear Pain	5 (2.2%)	6 (2.9%)
Eructation	4 (1.7%)	0
Diarrhea	4 (1.7%)	2 (1.0%)
Dizziness	4 (1.7%)	3 (1.4%)
Incision Site Reaction	4 (1.7%)	2 (1.0%)
Asthma	4 (1.7%)	3 (1.4%)
Device Site Reaction	4 (1.7%)	0
Device Site Pain	4 (1.7%)	2 (1.0%)
Migraine Headache	4 (1.7%)	2 (1.0%)

It is important to note that subjects often had comorbid illnesses and almost all study subjects were also receiving antidepressant and other drugs that could have contributed to these events.

9.1.1.2.2. Discontinuation due to adverse events

In the feasibility (D-01) study, no discontinuations were related to adverse events attributed to VNS Therapy or the implant procedure. By the time all continuing subjects in the pivotal (D-02) study had at least 1 year of VNS Therapy, 3% (8/235) of the subjects had discontinued VNS Therapy for an adverse event-related reason. The reasons for these 8 discontinuations included 1 case each of suicide, implant-related infection necessitating device removal, hoarseness, lightheadedness, post-operative pain, chest and arm pain, sudden death (of unknown cause), and worsening depression (reported by the investigator as an adverse event rather than as lack of efficacy).

9.1.1.3 Serious adverse events (SAEs)

9.1.1.3.1. SAEs

The SAEs described in this section are based on investigator reports from the pivotal (D-02) study from study initiation through the data cutoff date for submission; the data cutoff date included the entire period of evaluation for subjects who did not complete 12 months of VNS Therapy and included a minimum of 12 months of evaluation during VNS Therapy for all subjects who continued the study for 12 months or longer.

During the pivotal (D-02) study, 12 SAEs were considered related to the implant procedure (wound infection, asystole, bradycardia, syncope, abnormal thinking, vocal cord paralysis, aspiration pneumonia, voice alteration, device site reaction [2 reports], acute renal failure, and urinary retention). During the acute phase of the D-02 study, investigators did not report any SAE to be related to stimulation. During the long-term phase of the D-02 study, 8 SAEs were considered at least possibly related to stimulation (sudden death of unknown cause, syncope (2 reports), dizziness, a manic depressive reaction in a subject with bipolar disorder, hemorrhage GI, paresthesia, and an incident of worsening depression. Table 22 displays all the SAEs reported during the D-02 study prior to the data cutoff date, regardless of relationship to implantation or stimulation.

Table 22. Serious Adverse Events Reported in Study D-02, Regardless of Relationship to Implantation or Stimulation

Event	Acute (N=235)		Long Term (N=233)	
	Number of Events Treatment (N=119) / Sham (N=116)	Number of Subjects	Number of Events	Number of Subjects
Worsening Depression	5/7	11	62	31
Suicide Attempt	0	0	7	6
Syncope	0	0	4	3
Dehydration	1/1	2	1	1
Wound Infection	1/0	1	1	1
Cholecystitis	0/1	1	1	1
Gastrointestinal Disorder	0	0	2	2
Abnormal Thinking	1/0	1	1	1
Convulsion	0	0	2	2
Device Site Reaction	2/0	2	0	0
Pneumonia	0/1	1	0	0
Abdominal Pain	0	0	1	1
Accidental Injury	0	0	1	1
Chest Pain	0	0	1	1
Overdose	0	0	1	1
Peritonitis	0	0	1	1
Sudden Unexplained Death	0	0	1	1
Suicide	1/0	1	0	0
Surgical Procedure	0	0	1	1
Asystole	1/0	1	0	0
Bradycardia	1/0	1	0	0
Cholelithiasis	0	0	1	1
Constipation	0	0	1	1
Myasthenia	0/1	1	0	0
Confusion	1/0	1	0	0
Dizziness	0	0	1	1
Drug Dependence	0	0	1	1
Manic Depression	0	0	1	1
Somnolence	0	0	1	1
Vocal Cord Paralysis	0/1	1	0	0
Breast Cancer	0	0	1	1
Aspiration Pneumonia	1/0	1	0	0
Voice Alteration	0/1	1	0	0
Acute Renal Failure	0/1	1	0	0
Enlarged Uterine Fibroid	0	0	1	1
Urinary Retention	1/0	1	0	0

9.1.1.3.2. Deaths

Four deaths occurred during the pivotal (D-02) study: one after the subject had given consent, but before the subject was implanted; the second, a suicide; the third, a death of unknown cause; and the fourth, a subject who developed multi-organ failure.

9.1.1.3.3. Unanticipated adverse device effects

Two events in the pivotal (D-02) study met criteria for an unanticipated adverse device effect (UADE)—see *Glossary* for definition. Both these events were non-specific complications of surgery related to the implant procedure and occurred before stimulation began. One UADE was an episode of acute renal failure thought to be secondary to antibiotic administration, and the other was an episode of altered mental status thought to be due to perioperative narcotic administration.

9.1.1.4 Safety considerations specific to depressed patients

Two specific safety concerns in the use of all antidepressant therapies are the precipitation of manic or hypomanic episodes and the possible effect of antidepressant therapy on suicidal ideation and behavior.

9.1.1.4.1. Antidepressant treatments and manic or hypomanic reaction

Although patients with bipolar disorder experience manic episodes as the cardinal feature of their disorder, effective antidepressant therapies themselves can occasionally precipitate a manic or hypomanic episode. Antidepressant therapies can also occasionally precipitate a manic or hypomanic episode in patients without a prior history of mania who are being treated for a major depressive episode.

In the pivotal (D-02) study, 6 hypomanic or manic reactions were identified according to DSM IV criteria or the Young Mania Rating Scale (YMRS). Five were observed in subjects with a known history of prior hypomanic or manic episodes. One of the events was considered serious and the subject was hospitalized.

9.1.1.4.2. Suicidal ideation, suicide attempts, suicide, and worsened depression

Suicidal ideation was analyzed by examining the HRSD₂₄ Item 3 scores. At 12 months of VNS Therapy, 90% of the subjects in the pivotal (D-02) study showed either improvement (56%) or no change (34%) in their Item 3 scores. During the acute D-02 study, 2.6% of the sham subjects and 1.7% of the stimulation subjects increased their Item 3 score by 2 or more points, indicative of an increase in suicidal ideation. During the long-term D-02 phase, 2.8% of the subjects had an increase in their Item 3 score by at least 2 points at 12 months compared to baseline. In a non-randomized control group of subjects treated with standard antidepressant therapies without VNS Therapy (the D-04 study population), 1.9% of the subjects had an increase of at least 2 points. Based on the occurrence of any increase in Item 3 score from baseline to 12 months, 10% of the D-02 subjects had an increase compared to 11% of the D-04 population. Conversely, 27% of the D-02 subjects decreased their score by at least 2 points at 12 months compared to baseline, whereas only 9% of the D-04 subjects did.

Suicide attempts and completed suicides in the D-02 and D-04 studies are shown in Table 23. As noted above, 1 subject committed suicide in the acute phase and 6 attempted suicide during the long-term phase of the D-02 study (N = 235). One of the 6 subjects noted in the long-term phase attempted suicide twice. Although safety data were not prospectively

collected for the D-04 study, the healthcare utilization form documented suicide attempts. Three suicide attempts were reported for the D-04 study through the first year of the study (N=124).

Table 23. Suicide Attempt and Suicide Rates

	Number of Patients	Patient Years	Suicide Attempts/ Patient Years	Suicide/ Patient Years
D-02	235	502	2.4%	0.2%
D-04	124	118	2.5%	0.0%

In the acute phase of the D-02 study, there were 12 reports of worsening depression, 5 in the stimulation group (5 of 119 subjects) and 7 in the sham group (7 of 116 subjects). One of the treatment-group reports occurred prior to stimulation initiation. Following acute phase exit and during the long-term phase of stimulation, 62 events were reported in 31 subjects. The number of episodes of worsening depression per subject ranged from 1 to 6. Although specific rates of worsening depression (and other safety endpoints) were not collected during the D-04 study, "hospitalizations for psychiatric illness," which might be a reasonable surrogate for worsening depression, were recorded. The rate of this event was 0.237 events per patient-year in the D-04 group compared to 0.293 events of worsening depression per patient-year in the D-02 group.

9.1.1.5 Adverse event (AE) relationship to VNS Therapy and duration of events

The pivotal (D-02) study investigators determined whether an adverse event (AE) was possibly, probably, or definitely related to implantation of, or stimulation by, the VNS Therapy generator and lead.

9.1.1.5.1 Adverse events related to implantation

Because all eligible study subjects in the pivotal (D-02) study were implanted with the VNS Therapy System device, no control was available to assess whether an adverse event was related to the surgery. Investigators, therefore, determined which adverse events were related to implantation. The events reported as related to implantation and occurring in at least 10% of the subjects who received VNS Therapy System implants in the pivotal (D-02) study were device site pain, device site reaction, incision pain, dysphagia, hypesthesia, pharyngitis, voice alteration, and incision site reaction. The complete list of implantation-related adverse events is shown in Table 24 and Table 25.

 **Note:** Although not seen as part of the pivotal (D-02) study, seroma formation is a potential implantation related adverse event.

Table 24. Implantation-Related Adverse Events Occurring in Greater Than or Equal To 5% of Subjects During the Acute Phase of the Pivotal (D-02) Study

	D-02 Acute Phase Incidence of Surgery- Related AEs (n=235)
Body as a Whole	
Incision Pain	36%
Device Site Pain	23%
Device Site Reaction	14%
Headache	8%
Neck Pain	7%
Pain	7%
Digestive System	
Dysphagia	11%
Nausea	9%
Nervous System	
Hypesthesia	11%
Paresthesia	6%
Respiratory System	
Voice Alteration	33%
Pharyngitis	13%
Dyspnea	9%
Cough Increased	6%
Skin and Appendages	
Incision Site Reaction	29%

Table 25. Implantation-Related Adverse Events Occurring in Less Than 5% of Subjects in Acute Phase - Pivotal (D-02) Study

Body as a Whole
Abdominal Pain, Allergic Reaction, Anaphylactic Reaction, Asthenia, Back Pain, Chest Pain, Chills, Fever, Infection, Injection Site Pain, Neck Rigidity, Photosensitivity Reaction, Surgical Injury, Viral Infection, Wound Infection
Cardiovascular System
Arrhythmia, Asystole, Bradycardia, Hemorrhage, Migraine, Palpitation, Syncope, Tachycardia
Digestive System
Anorexia, Constipation, Diarrhea, Dyspepsia, Flatulence, Gastrointestinal Disorder, Vomiting
Endocrine System
Thyroid Disorder
Hemic and Lymphatic System
Ecchymosis, Lymphadenopathy
Metabolic and Nutritional Disorders
Edema, Hyperglycemia, Peripheral Edema
Musculoskeletal
Arthralgia, Joint Disorder, Myalgia, Myasthenia

Nervous System
Abnormal Dreams, Agitation, Ataxia, Dizziness, Hypertonia, Insomnia, Nervousness, Neuralgia, Neuropathy, Thinking Abnormal, Tremor, Vasodilatation, Vocal Cord Paralysis
Respiratory System
Aspiration Pneumonia, Asthma, Atelectasis, Bronchitis, Hiccup, Hypoxia, Laryngismus, Laryngitis, Lung Disorder, Respiratory Disorder, Rhinitis, Sinusitis, Sputum Increased
Skin and Appendages
Application Site Reaction, Maculopapular Rash, Pruritus, Rash, Sweating
Special Senses
Ear Disorder, Ear Pain, Tinnitus
Urogenital
Acute Kidney Failure, Dysuria, Metrorrhagia, Urinary Retention

9.1.1.5.2. Duration of implant-related adverse events

As can be seen in Table 26, many of the individual incidences of the most common implantation-related AEs resolved within 30 days. Hypesthesia (generally described as a localized numbness) and voice alteration, however, tended to be more persistent in some individuals. For example, in 17 of 24 reports of implantation-related hypesthesia, the event continued beyond 3 months. Hypesthesia would be an expected side effect of nerve injury during surgery. The persistence of voice alteration in some subjects is difficult to assess because it could represent surgical injury to the innervation of the larynx, but vagus nerve stimulation itself can cause voice alteration.

Table 26. D-02 Acute Phase Duration of Treatment-Emergent Adverse Events Related to Implantation Reported by More Than 10% of Subjects

Body System	Preferred Term	Duration to Resolution of Event in Days by all Implanted Subjects					
		1 – 7 Days	8 – 14 Days	15 – 30 Days	31 – 60 Days	61-90 Days	>90 Days
		Total N = 235 through 30 days, 234 for 31 to 90, 233 for >90 days					
		Number within each box indicates number of subjects whose event resolved within the days shown (i.e., 27 subjects had the event of device site pain resolve within 7 days)					
Body as a Whole	Device Site Pain	27	4	9	9	3	4
	Device Site Reaction	5	5	8	9	2	8
	Incision Pain	28	18	21	10	3	6
Digestive System	Dysphagia	2	5	9	5	2	5
Nervous System	Hypesthesia	0	0	3	2	2	17
Respiratory System	Pharyngitis	10	8	10	2	0	1
	Voice Alteration	11	7	22	17	3	21
Skin and Appendages	Incision Site Reaction	19	16	24	16	2	14

9.1.1.5.3. Stimulation-related adverse events

Among AEs judged by investigators to be stimulation-related in the D-02 study acute phase treatment group, 7 events occurred at a frequency of 10% or greater: voice alteration (55%), cough increased (24%), dyspnea (19%), neck pain (16%), dysphagia (13%), laryngismus (11%), and paresthesia (10%).

Table 27 and Table 28 list stimulation-related adverse events that occurred during the acute phase of the pivotal (D-02) study.

Table 27. Stimulation-Related Adverse Events Occurring in Greater Than or Equal To 5% of Subjects in Treatment Versus Control, Acute Phase - Pivotal (D-02) Study

	D-02 Treatment (n=119)	D-02 Sham-control* (n=116)
Body as a Whole		
Incision Pain	6 (5%)	3 (3%)
Neck Pain	19 (16%)	1 (<1%)
Digestive System		
Dysphagia	15 (13%)	0 (0%)
Nausea	8 (7%)	1 (<1%)
Nervous System		
Paresthesia	12 (10%)	3 (3%)
Respiratory System		
Cough Increased	28 (24%)	2 (2%)
Dyspnea	23 (19%)	2 (2%)
Laryngismus	13 (11%)	0 (0%)
Pharyngitis	9 (8%)	1 (<1%)
Voice Alteration	65 (55%)	3 (3%)

***Note:** These subjects were not receiving stimulation during this phase.

Table 28. Stimulation-Related Adverse Events Occurring in Less Than 5% of Subjects in the Treatment Group, Acute Phase - Pivotal (D-02) Study

Body as a Whole
Asthenia, Chest Pain, Device Site Pain, Device Site Reaction, Headache, Neck Rigidity, Pain
Cardiovascular System
Migraine, Palpitation, Postural Hypotension, Syncope, Tachycardia
Digestive System
Anorexia, Constipation, Diarrhea, Dyspepsia, Eructation, Flatulence, Increased Appetite, Vomiting
Metabolic and Nutritional Disorders
Weight Gain
Musculoskeletal
Myalgia, Myasthenia
Nervous System
Abnormal Dreams, Agitation, Depression, Dizziness, Emotional Lability, Hypertonia, Hypesthesia, Insomnia, Manic Reaction, Nervousness, Sleep Disorder, Somnolence, Twitching, Vasodilatation
Respiratory System
Asthma, Hiccup, Respiratory Disorder, Rhinitis
Skin and Appendages
Incision Site Reaction
Special Senses
Ear Pain, Tinnitus
Urogenital
Amenorrhea

9.1.1.5.4. Stimulation-related events, long-term phase

Table 29 lists stimulation-related adverse events that occurred at an incidence of $\geq 5\%$ during the pivotal (D-02) study. These adverse events were observed over quarters of stimulation. Note that this table also includes observations after 24 months of treatment. Subjects are counted only once within each preferred descriptive term, e.g., neck pain, nausea, pharyngitis, and time interval. Table 30 lists stimulation-related adverse events that occurred at an incidence of $< 5\%$ during the long-term phase of the D-02 study.

Table 29. Stimulation-Related Adverse Events Occurring in Greater Than or Equal To 5% of Subjects by Time Intervals After Initiation of Stimulation - Pivotal (D-02) Study

	0-3 Mos. n=232	>3-6 Mos. n=225	>6-9 Mos. n=217	>9-12 Mos. n=209	>12-24 Mos. n=184
Body as a Whole					
Neck Pain	16%	11%	14%	13%	15%
Pain	6%	7%	5%	6%	5%
Headache	5%	4%	4%	3%	3%
Digestive System					
Dysphagia	13%	8%	7%	5%	5%
Nausea	6%	2%	2%	1%	1%
Nervous System					
Paresthesia	11%	7%	3%	4%	4%
Respiratory System					
Voice Alteration	59%	60%	58%	54%	52%
Cough Increased	24%	10%	8%	7%	4%
Dyspnea	14%	16%	15%	16%	14%
Laryngismus	10%	8%	8%	6%	5%
Pharyngitis	6%	4%	4%	5%	4%

Table 30. Stimulation-Related Adverse Events Occurring in Less Than 5% of Subjects, Long-Term Phase - Pivotal (D-02) Study

Body as a Whole
Abdominal Pain, Asthenia, Chest Pain, Device Site Pain, Device Site Reaction, Flu Syndrome, Incision Pain, Neck Rigidity, Sudden Unexplained Death, Viral Infection
Cardiovascular System
Bradycardia, Hypotension, Migraine, Palpitation, Postural Hypotension, Syncope, Tachycardia
Digestive System
Anorexia, Colitis, Constipation, Diarrhea, Dyspepsia, Eructation, Flatulence, Gastritis, Gastrointestinal Disorder, Increased Appetite, Vomiting
Metabolic and Nutritional Disorders
Weight Gain, Weight Loss
Musculoskeletal
Athralgia, Joint Disorder, Myalgia
Nervous System
Abnormal Dreams, Agitation, Amnesia, Anxiety, Confusion, Depression, Dizziness, Dry Mouth, Emotional Lability, Hypertension, Hypertonia, Hypesthesia, Insomnia, Manic Reaction, Manic Depressive Reaction, Nervousness, Sleep Disorder, Somnolence, Speech Disorder, Thinking Abnormal, Tremor, Twitching, Vasodilatation, Vocal Cord Paralysis
Respiratory System
Asthma, Hiccup, Respiratory Disorder, Rhinitis, Stridor
Skin and Appendages
Incision Site Reaction, Sweating
Special Senses
Amblyopia, Deafness, Ear Pain, Eye Pain, Tinnitus

Urogenital

Amenorrhea, Menstrual Disorder

9.1.1.5.5. Late-emerging adverse events

After the first 3 months of stimulation, the incidence of first-reported (new event types) stimulation-related adverse events did not exceed 1.3% of total study subjects for any event. See Table 31.

Table 31. Incidence of First Reported Stimulation-Related Adverse Events Experienced After 3 Months of VNS Therapy

Body System	COSTART Term	Treatment Group (N=117) N (%)	Delayed Treatment Group (N=116) N (%)	Total (N=233) N (%)
Body as a Whole	Back Pain	1 (<1%)	0	1 (<1%)
	Flu Syndrome	1 (<1%)	0	1 (<1%)
	Sudden Unexpected Death	1 (<1%)	0	1 (<1%)
	Viral Infection	1 (<1%)	0	1 (<1%)
Cardiovascular System	Hypotension	1 (<1%)	0	1 (<1%)
	Syncope	3 (3%)	0	3 (1%)
Digestive System	Colitis	2 (2%)	0	2 (<1%)
	Gastritis	2 (2%)	1 (<1%)	3 (1%)
Metabolic and Nutritional Disorders	Weight Gain	1 (<1%)	2 (2%)	3 (1%)
	Weight Loss	1 (<1%)	0	1 (<1%)
Musculoskeletal System	Arthralgia	0	1 (<1%)	1 (<1%)
	Joint Disorder	0	1 (<1%)	1 (<1%)
	Myalgia	0	1 (<1%)	1 (<1%)
Nervous System	Speech Disorder	0	1 (<1%)	1 (<1%)
	Vocal Cord Paralysis	0	1 (<1%)	1 (<1%)
Respiratory System	Stridor	1 (<1%)	0	1 (<1%)
Special Senses	Amblyopia	1 (<1%)	0	1 (<1%)
	Deafness	2 (2%)	0	2 (<1%)

Note: First reported stimulation-related AEs are defined as stimulation-related AEs that were reported after the first 3 months of VNS Therapy and for which no subject reported an AE that coded to that term during the first 3 months.

Note: AEs were coded using the COSTART 5 dictionary.

Note: Subjects were reported only once within each preferred term.

Note: Includes all AEs where relationship to stimulation was recorded as possible, probable, or definite.

9.1.1.5.6. Duration of stimulation-related events

Subjects who reported adverse events during the first 3 months of stimulation and continued to be observed during the next 9 months were evaluated by 3-month intervals for continuation or resolution of their events. The largest decreases were noted between the first and second quarters of stimulation. The most notable exception was voice alteration. During the first quarter, 135 of 209 subjects (65%) reported voice alteration. Of those 135 subjects, 90 continued to report it during the fourth quarter of stimulation. See Table 32.

Table 32. Duration of Early Stimulation-Related Events Through 1 Year (Study D-02)

VNS Therapy (N=209)				
	N Reporting Event During First 3 Mos. ¹	N (%) Continuing to Report Event During Succeeding Quarters ²		
Preferred Term	0–3 Mos.	3–6 Mos.	6–9 Mos.	9–12 Mos.
Voice Alteration	135	115 (85%)	101 (75%)	90 (67%)
Cough Increased	55	18 (33%)	15 (27%)	11 (20%)
Neck Pain	38	17 (45%)	19 (50%)	16 (42%)
Dyspnea	35	22 (63%)	18 (51%)	16 (46%)
Dysphagia	31	16 (52%)	10 (32%)	6 (19%)
Paresthesia	26	12 (46%)	6 (23%)	4 (15%)
Laryngismus	23	13 (57%)	9 (39%)	5 (22%)
Pharyngitis	14	3 (21%)	2 (14%)	2 (14%)
Nausea	13	3 (23%)	1 (8%)	2 (15%)

¹Entries are the number of subjects who experienced the AEs between implantation and 3 months.

²Number of subjects who continued to experience the same adverse event between months 3 and 6, months 6 and 9, and months 9 and 12.

Note: Subjects were counted only once within each preferred term and time interval.

9.1.1.6 *Severity of adverse events*

Investigators rated adverse events as mild, moderate, or severe according to the protocol definitions: mild events were transient and easily tolerated by the subject; moderate events caused discomfort and interrupted usual activities; severe events caused considerable interference with the subject's usual activities.

Most adverse events for the feasibility (D-01) study and pivotal (D-02) study were mild or moderate. Because the pivotal (D-02) study included a sham-control group, further analysis of severity rating was performed. After 3 months of treatment, there were 280 (43%) adverse events that were categorized as mild, 293 (45%) as moderate, and 73 (11%) as severe in the sham-control group. The active VNS Therapy group had 360 (47%) adverse events categorized as mild, 349 (45%) as moderate, and 61 (8%) as severe.

9.1.1.7 *VNS Therapy continuation rates*

Of the 295 subjects implanted during both the feasibility (D-01) and pivotal studies (D-02), 270 subjects (92%) were still receiving VNS Therapy at 12 months and 242 subjects (82%) were still receiving VNS Therapy at 24 months. This compares to 12- and 24-month continuation rates of 95% and 83%, respectively, for the subjects implanted in the epilepsy preapproval trials.

9.1.1.8 *Analysis of Medical Device Reports submitted to the FDA—Epilepsy indication for the VNS Therapy System from July 1, 1997 through October 8, 2004*

9.1.1.8.1 *Summary*

Once a medical device is approved for commercial distribution, the United States Food and Drug Administration (FDA) regulations require certain parties, including manufacturers of

medical devices, to report to the FDA deaths and serious injuries to which a device has or may have caused or contributed. The required report is referred to as a medical device report (MDR). The FDA Office of Biometrics and Surveillance analyzed all MDRs submitted for the VNS Therapy System from July 1, 1997 through October 8, 2004. During this period, the VNS Therapy System had a single approved indication, epilepsy. The analysis included 2,887 reports, 2,453 of which were reported from sites within the United States. By the end of the period analyzed, there were 32,065 VNS Therapy device implants and 80,144 device-years of implant experience. It is important to emphasize that, although the events occurred during treatment with the VNS Therapy System, the submission of an MDR does not necessarily mean the product caused or contributed to the event being reported.

9.1.1.8.2. Deaths

A total of 524 deaths were reported to the FDA during the period from July 1, 1997 through October 8, 2004. By the end of the period, there were 32,065 VNS Therapy device implants and 80,144 device-years of implant experience. Of the 524 deaths, 102 (20%) were of an "unknown cause," including 24 deaths of unknown cause that occurred during sleep (5% of total deaths). Of those deaths with a reported cause, the following were the most common etiologies:

- Seizure disorder (152 reports; 29% of total deaths), including sudden unexplained death in epilepsy and status epilepticus
- Respiratory events (99 reports; 19% of total deaths), including pneumonia, pulmonary edema, and hypoxia
- Cardiac events (51 reports; 10% of total deaths), including cardiopulmonary arrest, infarction, and arrhythmias
- Neurovascular events (24 reports; 5% of total deaths), including stroke and cerebral hemorrhage
- Malignancy (19 reports; 3% of total deaths), including brain and colon
- Suicide (9 reports; 2% of total deaths)

9.1.1.8.3. Serious injuries

A total of 1,644 serious injuries were reported to the FDA during the period from July 1, 1997 through October 8, 2004. By the end of the period, there were 32,065 VNS Therapy device implants and 80,144 device-years of implant experience. The most frequently reported serious injury was infection (525 reports). Approximately 40% of these were known to have required device explantation. The second most common serious injury reported was increased seizure activity (324 reports). Others included:

- Vagus nerve injury (181 reports), including vocal cord paralysis (109) and hoarseness (71)
- Respiratory injuries (141 reports), including sleep apnea (33), dyspnea (50), and aspiration (14)
- Cardiac events (123 reports), including tachycardia, bradycardia, palpitations, hypertension, hypotension, syncope, and asystole
- Pain (81 reports), including chest and neck pain
- Gastrointestinal events (60 reports), including dysphagia (24) and weight loss (24)

- Depression (21 reports)

Of the 1,644 reports of serious injury, 694 (42%) were associated with subsequent device explantation in that subject.

9.1.1.8.4. Device malfunctions

A total of 708 device malfunctions were reported to the FDA during the period from July 1, 1997 through October 8, 2004. By the end of the period, there were 32,065 VNS Therapy device implants and 80,144 device-years of implant experience. Some of the most common malfunctions reported were high lead impedance (351), lead breakage (116), device failure (44), and device migration (20).

9.1.2 Pivotal and Pilot Studies—Effectiveness

9.1.2.1 Feasibility (D-01) study

The primary efficacy measure in the open-label feasibility (D-01) study was the percent of subjects responding (response was defined as a 50% or greater improvement in the HRSD₂₈ score). Of the 59 subjects with evaluable data, 18 (31%) responded at acute study exit, which was 12 weeks after implantation. Observation of subjects continued. After 1 year of adjunctive VNS Therapy, 25 of 55 subjects (45%) responded, and after 2 years, 18 of 42 (43%) responded. After 1 and 2 years of treatment, 27% and 21% of the subjects, respectively, were in remission (defined as HRSD₂₈ scores less than or equal to 10. Other measures of depressive symptoms (CGI, MADRS, BDI, IDS-SR) and quality of life (MOS-36) supported the HRSD₂₈ scores.

9.1.2.2 Pivotal (D-02) study

The pivotal (D-02) study of VNS Therapy consisted of both an acute and a long-term phase to collect data regarding the safety and efficacy of VNS Therapy as an adjunctive treatment for persons with chronic or recurrent treatment-resistant depression.

9.1.2.2.1. Pivotal D-02 study, acute phase

The acute phase was a 12-week (after implantation), double-blind, randomized, parallel-group sham treatment-controlled, multi-center study. Subjects were assigned randomly to either the treatment (stimulation) group or control (sham) group and results of these 2 groups were compared. All subjects in both groups meeting the eligibility criteria for participation in the study were implanted with the VNS Therapy pulse generator and VNS Therapy lead. The VNS Therapy System remained OFF for 2 weeks after implantation to allow for recovery from surgery. Most subjects in the pivotal (D-02) study were being treated with 1 or more antidepressant medications at the time of enrollment. Medications were to remain constant at the pre-implant baseline dosages throughout the acute phase for both the treatment and sham-control groups.

Sham Control: Sham-control group subjects were treated the same as the treatment group, except that the output current of the device remained at 0.0 mA so that it did not deliver stimulation during the acute phase.

Treatment Group: Two weeks after implant, stimulation was initiated for the treatment group. Over the next 2 weeks, parameters were adjusted to subject tolerance, then remained

constant for the rest of the acute phase (8 weeks). Decreases in stimulation parameters were permitted to accommodate subject tolerance.

9.1.2.2.2. Pivotal (D-02) study, long-term phase

All pivotal (D-02) study subjects who completed the acute phase were eligible to continue into the long-term extension phase, during which all subjects received active VNS Therapy. During the first 10 weeks of the extension phase, sham-control subjects (also referred to as the delayed treatment group for the long-term phase), received stimulation parameter adjustments. Weekly or every other week clinic visits and assessments were identical to those experienced by the treatment group during the acute phase. Otherwise, the protocol specified monthly clinic visits for both groups through 12 months of active VNS Therapy. Various assessments, including depression ratings, were performed throughout this period. During the long-term extension phase, investigational site programmers were allowed to adjust stimulation parameters as clinically indicated. Additionally, concomitant antidepressant treatments could be added, removed, or adjusted as clinically indicated.

9.1.2.3 Comparative assessments

Outcomes from a non-randomized comparative study (D-04) were compared with the long-term outcomes in study D-02. D-04 was a long-term, prospective, observational study to collect data regarding usual standard-of-care for treatment-resistant chronic or recurrent depression in persons who were experiencing a major depressive episode at the time of admission. Clinical (depression assessments) and quality of life outcomes were assessed at baseline, 3, 6, 9, and 12 months.

9.1.2.3.1. Concomitant therapies

Subjects enrolled in the comparative (D-04) study met the same enrollment criteria regarding chronicity or recurrence of depression, previous treatment failures, and severity of depression as subjects in the pivotal (D-02) study. Because the study was observational in nature, the protocol did not specify therapies for the treatment of depression; rather the physician managing the study subject's depression selected therapy according to clinical judgment. Thus antidepressant therapy in the comparative (D-04) study comprised "standard of care" treatment (also known as "treatment as usual"). The entire range of treatment options available for the comparative (D-04) study subjects was also available to the pivotal (D-02) study subjects as concomitant treatment to their VNS Therapy. Thus subjects in both the long-term pivotal (D-02) extension and the comparative (D-04) study received standard-of-care treatment; however, only the pivotal (D-02) study subjects received VNS Therapy.

9.1.2.3.2. Comparison of D-02 and D-04 study populations

The comparative (D-04) study was conducted at 13 investigational sites, 12 of which were also pivotal (D-02) study sites. The similarities in the key inclusion criteria and study sites provide a basis to expect that the demographic and disease characteristics of both groups would be comparable, which was confirmed by the results of the analyses conducted to examine the comparability. The D-04 subjects provided a comparison group for the pivotal (D-02) study subjects at 12 months. See Table 33.

Table 33. Description of Subjects in Pivotal (D-02) and Comparative (D-04) Studies

Parameter	Statistic	D-02 (N=205)	D-04 (N=124)
Age (years)	Mean	46.3	45.5
Male	N (%)	74(36)	39(31)
Female	N (%)	131(64)	85(69)
Caucasian	N (%)	198(97)	111(90)*
African-American	N (%)	3(1)	5(4)
Hispanic	N (%)	3(1)	2(2)
Unipolar	N (%)	185(90)	109(88)
Bipolar	N (%)	20(10)	15(12)
Recurrent	N (%)	161(87)	93(85)
Single Episode	N (%)	24(13)	16(15)
Length of Current MDE (mos)	Mean (S.D.)	49.9(52.1)	68.6(91.5)
# Failed Trials in Current MDE	Mean (S.D.)	3.5(1.3)	3.5(1.3)
Received ECT Lifetime	N(%)	108(53%)	32(26%)*
Received ECT, Current MDE	N(%)	72(35%)	15(12%)*
Duration of Illness (yrs)	Mean (S.D.)	25.5(11.9)	25.8(13.2)
Lifetime episodes of Depression*			
0-2	N(%)	50(24)	31(25)
3-5	N(%)	69(34)	36(29)
6-10	N(%)	56(27)	18(15)
>10	N(%)	19(9)	32(26)
No Suicide Attempts in Lifetime	N(%)	140(68)	80(65)
Treatment induced (hypo) mania	N(%)	16(8)	6(5)
Hospitalizations for Depression	Mean (S.D)	2.7(5.4)	2.1(2.9)
ECT Treatment Within past 2 yrs	N(%)	54(26)	19(15)

*P<.05

This comparison analyzed evaluable populations of 205 adjunctive VNS Therapy subjects (D-02) and 124 usual standard-of-care subjects (D-04). Groups were well matched, with similar demographic, psychiatric, and mood disorder treatment histories. The only relevant significant differences between groups were previous ECT history (with higher usage of ECT found in the D-02 group) and number of lifetime episodes of depression (with a higher percentage of the D-04 group reporting >10 lifetime episodes). These differences were handled within the efficacy analysis by use of a propensity adjustment.

9.1.2.4 Data analysis: D-02 and D-04 studies

9.1.2.4.1 Pivotal (D-02) study

The primary efficacy variable for both the acute and the long-term phases of the pivotal (D-02) study was the Hamilton Rating Scale for Depression-24 item (HRSD₂₄). For the acute-phase analysis, the HRSD₂₄ response rate (percentage of subjects with a ≥50% improvement from baseline to 3 months, acute phase exit) was compared between the treatment and the sham-control groups. For the long-term phase, a linear regression model was used to assess the

changes in HRSD₂₄ raw scores. Secondary efficacy analyses included within and between-group comparisons of 1) the Inventory of Depressive Symptomatology-Self Report (IDS-SR), 2) the Clinical Global Impressions (CGI), 3) the Montgomery-Asberg Depression Rating Scale (MADRS), and 4) the Medical Outcome Survey 36-Item Short Form Health Survey (MOS SF-36).

9.1.2.4.2. Comparative (D-04) study

The primary efficacy variable for the D-02 and D-04 comparative analysis was the IDS-SR (raw scores). Multiple assessments with the IDS-SR allowed use of a linear regression model for the analysis. The HRSD₂₄ was used as a secondary assessment variable to analyze differences in response rates and raw score changes between subjects in the pivotal (D-02) and comparative (D-04) studies. Subjects in the comparative (D-04) study were assessed with the HRSD₂₄ only at baseline and 12 months.

Secondary analyses included IDS-SR average change, IDS-SR response, IDS-SR remission, IDS-SR sustained response, and HRSD₂₄ remission. Other secondary analyses included the CGI response.

9.1.2.4.3. Propensity scores

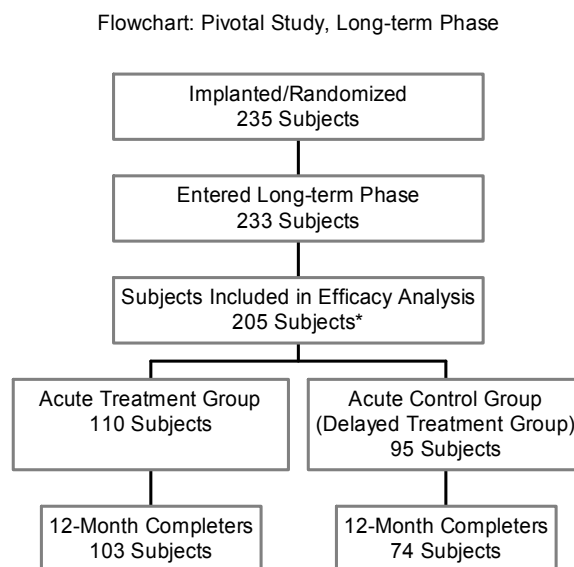
Propensity scores were calculated for the pivotal (D-02) study and comparative (D-04) study groups and used in the linear regression analysis to address the potential impact of baseline differences on differences in outcome between the 2 groups. Propensity scores provide a scalar summary of the covariate information (e.g., age, number of prior depressive episodes, etc.). They are not limited by the constraints of traditional methods of adjustment, which can only use a limited number of covariates for adjustment.

9.1.2.4.4. Responder rate

Response was prospectively defined as a $\geq 50\%$ improvement from baseline for the IDS-SR, HRSD₂₄, and MADRS ratings and as a score of much or very much improved for the CGI improvement rating. Remission (complete response) was prospectively defined as an HRSD₂₄ score of ≤ 9 , a MADRS score of ≤ 10 , or an IDS-SR score ≤ 14 .

All statistical analyses were performed using the updated SAS version 8.2. All statistical tests were 2-sided and performed at the 0.050 level of significance. No adjustments were made for multiple outcome measures.

Figure 47. Pivotal Study, Long-Term



*28 subjects did not qualify for Efficacy Analysis:
 - 21 sham-control subjects did not have required HRSD₂₄ score ≥ 18 at acute phase exit
 - 4 subjects did not have long-term phase efficacy assessments
 - 3 subjects did not meet continuation criteria for acute phase

9.1.2.5 **Results: Pivotal study (D-02)**

Figure 47 provides a flow chart of subjects from the acute phase through the long-term phase of the pivotal (D-02) study. Information describing subjects in the pivotal (D-02) and comparative (D-04) studies is presented in Table 33.

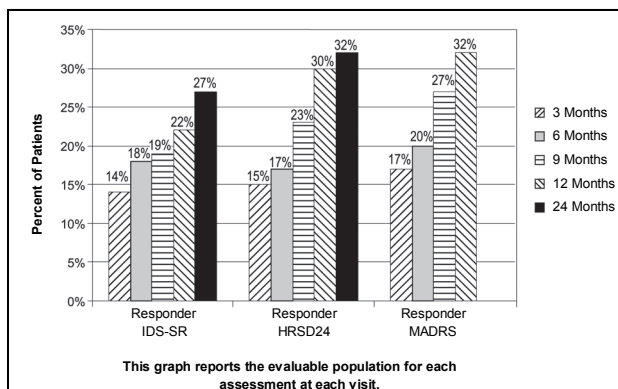
9.1.2.5.1 **Results: acute phase, pivotal (D-02) study**

In the primary efficacy measure, HRSD₂₄ response rate, (the percentage of subjects achieving a $\geq 50\%$ improvement in HRSD₂₄ total score from baseline to acute phase exit), 15% of the treatment group and 10% of the sham-control group were responders ($P=.238$). Analyses using a secondary efficacy parameter, the IDS-SR, did show a statistically significant advantage for VNS Therapy over sham treatment: 17% response versus 7% response ($P=.032$) using the last observation carried forward (LOCF) method.

9.1.2.5.2 **Results: long-term phase, pivotal study (D-02)**

During long-term adjunctive VNS Therapy, the D-02 subjects exhibited statistically significant and clinically meaningful improvement. The primary analysis found statistically significant improvement from baseline in HRSD₂₄ scores averaged over 12 months ($P<.001$). Additionally, clinical significance was shown, using the HRSD₂₄, IDS-SR, MADRS, and CGI (Figure 48 and Figure 49, evaluable population, and Table 34, 12-month completer population).

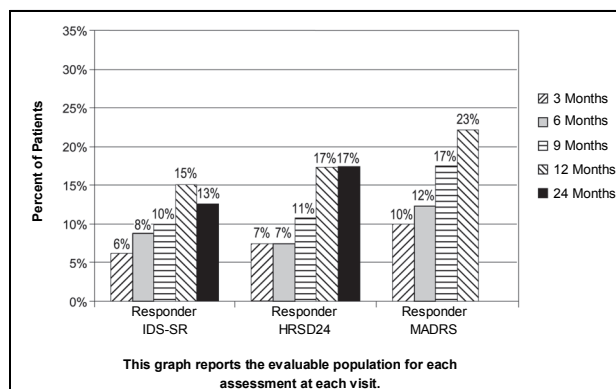
Figure 48. Responder Quarterly Results for D-02 Evaluable Subjects



The number of evaluable subjects in each of the above analyses is as follows:

Mos	IDSSR	HRSD	MADRS
3	203	205	205
6	192	197	197
9	185	186	196
12	180	181	181
24	157	157	N/A

Figure 49. Remitter Quarterly Results for D-02 Evaluable Subjects



The number of evaluable subjects in each of the above analyses is as follows:

Mos	IDSSR	HRSD	MADRS
3	203	205	205
6	192	197	197
9	185	186	196
12	180	181	181
24	157	157	N/A

Table 34. Responders, Remitters, and Percent Change Pivotal (D-02) Study, 12-Month Completer Population

	HRSD ₂₄ [*]	IDS-SR [†]	MADRS [‡]
	12-Month Visit	12-Month Visit	12-Month Visit
Responders – N (%)			
Treatment	34/103 (33%) ²	25/102 (25%)	34/103 (33%) ²
Delayed treatment	18/71 (25%)	13/71 (18%)	22/71 (31%) ¹
All 12-Month Completers	52/174 ^a (30%) ³	38/173 (22%) ¹	56/174 (32%) ³
Remitters – N (%)			
Treatment	19/103 (18%) ²	16/102 (16%) ¹	25/103 (24%) ²
Delayed treatment	10/71 (14%)	10/71 (14%)	16/71 (23%) ¹
All 12-Month Completers	29/174 (17%) ²	26/173 (15%) ²	41/174 (24%) ³
Mean Percent Change from Baseline			
Treatment	31.9% ³	27.8% ³	32.9% ³
Delayed treatment	26.5% ³	17.3% ³	26.3% ³
All 12-Month Completers	29.7% ³	23.5% ³	30.2% ³

¹ $P < .05$; ² $P < .01$; ³ $P < .001$; Response and Remitter used the Exact McNemar's test compared with 3 months; Percent Change used the paired t-test (change from pre-stimulation baseline).

^{*} Three subjects did not have 12-month HRSD₂₄ assessments. (These 3 subjects did have 11-month assessments.)

[†] One subject did not have a baseline IDS-SR assessment and several others did not have 12-month assessments, which accounts for the varying Ns in the comparison of HRSD₂₄ with IDS-SR data.

[‡] Two delayed-treatment subjects did not have 12-month MADRS assessments.

9.1.2.5.3. Quality of life assessment

The observed improvement in depression among subjects in the pivotal (D-02) study long-term phase was supported by improved quality of life as measured by the MOS SF-36. Significant improvement was observed in several of the MOS SF-36 subscales: Vitality, Social Functioning, Role Functioning – Emotional, Mental Health ($P < .01$).

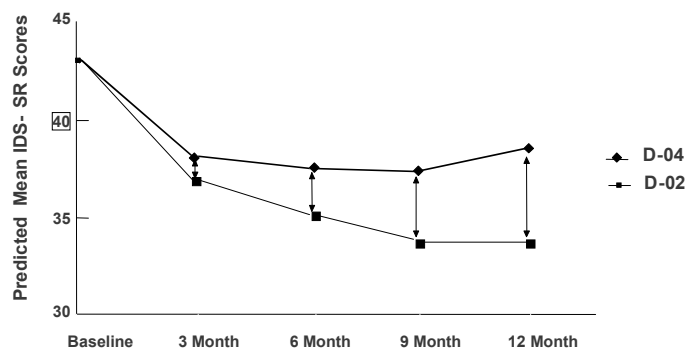
9.1.2.6 Results: Comparison of D-02 and D-04 studies

The D-04 study provided a control group of similarly ill subjects who received usual standard-of-care therapies for 12 months but were not implanted with the VNS Therapy device. See Table 33.

9.1.2.6.1. Primary effectiveness outcome

The primary and secondary analyses comparing subjects treated with VNS Therapy plus usual standard-of-care (pivotal, D-02) with subjects treated with usual standard-of-care alone (comparative, D-04) showed that adjunctive VNS Therapy produced statistically significantly greater improvement in depressive symptoms over 1 year of treatment. The primary efficacy analysis, a repeated measures linear regression analysis of the IDS-SR over 1 year, showed a statistically significant ($P < .001$ evaluable; $P < .001$ intent to treat) difference favoring adjunctive VNS Therapy. See Figure 50.

Figure 50. Comparison of IDS-SR Scores of Pivotal (D-02) Versus Comparative (D-04) Study Subjects by Quarter (Repeated Measures Linear Regression Analysis), Evaluable Population



	B/L	3 mos	6 mos	9 mos	12 mos
Mean D-04 Scores	43.0 (N=124)	38.1 (N=120)	37.5 (N=119)	37.3 (N=116)	38.5 (N=112)
Mean D-02 Scores	43.0 (N=201)	36.9 (N=200)	35.1 (N=195)	33.7 (N=183)	33.7 (N=177)
Predicted Mean Difference	0	-1.2	-2.4	-3.6	-4.8
Actual Mean Difference	-0.9	-4.6	-4.1	-5.0	-6.6

9.1.2.6.2. Secondary analyses

Additionally, the following secondary analyses were statistically significant and showed adjunctive VNS Therapy improved depressive symptoms more than usual standard-of-care alone after 12 months of therapy. See Figure 51 and Figure 52.

Figure 51. Secondary Analyses: Categorical Outcomes at 12 Months (Evaluable Observed Analysis)

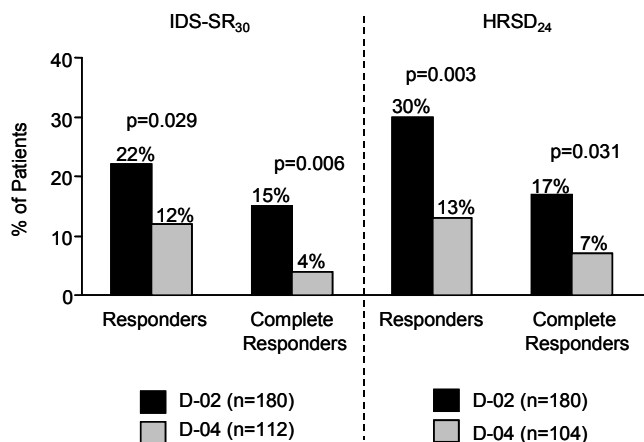
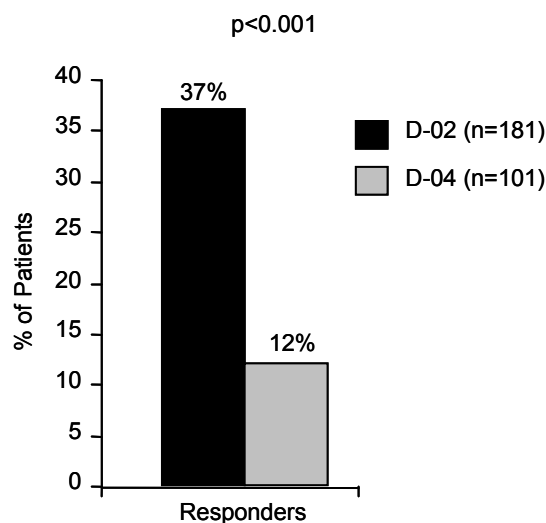


Figure 52. Secondary Analyses: CGI-I Categorical Outcome at 12 Months (Evaluable Observed Analysis)

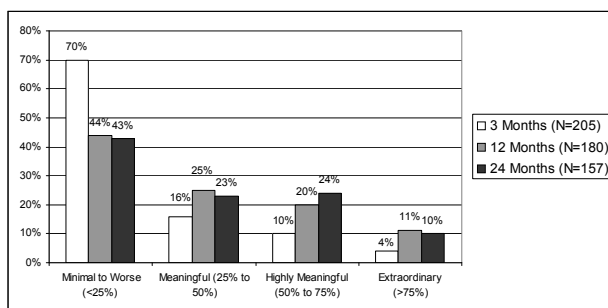


9.1.2.7 Clinical benefit over time

To explore whether these subjects were receiving benefit that was not fully reflected in the response rates, they were assigned to categories according to “clinical benefit.” Clinical benefit was prospectively defined as extraordinary ($\geq 75\%$ improvement in HRSD₂₄), highly meaningful (50% to $<75\%$), meaningful (25% to $<50\%$), minimal (0% to $<25\%$), and worsened (less than 0%). This scale is consistent with studies in many chronic illnesses that define less than a 50% improvement as a clinically meaningful response (e.g., schizophrenia, obsessive compulsive disorder).

As shown in Figure 53, clinical benefit increased over time. The percent of subjects realizing at least a meaningful clinical benefit at 12 months was significant when compared to those experiencing a similar benefit after 3 months (Stuart-Maxwell test, $P<.001$).

Figure 53. Clinical Benefit After 3, 12, and 24 Months; D-02 Evaluable Population; HRSD₂₄



The subjects realizing at least a meaningful clinical benefit after 12 months of adjunctive VNS Therapy included subjects who sustained their 3-month meaningful or greater benefit and those who had minimal to no 3-month benefit and accrued at least a meaningful benefit after 12 months. Of the 56 subjects who had at least a meaningful benefit at 3 months, 41 (73%) continued to have at least a meaningful benefit at 12 months and 34 (61%) of these same 56 subjects had at least the *same* level of clinical benefit after 12 months of adjunctive VNS

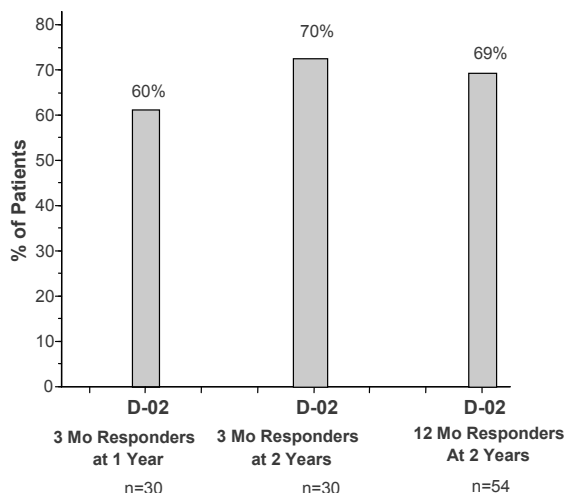
Therapy as they did after 3 months. Of the 118 subjects who realized minimal-to-worse clinical benefit after 3 months of adjunctive VNS Therapy, 56 (47%) had at least a meaningful benefit after 12 months of adjunctive VNS Therapy.

A majority (56%) of evaluable subjects treated with adjunctive VNS Therapy realized at least a meaningful clinical benefit after 12 months of treatment. After 24 months of VNS Therapy, 57% of evaluable subjects realized at least a meaningful clinical benefit.

9.1.2.8 ***Maintaining response (2-year data)***

An analysis of subjects having an initial $\geq 50\%$ reduction in HRSD score at the designated “early” visit (3 months or 12 months) and then maintaining at least a $\geq 40\%$ reduction at the later visit (1 or 2 years), was performed for the D-02 Study. Data are presented below in a bar graph (Figure 54), with each bar showing the percent of subjects that maintained their early response at the later observation.

Figure 54. Maintenance of Adjunctive VNS Therapy Response (% of HRSD₂₄ Responders who Maintained Response at 1 and 2 Years)



When IDS data were used instead of HRSD data, similar results were observed (61% of 3-month responders were also responders at 12 months, 57% of 3-month responders were also responders at 24 months, and 85% of 12-month responders were also responders at 24 months). By contrast, no D-04 3-month responder maintained that response at the 12-month observation.

9.1.2.9 ***Standard-of-care antidepressant treatments during the long-term phase of study D-02 and during study D-04***

9.1.2.9.1. **Electroconvulsive therapy**

Electroconvulsive therapy (ECT) use was similar among the pivotal (D-02) and comparative (D-04) study subjects (7% and 6%, respectively) during the first year of observation.

9.1.2.9.2. **Antidepressant drugs and response**

Antidepressant drug use was significantly greater among pivotal (D-02) study subjects who were non-responders and comparative (D-04) study subjects overall than among the pivotal (D-02) study subjects who achieved a response ($P < .01$). During the 12 months, 77% of the

pivotal (D-02) study non-responders and 81% of all comparative (D-04) study subjects either added a new antidepressant treatment or increased an existing antidepressant dose by an antidepressant resistance rating (ARR) level of 1 or more. By contrast, only 56% of the pivotal (D-02) study subjects who were responders to VNS Therapy either added a new antidepressant treatment or increased an existing antidepressant dose by an antidepressant resistance rating (ARR) level of 1 or more.

For the evaluable group at 12-months, 61 subjects were responders while 144 subjects were non-responders (N=205). On a percentage basis twice as many pivotal (D-02) study responders had no ARR changes or removed or decreased medications by at least 1 ARR level or were not taking medications as compared to the non-responders (44% versus 23%, respectively).

9.1.2.9.3. Medication censoring analyses

Additional medication censoring analyses were performed using the D-02 and the D-02 versus D-04 repeated measures linear regression methods to evaluate further the potential effect of medication changes. This censoring approach used a missing data paradigm to calculate the D-02 results that would have been observed under conditions where no intercurrent changes in medications would have occurred in the D-02 group. The approach censors the D-02 IDS-SR scores after the point at which a subject had a significant medication increase (ARR increase) or ECT treatment, and the last pre-censored score is carried forward and used for subsequent assessment periods. The censoring had the effect of truncating the VNS treatment benefit from 12 months to an average of 7 months. In the D-02 censored analysis, the average HRSD₂₄ change from baseline was -0.25 points per month in the repeated measures linear regression ($P<.001$).

The D-02 censored versus D-04 IDS-SR repeated measures linear regression comparison was an asymmetric comparison of the VNS group treated for 7 months with VNS plus no changes from baseline treatments versus the D-04 group treated for a full 12 months with unlimited standard-of-care treatments (no censoring was performed on the D-04 data). The results of the censoring analysis approached but did not reach statistical significance in the comparison of the D-02 group with the D-04 group ($P=.052$; 95% CI -0.37, 0.00) for the evaluable population.

9.1.3 Bibliography

A bibliography of animal, clinical, and mechanism of action studies is available from LivaNova on request.

9.2 Post-Approval (D-21) Study

The post-approval dosing study, hereafter referred to as D-21, was a condition of approval required by the US Food and Drug Administration.

9.2.1 Summary of D-21 Post-Approval Study Methods

9.2.1.1 Study objective

The objective of this study was to compare the safety and effectiveness of adjunctive vagus nerve stimulation therapy (VNS Therapy) administered at different levels of output current for the treatment of patients with treatment-resistant depression (TRD).

9.2.1.2 *Study design*

The D-21 study was a multicenter, double-blind, randomized comparison of VNS Therapy using 3 different amounts of electrical charge. Eligible men and women over the age of 18 years, with a diagnosis of chronic or recurrent depression that had not been responsive to multiple treatment attempts, were enrolled in the study and received a VNS Therapy System implant. The study was designed to evaluate the relative safety and effectiveness of different output currents. Following implantation, patients were randomized to one of the following 3 treatment dose groups based on target settings of (output current, pulse width):

- Low Dose: 0.25 mA, 130 μ sec
- Medium Dose: 0.5-1.0 mA, 250 μ sec
- High Dose: 1.25-1.5 mA, 250 μ sec

Following a satisfactory post-operative recovery period (generally 2 weeks), each patient received VNS Therapy initially programmed as follows:

- Output current 0.25 mA
- Frequency 20 Hz
- ON time 30 seconds
- OFF time 5 minutes

Over a 4-week period, the programmer titrated the output current to achieve the target setting for the patient's randomly assigned dose group. Other stimulation parameter settings remained constant during the titration period and for the first 16 weeks of treatment following the titration period.

During the initial 4-week titration phase, each patient underwent weekly programming sessions using a programming guide that maintained the blind to dose group assignment. The goal of the titration period for the Medium Dose and High Dose groups was to reach an output current setting corresponding to the upper end of the range (i.e., 1.0 or 1.5 mA, respectively) at the end of the titration period. When a patient experienced comfort/tolerability issues within the target output current despite multiple attempts, the programmer decreased the pulse width to 130 μ sec to improve patient tolerability. The programmer's objective was to ensure that all patients reached the lower output current settings within the range for their treatment group (e.g., 1.25 mA for the High Dose group). If the minimum treatment group target output current for any patient was not achieved during the initial 4-week titration period, titration was extended by 2 weeks.

To the greatest extent possible and consistent with patient welfare, the investigator refrained from adding, discontinuing, or changing the intensity of other (i.e., non-VNS) antidepressant or mood stabilizer treatments, including non-pharmacologic treatments, before study week 22 (study visit 11). After study week 22 (study visit 11), the investigator directed the programmer to increase the output current if clinically warranted. To help preserve the study blind, the maximum increase allowed for any patient after study week 22 was 0.75 mA. The pulse width in the Low Dose group was allowed to be increased to 250 μ sec after study week 22. Patients were followed for an additional 28 weeks for a total of 50 weeks post implant. During this period, concomitant antidepressant treatment changes were allowed and were guided according to the treatment algorithm defined in the clinical protocol.

9.2.1.3 Data source

The population data sets used in this study is represented below in Table 35. Twenty-one patients implanted and analyzed in the safety population were not included in the ITT population due to either an IDS-C less than or equal to 30, or absence of post-stimulation assessments. A total of 310 patients were included in the ITT population. A description of the ITT patient populations characteristics are described below in Table 36.

Table 35. Data Sets Analyzed

Population	Stimulation Treatment Group			
	Low n (%)	Medium n (%)	High n (%)	Total n (%)
Safety Population	111 (100.0)	107 (100.0)	113 (100.0)	331 (100.0)
<i>Excluded for ITT</i>	9 (8.1)	6 (5.6)	6 (5.3)	21 (6.3)
Baseline IDS-C $\leq$ 30	8 (7.2)	6 (5.6)	5 (4.4)	19 (5.7)
No post-stimulation assessments	1 (0.9)	0	1 (0.9)	2 (0.6)
Protocol defined ITT Population	102 (91.9)	101 (94.4)	107 (94.7)	310 (93.7)

Table 36. Description of Patients in the Post-approval (D-21) Study (ITT Population)

Parameter	Statistic	Low (N=102)	Medium (N=101)	High (N=107)	Total (N=310)
Age (years)	Mean	49.1	47.2	47.4	47.9
Male	N (%)	34(33)	32(32)	34(32)	100(32)
Female	N (%)	68(67)	69(68)	73(68)	210(68)
Caucasian	N (%)	97(95)	96(95)	104(97)	297(96)
African-American	N (%)	2(2)	3(3)	3(3)	8(3)
Hispanic	N (%)	0	1(1)	0	1(0.3)
Asian	N (%)	3(3)	1(1)	0	4(1.3)
Unipolar	N (%)	82(80)	81(80)	81(76)	244(79)
Bipolar	N (%)	20(20)	20(20)	26(24)	66(21)
Recurrent	N (%)	76(75)	71(70)	66(62)	213(69)
Single Episode	N (%)	6(6)	10(10)	15(14)	31(10)
Duration of Illness (yrs)	Mean (SD)	29.8(12.1)	26.3(10.9)	27.0(12.1)	27.7(11.8)
Length of Current MDE (mos)	Mean (SD)	106.7(122.8)	106.1(107.3)	111.3(146.3)	108.1(126.5)
Lifetime episodes of Depression					
0-2	N(%)	16(16)	17(17)	28(26)	61(20)
3-5	N(%)	23(23)	21(21)	22(21)	66(21)
6-10	N(%)	13(13)	16(16)	18(17)	47(15)
>10	N(%)	50(49)	46(46)	39(36)	135(44)
Treatment induced mania and (hypo) mania	N(%)	11(11)	9(9)	6(6)	26(8)
Number Suicide Attempts in Lifetime	N(%)	56(55)	44(44)	41(39)	141(46)
Number Suicide Attempts in Current Episode	N(%)	7(7)	4(4)	6(6)	17(6)

Parameter	Statistic	Low (N=102)	Medium (N=101)	High (N=107)	Total (N=310)
Hospitalizations for Mood Disorders	Mean (SD) [n]	4.0(5.1) [102]	3.9(6.1) [101]	2.8(3.3) [106]	3.6(4.9) [309]
Received ECT Lifetime	N(%)	61(60)	53(53)	62(58)	176(57)
# Failed Mood Disorder Treatments Lifetime	Mean (SD)	14.1(6.7)	15.3(8.2)	14.0(5.5)	14.5(6.9)
Failed Mood Disorder Treatments Lifetime					
2-3	N(%)	2(2)	0	0	2(0.6)
4-5	N(%)	1(1)	3(3)	3(3)	7(2)
>6	N(%)	99(97)	98(97)	104(97)	301(97)

Treatment groups were well matched, with similar demographics, psychiatric, and mood disorder treatment histories. Patients in this study had experienced numerous unsuccessful treatments for their major depressive disorder (MDD) or bipolar disease. All patients in the study had experienced failed treatments for mood disorders, and the majority had experienced 6 or more unsuccessful treatments (97.1%). Fifty seven percent of patients experienced ECT prior to enrollment and 46% had previously attempted suicide.

9.2.1.4 **Key study endpoints**

9.2.1.4.1. **Primary endpoint**

The primary effectiveness endpoint was the Inventory of Depressive Symptomatology-Clinician Rated (IDS-C) 30-item mean change from baseline over weeks 10, 14, 18, and 22 (acute treatment phase). Treatment groups were compared over the acute treatment phase using a Mixed Model Repeated Measures (MMRM) analysis of covariance (ANCOVA).

9.2.1.4.2. **Secondary endpoints**

Results for the secondary endpoints, including mean change, mean percent change, and the percent responders for the IDS-C, Quick Inventory of Depressive Symptomatology-Clinician Rated (QIDS-C), Montgomery-Åsberg Depression Rating Scale (MADRS), Inventory of Depressive Symptomatology Self-Rated (IDS-SR), and for Clinical Global Impressions Improvement (CGI-I) scales.

9.2.1.5 **Total number of enrolled study sites and subjects, follow-up rate**

There were a total of 331 patients at 29 sites actually enrolled and implanted with the VNS Therapy System, all of whom were included in the safety data-set. Of the 331 patients implanted with the VNS Therapy System during the post-approval (D-21) study, 298 patients (90%) were still receiving VNS Therapy at the end of the study (50 weeks).

Table 37 provides a matrix for patient follow-up visits for the Safety population within the D-21 study. Patient follow-up visit compliance in the D-21 study was above 99%.

Table 37. Patient Visit Matrix (Safety population [n=331])

	BL Visit	10 Wks Visit	14 Wks Visit	18 Wks Visit	22 Wks Visit	26 Wks Visit	32 Wks Visit	38 Wks Visit	44 Wks Visit	50 Wks Visit	Overall
Expected Visits	331	328	323	319	316	313	309	306	300	298	3143
Visit Assessments Completed	331 (100.00)	324 (98.78)	321 (99.38)	313 (98.12)	316 (100.00)	311 (99.36)	304 (98.38)	304 (99.35)	296 (98.67)	298 (100.00)	3118 (99.20)
Visit Assessments Not Done	0 (0)	4 (1.22)	2 (0.62)	6 (1.88)	0 (0)	2 (0.64)	5 (1.62)	2 (0.65)	4 (1.33)	0 (0)	25 (0.80)
Patient Follow-up Visit Compliance*	100	98.78	99.38	98.12	100	99.36	98.38	99.35	98.67	100	99.2

*Patients' completion of visit based on IDS-C or MADRS assessments.

If the actual visit date falls between the Protocol defined visit window, then Patient considered to be the visit completer.

If the patient has not completed at least the IDS-C or MADRS assessments, then the patient has missed the visit.

9.2.1.6 Study visits and length of follow-up

Patients in the D-21 study had 16 total study visits over the 54 weeks of the trial and were seen twice prior to implantation for baseline evaluation. These pre-study visits occurred 4-6 weeks and 1 week before implantation. Two weeks after implantation patients were randomized to their assigned dosing group and seen weekly, for up to five weeks, for VNS Therapy device titration. Acute treatment evaluations occurred starting 10 weeks after implantation, and patients were seen monthly from weeks 10 through 22. During that time all medications and VNS Therapy parameters were to be unchanged. Upon completion of the acute phase, 22 weeks post implant, patients were followed monthly up to week 50 during the long-term treatment phase.

9.2.2 Summary of D-21 Post-Approval Study Results

9.2.2.1 Final safety findings

9.2.2.1.1 Device performance

Device complications occurred in 6.3% of patients. Communication difficulties were the most commonly reported difficulty (2.4%), but programming anomalies (1.5%), high lead impedance (1.2%), problems with troubleshooting (0.6) were also reported.

9.2.2.1.2 Adverse Events

9.2.2.1.2.1 Events Reported

The safety criteria for evaluation were the Adverse Events (AE) Record and Frequency, Intensity and Burden of Side Effects Rating (FIBSER). These were observational data and epidemiological in nature. No conclusion can be drawn regarding a dose effect on adverse events, but the findings are consistent with other VNS Therapy trials. The number (and percentage) of patients reporting an adverse event during the 0-50 week period in the post-approval (D-21) study is depicted in Table 38 for adverse events occurring in at least 5% of patients in any of the 3 treatment groups. Adverse events were coded using the COSTART dictionary 5th edition. Note that some patients may have reported multiple events.

Although no statistical testing was performed, several numerical trends seemed to emerge. During the Acute Phase, the Low Dose group had fewer side effects up to week 22 (the Acute Phase) as compared with the Medium and High Dose groups. After week 22, this advantage diminished and all groups were similar.

Table 38. Adverse Events Reported in at Least 5% of Total Patients During VNS Therapy D-21 Study from Baseline to Week 50

	Treatment Group			
	Low n (%) (N=111)	Medium n (%) (N=107)	High n (%) (N=113)	Total n (%) (N=331)
Voice Alteration	71 (64.0)	82 (76.6)	86 (76.1)	239 (72.2)
Dyspnea	33 (29.7)	36 (33.6)	38 (33.6)	107 (32.3)
Pain	28 (25.2)	30 (28.0)	47 (41.6)	105 (31.7)
Paresthesia	31 (27.9)	35 (32.7)	39 (34.5)	105 (31.7)
Incision Pain	24 (21.6)	33 (30.8)	27 (23.9)	84 (25.4)
Cough Increased	27 (24.3)	28 (26.2)	28 (24.8)	83 (25.1)
Headache	19 (17.1)	21 (19.6)	21 (18.6)	61 (18.4)
Depression	25 (22.5)	14 (13.1)	21 (18.6)	60 (18.1)
Pharyngitis	19 (17.1)	19 (17.8)	19 (16.8)	57 (17.2)
Hypertonia	22 (19.8)	17 (15.9)	17 (15.0)	56 (16.9)
Pain Neck	12 (10.8)	14 (13.1)	20 (17.7)	46 (13.9)
Dysphagia	10 (9.0)	17 (15.9)	18 (15.9)	45 (13.6)
Nasopharyngitis	16 (14.4)	17 (15.9)	12 (10.6)	45 (13.6)
Incision Site Reaction	18 (16.2)	11 (10.3)	13 (11.5)	42 (12.7)
Nausea	15 (13.5)	15 (14.0)	9 (8.0)	39 (11.8)
Anxiety	13 (11.7)	12 (11.2)	13 (11.5)	38 (11.5)
Insomnia	12 (10.8)	12 (11.2)	12 (10.6)	36 (10.9)
Device Site Reaction	16 (14.4)	8 (7.5)	9 (8.0)	33 (10.0)
Device Site Pain	13 (11.7)	2 (1.9)	12 (10.6)	27 (8.2)
Sinusitis	7 (6.3)	8 (7.5)	10 (8.8)	25 (7.6)
Pain Chest	7 (6.3)	5 (4.7)	11 (9.7)	23 (6.9)
Pain Ear	4 (3.6)	4 (3.7)	11 (9.7)	19 (5.7)
Gastrointestinal Disease	4 (3.6)	4 (3.7)	10 (8.8)	18 (5.4)
Injury Accident	8 (7.2)	6 (5.6)	4 (3.5)	18 (5.4)
Constipation	1 (0.9)	10 (9.3)	6 (5.3)	17 (5.1)
Tremor	5 (4.5)	7 (6.5)	5 (4.4)	17 (5.1)

Table 39. Adverse Events Reported in less than 5% of Total Patients During VNS Therapy D-21 Study out to Week 50

Body as a Whole
Allergic Reaction, Asthenia, Cellulitis, Chills, Cyst, Death, Endometrial Adenocarcinoma, Fever, Hemorrhage, Homicidal Ideation, Infection, Neck Rigid, Overdose, Overdose Intentional, Pain Abdomen, Pain Back, Pain Pelvic, React Unevaluable, Suicide, Suicide Attempt, Wound Infection
Cardiovascular System
Sinus Bradycardia, ECG Abnormal, Hemorrhage, Hypertension, Migraine, Palpitation, Syncope, Tachycardia, Thrombophlebitis Deep
Digestive System
Abscess Periodontal, Anorexia, Appetite Increased, Cholecystitis, Cholelithiasis, Colitis, Diarrhea, Dry Mouth, Dyspepsia, Eructation, Gastric Reflux, Gastroenteritis, Incontinence Fecal, Melena, Nausea Vomit, Pain Biliary, Pancreas Disease, Proctitis, Tooth Disease, Ulcer, Viral Infection, Vomiting
Endocrine System
Adrenal Disease, Carcinoma, Hypothyroidism, Parathyroidism Disease, Thyroid Disease
Hemic System
Anemia, Ecchymosis
Metabolic System
Alcohol Intolerance, Anemia B12 Deficiency, Avitaminosis, Dehydration, Diabetes Mellitus, Edema, Edema General, Edema Peripheral, Gout, Hypercholesteremia, Hyperglycemia, Hypocalcemia, Hypoglycemia, Hypokalemia, SGOT Increased, Weight Decreased, Weight Increased
Musculoskeletal System
Arthralgia, Arthritis, Bone Disease, Bone Fraction Spontaneous, Joint Disease, Myalgia, Myasthenia, Osteoporosis, Spasm General, Tendon Disease, Tenosynovitis
Nervous System
Agitation, Akathisia, Amnesia, Ataxia, Confusion, Convulsion, Diplopia, Dizziness, Dream Abnormal, Dystonia, Emotion Lability, Hallucination, Hyperkinesia, Hypersthesia, Hypesthesia, Hypokinesia, Libido Decreased, Manic Depress Reaction, Manic Reaction, Movement Disease, Nervousness, Neuralgia, Neuritis, Neurosis, Paralysis Vocal Cord, Paresthesia Circumoral, Reflexes Decreased, Saliva Increased, Self Injurious Behavior, Sleep Disease, Somnolence, Speech Disease, Twitch, Vasodilation, Vertigo
Respiratory System
Apnea, Asthma, Bronchitis, Emphysema, Laryngismus, Laryngitis, Lung Disease, Pneumonia, Respiratory Disease, Rhinitis, Stridor
Skin and Appendages System
Acne, Carcinoma Skin, Herpes Zoster, Hypertrophy Skin, Nodule Skin, Rash, Skin Disease, Sweat, Ulcer Skin
Special Senses System
Amblyopia, Dry Eye, Eye Disease, Hemorrhage Eye, Lacrimation Disease, Miosis, Otitis Externa, Otitis Media, Pain Eye, Parosmia, Ptosis, Taste Perversion, Tinnitus, Vision Abnormal
Urogenital System
Amenorrhea, Anorgasmia, Dysuria, Ejaculation Abnormal, Hemorrhage Vaginal, Incontinence Urinary, Infect Urinary Tract, Kidney Calculus, Kidney Function Abnormal, Kidney Polycystic, Mastitis, Menorrhagia, Neoplasm Breast, Nephropathy Toxic, Pain Kidney, Prostatic Disease, Spasm Bladder, Urinary Frequency, Urinary Retention, Vaginitis

9.2.2.1.2.2. Discontinuation due to adverse events

Two patients discontinued due to an adverse event:

- One patient in the Medium Dose group experienced incision site pain, fatigue and insomnia during the Acute Phase, none of which were considered Serious Adverse Events; and

- One patient in the Low Dose group experienced headache, anxiety/agitation and device site pain /pulse generator movement during the Long-term Phase. Device site pain was considered a Serious Adverse Event and the generator was subsequently explanted.

9.2.2.1.3. Serious adverse events (SAEs)

9.2.2.1.3.1. SAEs

Ten patients had a total of 12 SAEs that were considered definite, possibly or probably related to the implant procedure (dyspnea, chest pain, incision site reaction, device site reaction, hemorrhage, device site pain and infection [6 patients]).

Seven patients had a total of 8 SAEs that were considered possibly related to stimulation (suicide attempt, abdominal pain, manic reaction [2 patients] and depression [4 patients]).

Table 40 displays all the SAEs reported during the D-21 study, regardless of relationship to implantation or stimulation.

Table 40. Study D-21, Serious Adverse Events Reported Regardless of Relationship to Implantation or Stimulation

	Treatment Group			Total n (%) (N=331)
	Low n (%) (N=111)	Medium n (%) (N=107)	High n (%) (N=113)	
Depression	8 (7.2)	6 (5.6)	4 (3.5)	18 (5.4)
Suicide Attempt*	7 (6.3)	1 (0.9)	4 (3.5)	12 (3.6)
Death	3 (2.7)	0	1 (0.9)	4 (1.2)
Pain Abdomen	0	2 (1.9)	1 (0.9)	3 (0.9)
Pain Chest	1 (0.9)	1 (0.9)	1 (0.9)	3 (0.9)
Anxiety	0	2 (1.9)	0	2 (0.6)
Infection	1 (0.9)	0	1 (0.9)	2 (0.6)
Manic Reaction	0	2 (1.9)	0	2 (0.6)
Pneumonia	1 (0.9)	0	1 (0.9)	2 (0.6)
Reaction Unevaluable	1 (0.9)	1 (0.9)	0	2 (0.6)
Suicide	1 (0.9)	0	1 (0.9)	2 (0.6)
Appetite Increased	1 (0.9)	0	0	1 (0.3)
Asthma	1 (0.9)	0	0	1 (0.3)
Sinus Bradycardia	0	0	1 (0.9)	1 (0.3)
Carcinoma	0	1 (0.9)	0	1 (0.3)
Carcinoma Skin	0	1 (0.9)	0	1 (0.3)
Constipation	0	1 (0.9)	0	1 (0.3)
Device Site Pain	1 (0.9)	0	0	1 (0.3)
Device Site Reaction	0	0	1 (0.9)	1 (0.3)
Dyspnea	0	1 (0.9)	0	1 (0.3)
Endometrial Adenocarcinoma	1 (0.9)	0	0	1 (0.3)
Gastric Reflux	1 (0.9)	0	0	1 (0.3)
Gastroenteritis	1 (0.9)	0	0	1 (0.3)

	Treatment Group			Total n (%) (N=331)
	Low n (%) (N=111)	Medium n (%) (N=107)	High n (%) (N=113)	
Hallucination	1 (0.9)	0	0	1 (0.3)
Hemorrhage	1 (0.9)	0	0	1 (0.3)
Homicidal Ideation	0	1 (0.9)	0	1 (0.3)
Incision Site Reaction	0	0	1 (0.9)	1 (0.3)
Injury Accident	1 (0.9)	0	0	1 (0.3)
Kidney Calculus	0	1 (0.9)	0	1 (0.3)
Kidney Function Abnormal	1 (0.9)	0	0	1 (0.3)
Nephropathy Toxic	1 (0.9)	0	0	1 (0.3)
Overdose Intentional	0	1 (0.9)	0	1 (0.3)
Pain	1 (0.9)	0	0	1 (0.3)
Pain Back	0	1 (0.9)	0	1 (0.3)
Pain Neck	0	0	1 (0.9)	1 (0.3)
Pancreas Disease	0	0	1 (0.9)	1 (0.3)
Parathyroid Disease	0	1 (0.9)	0	1 (0.3)
Syncope	0	1 (0.9)	0	1 (0.3)
Thrombophlebitis Deep	1 (0.9)	0	0	1 (0.3)

* Three additional suicide attempts in the low dose group were not classified as serious adverse events by the investigators or in the final study report.

9.2.2.1.3.2. **Deaths**

There were 6 patient deaths during the study, 4 in the Low Dose Group, and 2 in the High Dose Group. One death in each of the Low and High Dose Groups was considered to be suicide. None of these deaths or suicides were judged by the investigator to be related to implantation or stimulation. One patient committed suicide by ingesting lethal amounts of multiple drugs, a second patient overdosed on prescription drugs and antidepressants. Two deaths were due to cardiovascular disease, a third was due to complications of bariatric surgery, and a fourth was a motor vehicle accident.

9.2.2.1.3.3. **Unanticipated adverse device effects**

No unanticipated adverse device effects (UADEs) were reported in the post-approval (D-21) study.

9.2.2.1.4. **Safety considerations specific to depressed patients**

Two specific safety concerns in the use of all antidepressant therapies are the precipitation of manic or hypomanic episodes and the possible effect of antidepressant therapy on suicidal ideation and behavior.

9.2.2.1.4.1. **Antidepressant treatments and manic or hypomanic reaction**

Although patients with bipolar disorder experience manic episodes as the cardinal feature of their disorder, effective antidepressant therapies themselves can occasionally precipitate a manic or hypomanic episode. Antidepressant therapies can also occasionally precipitate a manic or hypomanic episode in patients without a prior history of mania who are being treated for a major depressive episode.

9.2.2.1.4.2. **Manic and hypomanic reactions**

Six manic reactions were identified, 2 in the Low Dose Group, 3 in the Medium Dose Group, and 1 in the High Dose Group. Two manic reactions in the Medium Dose Group were considered to be serious and both were hospitalized.

While there was no proactive assessment of hypomania in the D-21 study, such as the Young Mania rating Scale (YMRS), no events of hypomania or Adverse Events associated with hypomania were reported during the study.

9.2.2.1.4.3. **Suicide attempts, suicide, and worsening depression**

Fifteen patients attempted suicide during the study (10 in the Low Dose Group, 1 in the Medium Dose Group, and 4 in the High Dose Group), and 2 patients completed suicide (1 in each of the Low Dose and High Dose Groups). One patient committed suicide by ingesting lethal amounts of prescription drugs, a second patient overdosed on prescription drugs and antidepressants. Sixty patients reported worsening depression (21 in the Low Dose Group, 14 in the Medium Dose Group, and 21 in the High Dose Group), See Table 41.

Table 41. Suicide Attempts and Completed Suicides by Patient Years*

	Number of Patients	Patient Years	Suicide Attempts (Rate per 100 Patient Years)	Suicides (Rate per 100 Patient Years)	Worsening Depression (Rate per 100 Patient Years)
D-21 Total	331	304	15 (4.9%)	2 (0.7%)	60 (19.7%)
Low dose group	111	100	10 (10.0%)	1 (1.0%)	25 (25.0%)
Medium dose group	107	98	1 (1.0%)	0.0%	14 (14.3%)
High dose group	113	106	4 (3.8%)	1 (0.9%)	21 (19.8%)

*Safety Population

9.2.2.1.5. **Adverse event (AE) relationship to VNS Therapy**

Investigators determined whether an adverse event (AE) was possibly, probably, or definitely related to implantation or stimulation by the VNS Therapy pulse generator and lead.

9.2.2.1.5.1. **Adverse events related to implantation**

Because all eligible study patients were implanted with the VNS Therapy System device, no control was available to assess whether an adverse event was related to the implant procedure. Investigators, therefore, determined which adverse events were related to implantation. Table 42 and Table 43 present adverse events that were judged as “definitely related” in the study. Implant related AEs were reported by investigators for 138 patients in the D-21 study, and were evenly distributed across treatment groups. Multiple AEs were reported for some patients. The most frequently reported “definitely related” AEs were incision pain, incision site reaction, and voice alteration, all considered to be expected based on results of previous clinical trials and on clinical experience.

Table 42. Study D-21, Implantation-Related Adverse Events Occurring in at Least 5% of Patients

	Treatment Group			Total n (%) (N=331)
	Low n (%) (N=111)	Medium n (%) (N=107)	High n (%) (N=113)	
Incision Pain	20 (18.0)	23 (21.5)	19 (16.8)	62 (18.7)
Incision Site Reaction	15 (13.5)	6 (5.6)	10 (8.8)	31 (9.4)
Voice Alteration	7 (6.3)	13 (12.1)	5 (4.4)	25 (7.6)
Pain	5 (4.5)	5 (4.7)	10 (8.8)	20 (6.0)

Table 43. Study D-21, Implantation-Related Adverse Events Occurring in Less Than 5% of Patients

Body as a Whole
Allergic Reaction, Device Site Pain, Device Site Reaction, Hemorrhage, Infection, Injury Accident, Neck Rigid, Pain Chest, Pain Neck, Wound Infection
Cardiovascular System
Hemorrhage, Tachycardia, Thrombophlebitis Deep
Digestive System
Dysphagia, Nausea, Nausea Vomiting, Vomiting
Hemic System
Ecchymosis
Metabolic System
Edema
Nervous System
Agitation, Hypersthesia, Hypertonia, Hypesthesia, Hypokinesia, Paralysis Vocal Cord, Paresthesia
Respiratory System
Asthma, Cough Increased, Dyspnea, Pharyngitis
Skin and Appendages System
Hypertrophy Skin

9.2.2.1.5.2. Stimulation-related adverse events

Table 44 and Table 45 present AEs judged by investigators as “definitely related” to stimulation in the D-21 study, by body system. These occurred in 154 subjects (46.5% of the study population). The most commonly reported AEs were voice alteration (34.1%), paresthesia (10%), cough (8.8%), dyspnea (7.3%), and pain (6.9%). All were considered to be expected based on results of previous clinical trials and on clinical experience. A dose-response trend

was noted in these AEs: 36% in the Low Dose group, 49.5% in the Medium Dose group, and 54% in the High Dose group.

Table 44. Study D-21, Stimulation-Related Adverse Events Occurring in at Least 5% of Patients

	Treatment Group			
	Low n (%) (N=111)	Medium n (%) (N=107)	High n (%) (N=113)	Total n (%) (N=331)
Voice Alteration	31 (27.9)	40 (37.4)	42 (37.2)	113 (34.1)
Paresthesia	10 (9.0)	7 (6.5)	16 (14.2)	33 (10.0)
Cough Increased	5 (4.5)	14 (13.1)	10 (8.8)	29 (8.8)
Dyspnea	5 (4.5)	8 (7.5)	11 (9.7)	24 (7.3)
Pain	2 (1.8)	5 (4.7)	16 (14.2)	23 (6.9)

Table 45. Study D-21, Stimulation-Related Adverse Events Occurring in Less Than 5% of Patients

Body as a Whole
Asthenia, Device Site Pain, Device Site Reaction, Headache, Incision Pain, Neck Rigid, Pain Chest, Pain Neck
Digestive System
Dyspepsia, Dysphagia, Vomiting
Musculoskeletal System
Myalgia, Spasm General
Nervous System
Anxiety, Confusion, Depression, Hypertonia, Insomnia, Neuralgia, Speech Disease
Respiratory System
Laryngismus, Pharyngitis
Special Senses System
Pain Ear

9.2.2.2 Final Effectiveness Findings

9.2.2.2.1 Primary Endpoint

The primary effectiveness endpoint was the mean IDS-C 30-item change from baseline over weeks 10, 14, 18, and 22 (acute treatment phase). Treatment groups were compared over the acute treatment phase (Low vs. Medium; Low vs. High) using a Mixed Model Repeated Measures (MMRM) analysis of covariance (ANCOVA). Figure 55 represents IDS-C change from baseline over weeks 10, 14, 18, and 22 (acute treatment phase) and weeks 26, 32, 38, 44 and 50 (long-term treatment phase). Figure 56 represents IDS-C mean score from baseline over weeks 10, 14, 18, and 22 (acute treatment phase) and weeks 26, 32, 38, 44 and 50 (long-term treatment phase). In this randomized study, the primary endpoints did not demonstrate statistical significance (Low vs. Medium stimulation group: $P=.8131$; Low vs. High stimulation group: $P=.8027$). While improvement was noted in the within-group comparisons over time, this comparison does not preclude improvement for reasons other than device effect. Other

reasons for improvement may include the natural course of the disease, effectiveness of other concurrent treatments (e.g., medications), regression to the mean, and placebo effects.

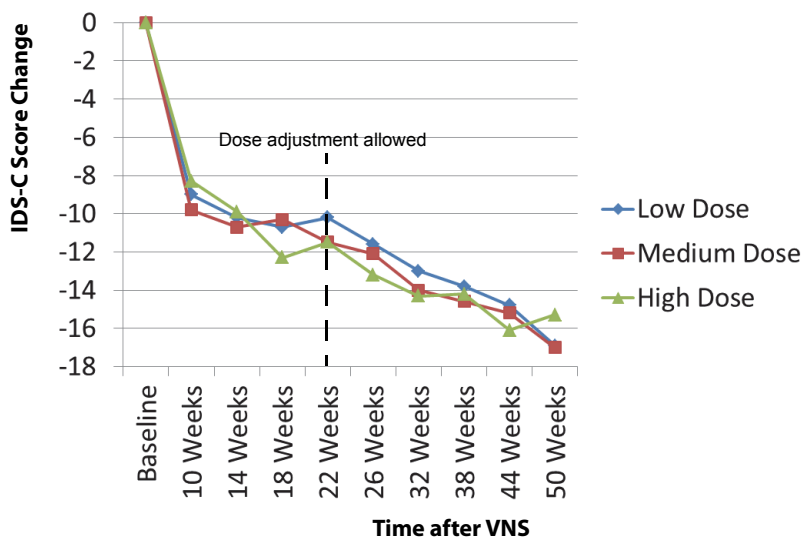
9.2.2.2.2. Secondary endpoints

Results for the secondary endpoints, including mean change, mean percent change, and the percent responders for the IDS-C, QIDS-C, MADRS, IDS-SR, and CGI-I scales did not demonstrate a statistical difference between the three groups. They did demonstrate a consistent and statistically significant improvement compared to baseline over time across the 3 treatment groups for both the Acute (week 22) and Long-term (week 50) phases of the study (except for the Acute Phase IDS-SR percent responders using a Mixed Model Repeated Measures [MMRM] analysis of covariance [ANCOVA]). The consistency of response across scales for the IDS-SR, IDS-C, QIDS-C, MADRS and CGI-I at week 22 is represented in Figure 57, and week 50 is represented in Figure 58.

Table 46. Mean Change IDS-C (ITT Population)

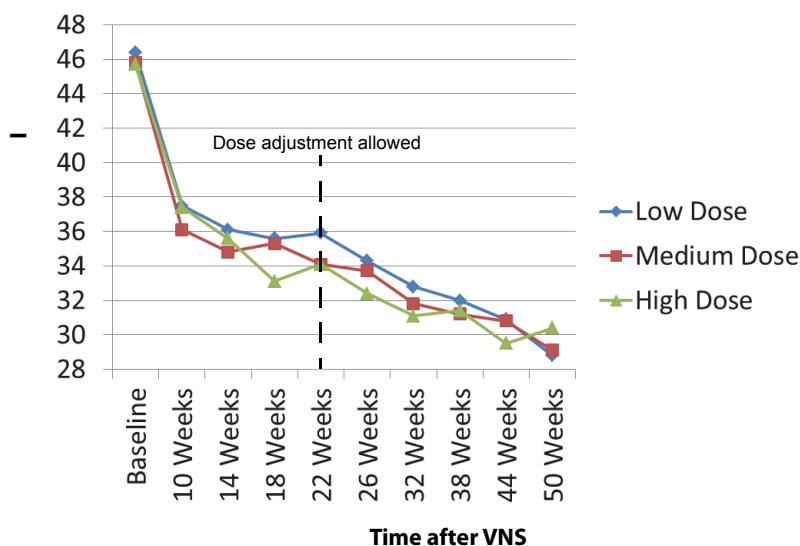
Time	Stimulation Treatment Group					
	Low		Medium		High	
	n	mean (SD)	n	mean (SD)	n	mean (SD)
Baseline	102	46.4 (8.0)	101	45.8 (7.5)	107	45.7 (8.0)
Post Treatment	n	mean (SD) change	n	mean (SD) change	n	mean (SD) change
Week 10	101	-9.0 (10.4)	100	-9.8 (10.3)	106	-8.3 (11.0)
Week 14	101	-10.2 (11.4)	96	-10.7 (12.3)	105	-9.9 (11.6)
Week 18	97	-10.7 (10.2)	97	-10.3 (12.7)	101	-12.3 (11.2)
Week 22	97	-10.2 (11.4)	97	-11.5 (12.9)	105	-11.5 (13.4)
Week 26	96	-11.6 (11.1)	94	-12.1 (12.9)	104	-13.2 (13.2)
Week 32	93	-13.0 (11.9)	93	-14.0 (13.6)	102	-14.3 (12.7)
Week 38	90	-13.8 (12.4)	93	-14.6 (14.0)	104	-14.2 (13.4)
Week 44	89	-14.8 (14.1)	92	-15.2 (13.0)	99	-16.1 (14.0)
Week 50	89	-16.9 (13.9)	90	-17.0 (14.6)	102	-15.3 (12.9)

Figure 55. IDS-C change from baseline over weeks 10, 14, 18, and 22 (acute treatment phase) and weeks 26, 32, 38, 44 and 50 (long-term treatment phase)*



*While improvement was noted in the within group comparisons over time, this comparison does not preclude improvement for reasons other than device effect. Reasons for improvement may include the natural course of the disease, effectiveness of other concurrent treatments (e.g., medications), regression to the mean, and placebo effects.

Figure 56. IDS-C mean score at baseline and over weeks 10, 14, 18, and 22 (acute treatment phase) and weeks 26, 32, 38, 44 and 50 (long-term treatment phase)



Note: While improvement was noted in the within group comparisons over time, this comparison does not preclude improvement for reasons other than device effect. Reasons for improvement may include the natural course of the disease, effectiveness of other concurrent treatments (e.g., medications), regression to the mean, and placebo effects.

Figure 57. Response rates for IDS-SR, IDS-C, QIDS-C, MADRS and CGI-I at 22 Weeks (ITT Population)

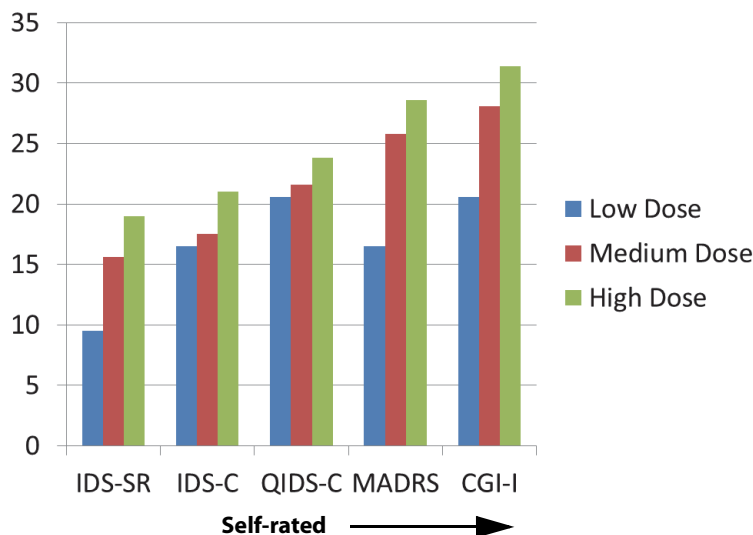
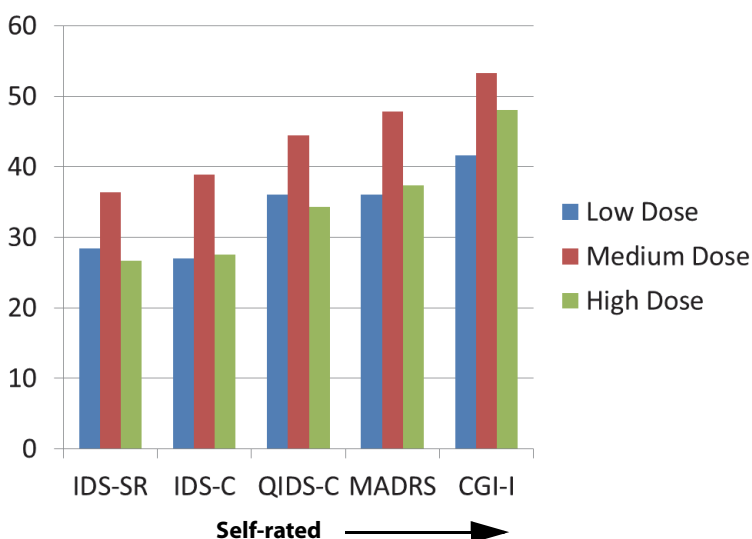


Figure 58. Response rates for IDS-SR, IDS-C, QIDS-C, MADRS and CGI-I at 50 Weeks (ITT Population)



9.2.2.2.3. Response and remission rates for Inventory of Depressive Symptomatology Clinician Rated Version (IDS-C)

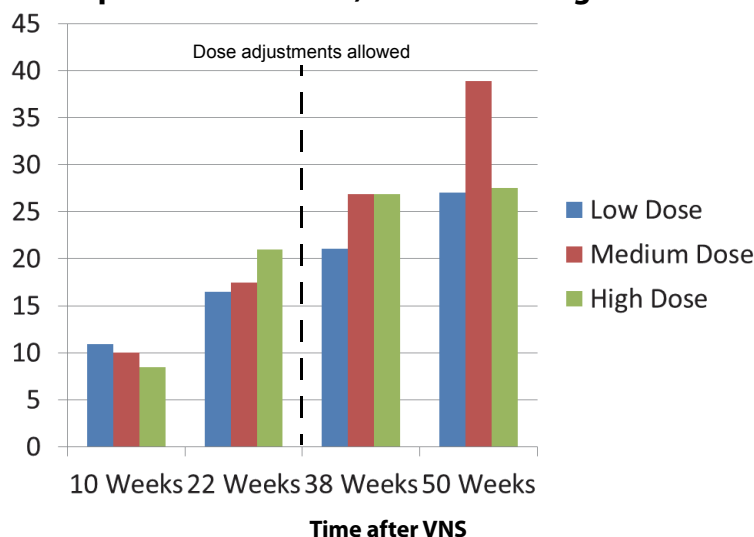
Response was defined as $\geq 50\%$ improvement from baseline. Remission was defined as a score of < 14 on the IDS-C. No statistically significant differences were noted between the 3 treatment groups for patients experiencing response in the Acute phase or the Long-term phase. Response rates using a Mixed Model Repeated Measures (MMRM) analysis of covariance (ANCOVA) on the IDS-C showed statistical improvement over time following initiation of stimulation for all treatment groups in the Acute (week 22) phase, and the Long-term (week 50) phase. Figure 59 represents the IDS-C response rates across the Acute (week 22) and Long-term (Week 50) phases of the study. While improvement was noted in the within group comparisons over time, this comparison does not preclude improvement for reasons other than device effect. Reasons for improvement may include the natural course of the disease,

effectiveness of other concurrent treatments (e.g., medications), regression to the mean, and placebo effects.

Table 47. Response Rates IDS-C (ITT Population)

Time Post Treatment	Stimulation Treatment Group							
	Low		Medium		High		Total	
	n	n (%) yes	n	n (%) yes	n	n (%) yes	n	n (%) yes
Week 10	101	11 (10.9)	100	10 (10.0)	106	9 (8.5)	307	30 (9.8)
Week 22	97	16 (16.5)	97	17 (17.5)	105	22 (21.0)	299	55 (18.4)
Week 38	90	19 (21.1)	93	25 (26.9)	104	28 (26.9)	287	72 (25.1)
Week 50	89	24 (27.0)	90	35 (38.9)	102	28 (27.5)	281	87 (31.0)

Figure 59. Response Rates IDS-C, Acute and Long-term Phase (ITT Population)

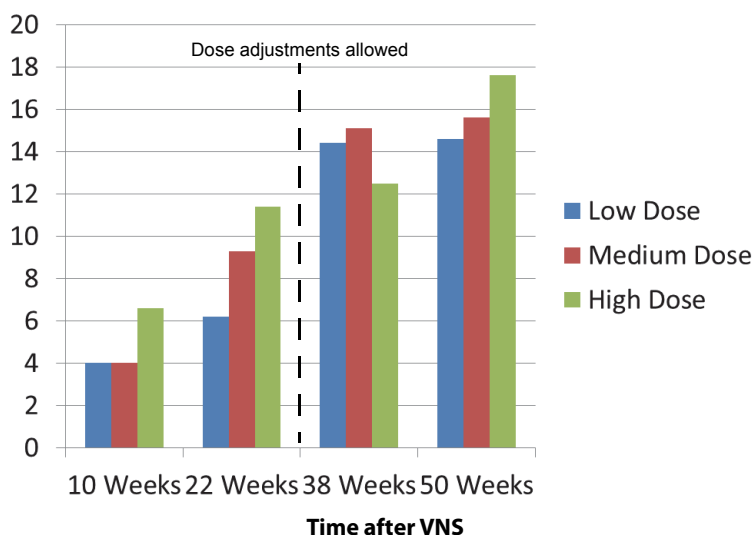


No statistically significant differences were noted between the 3 treatment groups for patients experiencing remission in the Acute phase or the Long-term phase. However, remission rates showed time-trend increase over the weeks following initiation of stimulation across the 3 treatment groups in the Acute phase (week 22) that was statistically significant using a Mixed Model Repeated Measures (MMRM) analysis of covariance (ANCOVA). Figure 60 represents IDS-C remission rates across the Acute (week 22) and Long-term (week 50) phases of the study. While improvement was noted in the within group comparisons over time, this comparison does not preclude improvement for reasons other than device effect. Reasons for improvement may include the natural course of the disease, effectiveness of other concurrent treatments (e.g., medications), regression to the mean, and placebo effects.

Table 48. Remission Rates IDS-C (ITT Population)

Time Post Treatment	Stimulation Treatment Group							
	Low		Medium		High		Total	
	n	n (%) yes	n	n (%) yes	n	n (%) yes	n	n (%) yes
Week 10	101	4 (4.0)	100	4 (4.0)	106	7 (6.6)	307	15 (4.9)
Week 22	97	6 (6.2)	97	9 (9.3)	105	12 (11.4)	299	27 (9.0)
Week 38	90	13 (14.4)	93	14 (15.1)	104	13 (12.5)	287	40 (13.9)
Week 50	89	13 (14.6)	90	14 (15.6)	102	18 (17.6)	281	45 (16.0)

Figure 60. Remission Rate IDS-C, Acute and Long-term Phase (ITT Population)



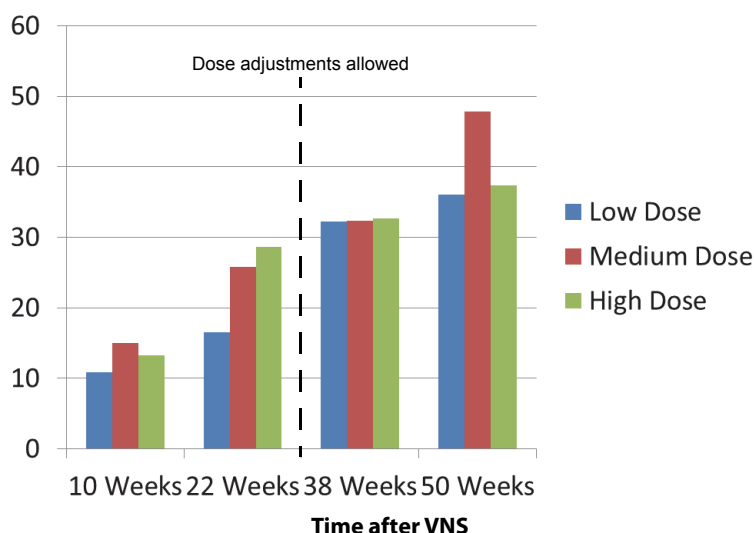
9.2.2.2.4. Response and remission rates for Montgomery-Åsberg Depression Rating Scale (MADRS)

Response was defined as $\geq 50\%$ improvement from baseline. Remission was defined as a score of ≤ 9 on the MADRS. No statistically significant differences were noted between the 3 treatment groups for patients experiencing response in the Acute phase or the Long-term phase. Response rates time-trends showed increase on the MADRS over all weeks following initiation of stimulation for all treatment groups in both the Acute and Long-term phases using a Mixed Model Repeated Measures (MMRM) analysis of covariance (ANCOVA). Figure 61 represents MADRS response rates across the Acute (week 22) and Long-term (week 50) phases of the study. While improvement was noted in the within group comparisons over time, this comparison does not preclude improvement for reasons other than device effect. Reasons for improvement may include the natural course of the disease, effectiveness of other concurrent treatments (e.g., medications), regression to the mean, and placebo effects.

Table 49. Response Rates MADRS (ITT Population)

Time Post Treatment	Stimulation Treatment Group							
	Low		Medium		High		Total	
	n	n (%) yes	n	n (%) yes	n	n (%) yes	n	n (%) yes
Week 10	101	11 (10.9)	100	15 (15.0)	106	14 (13.2)	307	40 (13.0)
Week 22	97	16 (16.5)	97	25 (25.8)	105	30 (28.6)	299	71 (23.7)
Week 38	90	29 (32.2)	93	30 (32.3)	104	34 (32.7)	287	93 (32.4)
Week 50	89	32 (36.0)	90	43 (47.8)	102	38 (37.3)	281	113 (40.2)

Figure 61. Response Rates MADRS, Acute and Long-term Phase (ITT Population)

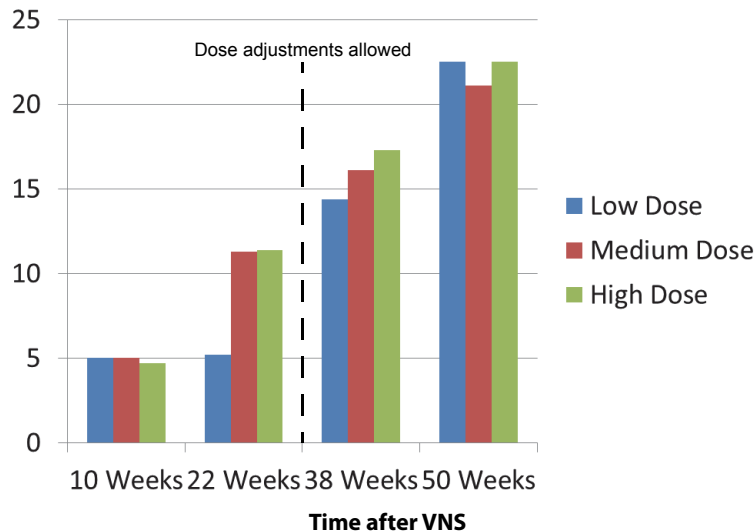


Although no statistically significant improvements over time were noted for remission rates in the Acute phase, remission rates improved over the weeks following initiation of stimulation for all treatment groups in the Long-term phase using a Mixed Model Repeated Measures (MMRM) analysis of covariance (ANCOVA). Figure 62 represents MADRS remission rates across the Acute (week 22) and Long-term (week 50) phases of the study. While improvement was noted in the within group comparisons over time, this comparison does not preclude improvement for reasons other than device effect. Reasons for improvement may include the natural course of the disease, effectiveness of other concurrent treatments (e.g., medications), regression to the mean, and placebo effects.

Table 50. Remission Rates MADRS (ITT Population)

Time Post Treatment	Stimulation Treatment Group							
	Low		Medium		High		Total	
	n	n (%) yes	n	n (%) yes	n	n (%) yes	n	n (%) yes
Week 10	101	5 (5.0)	100	5 (5.0)	106	5 (4.7)	307	15 (4.9)
Week 22	97	5 (5.2)	97	11 (11.3)	105	12 (11.4)	299	28 (9.4)
Week 38	90	13 (14.4)	93	15 (16.1)	104	18 (17.3)	287	46 (16.0)
Week 50	89	20 (22.5)	90	19 (21.1)	102	23 (22.5)	281	62 (22.1)

Figure 62. Remission Rates MADRS, Acute and Long-term Phase (ITT Population)



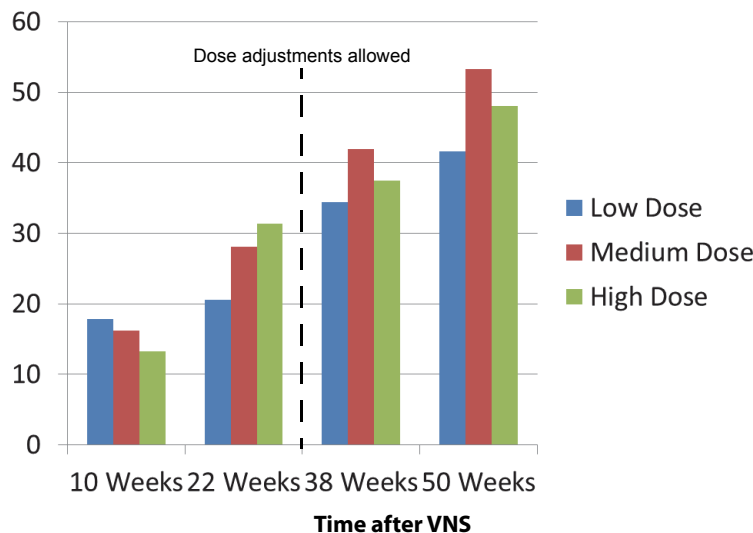
9.2.2.2.5. Response rates for Clinical Global Impressions Improvement Scale (CGI-I)

No statistically significant differences were noted between the 3 treatment groups for patients experiencing response in the Acute phase or the Long-term phase. The CGI-I (ordinal scale) rates the improvement in a patient's condition compared with the condition at admission to the study. For the purposes of this analysis, the patient was considered a responder if they were rated 'very much improved' (at least 85% improvement) and 'much improved' (at least 60% improvement). Response rates showed statistically significant time-trend improvement over the weeks following initiation of stimulation for all treatment groups and for both the Acute and Long-term phases of the study using a Mixed Model Repeated Measures (MMRM) analysis of covariance (ANCOVA). Figure 63 represents CGI-I responder rates across the Acute (week 22) and Long-term (week 50) phases of the study. While improvement was noted in the within group comparisons over time, this comparison does not preclude improvement for reasons other than device effect. Reasons for improvement may include the natural course of the disease, effectiveness of other concurrent treatments (e.g., medications), regression to the mean, and placebo effects.

Table 51. Response Rates CGI Global Improvement, Acute Phase ITT Population

Time	Stimulation Treatment Group							
	Low		Medium		High		Total	
	n	mean (SD)	n	mean (SD)	n	mean (SD)	n	mean (SD)
Response	n	n (%) yes	n	n (%) yes	n	n (%) yes	n	n (%) yes
Week 10	101	18 (17.8)	99	16 (16.2)	106	14 (13.2)	306	48 (15.7)
Week 22	97	20 (20.6)	96	27 (28.1)	105	33 (31.4)	298	80 (26.8)
Week 38	90	31 (34.4)	93	39 (41.9)	104	39 (37.5)	287	109 (38.0)
Week 50	89	37 (41.6)	90	48 (53.3)	102	49 (48.0)	281	134 (47.7)

Figure 63. Response Rates CGI Global Improvement, Acute and Long-term Phase (ITT Population)



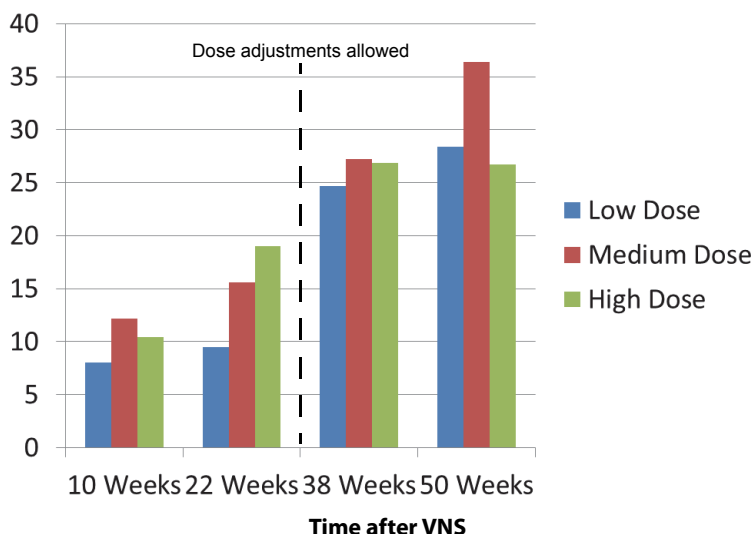
9.2.2.2.6. Response and remission rates for Inventory of Depressive Symptomatology Self-Rated (IDS-SR)

No statistically significant differences were noted between the 3 treatment groups for patients experiencing response in the Acute phase or the Long-term phase. Response rates using a Mixed Model Repeated Measures (MMRM) analysis of covariance (ANCOVA) improved over the weeks following initiation of stimulation for all treatment groups in the Long-term phase. Figure 64 represents IDS-SR responder rates across the Acute (week 22) and Long-term (week 50) phases of the study. While improvement was noted in the within group comparisons over time, this comparison does not preclude improvement for reasons other than device effect. Reasons for improvement may include the natural course of the disease, effectiveness of other concurrent treatments (e.g., medications), regression to the mean, and placebo effects.

Table 52. Response Rates IDS-SR (ITT Population)

Time Post Treatment	Stimulation Treatment Group							
	Low		Medium		High		Total	
	n	n (%) yes	n	n (%) yes	n	n (%) yes	n	n (%) yes
Week 10	100	8 (8.0)	98	12 (12.2)	106	11 (10.4)	304	31 (10.2)
Week 22	95	9 (9.5)	96	15 (15.6)	105	20 (19.0)	296	44 (14.9)
Week 38	89	22 (24.7)	92	25 (27.2)	104	28 (26.9)	285	75 (26.3)
Week 50	88	25 (28.4)	88	32 (36.4)	101	27 (26.7)	277	84 (30.3)

Figure 64. Response Rates IDS-SR, Acute and Long-term Phase (ITT Population)

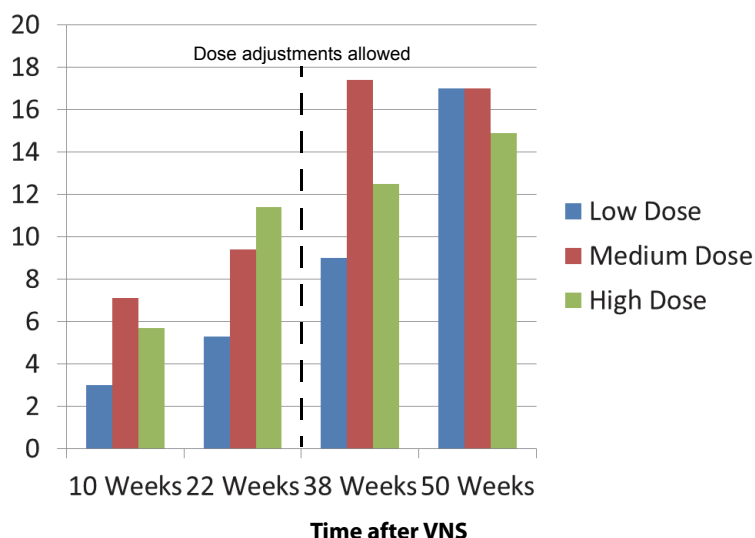


No statistically significant differences were noted between the 3 treatment groups for patients experiencing remission in the Acute phase or the Long-term phase. As well, no statistically significant improvements over time were noted for remission rates in the Acute phase. Remission rates improved over the weeks following initiation of stimulation for all treatment groups in the Long-term phase using a Mixed Model Repeated Measures (MMRM) analysis of covariance (ANCOVA). Figure 65 represents IDS-SR remission rates across the Acute (week 22) and Long-term (week 50) phases of the study. While improvement was noted in the within group comparisons over time, this comparison does not preclude improvement for reasons other than device effect. Reasons for improvement may include the natural course of the disease, effectiveness of other concurrent treatments (e.g., medications), regression to the mean, and placebo effects.

Table 53. Remission Rates IDS-SR (ITT Population)

Time Post Treatment	Stimulation Treatment Group							
	Low		Medium		High		Total	
	n	n (%) yes	n	n (%) yes	n	n (%) yes	n	n (%) yes
Week 10	100	3 (3.0)	98	7 (7.1)	106	6 (5.7)	304	16 (5.3)
Week 22	95	5 (5.3)	96	9 (9.4)	105	12 (11.4)	296	26 (8.8)
Week 38	89	8 (9.0)	92	16 (17.4)	104	13 (12.5)	285	37 (13.0)
Week 50	88	15 (17.0)	88	15 (17.0)	101	15 (14.9)	277	45 (16.2)

Figure 65. Remission Rates IDS-SR, Acute and Long-term Phase (ITT Population)



9.2.2.2.7. Long-term data (1-year)

To evaluate the long-term durability of the improvements in depression scores observed with adjunctive VNS Therapy, an analysis of durable effectiveness was performed using 2 clinician rated scales (IDS-C and MADRS). The durable effectiveness is defined as the percentage of Acute Phase responders (week 22) who were also responders at the end of the Long-term (week 50) phase. For both the IDS-C and MADRS results (Table 54), the Medium and High Dose groups exhibited high rates of durable effectiveness (88.2% and 92% for the Medium group, and 81.8% and 76.7% for the High Dose group). Of note is that durable effectiveness in the Low Dose group was lower than the Medium or High Dose groups for both the IDS-C (43.8%) and MADRS (68.8%).

Table 54. Proportion of Patients with a Sustained IDS-C or MADRS Response

Parameter	Stimulation Treatment Group			
	Low N=111	Medium N=107	High N=113	Total N=331
IDS-C				
N with at least 50% improvement Acute Phase	16	17	22	55
N (%) at least 50% improvement week 50	7 (43.8)	15 (88.2)	18 (81.8)	40 (72.7)
MADRS				
N with at least 50% improvement Acute Phase	16	25	30	71
N (%) at least 50% improvement week 50	11 (68.8)	23 (92)	23 (76.7)	57 (80.3)

9.2.2.3 Overall D-21 study conclusions

The D-21 study intended to compare the safety and effectiveness of 3 different doses of adjunctive VNS Therapy for the treatment of patients with TRD, where dose is defined as the magnitude of output current. It represented the first systematic investigation of a VNS Therapy dose response relationship employing a fixed-dose classification group (Low, Medium and High) design. The study was designed to evaluate the relative safety and effectiveness of different output currents.



Note: For intended use/indications, see the *Introduction to the VNS Therapy System* chapter.

The primary endpoint did not show statistically significant differences between the 3 assigned dose groups. Improvement over the weeks following the initiation of stimulation was noted for all treatment groups combined. While improvement was noted in the within-group comparisons over time, this comparison does not preclude improvement for reasons other than device effect. Reasons for improvement may include the natural course of the disease, effectiveness of other concurrent treatments (e.g., medications), regression to the mean, and placebo effects. In addition, not all the patients in a given treatment cohort were sequestered into distinct VNS parameter settings. According to the majority of effectiveness measures, the percentage of patients experiencing a positive treatment effect at week 22 (End of the Acute Phase) was numerically higher in the High Dose and Medium Dose groups compared with the Low Dose group. Despite the inherent variability expected when using multiple clinician and patient rating scales, improvement over time was consistent for all scales used and for both Acute and Long-term Phases of the study. This finding is consistent with prior VNS Therapy studies showing a significant improvement in depression symptoms with long-term treatment. This is particularly important when the severity of illness in the patient population is considered. Individuals enrolled in this study had experienced a large number of unsuccessful therapies (pharmaceutical), past suicide attempts, psychiatric related hospitalizations, and non-pharmacological treatments including ECT. Importantly, most of the patients (76.7% to 92%) who met response criteria while receiving the Medium and High Doses of VNS Therapy during the Acute Phase of the study continued to meet response criteria after 50 weeks of treatment. Patients in the Low Dose group experienced a lower sustained response than the Medium or High Dose groups for both the IDS-C (43.8%) and MADRS (68.8%), but this may still represent a higher level of sustained response than typically expected in this patient population. Previous studies of patients experiencing treatment-resistant depression have shown low rates of sustained response (high relapse rates) after ECT, and after pharmaceutical intervention.

Although the results of this study do not provide support for a dose-response effect or of general effectiveness, given the lack of a sham control, the lack of statistically significant mean differences between the dose groups for the effectiveness outcomes does not preclude the possibility that some individual patients might benefit from the Medium or High dose of VNS Therapy used in this study. Indeed, there were consistent numerical trends for increased improvement in response and remission rates that favored the Medium or High dose over the Low dose. Likewise, as outlined earlier, sustained response was more likely in the Medium or High doses groups than in the Low dose group.

Balancing the benefit and risk of the various doses utilized in this study, the study results reported here suggest that it is reasonable to initiate VNS Therapy treatment by titrating dose to an output current of 1.0 to 1.25 mA with a pulse width of 250 μ sec. Individual patients may respond differently to output current settings and require adjustment of the implanted devices parameter settings. Output currents up to 2.5mA were used in the D-21 trial. Individual patient tolerability should be considered during the dosing process; higher output current may be associated with less tolerability and time to adapt to increased dose should be

considered. Additional information regarding mortality¹ and effectiveness² can be found in peer-reviewed literature.

9.2.2.3.1. Study Strengths

- The randomized design of this trial is more advantageous than mere clinical observation of patients over time.
- The D-21 post-approval study was one of the largest studies ever conducted to include a patient population with this level of severe treatment-resistant major depressive disorder.

9.2.2.3.2. Study Weaknesses

- Simplification of VNS Therapy dose to 3 discrete levels, may have been a design feature that inadvertently obscured any possible relationship between stimulation dose and effectiveness.
- A difference in tolerability was noted amongst the treatments arms. Patients in the High group were less likely to receive the target dose at both the 10 week and 22 week time points (74.3% at week 10, and 72.6% at week 22), compared with the Low (88.3% at week 10, and 85.6% at week 22) and Medium (85% at week 10, and 87.9% at week 22) groups. This resulted in the actual charge delivered in the High dose group being closer to the Medium dose than desired.
- In the acute phase of the trial, the execution of the study did not adequately separate the active treatment groups into 3 discrete, non-overlapping groups. Our current knowledge (unknown at the time of the study design) indicates that the relationship between net charge delivered over time and antidepressant outcome is complex. In addition, not all patients in a given cohort were isolated into distinct VNS parameter settings.
- The effect of duty cycle (ON time as compared to OFF time) of the device may also play an important role in reduction of symptomatology for patients with TRD. Duty cycle is known to have an impact upon seizure reduction in some patients treated with VNS Therapy for epilepsy. This variable was not considered as part of the study.
- Subjects in this double-blind study were asked to identify the treatment group to which they had been assigned. 73% of subjects in the Low Dose group correctly identified their treatment group. However, subjects in the Medium Dose and High Dose groups were unable to accurately identify the group to which they had been assigned (48.6% and 24.8%, respectively). These data may indicate imperfect blinding which may have biased the results.
- A "Per Protocol Analysis" might have been favored over the "Intention to Treat Analysis" reported here.

¹ Bryan Olin, Amara K. Jayewardene, Mark Bunker, Francisco Moreno. "Mortality and Suicide Risk in Treatment-Resistant Depression: An Observational Study of the Long-Term Impact of Intervention." PLOS One. Article is available on-line: <http://dx.plos.org/10.1371/journal.pone.0048002>

² Scott T. Aaronson, Linda L. Carpenter, Charles R. Conway, et al. "Vagus nerve stimulation therapy randomized to different amounts of electrical charge for treatment-resistant depression: Acute and chronic effects." Brain Stimulation. Article is available on-line: <http://dx.doi.org/10.1016/j.brs.2012.09.013>

9.2.3 Bibliography

A bibliography of animal, clinical, and mechanism of action studies is available from LivaNova on request.

9.3 Post-Approval (D-23) Study

The post-approval registry study was a condition of approval required by the US Food and Drug Administration.

9.3.1 Summary of D-23 Post-Approval Registry Study Methods

9.3.1.1 Study objective

The objective of this registry was to follow the clinical course and outcome for patients with Treatment Resistant Depression (TRD) treated with and without adjunctive VNS Therapy.

9.3.1.2 Study design and study population

The D-23 study was a long-term, prospective, observational, multi-center patient outcome registry to collect data in patients with TRD who were in a Major Depressive Episode (MDE) and follow them for 60 months.

All sites were to first recruit patients who had agreed to have adjunctive VNS Therapy. This was to be defined by one of the following:

1. The patient was being evaluated for surgery or anesthesia to undergo implantation
2. The patient had signed surgical or anesthesia consent
3. The patient had a scheduled date for the surgical implantation procedure
4. Sites attempted to recruit patients who had completed the D-21 Dosing Study (Please see above for study description).

VNS Therapy-treated patients, henceforth known as the VNS population, were to be followed for 60 months from the date of first treatment stimulation.

Sites were also asked to enroll patients with TRD who would not be implanted with the VNS Therapy System and would act as a concurrent control group. All patients not receiving VNS Therapy, henceforth known as the Treatment-As-Usual (TAU) population, were to be followed for 60 months after baseline.

The registry was to enroll a minimum of 500 patients with TRD treated with adjunctive VNS Therapy and a minimum of 300 patients with TRD who were not treated with VNS Therapy (TAU group). Patients that completed the D-21 VNS Therapy Study were also eligible to enroll into the TRD Registry and were included within the VNS Therapy treatment group.

Patients' current depressive episodes were assessed by the registry physician using DSM-IV-TR criteria and confirmed by the Mini International Neuropsychiatric Interview (MINI). For purposes of this registry, TRD was defined as a chronic depression at least 2 years in duration or a recurrent depression which included at least 3 lifetime episodes including the current MDE; and an inadequate response to 4 or more adequate antidepressant treatments. Adequate antidepressant treatment was to be defined as dosage per Physicians' Desk Reference labeling for a minimum of 4 weeks.

9.3.1.3 Data source

The population data sets used in this study are represented below in Table 55. Three groups are analyzed: VNS Therapy D-23 only patients; VNS Therapy D-23 + D-21 patients; and TAU patients. Twenty-nine patients in the safety population were excluded from the ITT because they withdrew from the study prior to visit 3 (First visit post-baseline assessment).

Table 55. Data Sets Analyzed

	VNS Therapy D-23 Original N	VNS Therapy D-21 Rollover N	VNS Therapy D-23 + D-21 N	TAU N	Overall N
Safety Population	335	159	494	301	795
ITT Population*	330	159	489	276	765

* ITT = Intent-to-treat

Table 56 presents demographics by group. Demographic characteristics were similar across the groups. Approximately 70% of patients in the study were women, 94.6% of patients were Caucasian, and mean age was 49.3 years.

Table 56. Demographics — Safety Population

Parameter	VNS Therapy D-23 Original n (%)	VNS Therapy D-21 Rollover n (%)	VNS Therapy D-23 + D-21 n (%)	TAU n (%)	Overall n (%)
Gender [n()]	335	159	494	301	795
Male	104 (31.0%)	40 (25.2%)	144 (29.1%)	90 (29.9%)	234 (29.4%)
Female	231 (69.0%)	119 (74.8%)	350 (70.9%)	211 (70.1%)	561 (70.6%)
Ethnic Origin	335	159	494	301	795
Caucasian	323 (96.4%)	155 (97.5%)	478 (96.8%)	274 (91.0%)	752 (94.6%)
African-American	4 (1.2%)	4 (2.5%)	8 (1.6%)	6 (2.0%)	14 (1.8%)
Hispanic	3 (0.9%)		3 (0.6%)	19 (6.3%)	22 (2.8%)
Asian	2 (0.6%)		2 (0.4%)	1 (0.3%)	3 (0.4%)
Other	3 (0.9%)		3 (0.6%)	1 (0.3%)	4 (0.5%)
Age at baseline (Years)					
N	335	159	494	301	795
Mean (SD)	48.9 (10.41)	48.9 (9.51)	48.9 (10.12)	49.9 (11.07)	49.3 (10.49)

Table 57 presents baseline characteristics. Based on characteristics of depression the groups were well matched. Patients had been suffering from depression for almost 20 years before entering the study, had multiple episodes of depression and having tried and failed on average, 7.9 treatments.

Table 57. Baseline Patient and Disease Characteristics — Safety Population

Parameter	VNS Therapy D-23 Original	VNS Therapy D-21 Rollover	VNS Therapy D-23 + D-21	TAU	Overall
Age at initial onset of depression (years)					
N	334	159	493	301	794
Mean (SD)	20.8 (12.13)	21.1 (11.11)	20.9 (11.80)	21.1 (11.40)	21.0 (11.64)
Age at initial diagnosis of depression (years)					
N	334	159	493	301	794
Mean (SD)	29.0 (10.95)	28.6 (10.50)	28.9 (10.79)	29.5 (11.89)	29.1 (11.22)
Lifetime episodes of depression diagnosed*					
N	335		335	301	636
Mean (SD)	14.9 (24.14)		14.9 (24.14)	12.0 (23.86)	13.5 (24.03)
Number of failed treatments [†]					
N	335	159	494	301	795
Mean (SD)	8.0 (3.05)	8.6 (3.74)	8.2 (3.30)	7.3 (2.92)	7.9 (3.19)
Suicide attempts in lifetime					
N	335	159	494	301	795
Mean (SD)	2.1 (4.38)	1.2 (2.95)	1.8 (3.99)	1.2 (2.41)	1.6 (3.49)

* For D-21 Rollover patients the Lifetime episodes of Depression Diagnosed variable captured as a categorical variable.

[†] For D-21 Rollover patients, data not captured as part of the D-21 Study Protocol.

Table 58 presents the primary diagnosis of the current MDE at baseline.

Table 58. Primary Diagnosis of Current Major Depressive Episode — Safety Population

	VNS Therapy D-23 Original N	VNS Therapy D-21 Rollover N	VNS Therapy D-23 + D-21 N	TAU N	Overall N
Diagnosis Disorder N	335	159	494	301	795
	n (%)	n (%)	n (%)	n (%)	n (%)
Major Depressive Disorder, Recurrent, Moderate Severity	42 (12.5%)	21 (13.2%)	63 (12.8%)	69 (22.9%)	132 (16.6%)
Major Depressive Disorder, Recurrent, Severe Without Psychotic Features	135 (40.3%)	90 (56.6%)	225 (45.5%)	95 (31.6%)	320 (40.3%)
Major Depressive Disorder, Single Episode, Moderate Severity	12 (3.6%)	4 (2.5%)	16 (3.2%)	30 (10.0%)	46 (5.8%)
Major Depressive Disorder, Single Episode, Severe Without Psychotic Features	49 (14.6%)	7 (4.4%)	56 (11.3%)	36 (12.0%)	92 (11.6%)
Bipolar I Disorder, Most Recent Episode Depressed, Moderate Severity	19 (5.7%)	6 (3.8%)	25 (5.1%)	21 (7.0%)	46 (5.8%)
Bipolar I Disorder, Most Recent Episode Depressed, Severe Without Psychotic Features	46 (13.7%)	16 (10.1%)	62 (12.6%)	12 (4.0%)	74 (9.3%)
Bipolar II Disorder, Most Recent Episode Depressed	32 (9.6%)	15 (9.4%)	47 (9.5%)	38 (12.6%)	85 (10.7%)

9.3.1.4 Key study endpoints

9.3.1.4.1. Primary endpoint

The primary effectiveness endpoint was response rate based on the Montgomery-Åsberg Depression Rating Scale (MADRS): a comparison of the proportion of responders for patients in the VNS Therapy-treated group versus the proportion of responders for patients in the TAU group. Response rate was a binary variable (Responder = Yes/No). A patient was a responder if they experienced at least a 50% improvement in the MADRS score from baseline to the corresponding post-baseline visit assessment. (Note: Baseline date for VNS Therapy treated patients is the date of first device stimulation and Visit 2 for TAU treated patients). Binary response data from missing visits were imputed for the primary endpoint utilizing a Markov chain Monte Carlo (MCMC) technique to create 101 imputed datasets. For each missing visit, the overall imputed response was assigned based on the majority response across the 101 imputed values; therefore, the final overall imputed dataset has no missing data. A Mixed Model Repeated Measures (MMRM) method was then utilized for repeated binary data with propensity quintile classification as a covariate in the model to determine if response rates were different in VNS Therapy treated versus TAU treated patients during the 60 month follow-up.

Additional supplementary sensitivity analyses (cumulative proportion of subjects achieved first time response and time until first response) to the primary analysis were conducted.

9.3.1.4.2. Secondary endpoints

In addition to response rate based on MADRS (primary endpoint), additional secondary endpoints for MADRS included duration of response, remission, and duration of remission. Similarly, Clinical Global Impression-Improvement (CGI-I) and the Quick Inventory of Depressive Symptomatology-Self Report (QIDS-SR) were used to measure response rate, time until first response and duration of response. Remission was measured with Streamlined Longitudinal Interval Continuation Evaluation-Condensed Psychiatric Status Ratings (SLICE-C-PSR) and QIDS-SR which included assessments of time until remission and duration of remission (QIDS-SR only). Finally, clinically important Quality of Life and Depression symptomatology measures were summarized comparing the 2 treatment arms on questionnaires response scores (i) Quality of Life, Enjoyment and Satisfaction Questionnaire - Short Form (Q-LES-Q SF) and (ii) Arizona Sexual Experience questionnaire (ASEX).

9.3.1.5 Total number of enrolled study sites and subjects, follow-up rate

A total of 61 sites were activated to enroll patients into the D-23 study. Of these 61 sites, 55 sites enrolled patients. Out of 878 patients screened during the enrollment period, a total of 795 patients were evaluated and included in the Safety Population (SP). At the end of the study, 437 patients (55%) had completed the study per protocol and 358 patients (45%) had withdrawn from the study at different visit time-points post-baseline for various reasons (e.g. withdrew consent, lost to follow up (LTF), death, study site closure, etc.) The rate of early exits was high for both the VNS group (39%) and the TAU group (54%). The reasons for early termination were similar for both treatment groups. Of the 795 patients included, 494 patients were treated with VNS Therapy by either enrolling directly into D-23 study (335 patients) or rolling over from the D-21 study (159 patients). Of the 335 patients enrolling directly into the D-23, 177 patients completed the study per the protocol while 122 of the 159 who rolled over from D-21 study completed the protocol.

Table 59 and Table 60 provide a matrix for patient follow-up visits for the safety population within the D-23 study.

Table 59. Patient Accounting by Treatment and Visit (VNS D-23 + D-21 Safety Population)

Post-Baseline Visit														
Group	BL	3	6	9	12	18	24*	30	36	42	48	54	60	Overall
Expected Visits	494	489	480	470	461	292	289	299	313	333	334	319	300	4,873
Visit Assessments Completed [†]	494 (100%)	487 (99.6%)	477 (99.4%)	467 (99.4%)	459 (99.6%)	290 (99.3%)	284 (98.3%)	291 (97.3%)	308 (98.4%)	328 (98.5%)	332 (99.4%)	317 (99.4%)	299 (99.7%)	4,833 (99.17%)
Visit Assessments Not Done		2 (0.4%)	3 (0.6%)	3 (0.6%)	2 (0.4%)	2 (0.7%)	5 (1.7%)	8 (2.7%)	5 (1.6%)	5 (1.5%)	2 (0.6%)	2 (0.6%)	1 (0.3%)	

* One Patient is counted as a completer after 2 year participation as per the TRD Registry Protocol

[†] Same as expected visits excluding patients with missing data at visit

Table 60. Patient Accounting by Treatment and Visit (TAU Group Safety Population)

Group	BL	Post-Baseline Visit												Overall
		3	6	9	12	18	24*	30	36	42	48	54	60	
Expected Visits	301	276	260	242	224	201	185	172	168	160	149	143	138	2,619
Visit Assessments Completed [†]	301 (100%)	276 (100%)	259 (99.6%)	238 (98.3%)	219 (97.8%)	195 (97.0%)	172 (93.0%)	158 (91.9%)	166 (98.8%)	159 (99.4%)	148 (99.3%)	142 (99.3%)	137 (99.3%)	2,570 (98.1%)
Visit Assessments Not Done			1 (0.4%)	4 (1.7%)	5 (2.2%)	6 (3.0%)	13 (7.0%)	14 (8.1%)	2 (1.2%)	1 (0.6%)	1 (0.7%)	1 (0.7%)	1 (0.7%)	

* One Patient is counted as a completer after 2 year participation as per the TRD Registry Protocol

[†] Same as expected visits excluding patients with missing data at visit

Table 61 shows the patient disposition for all patients in D-23.

Table 61. Patient Disposition — Safety Population

	VNS Therapy D-23 Original N	VNS Therapy D-21 Rollover N	VNS Therapy D-23 + D-21 N	TAU N	Overall N
Total enrolled					878
Enrolled but not treated					83
Enrolled and treated	335	159	494	301	795
Completed study per protocol	177	122	299	138	437
Early withdrawals	158	37	195	163	358
Primary Reasons for Early Study Termination	VNS Therapy D-23 Original N (%)	VNS Therapy D-21 Rollover N (%)	VNS Therapy D-23 + D-21 N (%)	TAU N (%)	Overall N (%)
Patient withdrew consent	44 (13.1%)	11 (6.9%)	55 (11.1%)	37 (12.3%)	92 (11.6%)
Patient non-compliance	32 (9.6%)	8 (5.0%)	40 (8.1%)	39 (13.0%)	79 (9.9%)
Patient did not meet criteria		3 (1.9%)	3 (0.6%)	1 (0.3%)	4 (0.5%)
Participating physician decision	4 (1.2%)		4 (0.8%)	7 (2.3%)	11 (1.4%)
Death	7 (2.1%)		7 (1.4%)	8 (2.7%)	15 (1.9%)
Other	71 (21.2%)	15 (9.4%)	86 (17.4%)	71 (23.6%)	157 (19.7%)

9.3.1.6

Study visits and length of follow-up

Patients enrolling directly into the D-23 study had a total of 14 clinic study visits plus at least one telephone Central Rater Group (CRG) visit per clinic visit beginning at Baseline for a total of 13 telephone visits. Study visits occurred every 3 months for the first year and every 6 months for the remaining 4 years over the 10 year trial period. D-23 patients selected their treatment arm (VNS Therapy or TAU) at the screening visit. Patients electing to receive VNS Therapy were followed for 60 months from the date of first VNS stimulation while patients electing TAU were followed for 60 months from Baseline.

Patients rolling over from the D-21 study into the D-23 registry began study procedures no earlier than 1 year into the D-23 follow-up procedures (visit 6, month 12).

9.3.2 Summary of D-23 Post-Approval Study Results

9.3.2.1 *Final safety findings*

9.3.2.1.1. FIBSER -- Impairment Baseline Treatment Assignment

The Frequency, Intensity and Burden of Side Effects Rating Scale (FIBSER) was developed to document 3 domains of side effects in patients treated in the STAR*D project (Wisniewski, Rush et al. 2006). Although it does not measure the impact of specific side effects, it does measure 3 domains of impact: frequency, intensity, and burden of the side effects. Its brevity makes it a useful tool for routine clinical practice. The FIBSER is a 3-item scale and is rated from 0, No side effects to 6, present all of the time. Categorization of FIBSER scores is as follows: Acceptable (0-2); Moderate (3-4); Unacceptable (5-6) (Wisniewski, Rush et al. 2006). Total scores range from 0 (best score) to 18 (worst score).

The FIBSER Scale was used to assess the overall impact of medication side effects. In regards to depression medications taken within the past week prior to the assessment, the scale measures the frequency of side effects; the intensity of side effects; and the degree to which they interfere with day-to-day function. As seen in Figure 66, Figure 67, and Figure 68, the percent of patients experiencing a score of 5 or 6 (least favorable) on the frequency, intensity and burden subscales, respectively, are similar in all groups. As well, they all decrease over time.

Overall FIBSER percentages for “no side effects” in frequency, intensity and burden were slightly higher (more favorable) in the TAU patients throughout the long-term follow up interval. Notable differences occurred at the 24 month visit, where patients treated with VNS Therapy plus medications reported higher (more favorable) percentage with “no side effects” on frequency and impairment. Additionally, the percentage of patients experiencing a score of 5 or 6 (least favorable) on the frequency, intensity and burden subscales, respectively, are similar in all groups and decrease over time. These data show that adding VNS Therapy does not impart a clinically greater side effect burden than what is seen with medication (TAU) treatment alone, and VNS Therapy is tolerable with multiple adjunctive medication regimens.

Figure 66. Percentage of Patients Reporting “Frequent” in 5-6 Scale Based on FIBSER Assessment (Safety Population)

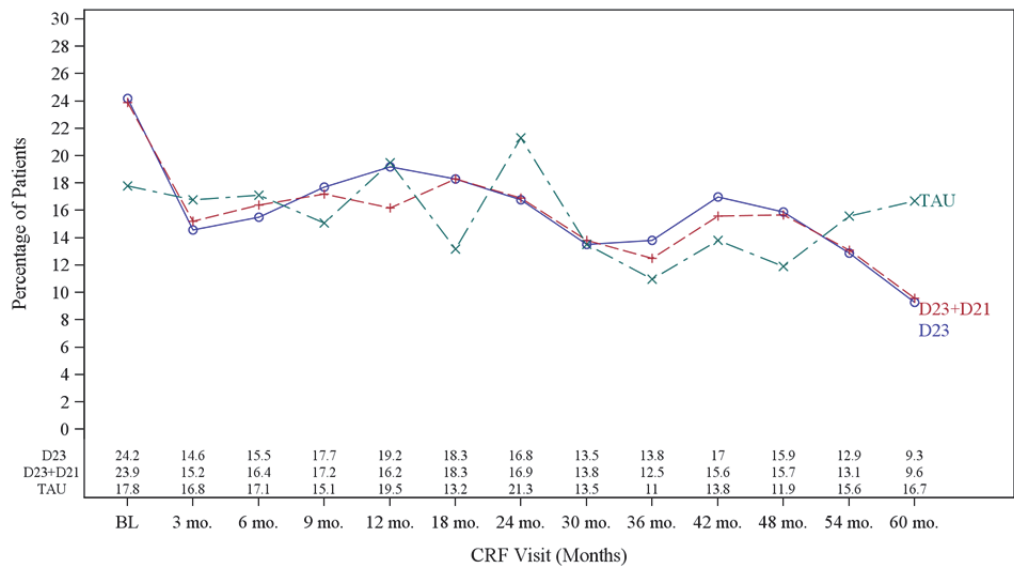


Figure 67. Percentage of Patients Reporting “Intense” in 5-6 Scale Based on FIBSER Assessment (Safety Population)

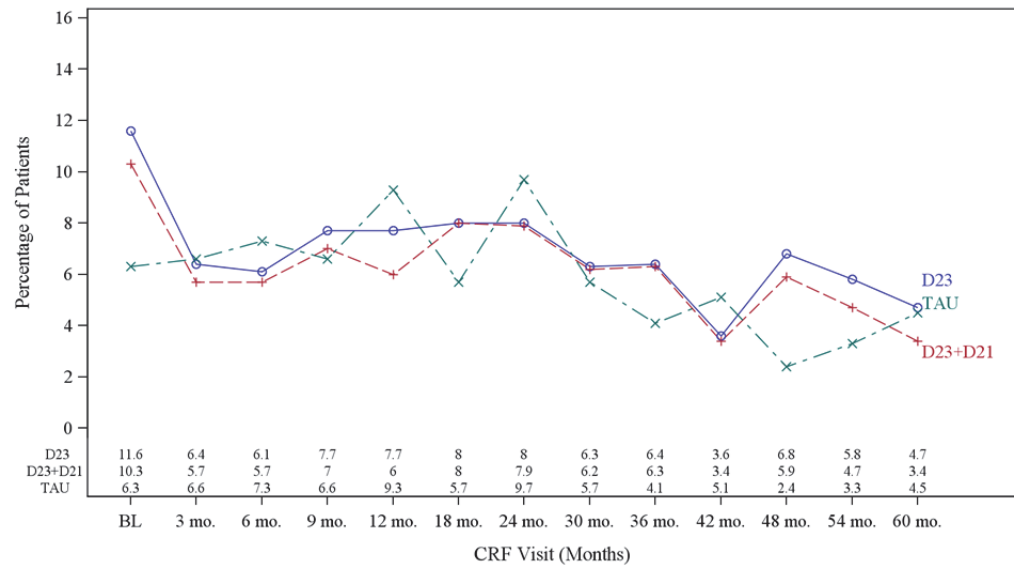
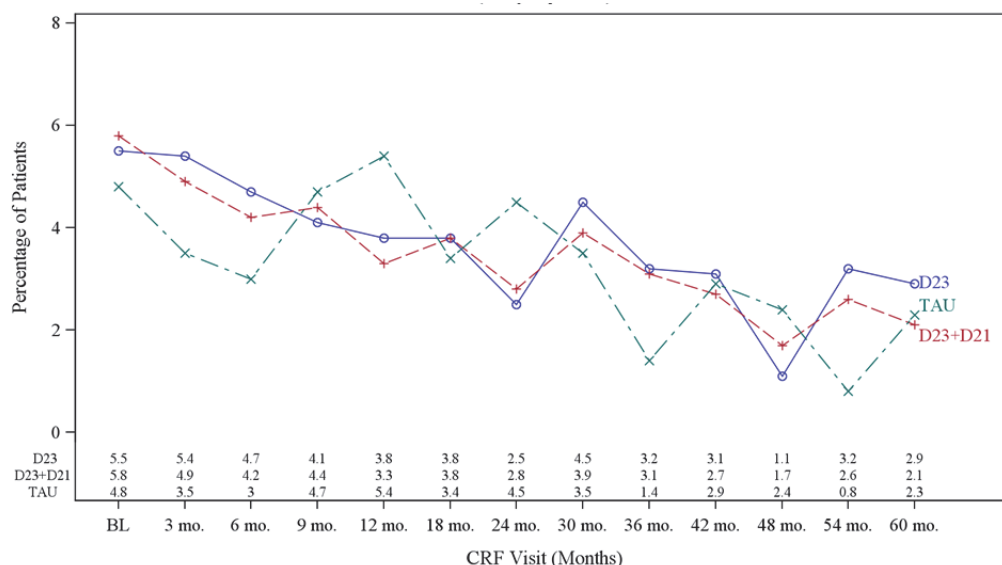


Figure 68. Percentage of Patients Reporting “Burden of Side Effects” in 5-6 Scale Based on FIBSER Assessment (Safety Population)



9.3.2.1.2. Suicidality

Suicidality was assessed using 3 outcome variables:

- MADRS Item 10 score of ≥ 4 (“Probably better off dead”)
- QIDS-SR Questionnaire Item 12 score of ≥ 2 (“I think of suicide or death several times a week for several minutes”)
- Assessment of Suicidality (ASO) Clinician (Investigator) Response = “Yes” if patient made a suicidal gesture or attempt since the last visit.

To account for the presence of three sets of statistical analyses, a multiplicity adjustment was performed on the P values to account for $K=3$ tests. Because of the dependence of the D23 and D23 + D21 groups, further adjustments for testing both the D23 and D23 + D21 groups against TAU were not required. As such, the following two-sided Tukey, Ciminera, and Heyse (TCH) adjustment was applied to each assessment's P value with $K=3$, which takes into account simultaneous testing of correlated hypotheses (Zhang et al., 1997):

$$p_{TCH} = 1 - (1 - p)^{\sqrt{K}}$$

For Suicidality measures based on MADRS Item 10 score (Investigator), QIDS-SR Item 12 score (Patient), and Assessment of Suicidality (Investigator), for VNS D-23 (MADRS adjusted P value: 0.002, QIDS-SR adjusted P value: $<.001$, AOS adjusted P value: 0.0497) and for VNS D-23 + D-21 (QIDS-SR adjusted P value: 0.006) treatment groups, the results shows a statistically significant reduction in the suicidality profile as compared to the TAU treatment group for all comparisons except the comparison between the VNS D-23 + D-21 (adjusted P value: 0.099) and TAU treatment group for the suicidality measure based on the MADRS Item 10 score.

Table 62, Table 63, and Table 64 list odds ratios and P values for statistical comparisons between groups. Odds Ratios were calculated for comparison of suicidality measures between VNS patients and TAU patients. An odds ratio > 1 favors VNS treatment over TAU.

Table 62. Summary of 'Suicidal Ideation' based on MADRS Item 10 Score by Treatment through 60 Months Post Baseline (Generalized Linear Mixed Repeated Measures Model)

VNS D-23 vs TAU		
Odds Ratio (SE)	Odds Ratio 95% CI	Adjusted P value
2.6553 (1.0854)	(1.4742, 4.7826)	0.0019
VNS D-23 + D-21 vs TAU		
Odds Ratio (SE)	Odds Ratio 95% CI	Adjusted P value
1.6667 (0.5923)	(0.9824, 2.8277)	0.0985

Table 63. Summary of 'Suicidal Ideation' based on QIDS-SR Item 12 Score by Treatment through 60 Months Post baseline

VNS D-23 vs TAU		
Odds Ratio (SE)	Odds Ratio 95% CI	Adjusted P value
2.8314 (1.0439)	(1.6436, 4.8774)	0.0002
VNS D-23 + D-21 vs TAU		
Odds Ratio (SE)	Odds Ratio 95% CI	Adjusted P value
2.1099 (0.7011)	(1.2777, 3.4840)	0.0061

Table 64. Summary of 'Suicidal Ideation' based on Assessment of Suicidality Measure

VNS D-23 vs TAU		
Odds Ratio (SE)	Odds Ratio 95% CI	Adjusted P value
2.0368 (0.9284)	(1.0757, 3.8565)	0.0497

Note: Assessment of Suicidality data was collected by the Central Rater Group (CRG) at Baseline, while at Follow-up data was collected by Clinician. Since Assessment of Suicidality data was not collected at Baseline For D-21 rollover patients; change from baseline analysis is not performed.

Based on the repeated measures analyses conducted, the Odds Ratio was greater than 1 in favor of the VNS Therapy treatment groups as compared to TAU treatment group in all suicidality outcome measures. The analyses demonstrated that treatment with adjunctive VNS Therapy (VNS D-23 and VNS D-23 + D-21 treatment groups) has a greater reduction in the average suicidality profile through sixty months post baseline as compared to the TAU treatment group only, after adjustment of the baseline risk.

9.3.2.1.3. Adverse events

Adverse event information was not collected as part of the D-23 study, but physicians were instructed to notify LivaNova of all potential Medical Device Reporting (MDR) events that occurred among the D-23 study participants receiving VNS Therapy.

MDRs were not reported for any patients in the TAU group since they were not implanted with a VNS Therapy system.

A total of 308 MDR events for 166 patients were reported to LivaNova regarding VNS Therapy patients enrolled in the D-23 study. None of the reported events were unanticipated.

After examination of MDR reportable events related to the disease state of patients under study, the following was observed. Of the 85 events regarding suicidality, 53 suicidal gestures were reported in 39 patients, 17 suicide attempts were reported by 15 patients, and 15 suicidal ideations were reported by 15 patients. Of the 39 patients who reported suicidal gestures, 28 patients reported the event 1 time, 8 patients reported the event 2 times, and 3 patients reported the event 3 times. Of the 21 events regarding worsening depression, 19 separate patients reported this occurrence. The 308 MDR events shown in Table 65 are similar to events reported with use of the VNS Therapy system in other patient populations.

Table 65. MDR Reportable Events (Implanted Patients Only)

MDR Reportable Events (D-23 + D-21 Rollovers)	Original Registry (N=336)*	D-21 Rollovers (N=159)†	Total (N=495)*
Suicidality	62	23	85
Suicide attempt (17)	9	8	17
Ideation (15)	8	7	15
Gesture (53)	45	8	53
Software error	42	14	56
High impedance (22) /Lead fracture (16)	(17)/(11)	(5)/(5)	38
Worsening depression	10	11	21
Pain	9	2	11
Migration of generator (6)/ Migration of electrode (1)/Lead pulling sensation (2)	(5)/(0)/(2)	(1)/(1)/(0)	9
Infection	5	3	8
Vocal cord paralysis	1	5	6
Dyspnea	3	2	5
Failure to program	2	3	5
Arrhythmia	1	3	4
Death - autopsy report not received (1)/ accidental (2)/ suicide (1)	4	0	4
Manic episode	1	3	4
Painful stimulation	0	4	4
Short circuit condition	1	3	4
Anxiety	2	1	3
Erratic stimulation (3)	3	0	3
Syncope	2	1	3
Altered stimulation (2)	2	0	2
Cancer (breast)	2	0	2
Insertion difficulty	1	1	2
Lack of effectiveness	2	0	2
Nausea	1	1	2
Vomiting	1	1	2
Apnea	1	1	2
Asthma	0	1	1

MDR Reportable Events (D-23 + D-21 Rollovers)	Original Registry (N=336)*	D-21 Rollovers (N=159) [†]	Total (N=495)*
Constipation	0	1	1
Corrosion of lead coil	2	2	4
Dehiscence	1	0	1
Dyspepsia	1	0	1
Embolism	0	1	1
Extrusion	1	0	1
Fatigue	1	0	1
General performance issue	1	0	1
Mechanical Failure	1	1	2
Myocardial infarction	1	0	1
Nerve damage	0	1	1
Protrusion	1	0	1
Scarring	1	0	1
Sepsis	1	0	1
Urinary retention	0	1	1
Other (not specified) [‡]	1	0	1
Total	206	102	308

* Includes one TAU patient (1034-010) who was implanted following Visit 3

[†] All D-21 MDRs were previously reported to the FDA during the D-21 study

[‡] One event of "Not Specified (Cardiomyopathy)" was updated to "Not Specified"

9.3.2.1.4. D-23 study: mortality

Table 66 presents D-23 study results for all-cause mortality and suicidality. Seven patients in the VNS Therapy group (1.4% or 3.53/1000 person years) and 8 patients in the TAU group (2.9% or 8.63/1000 person years in the TAU group) died during the study. Two patients in each group completed suicides during the study (0.4% or 1.01/1000 person years in the VNS Therapy group and 0.7% or 2.20/1000 person years in the TAU group).

Table 66. D-23 Study: Mortality

Variable	VNS (N=494)	TAU (N=301)
Total number of deaths	7	8
Patient year exposure	1985.083	926.493
All-cause mortality/1000 person years	3.53	8.63
Suicides	2	2
Suicides/1000 person years	1.01	2.20

9.3.2.1.5. Safety conclusions

A total of 308 MDR events for 166 patients were reported to LivaNova by the patient population receiving VNS Therapy. None of the reported events were unanticipated. The percentage of patients experiencing a score of 5 or 6 (least favorable) on the frequency,

intensity and burden subscales of the FIBSER, respectively, are similar in all groups and decrease over time for all groups.

Patients treated with VNS Therapy, experienced a statistically greater reduction in suicide risk over time as compared with patients treated with TAU.

All-cause mortality and completed suicide rates were lower for the VNS Therapy group as compared with patients in the TAU group. Eight patients in the TAU group and 7 patients in the VNS Therapy group died during the trial. Of these, 4 deaths were considered suicides, 2 in the TAU group and 2 in the VNS group. None of the TAU patients were considered responders at last recorded visit. Three of the deceased VNS Therapy patients were considered responders at last recorded visit (deaths attributed to accidental overdose of opiates; suicide; and homicide).

These data re-confirm earlier studies regarding the safety of VNS Therapy for treatment of patients with treatment-resistant major depressive disorder.

9.3.2.2 **Final Effectiveness Findings**

9.3.2.2.1. **Primary Endpoint**

The primary effectiveness endpoint was response rate based on MADRS: a comparison of the proportion of responders for patients in the VNS Therapy-treated group versus the proportion of responders for patients in the TAU group. A statistically significant higher response rate was observed in the VNS Therapy treated versus TAU treated patient population (32.5% for the VNS: D-23 + D-21 group versus 13.6% for the TAU group; $P < .001$). Also, a statistically significant higher response rate was seen in the VNS Therapy D-23 only group than in the TAU group (30.5% versus 13%, respectively, $P < .001$) (Table 67).

Table 67. Mixed Model Repeated Measure (MMRM) Analysis on MADRS Comparison of Response Rate (VNS D-23 + D-21, VNS D-23, and TAU Patients - ITT Population)

Treatment Group	Least Square Mean (Std Err)	95pct CI (Lower, Upper)	P Value
TAU	13.6 (0.7)	(12.4, 14.9)	<.0001
VNS (D-23 + D-21)	32.5 (0.7)	(31.2, 33.9)	
TAU	13.0 (0.6)	(11.9, 14.3)	<.0001
VNS (D-23)	30.5 (0.8)	(29.0, 32.0)	

Two additional supplementary sensitivity analyses to the primary analysis were conducted; Cumulative proportion of subjects achieving first time response at post-baseline visit assessment, and Time until first response.

Cumulative proportion of subjects achieved first time response by post-baseline visit assessment

Figure 69 presents cumulative observed response rate by post-baseline visit through the 60 month follow up period, based on MADRS (percent cumulative responders are shown above the X-axis). Running a logistic regression model adjusted to propensity quintiles we find statistically significant greater response rates in the VNS Therapy population when comparing the cumulative response rate at 60 months (67.6% for the VNS Therapy D-23 + D-21 group

versus 40.9% for the TAU patients, $P<.001$). Also, similar statistically significant greater response rates were observed when comparing the VNS Therapy D-23 only group to the TAU group (63.7% versus 40.9%, respectively, $P<.001$). (Table 68)

Figure 69. Cumulative, Observed First Time Responder Rate by Post-Baseline Visit Assessment by Treatment Group: MADRS (VNS D-23 + D-21, D-23, and TAU Patients (ITT Population)

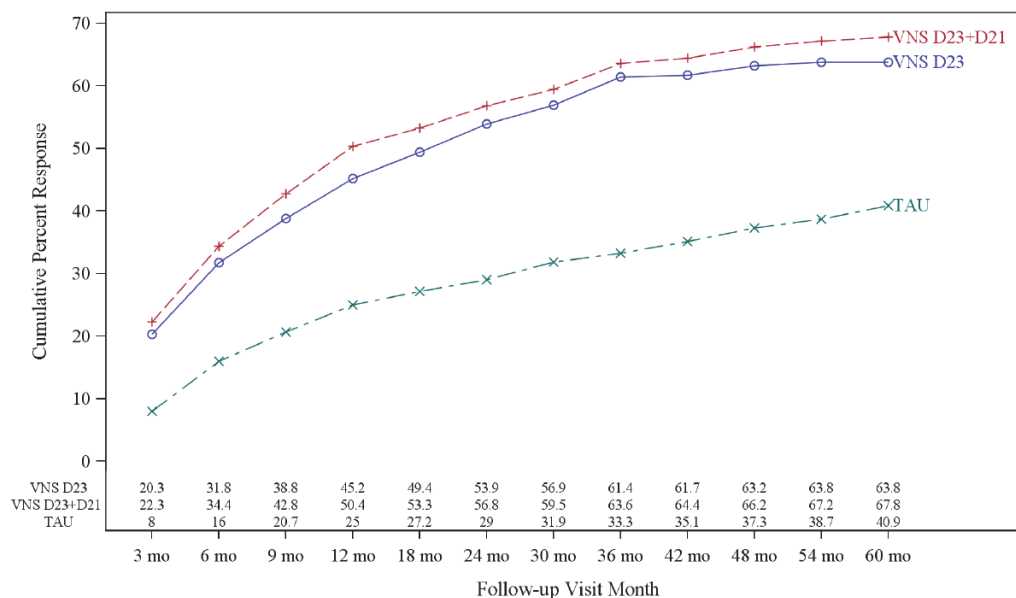


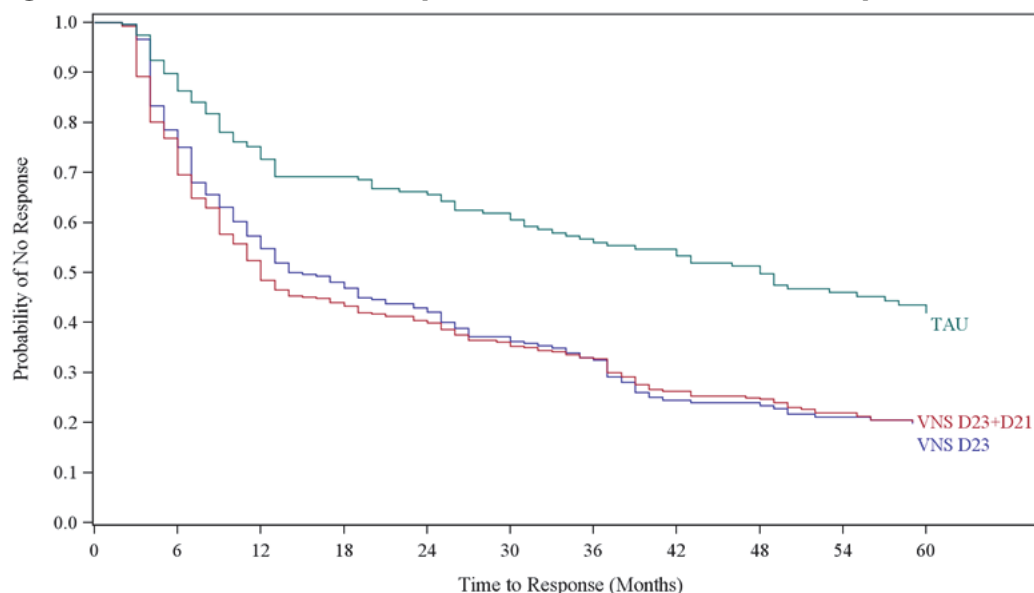
Table 68. Overall Propensity Quintiles Adjusted Response Rate Comparison for VNS D-23 + D-21, VNS D-23, and TAU Patients by Post-Baseline Visit Assessment Based on MADRS (ITT Population)

Parameter	Statistic	VNS D-23+D-21	VNS D-23	TAU
Overall MADRS Response Rate	n (%)	331 (67.6)	211 (63.7)	113 (40.9)
95pct CI (Overall Response Rate)	(%,%)	(63.4, 71.7)	(58.6, 68.9)	(35.4, 47.1)
P value (vs. TAU)		<.0001	<.0001	

Time until First Response Based on MADRS

Time until first response was computed as date the first time response of $\geq 50\%$ improvement from baseline in MADRS total score was observed minus baseline date. Figure 70 presents the time until first response by post-baseline visit assessment based on MADRS. Median time to first response was statistically significantly shorter for patients in the VNS Therapy D-23 + D-21 group versus patients in the TAU group (12 months for the VNS Therapy D-23 + D-21 group versus 48 months for the TAU group, $P<.001$). This was also seen in patients in the VNS Therapy D-23 only group versus patients in the TAU group (15 months for the VNS Therapy D-23 + D-21 group versus 48 months for the TAU group, $P<.001$).

Figure 70. Time to First Response Based on MADRS (ITT Population)



Note: *P* value: <0.001 for D-23 + D-21 and D-23 vs TAU

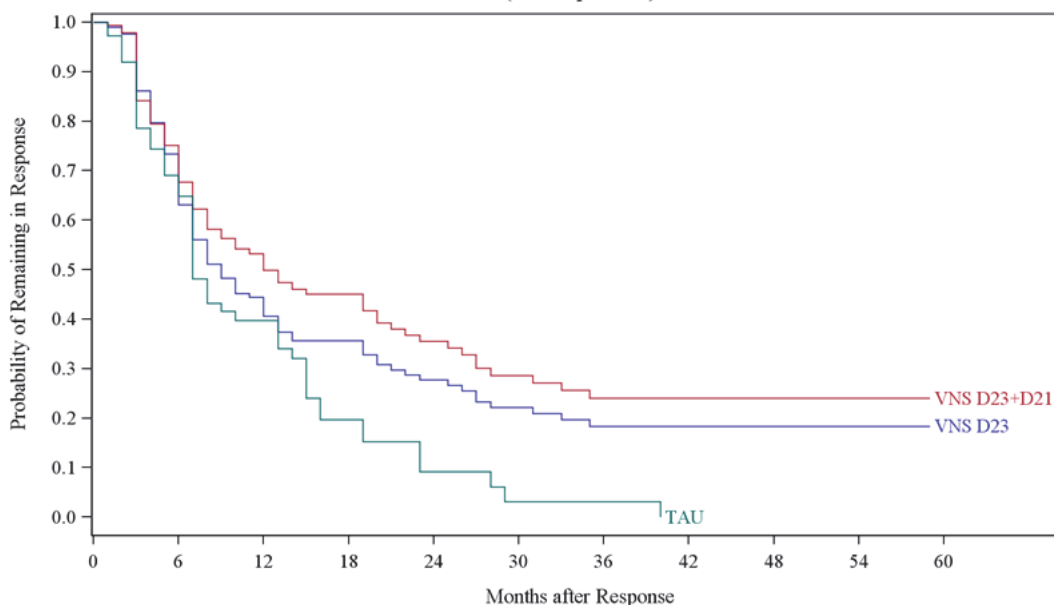
9.3.2.2.2. Secondary Endpoints

9.3.2.2.2.1. MADRS

Duration of response based on MADRS (ITT Population)

Duration of response was computed as the difference between the first recorded date post-baseline that response is achieved (MADRS $\geq 50\%$) and the first date at which MADRS response decreased to < 40%. Figure 71 presents duration of response for the VNS Therapy D-23 + D-21 group versus patients in the VNS Therapy D-23 only patients versus TAU patients based on MADRS scores. Patients in the VNS Therapy D-23 + D-21 group had a statistically significantly longer median duration of response than patients in the TAU group (12 months for the VNS Therapy D-23 + D-21 group versus 7 months for the TAU patients, $P=.017$). Patients in the VNS Therapy D-23 only group versus patients in the TAU group did not show a statistical difference.

Figure 71. Duration of Response Based on MADRS (ITT Population)



Note: *P* value: 0.017 for D-23 + D-21 vs. TAU; not statistically significant for D-23 vs. TAU

First Time Remitters by Post-Baseline Visit Assessment by Treatment Group Based on MADRS (ITT Population)

Remission is a binary outcome response variable (Yes/No In remission) defined as MADRS total score ≤ 9 at post-baseline visit assessment. An MMRM method was used for repeated binary data with propensity quintile classification as a covariate in the model to determine if remission rates were different in the VNS Therapy treated population versus the TAU population during the 60 month follow-up. A statistically significant higher remission rate was observed in the VNS Therapy treated population versus the TAU treated population (19.8% for the VNS: D-23 + D-21 group versus 8.1% for the TAU group; $P < .001$). Also, a statistically significant greater remission rate was seen in the VNS Therapy D-23 only group than in the TAU group (19.5% versus 8.5%, respectively, $P < .001$) (Table 69).

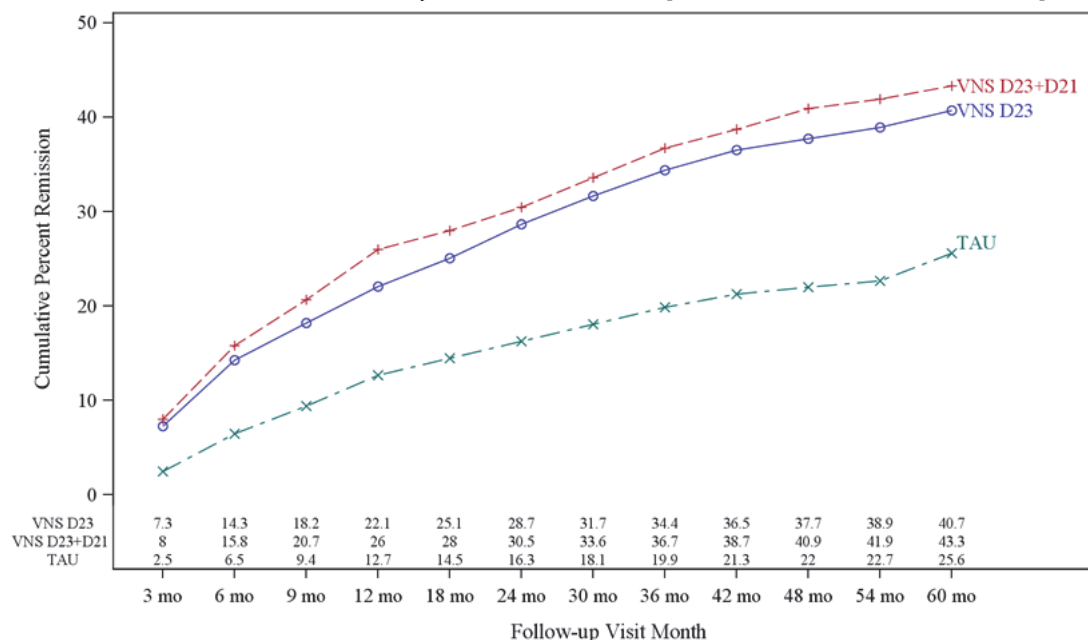
Table 69. Mixed Model Repeated Measure (MMRM) Analysis on MADRS Remitters Propensity Score Quintile as a Covariate (VNS D-23 + D-21, VNS D-23, and TAU Patients - ITT Population)

Treatment Group	Least Square Mean (Std Err)	95pct CI (Lower, Upper)	<i>P</i> Value
TAU	8.1 (0.7)	(6.9, 9.5)	<.0001
VNS (D-23 + D-21) + TAU	19.8 (0.8)	(18.4, 21.3)	
TAU	8.5 (0.7)	(7.3, 10.0)	<.0001
VNS (D-23) + TAU	19.5 (0.9)	(17.9, 21.3)	

Figure 72 presents the cumulative observed first time remitter rate in the VNS Therapy D-23 + D-21 group versus first time remitters in the VNS Therapy D-23 only group versus first time remitters in the TAU group, by month post-baseline, based on MADRS scores. Running a logistic regression model adjusted to propensity quintiles we find patients in the VNS Therapy D-23 + D-21 group were statistically significantly more likely to experience remission than

patients in the TAU group (43.3% for patients in the VNS Therapy D-23 + D-21 group, versus 25.7% in the TAU patients, $P<.001$). VNS Therapy D-23 only patients were statistically significantly more likely to experience remission than patients in the TAU group (40.8% for patients in the VNS Therapy D-23 group, versus 25.7% in the TAU patients, $P<.001$) (Table 70).

Figure 72. Cumulative, Observed First Time Remitter Rate by Post-Baseline Visit Assessment by Treatment Group Based on MADRS (ITT Population)



Note: P value: $<.001$ for D-23 + D-21 and D-23 vs TAU

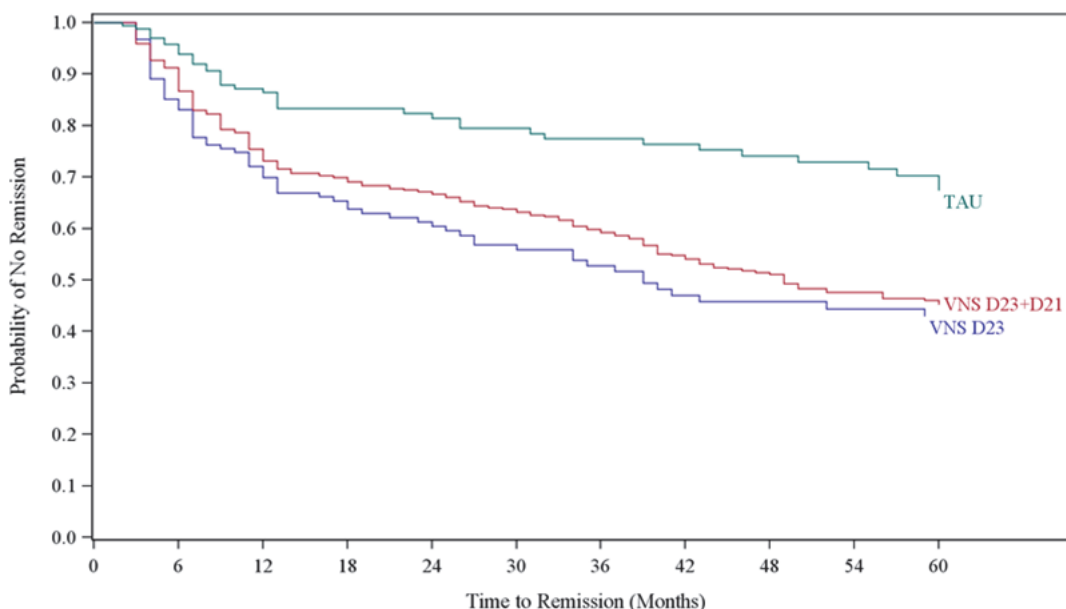
Table 70. Overall Propensity Quintiles Adjusted Remission Rate Comparison for VNS D-23 + D-21, VNS D-23, and TAU Patients Based on MADRS (ITT Population)

Parameter	Statistic	VNS D-23 + D-21	VNS D-23	TAU
Overall MADRS Remission Rate	n (%)	212 (43.3)	135 (40.8)	71 (25.7)
95pct CI (Overall Remission Rate)	(%,%)	(38.9, 47.7)	(35.5, 46.1)	(20.7, 31.1)
P value (vs. TAU)		$<.0001$	$<.0001$	

Time until First Remission Based on MADRS (ITT Population)

Figure 73 presents the time until first time remission for patients in the VNS Therapy D-23 + D-21 group versus patients in the VNS Therapy D-23 only group versus TAU patients, by visit month post-baseline, based on MADRS scores. As noted above, patients in the VNS Therapy D-23 + D-21 group were statistically significantly more likely to experience remission than patients in the TAU group ($P<.001$). The median time until remission was 49 months for patients in the VNS Therapy D-23 + D-21 group. Similarly, patients in the VNS Therapy D-23 only group were statistically significantly more likely to experience remission than patients in the TAU group ($P<.001$). The median time until remission was 45 months for patients in the VNS Therapy D-23 only group

Figure 73. Time to First Remission Based on MADRS (ITT Population)



Note: *P* value: <.001 for D-23+D-21 and D-23 vs TAU

Duration of Remission, Based on MADRS (ITT Population)

Duration of remission was computed as the recorded date of first recurrence/relapse (MADRS score ≥ 20) minus the recorded date of first achieved remission (MADRS score ≤ 9). Duration for patients that experienced remission was longer for patients in the VNS Therapy D-23 + D-21 group versus patients in the TAU group, based on MADRS scores (40 months for the VNS Therapy D-23 + D-21 group versus 19 months for TAU group, but did not reach statistical significance ($P=.102$). The difference between scores of patients in the VNS Therapy D-23 and those of patients in the TAU group also did not reach statistical significance.

9.3.2.2.2. CGI-I

First Time Response by Post-Baseline Visit Assessment Based on CGI-I (ITT Population)

First time response based on CGI-I was analyzed as a binary variable. A patient who achieved a (CGI-I) rating "1" or "2" ("Very much improved" or "Much improved") at post-baseline visit assessment was considered as a responder (Yes = 1). A patient that did not achieve a CGI-I rating "1" or "2" ("Very much improved" or "Much improved") at post-baseline visit assessment was considered as a non-responder (No = 0). An MMRM method was used for repeated binary data with propensity quintile classification as a covariate in the model to determine if response rates were different in the VNS Therapy treated population versus the TAU treated population during the 60 month follow-up. The response rate for CGI-I is defined as a score of "Much Improved" or "Very Much Improved." A statistically significant greater response rate was observed in the VNS Therapy treated group versus the TAU treated group (51.2% for the VNS Therapy D-23 + D-21 group versus 21.1% for the TAU group; $P<.001$). Also, a statistically significant greater response rate was seen in the VNS Therapy D-23 only group than in the TAU group (50.3% versus 21.3%, respectively, $P<.001$) (Table 71).

Table 71. Mixed Model Repeated Measure (MMRM) Analysis on CGI-I Responders Propensity Score Quintile as a Covariate (VNS D-23 + D-21, VNS D-23, and TAU Patients - ITT Population)

Treatment Group	Least Square Mean (Std Err)	95pct CI (Lower, Upper)	P Value*
TAU	21.1 (1.0)	(19.3, 23.1)	<.0001
VNS (D-23 + D-21)	51.2 (0.9)	(49.5, 52.9)	
TAU	21.3 (1.0)	(19.4, 23.2)	<.0001
VNS (D-23)	50.3 (1.0)	(48.3, 52.3)	

*P value generated by performing MMRM analysis with Binary distribution and logit function in PROC GLIMMIX

Figure 74 presents cumulative, observed response rate by visit post-baseline, and a total (cumulative) rate for 60 months data, based on CGI-I. Adjusted with propensity quintiles model results are consistent with those based on MADRS and show a statistically significant greater response for patients in the VNS Therapy D-23 + D-21 group versus patients in the TAU group (75.9% Response Rate for the VNS Therapy D-23 + D-21 group versus 48.6% for the TAU group, $P<.001$). The response rate for patients in the VNS Therapy D-23 only group versus patients in the TAU group was also statistically significant based on the CGI-I (71.0% versus 48.6%, respectively, $P<.001$ (Table 72).

Figure 74. Cumulative, Observed First Time Responders by Post-Baseline Visit Assessment by Treatment Group Based on CGI-I (ITT Population)

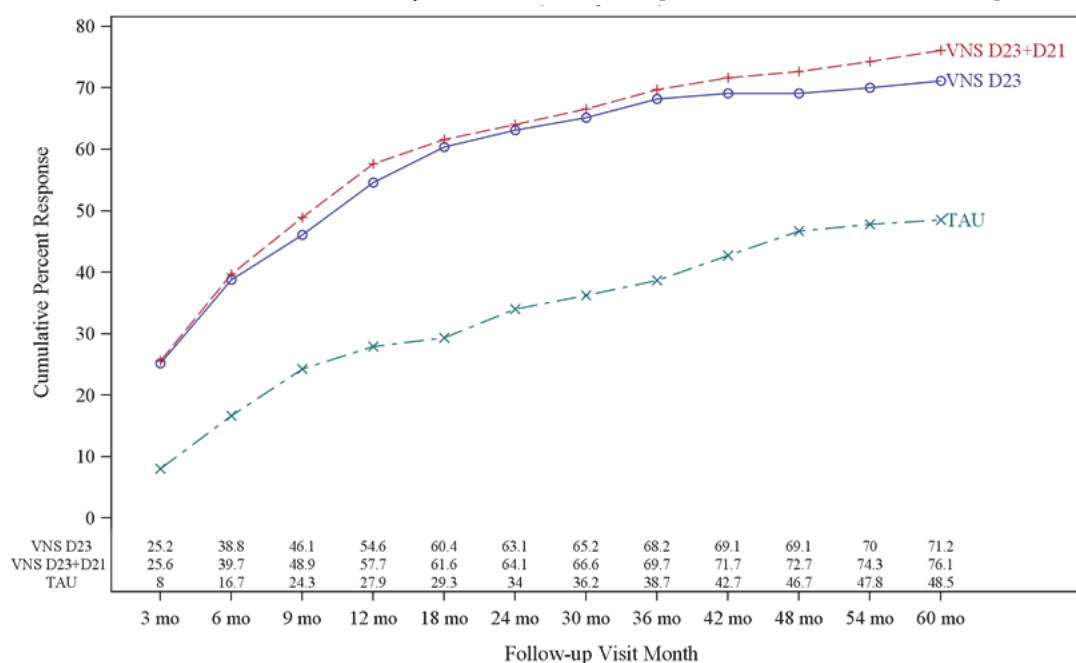


Table 72. Overall Propensity Quintiles Adjusted Response Rate Comparison for VNS D-23 + D-21, VNS D-23, and TAU Patients by Post-Baseline Visit Assessment Based on CGI-I (ITT Population)

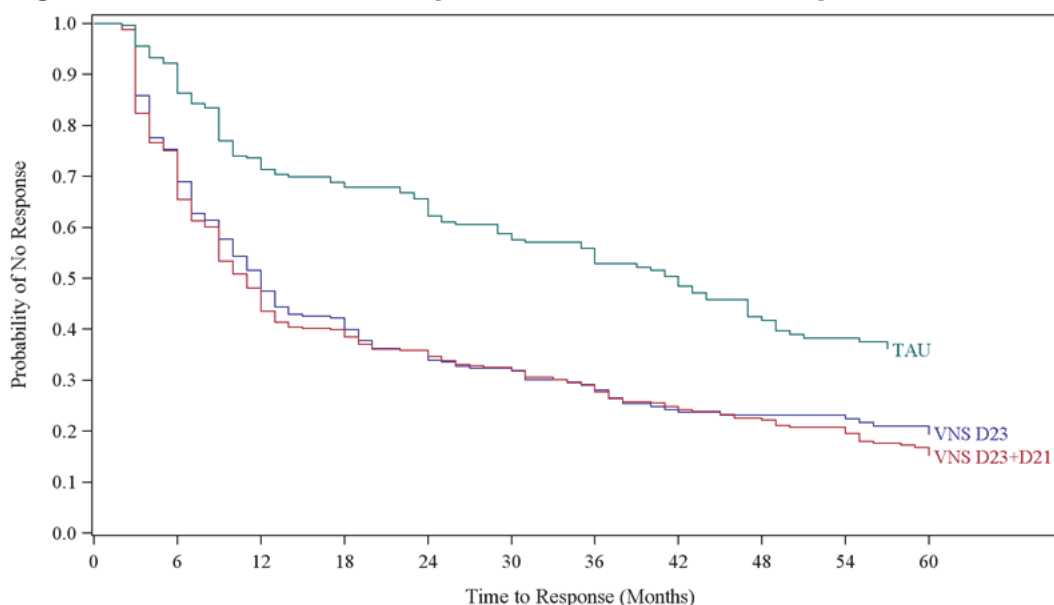
Parameter	Statistic	VNS D-23 + D21	VNS D-23	TAU
Overall CGI Response Rate	n (%)	372 (75.9)	235 (71.0)	134 (48.6)
95pct CI (Overall Response Rate)	(%,%)	(72.3, 79.9)	(66.3, 76.1)	(43.0, 54.8)
P value* (vs. TAU)		<.0001	<.0001	

*Logistic Regression, Chi Square Test

Time until First Response Based on CGI-I (ITT Population)

Time to first response based on CGI-I was computed as date the first time patient achieved CGI-I rating of 1 or 2 post-baseline (Note: Baseline date for VNS Therapy treated patients is the date of first device stimulation and Visit 2 for TAU treated patients). Figure 75 presents time to response for patients in the VNS Therapy D-23 + D-21 group versus patients in the VNS Therapy D-23 only group versus patients in the TAU group, by post-baseline visit assessment based on CGI-I scores. As noted above, patients in the VNS Therapy D-23 + D-21 group were statistically significantly more likely to experience response than patients in the TAU group ($P < .001$). Median time to first response was statistically significantly shorter for patients in the VNS Therapy D-23 + D-21 group versus patients in the TAU group (11 months for the VNS Therapy D-23 + D-21 group versus 42 months for the TAU group, $P < .001$). Similarly, patients in the VNS Therapy D-23 only group were statistically significantly more likely to experience remission than patients in the TAU group ($P < .001$). The median time to response was 12 months for patients in the VNS Therapy D-23 only group, and 42 months for patients in the TAU group.

Figure 75. Time to First Response Based on CGI (ITT Population)

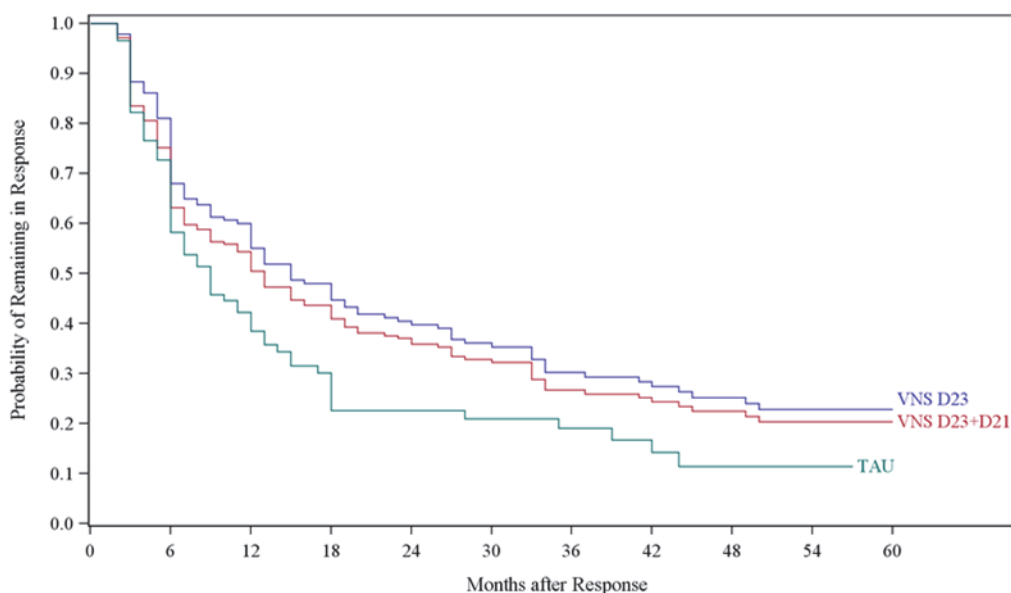


Note: P value: <.001 for D-23 + D-21(11 months) vs TAU (42 months) and <.001 for D-23 (12 months) vs TAU (42 months)

Duration of Response Based on CGI-I

Duration of response was computed as the difference between the first recorded date post-baseline that response was achieved (CGI-I rating of 1 or 2) and the first date at which CGI-I rating is greater than 2. Figure 76 presents duration of response for the VNS Therapy D-23 + D-21 group versus patients in the VNS Therapy D-23 only patients versus TAU patients based on CGI-I scores. Patients in the VNS Therapy D-23 + D-21 group had a statistically significantly longer duration of response than patients in the TAU group (13 months for the VNS Therapy D-23 + D-21 group versus 9 months for the TAU patients, $P=.031$). Patients in the VNS Therapy D-23 only group versus patients in the TAU group had a statistically significantly longer duration of response than patients in the TAU group (15 months for the VNS Therapy D-23+D-21 group versus 9 months for the TAU patients, $P=.034$).

Figure 76. Duration of Response Based on CGI (ITT Population)



Note: P value: 0.031 for D-23 + D-21(13 months) vs TAU (9 months) and P value: 0.034 for D-23 (15 months) vs TAU (9 months)

9.3.2.2.2.3. QIDS-SR

First Time Responders by Post-Baseline Visit Assessment by Treatment Group Based on QIDS-SR (ITT Population)

Response rate based on QIDS-SR was computed and summarized as the proportion (percent) of patients that achieved $\geq 50\%$ reduction from baseline in QIDS-SR total score at post-baseline visit assessment. A patient is considered a "Responder" (Yes = 1) the first time they achieved $\geq 50\%$ improvement from baseline in QIDS-SR total score at post-baseline visit assessment. A "Non-Responder" (No = 0) is any patient who did not achieve $\geq 50\%$ reduction from baseline in QIDS-SR score at post-baseline visit assessment. Statistically significant higher response rate was observed in the VNS Therapy treated population versus the TAU treated population (33.7% for the VNS: D-23 + D-21 group versus 17.0% for the TAU group; $P<.001$). Also, a statistically significant greater response rate was seen in the VNS D-23 only group than in the TAU group (33.1% versus 17.3%, respectively, $P<.001$) (Table 73).

Table 73. Mixed Model Repeated Measure (MMRM) Analysis on QIDS-SR Responders Propensity Score Quintile as a Covariate (VNS D-23 + D-21, VNS D-23, and TAU Patients - ITT Population)

Treatment Group	Least Square Mean (Std Err)	95pct CI (Lower, Upper)	P Value
TAU	17.0 (0.9)	(15.3, 18.9)	<.0001
VNS (D-23 + D-21)	33.7 (0.8)	(32.1, 35.3)	
TAU	17.3 (0.9)	(15.6, 19.1)	<.0001
VNS (D-23)	33.1 (1.0)	(31.3, 35.1)	

Cumulative proportion of subjects achieved first time response by post-baseline visit assessment based on QIDS-SR

Figure 77 presents cumulative, observed response rate by post-baseline visit assessment, and a total (cumulative) rate for 60 month data, based on QIDS-SR, a patient self-rated scale. Running a logistic regression model adjusted to propensity quintiles we find results are consistent with those based on MADRS and CGI-I, clinician-rated scales, and show a statistically significant greater response for patients in the VNS Therapy group versus patients in the TAU group (64.7% response rate for the VNS Therapy D-23 + D-21 group versus 41.7% for the TAU group, $P<.001$, and 61.0% for the VNS Therapy D-23 only group, $P<.001$ (Table 74).

Figure 77. Cumulative, Observed First Time Responder Rate by Post-Baseline Visit Assessment by Treatment Group Based on QIDS-SR (ITT Population)

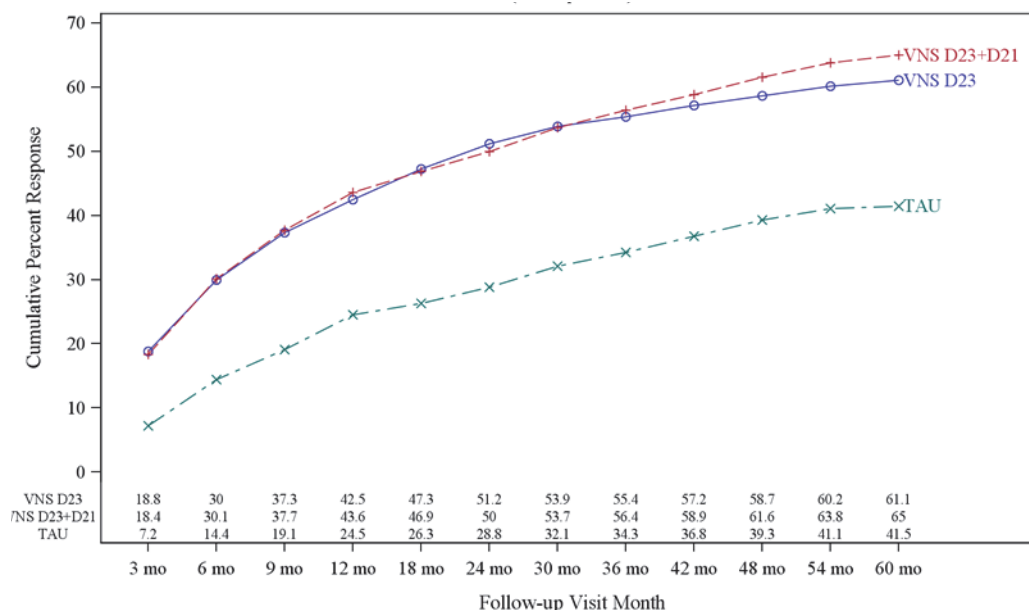


Table 74. Overall Propensity Quintiles Adjusted Response Rate Comparison for VNS D-23 + D21, VNS D-23, and TAU Patients Based on QIDS-SR (ITT Population)

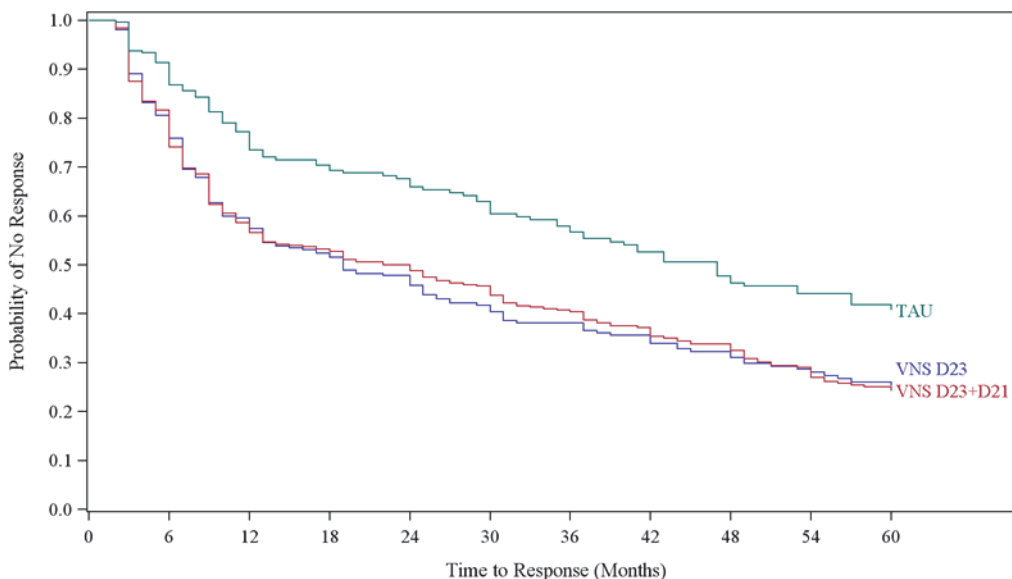
Parameter	Statistic	VNS D-23+D21	VNS D-23	TAU
Overall QIDS-SR Response Rate	n (%)	317 (64.7)	202 (61.0)	115 (41.7)
95pct CI (Overall Response Rate)	(%,%)	(60.7, 69.2)	(56.0, 66.5)	(35.9, 47.5)
P value*		<.0001	<.0001	

*Logistic Regression, Chi Square Test

Time to First Response Based on QIDS-SR (ITT Population)

Time to first response based on QIDS-SR was computed as date the first time improvement of $\geq 50\%$ from baseline in QIDS-SR total score was observed minus Baseline date. Figure 78 presents time to first response by post-baseline visit assessment based on QIDS-SR. Median time to first response was statistically significantly shorter for patients in the VNS Therapy group versus patients in the TAU group (24 months for the VNS Therapy D-23 + D-21 group versus 47 months for the TAU group, $P < .001$ and 19 months for the VNS Therapy D-23 only group versus 47 months for the TAU group, $P < .001$).

Figure 78. Time to First Response Based on QIDS-SR (ITT Population)



Note: P value: $< .001$ for D-23 + D-21 (24 months) and D-23 (19 months) vs TAU (47 months)

Duration of Response, Based on QIDS-SR (ITT Population)

Duration of response for QIDS-SR was computed as the difference between the first recorded date, post-baseline, that response was achieved (QIDS-SR $\geq 50\%$) and the first date at which QIDS-SR response decreased to $< 40\%$. Patients in the combined VNS Therapy D-23 + D-21 group had a numerically longer median duration of response for patients who responded than patients in the TAU group (10 months for the VNS Therapy D-23 + D-21 group versus 6 months for the TAU group), but this difference did not reach statistical significance ($P = .227$). Results for patients in the VNS Therapy D-23 only group versus patients in the TAU group were similar to each other.

First Time Remission by Post-Baseline Visit Assessment by Treatment Group Based on QIDS-SR (ITT Population)

Remission is a binary outcome response variable (Yes/No In-remission) defined as QIDS-SR total score ≤ 5 at post-baseline visit assessment. A statistically significant greater remission rate was observed in the VNS Therapy treated population versus the TAU treated population (16.2% for the VNS: D-23 + D-21 group versus 8.1% for the TAU group; $P < .001$). Also, a statistically significant greater remission rate was seen in the VNS D-23 only group than in the TAU group (15.4% versus 8.4%, respectively, $P < .001$) (Table 75).

Table 75. Mixed Model Repeated Measure (MMRM) Analysis on QIDS-SR Remission Propensity Score Quintile as a Covariate

Treatment Group	Least Square Mean (Std Err)	95pct CI (Lower, Upper)	P Value
TAU	8.1 (0.6)	(6.9, 9.4)	<.0001
VNS (D-23+D-21)	16.2 (0.7)	(15.0, 17.5)	
TAU	8.4 (0.6)	(7.2, 9.7)	<.0001
VNS (D-23)	15.4 (0.8)	(14.0, 16.9)	

Cumulative proportion of patients achieved first time remission by post-baseline visit assessment

Figure 79 presents cumulative, observed remission rates for patients in the VNS Therapy D-23 + D-21 group versus patients in the VNS Therapy D-23 only group versus patients in the TAU group, based on QIDS-SR scores. Running a logistic regression model adjusted to propensity quintiles we find patients in the VNS Therapy D-23 + D-21 group had a statistically significantly greater cumulative remission rate than patients in the TAU group (40.4% for the VNS Therapy D-23 + D-21 group versus 25.0% for the TAU group, $P < .001$). This was also seen in the comparison of patients in the VNS Therapy D-23 only group versus patients in the TAU group (37.5% versus 25.0%, respectively, $P < .001$) (Table 76).

Figure 79. Cumulative, Observed First Time Remission Rates by Post-Baseline Visit Assessment by Treatment Group Based on QIDS-SR (ITT Population)

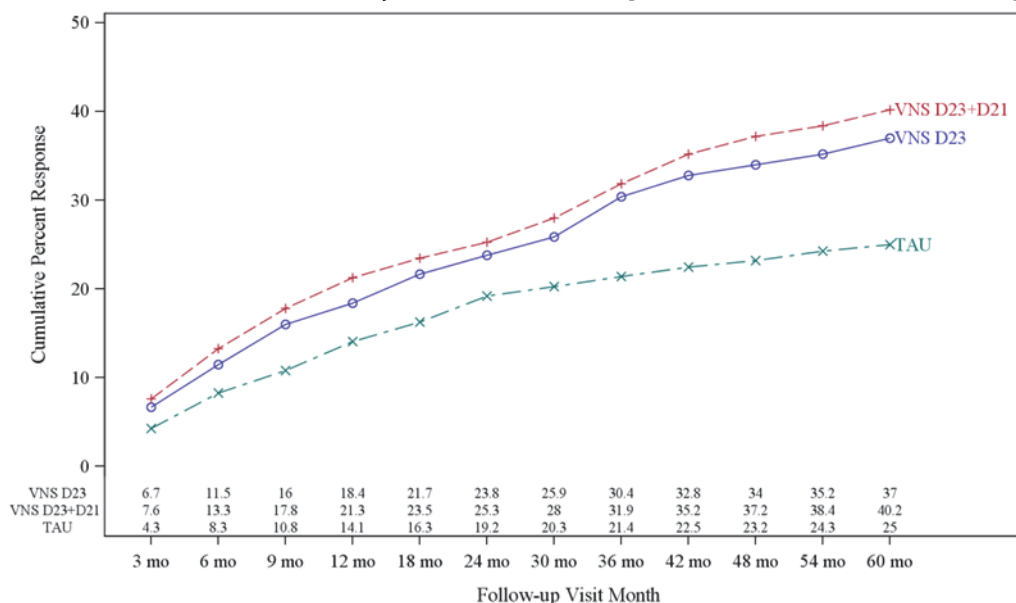


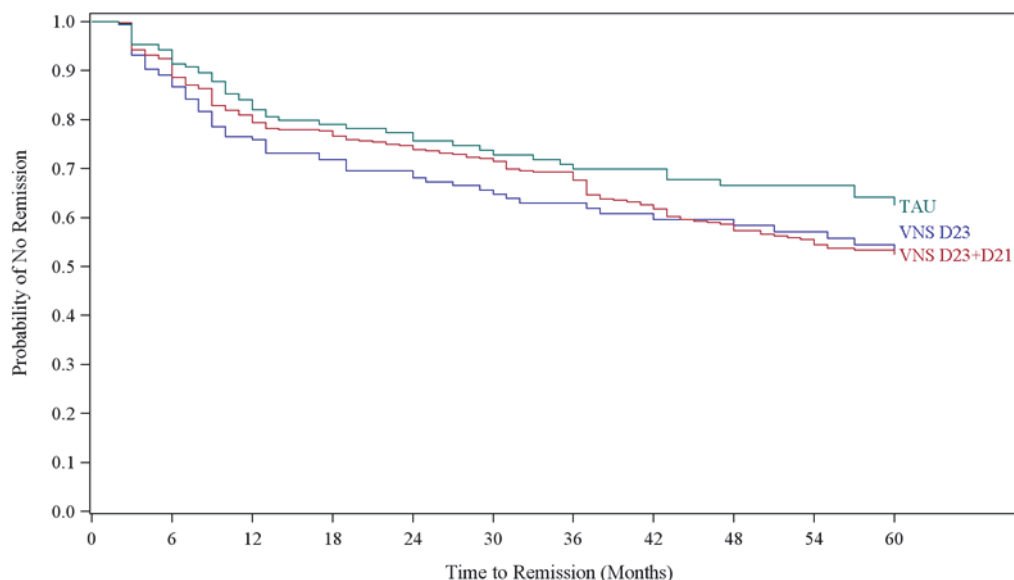
Table 76. Overall Propensity Quintiles Adjusted Remission Rate Comparison for VNS D-23 + D-21 and TAU Patients Based on QIDS-SR (ITT Population)

Parameter	Statistic	VNS D-23+D21	VNS D-23	TAU
Overall QIDS-SR Remission Rate	n (%)	198 (40.4)	124 (37.5)	69 (25.0)
95pct CI (Overall Remission Rate)	(%,%)	(36.2, 44.9)	(32.4, 42.8)	(19.9, 30.1)
P value		<.0001	<.0001	

Time to First Remission Based on QIDS-SR (ITT Population)

Time to first remission based on QIDS-SR was computed as date of first confirmed remission based on QIDS-SR. Figure 80 presents time to first remission for patients in the VNS Therapy D-23 + D-21 group versus patients in the VNS Therapy D-23 only group versus patients in the TAU group, based on QIDS-SR scores.

Figure 80. Time to First Remission Based on QIDS-SR (ITT Population)



Duration of Remission Based on QIDS-SR (ITT Population)

Duration of remission was computed as the recorded date of first recurrence/relapse minus the recorded date of first achieved remission. Remission Rate is proportion of patients that achieved a score of ≤ 5 post-baseline in QIDS-SR total score at visit month assessment post-baseline. Patients in the VNS Therapy D-23 + D-21 group had a longer median time until recurrence versus patients in the TAU group; however, this did not reach statistical significance (30 months for the VNS Therapy D-23 + D-21 group versus 18 months for the TAU group). No statistical difference was seen between patients in the VNS Therapy D-23 only group and patients in the TAU group.

9.3.2.2.2.4. SLICE-C-PSR

First Time Remission by Post-Baseline Visit Assessment by Treatment Group Based on SLICE-C-PSR (ITT Population)

Remission is a binary outcome variable (Yes/No In-remission). A patient is considered in remission (Yes=1) for current episodes if he/she has at least 8 consecutive weeks of no symptoms or minor symptoms as indicated by the SLICE-C-PSR of 1 (no symptoms) or 2 (1 or 2 symptoms to a mild degree). An MMRM method was utilized for repeated binary data with propensity quintile classification as a covariate in the model to determine if remission rates were different in the VNS Therapy treated population versus the TAU treated population during the 60 month follow-up. A statistically significant greater remission rate was observed in the VNS Therapy treated population versus the TAU treated population (15.6% for the VNS D-23 only group versus 8.1% for the TAU group; $P < .001$) (Table 77).

Table 77. Mixed Model Repeated Measure (MMRM) Analysis on SLICE-C-PSR Remission Propensity Score Quintile as a Covariate (VNS Therapy D-23+D-21 and TAU Patients - ITT Population)

Treatment Group	Least Square Mean (Std Err)	95pct CI (Lower, Upper)	P Value*
TAU	8.1 (0.6)	(6.9, 9.4)	<.0001
VNS + TAU	15.6 (0.6)	(14.5, 16.7)	

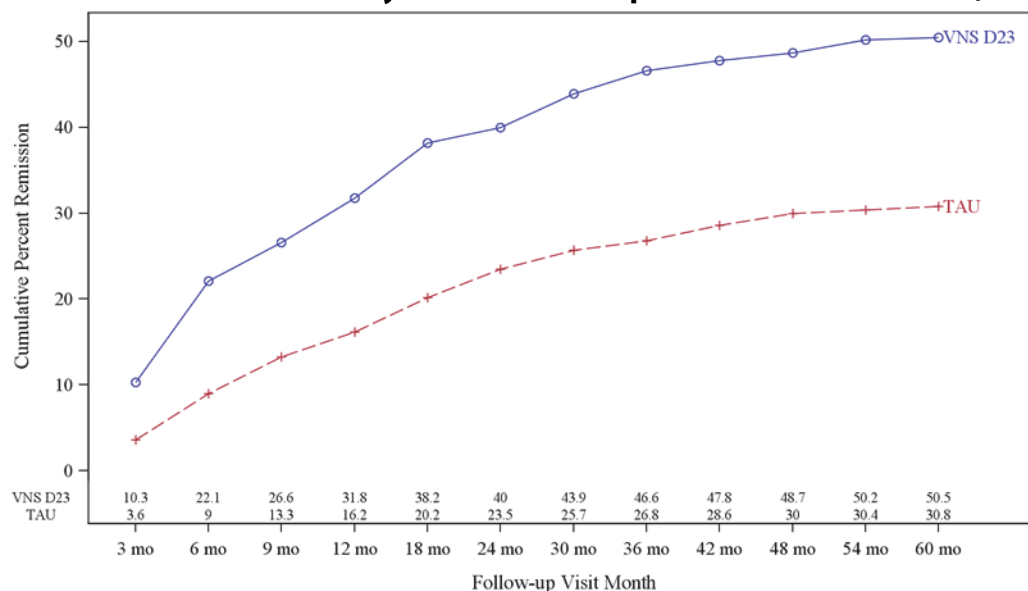
*P value generated by performing MMRM analysis with Binary distribution and logit function in PROC GLIMMIX

The VNS Therapy D-21 rollover data are not included through 18 months because SLICE was not an assessment used in the D-21 study.

Cumulative proportion of patients achieved first remission by post-baseline visit assessment.

Figure 81 presents cumulative, observed remission rates for patients in the VNS Therapy D-23 only group versus patients in the TAU group, by month post-baseline, based on SLICE-C-PSR scores. Propensity quintiles adjusted model results show patients in the VNS Therapy D-23 group had a greater cumulative remission rate than patients in the TAU group (50.5% for the VNS Therapy D-23 group versus 30.8% in TAU group, $P < .001$) (Table 78).

Figure 81. Cumulative, Observed First Time Remission by Post-Baseline Visit Assessment by Treatment Group Based on SLICE-C-PSR (ITT Population)



Note: P value: <.001 for D-23 (50.5%) vs TAU (30.8%)

Table 78. Overall Propensity Quintiles Adjusted Remission Rate Comparison for D-23 and TAU Patients Based on SLICE-C-PSR (ITT Population)

Parameter	Statistic	VNS D23	TAU	P Value
Overall SLICE-PSR Remission Rate	n (%)	167 (50.5)	85 (30.8)	<.0001
Overall Remission Rate 95pct CI	(%,%)	(45.1, 55.8)	(25.4, 36.2)	

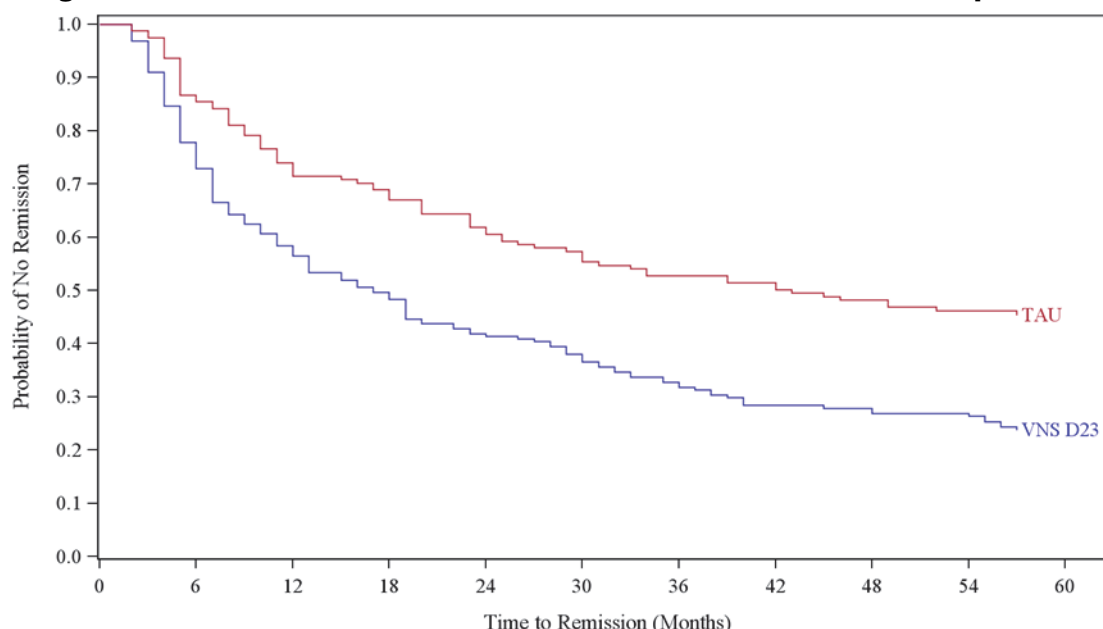
Time Until Remission Based on SLICE-C-PSR (ITT Population)

Time until Remission (TUR) was computed as date of first confirmed remission based on SLICE-C-PSR.

Figure 82 presents time to first remission for patients in the VNS Therapy D-23 only group versus patients in the TAU group, based on SLICE-C-PSR scores. Patients in the VNS Therapy D-23 group reach median remission at 17 months, which is statistically significantly sooner than patients in the TAU group, reaching at 43 months ($P < .001$).

The VNS Therapy D-21 rollover data are not included through 18 months as SLICE was not an assessment used in the D-21 study.

Figure 82. Time to First Remission Based on SLICE-C-PSR (ITT Population)



Note: P value: D-23 (17 months) vs TAU (43 months)

9.3.2.2.2.5. Quality of life outcomes

The data in this section includes analyses for the D-23 cohort only. No assessments for the VNS Therapy D-21 rollover group are included because the protocol for D-21 did not require collection of QOL outcomes assessments at Baseline.

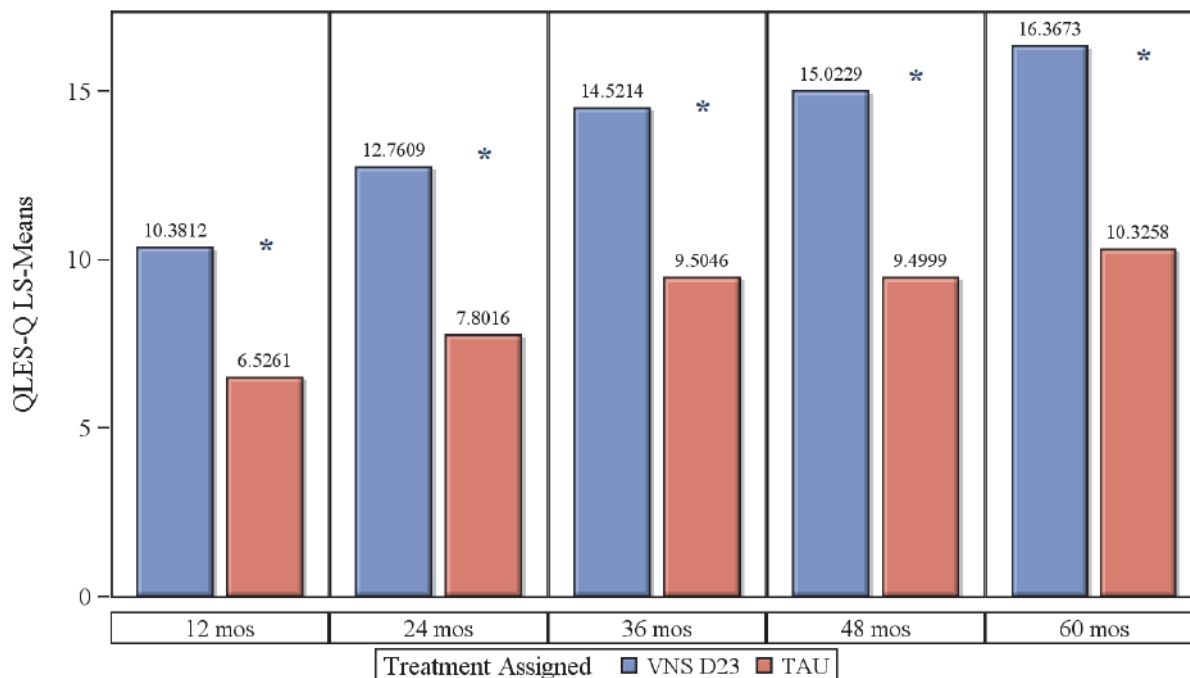
Quality of Life Enjoyment and Satisfaction Questionnaire - Short Form (Q-LES-Q SF)

The Q-LES-Q measures quality-of-life in key domains (Endicott, Nee et al. 1993). There are 2 forms of this instrument: the short form and the long form. The short form employs the 14 general activities included in the long form, as well as 2 global items. Five-point item scores (1 to 5) are aggregated, with higher scores indicative of greater enjoyment or satisfaction in each domain. The scoring of the Q-LES-Q-SF involves summing only the first 14 items to yield a raw total score. The last 2 items are not included in the total score but stand alone. The raw total score ranges from 14 (worst score) to 70 (best score).

As shown in Figure 83, statistically significant improvement in QOL as measured by least

square means was observed for patients in the VNS Therapy D-23 group versus patients in TAU group at 60 months ($P<.001$).

Figure 83. Least Square Means of Improvement from Baseline based on Quality of Life Enjoyment and Satisfaction Questionnaire (Q-LES-Q) VNS Therapy D-23 vs TAU (ITT Population)



The minimum important difference (MID) is the smallest improvement from baseline using Q-LES-Q needed for a clinically meaningful effect. The MID for anxiety was reported as 6.80 points by Wyrwich et al (Wyrwich, Harnam et al. 2011). The MID for bipolar I and II disorders was reported as 11.89 by Endicott et al (Endicott, Rajagopalan et al. 2007, Stevanovic 2011). Stevanovic et al. reported the MID for psychiatric patients presenting with a variety of diagnoses to be 8.95 (Stevanovic, 2011). No specific MID has been established for patients with TRD.

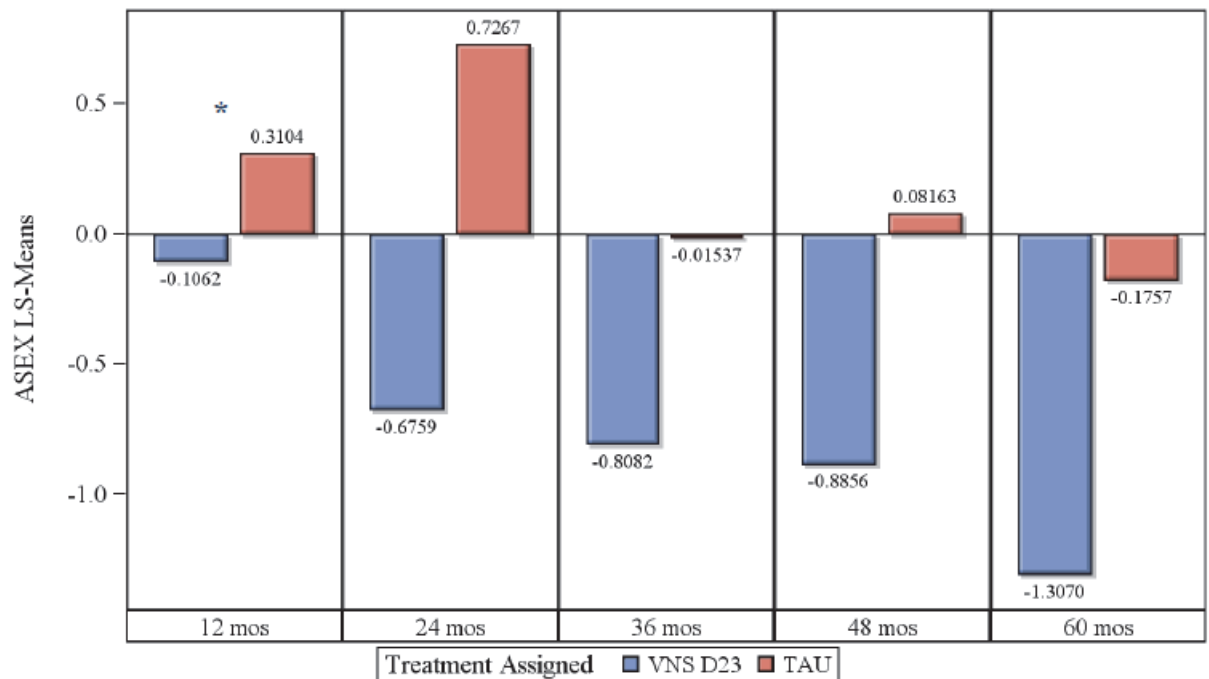
Arizona Sexual Experience (ASEX) questionnaire

The ASEX is a patient self-reported 6-point (1-6) scale used to assess 5 items that make up the core elements of sexual function: sexual drive; arousal; penile erection/vaginal lubrication; ability to reach orgasm; and satisfaction from orgasms. Scores ranged from 5 (best score) to 30 (worst score). Reduction in scores over time represents improvement in sexual function (McGahuey, Gelenberg et al. 2000).

Statistically significant improvement in sexual function was observed for patients in the VNS Therapy D-23 group as compared with patients in the TAU group ($P=.016$). Overall LS-mean change (SE) in VNS Therapy D-23 is -0.64 (0.22) and TAU is 0.22 (0.27) (Figure 84). At most time points analyzed sexual function worsened for patients in the TAU group and improved for patients in the VNS Therapy D-23 group, with the exception of 54 and 60 months where sexual function improved slightly for patients in the TAU group. This trend was similar when analyzed for women (Figure 85) and men (Figure 86) separately, although the difference between

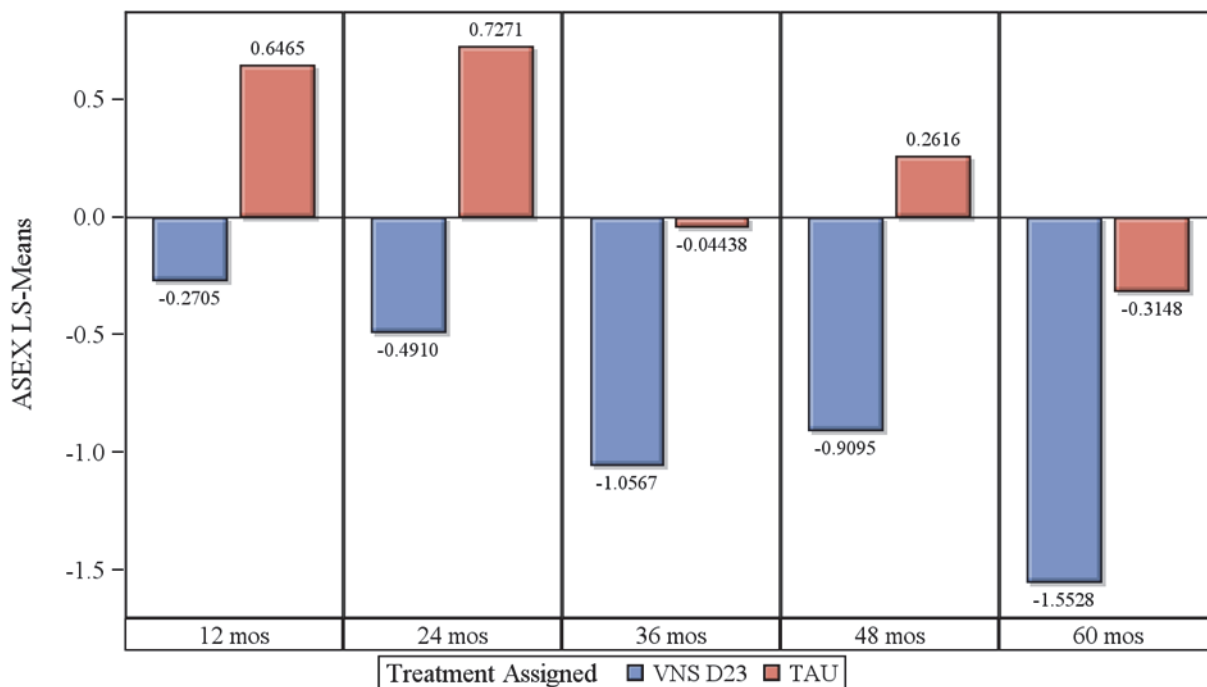
patients in the VNS Therapy D-23 group and patients in the TAU group reached statistical significance only for women ($P=.017$).

Figure 84. Least Squares Means of Improvement from Baseline based on Arizona Sexual Experience Scale (ASEX) VNS Therapy D-23 vs. TAU (ITT Population)



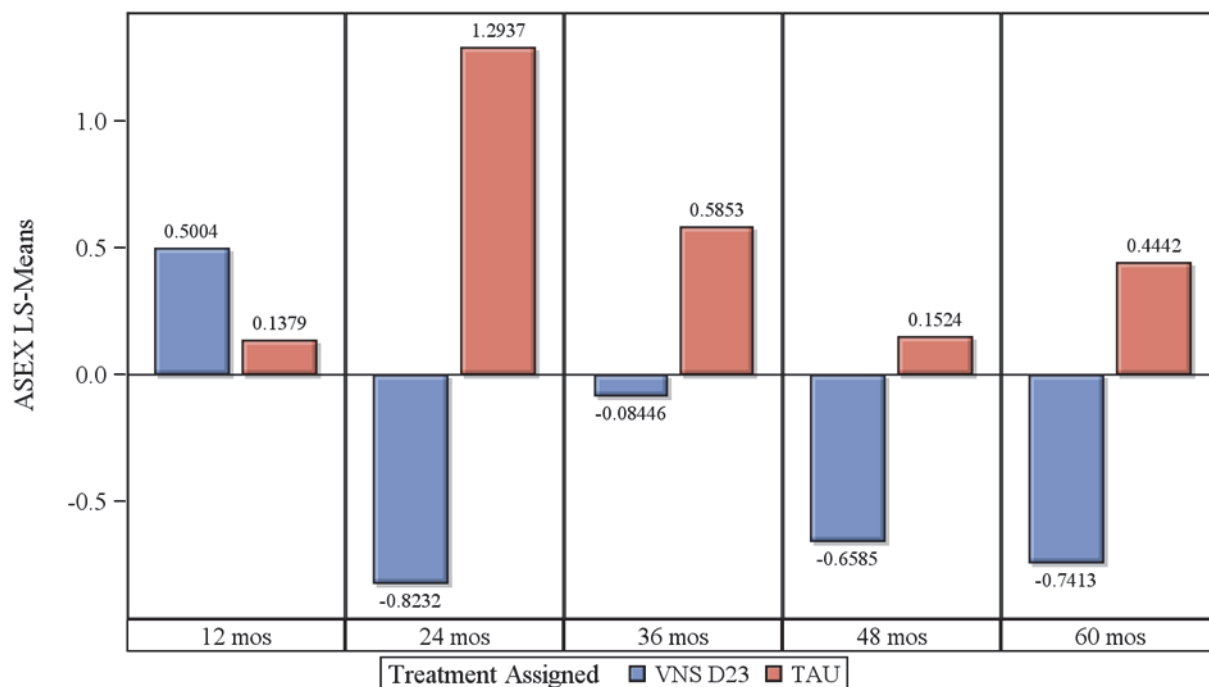
Treatment Group	Least Square Mean (Std Err)	95pct CI (Lower, Upper)	P Value
TAU	0.2243 (0.2707)	(-0.3072, 0.7557)	0.0157
VNS D-23	-0.6436 (0.2198)	(-1.0750, -0.2121)	

Figure 85. Least Squares Means of Improvement from Baseline for VNS Therapy D-23 FEMALE and TAU FEMALE Patients based on Arizona Sexual Experience Scale (ASEX) (ITT Population)



Treatment Group	Least Square Mean (Std Err)	95pct CI (Lower, Upper)	P Value
TAU	0.3781 (0.3427)	(-0.2952, 1.0513)	0.017
VNS D-23	-0.7526 (0.2908)	(-1.3240, -0.1813)	

Figure 86. Least Squares Means of Improvement from Baseline for VNS Therapy D-23 MALE and TAU MALE Patients based on Arizona Sexual Experience Scale (ASEX) (ITT Population)



9.3.2.2.3. Final effectiveness conclusions

The overall trend regarding response, remission, and duration are consistent across efficacy instruments and shows superiority in the VNS Therapy treated group compared to the TAU treated group (Table 79).

Table 79. Summary Table of Primary and Secondary Endpoints (MADRS, CGI-I, QIDS-SR, SLICE-C-PSR)

Instrument	Measure	VNS Treated	TAU	P Value
Primary Efficacy Endpoint				
MADRS	Response Rate	32.5%	13.6%	<.001
Secondary Analyses of Primary	Cumulative First Time Response Rate	67.6%	40.9%	<.001
	Time Until First Response	12 mo	48 mo	<.001
Secondary Efficacy Endpoints				
MADRS	Duration of Response	12 mo	7 mo	0.017
	Remission Rate	19.8%	8.1%	<.001
	Cumulative First Time Remission	43.3%	25.7%	<.001
	Duration of Remission	40 mo	19 mo	0.102
CGI-I	First Time Response	51.2%	21.1%	<.001
	Cumulative First Time Response Rate	75.9%	48.6%	<.001
	Time Until First Response	11 mo	42 mo	<.001
	Duration of Response	13 mo	9 mo	0.031

QIDS-SR	Response Rate	33.7%	17.0%	<.001
	Cumulative First Time Response Rate	64.7%	41.7%	<.001
	Time Until First Response	24 mo	47 mo	<.001
	Duration of Response	10 mo	6 mo	0.227
	Remission Rate	16.2%	8.1%	<.001
	Cumulative First Time Remission Rate	40.4%	25.0%	<.001
	Time Until First Remission	N/A*	N/A*	N/A*
	Duration of Remission	30 mo	18 mo	0.204
SLICE-C-PSR [†]	Remission Rate	15.6%	8.1%	<.001
	Cumulative First Time Remission Rate	50.5%	30.8%	<.001
	Time Until First Remission	17 mo	43 mo	<.001

*Estimate for Kaplan-Meier Median cannot be computed

[†]D-21 data are not included since SLICE was not an assessment used in D-2

Arizona Sexual Experience questionnaire response score (ASEX)

Statistically significant improvement in sexual function was observed for patients in the VNS Therapy D-23 group compared with the TAU patients. With the exception of the 54 and 60 month time points, sexual function worsened for patients in the TAU group and improved for patients in the VNS Therapy D-23 group for all time points analyzed. When separated by sex, the ASEX reached statistical significance for women in the VNS Therapy D-23 group versus patients in the TAU group.

Primary endpoints measured by multiple validated assessment tools - physician-rated and patient-rated, including symptomatology assessment, response rate, time to response and quality of life, consistently favored the VNS Therapy population compared to the TAU population. This finding has been consistent over the long-term management of the D-23 study.

9.3.2.3 Overall D-23 study conclusions

9.3.2.3.1. Study strengths

- The D-23 post-approval study represents the largest and longest conducted study to include a patient population with this level of severe treatment-resistant major depressive disorder treated with VNS Therapy and TAU
- The study used multiple and well established rating scales assessing both clinician ratings as well as patient ratings in a variety of domains, including depressive symptomatology, quality of life, safety and suicidality
- The study suggests consistent superiority for VNS Therapy over TAU across multiple ratings scales, both patient and clinician, and time points

9.3.2.3.2. Study weaknesses

- Non-randomized, naturalistic, observational study design
- Patients and treating clinicians were un-blinded to the treatment arms, which may introduce bias

- Inclusion of the D-21 rollover patients in the VNS + TAU arm, as the D-21 rollover patients who had a positive experience with VNS Therapy may have been more likely to participate in the registry
- Patients who have undergone a surgical procedure to treat their TRD may be more likely to report a positive impact upon their disease to justify their decision to undergo the surgery
- Larger dropout rate of patients in both arms of the study over time
- Medication changes were allowed during the study
- Only adverse events submitted via Medical Device Reporting guidelines were included in the study results
- Intended data points were not always collected

9.3.3 Bibliography

A bibliography of animal, clinical, and mechanism of action studies is available from LivaNova on request.

9.4 Patient Follow-Up

9.4.1 Guidelines for Patient Follow-Up

During the first few weeks after implantation, the patient should be seen to confirm wound healing and proper generator operation. The generator's output current for both the magnet and the programmed stimulation must be 0 mA for the first 14 days after implantation.

The VNS Therapy System is an adjunctive therapy to existing (prior to device implantation) antidepressant medications. Physicians are strongly encouraged **to keep all antidepressant medications stable for the first 3 months** of stimulation before attempting to reduce or change a patient's medication.

During initial programming, the output current should be programmed to start at nominal parameters (0 mA) and then be slowly increased in 0.25 mA increments until the patient feels the stimulation at a comfortable level. Patients who are receiving replacement generators should also be started at nominal parameters, with 0.25 mA-step increases to allow re-accommodation.

At each patient visit, the generator should be interrogated, using the appropriate version of the VNS Therapy programming software. After reprogramming and/or diagnostics testing, data should be printed out and filed. These data can be used for comparison with a patient's own records to evaluate the VNS Therapy System, to confirm proper VNS Therapy System functioning, and to assess the need for reprogramming.



Note: For instructions on printing out data, see the programming system physician's manuals.

The median output current used during the clinical studies was about 1 mA. Other standard treatment settings were 20 Hz, 500 μ sec pulse width, 30 sec ON time, and 5 min OFF time. There are no data to verify that these are optimal parameters.

There is no proven correlation at present between high output current (mAmps) and device effectiveness, nor is there a standard treatment level that needs to be achieved during

treatment ramping. VNS Therapy System treatment should not be uncomfortable, nor should it cause bothersome side effects. Patients should be observed for at least 30 minutes after the last stimulation adjustment to make certain that they are comfortable with programmed stimulation.

Although LivaNova recommends adjusting output current as necessary, there are no controlled data at this time to indicate that higher current levels are associated with better effectiveness. Patients whose depression is well controlled at follow-up should not have their settings changed unless they experience uncomfortable side effects.

The subsequent follow-up schedule and the nature of each examination should be determined by the physician on the basis of patient response to and tolerance of the implant. In all other respects, follow-up should be performed in accordance with the standard medical practice for patients with depression.

In the event intolerable adverse events are reported, physicians should always try reducing the output current (mA) as a means of eliminating or reducing the severity of an event. Additionally, physicians should instruct patients or caregivers on the application of the magnet to turn the generator off (output current 0 mA) if an adverse event becomes intolerable.

9.4.2 Individualization of Treatment

Patients should be started on stimulation at a low current output setting (0.25 mA), and the current should be increased gradually to allow accommodation to the stimulation. For patient comfort, the output current should be increased in 0.25 mA increments until reaching a tolerable level at which improvement in depression symptoms are seen. Physicians should appreciate that some patients will accommodate to stimulation levels over time and should therefore allow further increases (in 0.25 mA steps) in output current, if needed.

 **Note:** See the programming system physician's manuals.

Table 80 lists the stimulation parameters reported at 12 months of VNS Therapy in the pivotal (D-02) study.

Table 80. Stimulation Parameters at 12 Months of VNS Therapy in the Pivotal (D-02) Study

Stimulation Parameters*	Median Value at 12 Months	Range
Output current	1.0	0 to 2.25
Frequency	20 Hz	2 to 30 Hz
Pulse width	500 µsec	130 to 750 µsec
ON time	30 sec	7 to 60 sec
OFF time	5 min	0.3 to 180 min

*The magnet output current should be set to 0 mA.

Table 81 lists the stimulation parameters reported at week 50 of VNS Therapy in the Post-approval (D-21) study.

Table 81. Stimulation Parameters at Week 50 of VNS Therapy in the Post-approval (D-21) Study (Safety Population)

Stimulation Parameters*	LOW Median Value at week 50 (Range) N=97	MEDIUM Median Value at week 50 (Range) N=95	HIGH Median Value at week 50 (Range) N=103	TOTAL Median Value at week 50 (Range) N=295
Output current (mA)	1.0 (0.0-2.00)	1.25 (0.0-2.25)	1.50 (0.0-2.50)	1.25 (0.0-2.50)
Frequency (Hz)	20 (1-30)	20 (1-30)	20 (1-25)	20 (1-30)
Pulse width (µsec)	250 (130-500)	250 (130-250)	250 (130-500)	250 (130-500)
ON time (seconds)	30 (30-60)	30 (7-60)	30 (30-60)	30 (7-60)
OFF time (minutes)	5.0 (1-180)	5.0 (0-180)	5.0 (2-180)	5.0 (0-180)

*The magnet output current should be set to 0 mA.

9.4.3 Patient Counseling Information

In the event of uncomfortable adverse events, continuous stimulation, or other malfunction, the patient must be advised to hold or tape the magnet directly over the implanted generator to prevent additional stimulation. If patients or caregivers find this procedure necessary, they should immediately notify the patient's physician.

10 Epilepsy Information—Clinical Studies

10.1 Clinical Studies—Safety

 **Note:** For intended use/indications, see the *Introduction to the VNS Therapy System* chapter.

The VNS Therapy System was implanted in 454 patients in five clinical studies involving 611 devices (some patients had generator replacements). As of August 1996, total VNS Therapy exposure in these 454 patients was 901 device-years. Individual patient exposure averaged 24 months, with a range of eight days to 7.4 years.

A total of nine patients died during these five studies. One patient died from each of the following: thrombotic thrombocytopenic purpura, drowning, aspiration pneumonia, pneumonia, and renal failure associated with drug and alcohol ingestion. No cause of death was apparent for the other four deaths, which may be classified as sudden unexpected death in epilepsy (SUDEP). None of these deaths were attributed by the investigators to the VNS Therapy System.

10.1.1 Device Performance

The VNS Therapy System performed according to its specifications. Most device issues were communication difficulties resolved by repositioning the programming wand or replacing the programming wand batteries. One high lead impedance occurred requiring replacement; a lead break due to fatigue at the electrode bifurcation was noted. Most device complaints were resolved on the day of initial complaint.

10.1.2 Adverse Events Observed in Studies

Included among the five clinical trials were two randomized, blinded, active control trials (Study E03 and E05), which involved 314 patients and the implantation of 413 devices, yielding a total VNS Therapy System exposure (inclusive of long-term follow up) of 591 device years. These trials form the basis of the rates of observed adverse events.

Table 82 contains only a partial list of the more common and expected observed adverse events associated with the VNS Therapy System. A comprehensive listing of adverse events observed in studies is available by study from the Clinical Research department at LivaNova.

Table 82 reports the adverse events from these studies during the randomized phase (approximately a 14-week observation period) and randomized phase plus long-term follow up (> 3 months) through August 1996. The most common side effect associated with stimulation was hoarseness (voice alteration), which, depending on device settings, can be severe to barely perceptible. Hoarseness is reported to occur primarily during the ON period of stimulation.

Table 82. Observed Adverse Events

N=413 devices in 314 patients, 152 patients in the HIGH treatment group, 591 device years						
Randomized + Long-term Follow Up (> 3 Months) N=314 Patients, 591 Device-Years					Randomized Phase, HIGH Only, N=152 Pts	
Adverse Event (AE)	Number of Patients*	% of Patients[†]	Number of Events	Events/ Device- Year	Number of Patients	% of Patients
Serious AEs‡						
Surgically related	13	4.1	13	0.022	N/A	N/A
Stimulation related	4	1.2	4	0.007	1	0.7
Non-serious AEs						
Voice alteration	156	50	720	1.228	91	60
Increased coughing	129	41	456	0.772	57	38
Pharyngitis	84	27	182	0.308	36	24
Paresthesia	87	28	377	0.638	32	21
Dyspnea	55	18	55	0.093	32	21
Dyspepsia	36	12	98	0.166	22	15
Nausea	59	19	154	0.261	21	14
Laryngismus	10	3.2	30	0.051	9	5.9

* Number of patients reporting the event at least once.

† Percentage of patients reporting the event at least once.

‡ Included infection, nerve paralysis, hypesthesia, facial paresis, left vocal cord paralysis, left facial paralysis, left hemidiaphragm paralysis, left recurrent laryngeal nerve injury, urinary retention, and low-grade fever.

10.1.2.1 *Status epilepticus*

Valid estimates of the incidence of treatment-emergent status epilepticus among VNS Therapy System treated patients are difficult to obtain because Investigators participating in clinical trials did not all employ identical rules for identifying cases. At a minimum, two of 441 adult patients had episodes that could be described unequivocally as “status.” In addition, a number of reports were made of variably defined episodes of seizure exacerbation (for example, seizure clusters and seizure flurries).

10.1.2.2 *Rebound after stimulation was stopped*

Seizure frequency was monitored for one to four weeks after stimulation was stopped because of battery depletion in 72 instances (68 patients) in Study E03. Of these instances, 11 of the 72 (15%) **had a greater than 25 percent increase above baseline**, and 42 of the 72 (58%) had a greater than 25 percent decrease in seizure rate. The seizure rate increased by more than 1.5 standard deviations above baseline in 10 percent of instances (compared with the 7 percent expected).

10.1.3 Potential Adverse Events

Adverse events reported during clinical studies as statistically significant are listed below in alphabetical order:

- Ataxia (Loss of the ability to coordinate muscular movement)
- Dyspepsia (indigestion)
- Dyspnea (difficulty breathing, shortness of breath)
- Hypesthesia (impaired sense of touch)
- Increased coughing
- Infection
- Insomnia (inability to sleep)
- Laryngismus (throat, larynx spasms)
- Muscle movement or twitching generally associated with stimulation
- Nausea
- Pain
- Paresthesia (prickling of the skin)
- Pharyngitis (inflammation of the pharynx, throat)
- Voice alteration (hoarseness)
- Vomiting

Other potential adverse events possibly associated with surgery or stimulation include, but are not limited to, the following:

- Aspiration (fluid in the lungs)
- Blood clotting
- Choking sensation
- Damage to nerves or vasculature in the surgical area, including the carotid artery and jugular vein
- Device (generator and/or lead) migration or extrusion
- Dizziness
- Dysphagia (difficulty swallowing)
- Duodenal ulcer, gastric ulcer
- Ear pain
- Facial flushing (may be more likely in children aged 4-11 years)
- Facial paralysis, paresis
- Foreign body reaction to implants, including possible tumor formation
- Formation of fibrous tissue, pockets of fluid

- Heart rate and rhythm changes
- Hiccups
- Incision site pain
- Irritability
- Laryngeal irritation (sore, painful throat)
- Left hemidiaphragm paralysis
- Left recurrent laryngeal nerve injury
- Left vocal cord paralysis
- Low-grade fever
- Muscle pain
- Neck pain
- Nerve injury
- Painful or irregular stimulation
- Seroma
- Skin, tissue reaction
- Stomach discomfort
- Tinnitus (ringing in the ears)
- Tooth pain
- Unusual scarring at the incision site
- Urinary retention
- Vagus nerve paralysis
- Weight change / Loss of appetite (potential for increased risk in children and adolescents)
- Worsening of asthma and bronchitis



Caution: Patients who manipulate the generator and lead through the skin may damage or disconnect the lead from the generator and/or possibly cause damage to the vagus nerve.

10.2 Clinical Studies—Effectiveness

Five acute-phase clinical studies involving the VNS Therapy System have been conducted (see Table 83). These studies enrolled 537 patients, of whom 454 were implanted with the VNS Therapy System. A total of 611 devices were implanted, and patient exposure totaled 901 device-years, with an individual mean patient exposure of 24 months (ranging from eight days to 7.4 years). A total of 45 centers participated in these studies: 40 in the United States, 2 in Germany, and 1 each in Canada, Holland, and Sweden.

Note: References to “partial onset seizures” have been changed to “focal onset seizures” to align with updated official seizure type terminology; however, the functional definition of the condition remains the same.

Table 83. Description of Clinical Studies

All patients enrolled in all clinical studies, N=537						
Description of Clinical Studies						
	Longitudinal			Parallel		
Study	E01	E02	E04	E03	E05	Total
Type of study	pilot longitudinal	pilot longitudinal	open longitudinal	randomized parallel high/low	randomized parallel, high/low	-
Number of patients enrolled	11	5	133	126	262	537
Number of centers*	3	2	24	17	20	45
Reference period (baseline)	weeks 2 to 4	weeks 3 to 6	weeks -4 to 0	weeks -12 to 0	weeks -12 to 0	-
Seizure type	focal	focal	all types	focal	focal	-
Number of AEDs	1 to 2	1 to 2	not specified	0 to 3	1 to 3	-

* Total includes non-U.S. centers (Canada, Holland, Germany-2, and Sweden); several U.S. centers participated in more than one study.

10.2.1 Purpose

The purpose of the studies was to determine whether adjunctive use of optimal stimulation of the left vagus nerve could reduce seizure frequency in patients with refractory seizures.

10.2.2 Methods

In the two randomized, blinded, active control trials (E03 and E05), patients were randomly assigned to either of two treatment groups: HIGH (believed to be therapeutic) or LOW (believed to be less therapeutic). Patients enrolled in the study were seen every four weeks during the baseline period (weeks -12 to 0). Patients meeting eligibility were implanted with the generator and lead (see Table 84).

Two weeks after implantation, patients were randomized to the HIGH or LOW stimulation group, and the generator was activated. Patients in the HIGH groups received a higher frequency, greater pulse width, and higher duty cycle of stimulation. The randomized treatment period that followed activation of the generator lasted 14 weeks (the last 12 weeks of which were used in the efficacy analysis—the first two weeks for a treatment ramp-up period).

Table 84. Description of Patients

All patients implanted in all clinical studies, N=454						
Description of Patients						
	Longitudinal			Parallel		
Study	E01	E02	E04	E03	E05	Total
Number of patients implanted	11	5	124	115	199	454
Number of patients stimulated	10	5	123	115	198	451
Age in years (range)	32 (20–58)	33 (18–42)	24 (3–63)	33 (13–57)	33 (13–60)	32 (3–63)
Number of females (%)	4 (36%)	2 (40%)	57 (46%)	43 (37%)	104 (52%)	210 (46%)
Years with epilepsy (range)	22 (13–32)	20 (5–36)	17 (0.8–48)	21 (4–47)	23 (2–52)	21 (0.8–52)
Average number of AEDs	1.0	1.0	2.2	2.1	2.1	2.1
Median number of seizures per day at baseline	0.6	0.42	0.65	0.70 high/ 0.85 low	0.58 high/ 0.51 low	-

10.2.3

Results

The primary efficacy endpoint (percent reduction in seizure rate) was measured over 12 weeks (see Table 85). Adverse events were assessed at each patient visit.

Table 85. Principal Efficacy and Safety Results

All patients in efficacy analyses in all clinical studies, N=441						
Principal Efficacy Results						
	Longitudinal			Parallel		
Study	E01	E02	E04	E03	E05	Total
Number of patients in efficacy analysis	10	5	116	114	196	441
Median reduction in seizures/day	32%*	48%	22%*	23% high*/6% low	23% high [†] /21% low [†]	-
Mean reduction in seizures/day	24% [‡]	40%	7% [‡]	24% high [‡] /6% low	28% high [‡] /15% low [‡]	-
Difference in mean (high/low)	-	-	-	17% [§] (3%/31%)	13% (2%/23%)	-
% with > 50% response	30%	50%	29%	30% high/14% low	23% high/16% low	-
Principal Safety Results Through Long-term Follow Up						
Exposure (pt-yr)	45	20	245	456	135	901
SAEs [¶]	9%/-	0%/-	6%/-	5%/0%	7%/9%	-
Discontinued (LOE/AE) [#]	0/1	0/0	2/3	0/2	1/3	3/9
Number of explants ^{**}	2	2	15	9	5	33
Deaths SUDEP/total ^{††}	0/0	0/0	3/4	0/3	1/2	4/9

Within group broad analyses:

* P ≤0.05, by Wilcoxon sign rank.

† P <0.0001, by anova.

‡ P ≤0.05, by Student's t-test.

Between group broad analyses:

§ P ≤0.02, by Wilcoxon rank sum; P ≤0.02, by Student's t-test.

|| P <0.04, by aligned ranks test; P <0.02, by Student's t-test; P <0.03, by anova.

Safety information:

¶ SAEs = serious adverse events.

Discontinuing for lack of efficacy (LOE)/adverse events (AE) at one year, excluding deaths.

** Number of explants through August 1996, excluding deaths.

†† All deaths occurred by the long-term follow-up closing date of August 1996.

Figure 87 and Table 86, which follow, show the results from Study E05, the largest and most recent of the randomized, blinded, active control studies:

Figure 87. Change in Seizure Frequency, Patient Distribution

(With Corresponding Table)
All E05 patients completing effectiveness evaluation, N=196

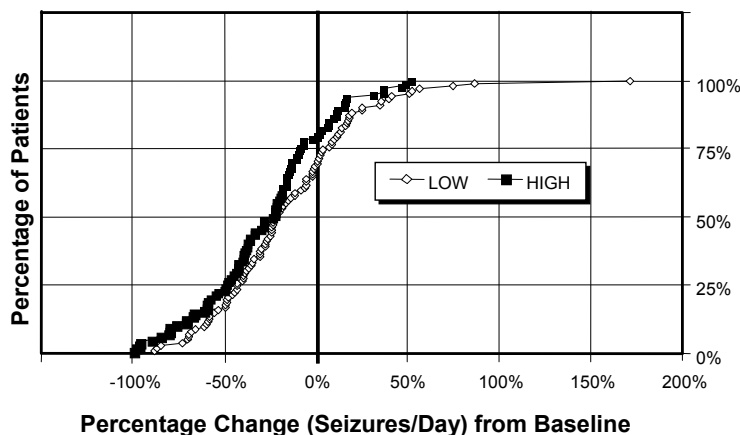


Table 86. Principal Effectiveness Statistics (E05)

All patients in E05 effectiveness analyses, N=196			
Percentage Change (Seizures/Day) from Baseline			
Statistics	High (94)	Low (102)	Difference
Median	-23%	-21%	N/A
25%, 75% Quartiles	-8.9%, -49%	4.0%, -43%	N/A
95% Confidence intervals	-35%, -21%	-23%, -7.7%	-23%, -2.3%
Range (min, max) Mean $\pm$ SD	-100%, 52%	-89%, 171%	-23%, -2.3%
	-28% $\pm$ 34%	-15% $\pm$ 39%	-13%* $\pm$ 37%

* Difference is statistically significant ($P < 0.05$) by analysis of variance ($P = 0.032$) and by Cochran-Mantel-Haenszel aligned ranks ($P = 0.040$).

Patient response to VNS Therapy was examined using statistical modeling (examining group characteristics) and an evaluation of individual patients. No useful predictors were found of an increase or a decrease in seizure frequency.

10.2.4

Conclusions

Patients with refractory focal onset seizures treated with HIGH VNS Therapy had a statistically significant decrease in frequency of seizures, compared with the baseline and compared with patients treated with LOW (active control) VNS Therapy. As indicated in Figure 87, most patients had a reduction in seizure frequency; some, however, had either no change or an increase in seizure frequency. The most common treatment-related adverse events were voice alteration and dyspnea. Treatment was well tolerated, with 97 percent (306 of 314) of the implanted patients continuing into the long-term follow-up phase of the study.

10.2.5

Long-term Data from Uncontrolled Follow Up

Long-term data (> 3 months' stimulation) were collected on all available E01 through E04 study patients (see Table 87). At the time the VNS Therapy System Premarket Approval

Application was considered by the U. S. Food and Drug Administration, long-term data on most Study E05 patients were not available. These long-term follow-up data are uncontrolled because they come from an open-label protocol in which both the antiepileptic drug medications and the VNS Therapy device settings were allowed to be changed.

Ninety-five percent (95%) of patients were continuing one year after their original implant; 82 percent were still receiving stimulation at two years; and 69 percent were receiving stimulation at three years. Some E04 patients had not yet had the opportunity to reach two or three years of stimulation and therefore were not used in the calculations. Additionally, 28 E03 patients were implanted outside the United States in countries that later received commercial approval, and data were available through one year of stimulation only.

Table 87. Patient Summary Chart

Patients continuing treatment as of 8/22/96					
Study	E01	E02	E03	E04	Total
No. of patients randomized/ stimulated	10	5	115	123	253
No. of patients entering long- term phase	10	5	113	123	251
No. of continuing patients being treated for up to 1 year/ No. started	10/10	5/5	111/115	112/121*	238/251
No. of continuing patients being treated for up to 2 years/ No. started	9/10	4/5	71/87†	58‡/70	142/172
No. of continuing patients being treated for up to 3 years/ No. started	7/10	3/5	57/87	21§/24	88/126

* Two E04 Study patients had not been implanted long enough to reach the one-year date after implantation.

† Twenty-eight (N=28) commercial European patients were excluded from follow up after one year of treatment because of the commercial release of the VNS Therapy System in those countries.

‡ As of 8/22/96, only 70 patients had been implanted long enough to reach the two-year treatment period; 58 of the 70 were continuing.

§ As of 8/22/96, only 24 patients had been implanted long enough to reach the three-year treatment period; 21 of the 24 were continuing.

Table 88 shows the number of patients included in the efficacy analysis. It is apparent from the table that not all continuing patients were used in the efficacy analysis. This difference was mostly because of missing data (some patients kept only sporadic records over the long term), although two patients were not used because they had lobectomy surgery, which affected their seizure rates.

Table 88. Patients Used for Efficacy Analysis

Study	E01	E02	E03	E04	Total
No. of patients randomized/stimulated	10	5	115	123	253
No. of patients entering long-term phase	10	5	113	123	251
No. of patients used in 1-year efficacy analysis/No. stimulated	10/10	4/5	102/111	86/112	202/238
No. of patients used in 2-year efficacy analysis/No. stimulated	8/9	2/4	51/71*	34/58†	95/142
No. of patients used in 3-year efficacy analysis/No. stimulated	4/7	2/3	49/57	0‡	55/67

* Of the 71 patients continuing, efficacy data were available for only 51.

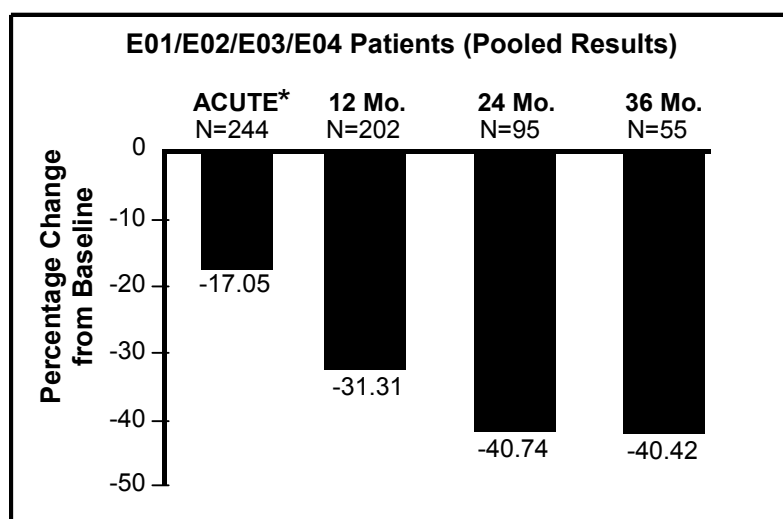
† Of the 58 patients, efficacy data were available for only 34.

‡ No data were available at the three-year time for the E04 patients.

10.2.5.1 Long-term results

Available long-term data from uncontrolled, open-label protocols during which antiepileptic drug and VNS Therapy device setting changes were allowed suggest improved efficacy over the first 24 months of treatment, with stabilization of this improvement after two years (see Figure 88). As evident from Table 88, these long-term data are limited at years two and three, with no patients being represented in the three-year analysis from Studies E04 or E05. There can be no assurances that the efficacy of the VNS Therapy treatment will continue to improve or will not decline over time, nor can there be assurances that additional long-term data will not reveal new adverse information presently unknown to LivaNova. However, currently available long-term data do not suggest an increase or a worsening of adverse events, or a decline in efficacy.

Figure 88. Median Percentage Change in Seizure Frequency



* The acute phase results include seizure frequencies of the E03 Study LOW stimulation group, which included one-half the E03 patients, N=57. Patients were permitted to change their AEDs during these long-term follow-up studies, and these changes may have contributed to the change in seizure frequency.

10.2.5.2 **Other information**

Unlike the two randomized studies, Study E04, an open-label safety study, included patients 12 years old and younger, and patients with generalized seizures. Sixteen patients under age 12, ranging from 3.6 to 12 years old, were evaluated. (Two additional patients had unevaluable seizure data.) These patients were found to have a 17.9 percent median decrease in seizures during the acute phase, with 31 percent of the patients experiencing a greater than 50 percent decrease.

Additionally, 25 patients with generalized seizures were evaluated. (Two additional patients had unevaluable seizure data.) These patients were found to have a 46.6 percent median decrease in seizures during the acute phase, with 44 percent experiencing a greater than 50 percent decrease. The E04 results (N=116 analyzed), including patients younger than 12 and those with generalized seizures, showed a 22 percent median decrease during the acute phase, with 29 percent of the patients experiencing a greater than 50 percent decrease.

The E04 results (N=86 analyzed), excluding patients younger than 12 and those with generalized seizures, showed an 18.3 percent median decrease in seizures during the acute phase, with 27.9 percent of the patients experiencing a greater than 50 percent decrease.

10.2.5.3 **Mechanism of action**

The precise mechanism(s) by which the VNS Therapy System exerts its anticonvulsant action is unknown. In animal models designed to examine anticonvulsant activity, vagus nerve stimulation prevented seizures or seizure spread in these models: maximum electroshock (MES), pentylenetetrazol (PTZ) tests, 3-mercaptopropionic acid (3-MPA), alumina gel, potassium penicillin, strychnine, and kindling. With the exception of the alumina gel model, vagus nerve stimulation did affect the heart and respiratory rate, which may have contributed to the alteration in seizure activity.

Localization of vagus-initiated activity in the brain has been observed through animal studies of *fos*¹ immunoreactivity, regional brain glucose metabolism, and positron emission tomography (PET) imaging in human patients.

An [¹⁵O] H₂O PET study in 10 patients demonstrated that vagus nerve stimulation by the VNS Therapy System does increase blood flow in the rostral medulla, right thalamus, and right anterior parietal cortex, and bilaterally in the hypothalamus, anterior insula, and inferior cerebellum. Decreases in blood flow were detected bilaterally in the hippocampus, amygdala, and posterior cingulate gyrus.

10.3 **Safety and Effectiveness in Pediatric Patients**

Pre-market clinical and post-market surveillance data were used to extrapolate the safety and effectiveness of VNS Therapy from adolescents and adults (≥ 12 years) to children (4-11 years) to support the approval of VNS Therapy in patients 4-11 years of age.

10.3.1 **Extrapolation Methods**

Extrapolation is the leveraging process whereby an indication for use of a device in a pediatric population can be supported by existing clinical data from a studied population. Extrapolation

¹ *Fos* is a nuclear protein that is expressed under conditions of high neuronal activity.

of the safety and effectiveness of VNS Therapy for patients 4-11 years of age was conducted using a Bayesian hierarchical model to estimate the 50% responder rate in patients 4-11 years of age following 12 months of treatment utilizing existing clinical data across several data sources. Data from a prospective, open label, post-approval study of all consecutive patients treated with VNS Therapy in Japan was the primary data source. Historical data from E03, E04, E05, and E06 were used as prior data in the Bayesian model. The Bayesian hierarchical model borrows data from prior studies to the degree the current study data matches the previous studies' data.

10.3.2 Data Sources

The following data sources were included in the analysis:

- Pre-market data from the E03, E04, and E05 clinical trials
- E06: Randomized, parallel group, comparative study to compare the efficacy of VNS Therapy to antiepileptic drug (AED) treatment in reducing the frequency of seizures in children (age 17 or less). The study was initiated in October 2004 and completed in January 2010.
- Japan Post-Approval Study (PAS): Prospective, open label, post-approval study of all consecutive patients treated with VNS Therapy in Japan. Patients were implanted between July 2010 and December 2012.
- LivaNova Post-Market Surveillance Database: Passively reported adverse events and device tracking data from patients implanted with the VNS Therapy System from November 1988 to September 2015. When assessing device relatedness, post-market data was restricted to reports starting in November 2006 when the post-market coding system was updated to include device relatedness.

10.3.3 Data Sets Analyzed

The safety population included all patients 4 years of age and older who underwent implantation with the VNS Therapy System who:

- participated in the E03, E04, E05, or E06 clinical trial,
- participated in the Japanese PAS (initial implants only), or
- had a record in the LivaNova Post-Market Surveillance database.

Clinical study data from 847 patients were included in the safety population. Of these, 13.8% (n=117) of patients were 4-11 years of age, 23.5% (n=199) of patients were 12-21 years of age, and 62.7% (n=531) were > 21 years of age. Post-market surveillance data with information on device relatedness for adverse events were available from 40,926 patients. Of these 18.9% of patients were 4-11 years (n=7,729), 22.9% (n=9,389) of patients were 12-21 years of age, and 58.2% (n=23,808) were > 21 years of age.

Patients in the efficacy population included all patients in the safety population with refractory focal onset seizures who had at least 1 seizure recorded at baseline. Patients who were only in the post-market surveillance database were excluded from the efficacy analysis. In total, clinical study data from 663 patients were included in the efficacy analysis. Of these, 582 patients had 12-month efficacy outcome data (n=54 patients 4-11 years, n=126 patients 12-21 years, and n=402 patients > 21 years of age).

Pediatric patients under 12 years of age participated in the E04, E06, and Japan PAS. The E03 and E05 studies consisted of patients > 12 years of age. Baseline characteristics by age group are reported in Table 89. Both groups had similar rates of prior brain or epilepsy surgery (35.0% of patients 4-11; 34.3% of patients ≥ 12), however pediatric patients 4-11 years of age in the safety population were more likely to have only generalized seizures (49.6% of patients 4-11; 16.2% of patients ≥ 12 years). Note that the efficacy evaluation is limited to patients with focal onset seizures.

Table 89. Demographics (Safety Population)

	4-11 Years	≥ 12 Years	Overall
Gender [n=(%)]			
N	117	730	847
% Female	45.3% (53/117)	44.1% (322/730)	44.3% (375/847)
Age (years)			
N	117	730	847
Average ± SD (Range)	8.4 ± 2.2 (4.0-11.9)	30.6 ± 11.5 (12.0-73.0)	27.6 ± 13.2 (4.0-73.0)
Age at Epilepsy Onset (years)			
N	117	719	836
Average ± SD (Range)	1.7 ± 1.9 (0.0-7.8)	10.5 ± 10.6 (0.0-63.7)	9.2 ± 10.4 (0.0-63.7)
Time to VNS Since Diagnosis (years)			
N	117	719	836
Average ± SD (Range)	6.7 ± 2.5 (1.2-11.6)	20.1 ± 10.6 (0.1-61.0)	18.2 ± 10.9 (0.1-61.0)
Prior Brain or Epilepsy Surgery			
N	117	727	844
% Prior Surgery	35.0% (41/117)	34.3% (249/727)	34.4% (290/844)
Baseline Seizure Type			
N	117	730	847
Focal Onset	49.6% (58/117)	83.7% (611/730)	79.0% (669/847)
Generalized Only	49.6% (58/117)	16.2% (118/730)	20.8% (176/847)
Zero Baseline Seizures	0.9% (1/117)	0.1% (1/730)	0.2% (2/847)

10.3.4 Primary Endpoints

10.3.4.1 Primary Safety Endpoint

The primary safety endpoint was the incidence rate of device-related treatment emergent adverse events through 12 months of treatment. Adverse event rates for patients 4-11 years were compared to that of patients 12-21 years (comparable with respect to physiological development) via a 95% confidence interval for the incidence rate ratio. Adverse events with statistically significant incidence rate ratios greater than 1 indicate that the incidence rate for

patients 4-11 years of age is greater than the incidence rate for patients 12-21 years of age. This analysis was performed separately for the shorter-term clinical study data and the longer term, larger post-market surveillance data to assess consistency of results.

10.3.4.2 Primary Efficacy Endpoint

The primary effectiveness endpoint is the proportion of patients 4-11 years of age in the Japan PAS with at least a 50% reduction in the frequency of seizures following 12 months of treatment. The pre-established efficacy threshold was set to a 30% responder rate with a corresponding 10% non-inferiority (NI) margin. A Bayesian hierarchical model was used to model the 12-month rates for each study.

10.3.5 Safety

Based on the clinical data, the overall incidence rate of device related treatment emergent adverse events was not different for patients 4-11 years of age compared to patients 12-21 years of age (Incidence Rate Ratio (IRR): 0.44, 95% CI: 0.20, 1.04). There were no device-related treatment emergent adverse events that had a statistically higher incidence rate in the 4-11 year age group when compared to 12-21 year age group. Two adverse events, myalgia and paresthesia, had statistically lower incidence rates in the 4-11 age group when compared to the 12-21 age group (Table 90).

Table 90. Primary Safety Analysis (Clinical Data)

Device-Related, Treatment-Emergent Adverse Events by Age based on Clinical Data (overall and statistically significant differences)					
Adverse Event	4-11 years (N=117 patients, 113 person years)		12-21 years (N=199 patients, 194 person years)		Incidence Rate Ratio* (IRR) (95% CI)
	Number of AE Reports	Incidence Rate/PY (95% CI)	Number of AE Reports	Incidence Rate/PY (95% CI)	
Overall Rate	75	66.6% (32.2%-145%)	293	151% (109%-207%)	0.44 (0.20-1.04)
Statistically Significant Difference in Incidence Rates (IRR <1)					
Myalgia	1	0.9% (0.0%-5.4%)	11	5.7% (0.0%-33.8%)	0.16 (0.00-0.90)
Paresthesia	1	0.9% (0.0%-5.1%)	23	11.9% (5.8%-22.1%)	0.07 (0.00-0.79)

Based on post-market surveillance data, the overall incidence rate of device related treatment emergent adverse events was lower for patients 4-11 years of age compared to patients 12-21 years of age (IRR: 0.82, 95% CI: 0.77, 0.88) (Table 91). Infection and extrusion of lead had a statistically greater incidence rate in patients 4-11 years of age. Younger patients may have a greater risk for wound infection when compared to adolescent and adult patients; therefore, the importance of monitoring for site infection as well as the avoidance of manipulation of the surgical site post implant in children should be stressed.

Table 91. Primary Safety Analysis (Post-Market Data)

Device-Related, Treatment-Emergent Adverse Events by Age based on Post-Market Data (overall and statistically significant differences)							
Adverse Event	4-11 years (N=7,729 patients, 31,220 person years)			12-21 years (N=9,389 patients, 37,647 person years)			Incidence Rate Ratio (95% CI)
	Number of AE Reports	Incidence Rate / Person Year (95% CI)	% of Total Reports	Number of AE Reports	Incidence Rate / Person Year (95% CI)	% of Total Reports	
Overall	1328	4.25% (4.03%, 4.49%)	100%	1948	5.17% (4.95%, 5.41%)	100%	0.82 (0.77, 0.88)
Statistically Significant Difference in Incidence Rates (IRR > 1)							
Infection	85	0.27% (0.22%, 0.34%)	6.40%	67	0.18% (0.14%, 0.23%)	3.44%	1.53 (1.11, 2.11)
Extrusion of Lead	15	0.05% (0.03%, 0.08%)	1.13%	5	0.01% (0.00%, 0.03%)	0.26%	3.62 (1.31, 9.95)
Statistically Significant Difference in Incidence Rates (IRR < 1)							
Painful Stimulation	83	0.27% (0.21%, 0.33%)	6.25%	200	0.53% (0.46%, 0.61%)	10.27%	0.50 (0.39, 0.65)
Pain	60	0.19% (0.15%, 0.25%)	4.52%	150	0.40% (0.34%, 0.47%)	7.70%	0.48 (0.36, 0.65)
Voice Alteration	66	0.21% (0.16%, 0.27%)	4.97%	122	0.32% (0.27%, 0.39%)	6.26%	0.65 (0.48, 0.88)
Stimulation Not Perceived	37	0.12% (0.08%, 0.16%)	2.79%	99	0.26% (0.21%, 0.32%)	5.08%	0.45 (0.31, 0.66)
Coughing	47	0.15% (0.11%, 0.20%)	3.54%	88	0.23% (0.19%, 0.29%)	4.52%	0.64 (0.45, 0.92)
Migration of Generator	12	0.04% (0.02%, 0.07%)	0.90%	48	0.13% (0.09%, 0.17%)	2.46%	0.30 (0.16, 0.57)
Dysphagia	14	0.04% (0.02%, 0.08%)	1.05%	40	0.11% (0.08%, 0.14%)	2.05%	0.42 (0.23, 0.78)
Cognitive Changes	16	0.05% (0.03%, 0.08%)	1.20%	35	0.09% (0.06%, 0.13%)	1.80%	0.55 (0.31, 1.00)
Erratic Stimulation Perceived	4	0.01% (0.00%, 0.03%)	0.30%	15	0.04% (0.02%, 0.07%)	0.77%	0.32 (0.11, 0.97)
Continuous Stimulation Perceived	3	0.01% (0.00%, 0.03%)	0.23%	13	0.03% (0.02%, 0.06%)	0.67%	0.28 (0.08, 0.98)
Syncopal	1	0.00% (0.00%, 0.02%)	0.08%	11	0.03% (0.01%, 0.05%)	0.56%	0.11 (0.01, 0.85)

10.3.6

Efficacy

Baseline seizure frequency for patients in the ITT population with 12-month efficacy data by age group is reported in Table 92. The median baseline seizure rate per month across all studies is 21.3 seizures/month. The median baseline seizure rate for patients 4-11 years of age (85.7 seizures/month) was higher when compared to adolescent and adult patients (30.4 and 17.4 seizures/month, respectively). Due to the differences in baseline seizure frequency rates by age group, this variable was evaluated as potential covariate in the hierarchical model;

however, there was no evidence that baseline seizure frequency is associated with the probability of being a responder (% seizure reduction $\geq 50\%$).

Table 92. Baseline Seizure Frequency

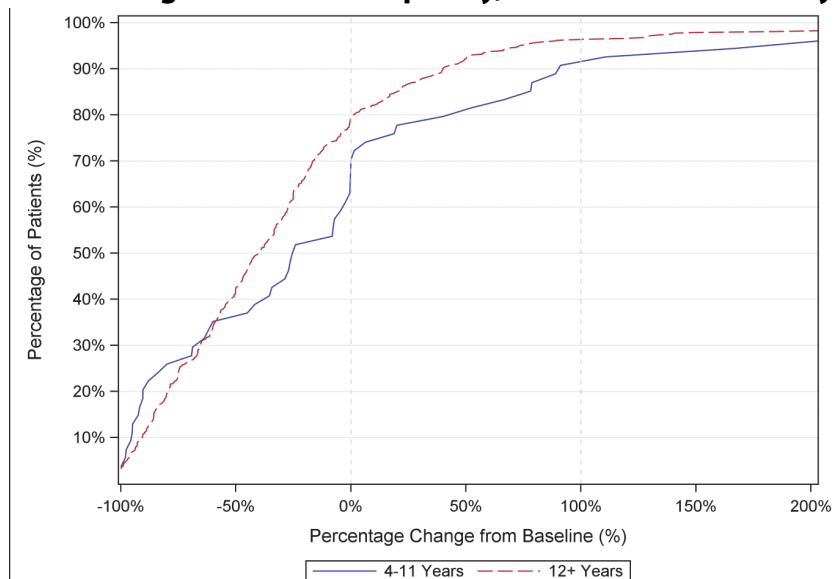
Baseline Seizure Frequency Per Month				
Age Group	4-11 Years	≥12 Years		Overall
N	54	528		582
Median	85.7	19.5		21.3
Age Group	4-11 Years	12-21 Years	≥22 Years	Overall
N	54	127	401	582
Median	85.7	30.4	17.4	21.3

Median percent reduction in seizure frequency from baseline to 12 months by age category is reported in Table 93. Following 12 months of treatment, there was no statistically significant difference in the median percent seizure reduction when comparing patients 4-11 years (-24.7%) with patients > 12 years of age (-40.4%) ($p=0.142$). Figure 89 shows the cumulative change in seizure frequency following 12 months of treatment for the ITT population by age group (4-11, ≥ 12 years).

Table 93. Median Percent Change in Seizure Frequency

Median percent change in seizure frequency at 12 months, by age group		
Age Group	4-11 Years	> 12 Years
N	54	528
Median	-24.7%	-40.4%
95% CI	-45.1% to 0%	-45.6 to -33.3%
p-value, Mann-Whitney	0.142	

Figure 89. Change in Seizure Frequency, Patient Distribution by Age Group



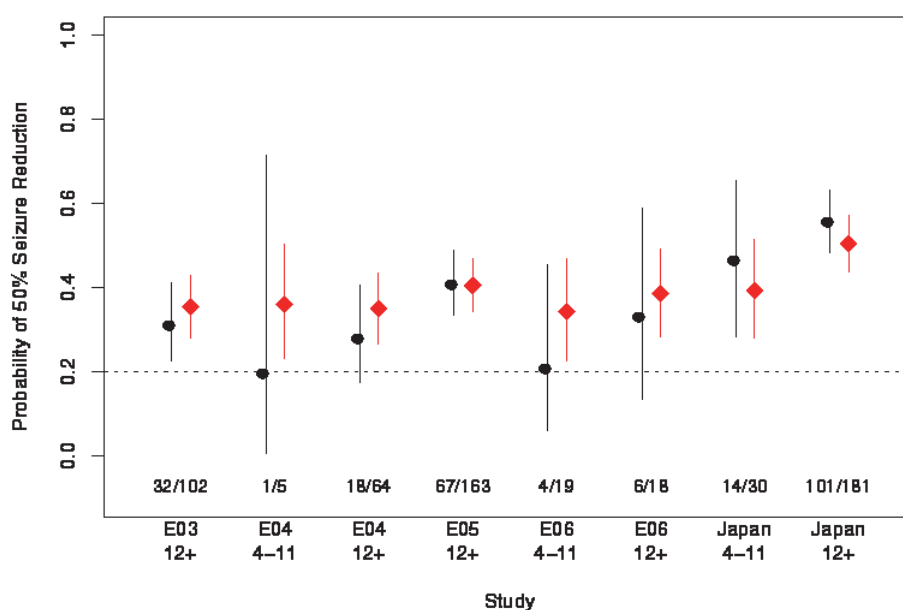
The Bayesian hierarchical model, which borrows from previous study data, estimated the 12 month responder rate for patients 4-11 years of age with focal onset seizures in the Japan PAS as 39% (95% credible interval: 28%, 52%). Since the entire 95% credible interval is greater than 20%, the primary efficacy endpoint was met. Across all studies, the Bayesian hierarchical model estimate for responder rate in patients 4-11 years of age is 37% (95% credible interval: 26%, 48%). (Table 94, Figure 90).

Table 94. Primary Efficacy Analysis

50% Response Rates at 12 Months						
Study	Age Group	Responders	Frequentist Estimate	95% Exact Binomial CI	Bayesian Estimate	95% Credible Interval
E-03	12+	32/102	31%	23-41%	35%	28-43%
E-04	4-11	1/5	20%	1-72%	36%	23-50%
E-04	12+	18/64	28%	18-41%	35%	27-44%
E-05	12+	67/163	41%	33-49%	41%	34-47%
E-06	4-11	4/19	21%	6-46%	34%	23-47%
E-06	12+	6/18	33%	13-59%	39%	28-49%
Japan	4-11	14/30	47%	28-66%	39%	28-52%
Japan	12+	101/181	56%	48-63%	50%	44-57%
Overall	4-11	19/54	35%	23-49%	37%	26-48%
Overall	12+	224/528	42%	38-47%	39%	33-46%

Note: VNS may be less effective in children who have previously undergone epilepsy surgery.

Figure 90. Probability of $\geq 50\%$ Seizure Reduction at 12 Months



Frequentist (black), Bayesian (red)

10.3.7 Overall Study Conclusions

The results of the extrapolation study demonstrate that VNS Therapy is a safe and effective treatment for the reduction of focal onset seizures in pediatric patients 4-11 years of age with refractory epilepsy. Based on the Bayesian hierarchical model, the 12 month responder rate for pediatric patients 4-11 years of age with focal onset seizures in the Japan PAS is 39% (95% credible interval: 28%-52%). There were no unanticipated adverse device effects observed in pediatric patients 4-11 years of age. However, infection and extrusion of lead had a statistically greater incidence rate in patients 4-11 years of age. Younger patients may have a greater risk for wound infection when compared to adolescent and adult patients; therefore, the importance of monitoring for site infection as well as the avoidance of manipulation of the surgical site post implant in children should be stressed. Otherwise, treatment-emergent adverse events in patients 4-11 years of age were consistent with patients ≥ 12 years of age treated with VNS Therapy and no new risks were identified.

10.4 Bibliography

A bibliography of animal and clinical studies is available from LivaNova on request.

11 ***Epilepsy Information—Patient Follow Up***

11.1 **Guidelines for Patient Follow Up**

11.1.1 **After Implantation**

During the first few weeks after implantation of new or replacement devices, the patient should be seen to confirm wound healing and proper generator operation. The generator's output current for the programmed stimulation in all modes must be 0 mA for the first 14 days after implantation.

The VNS Therapy System is an adjunctive therapy to existing (prior to device implantation) antiepileptic medications. Physicians are encouraged **to keep all antiepileptic medications stable for the first three months** of stimulation before attempting to reduce or change a patient's medication.

11.1.2 **Follow-up Visits**

11.1.2.1 ***Initial titration visits (ramping up VNS Therapy)***

During initial programming, the patient may be seen more frequently to make adjustments in therapy until a target level (i.e., adequate seizure control with minimal side effects) is reached. Once stimulation is ready to be programmed ON, the output current should be slowly increased in 0.25 mA increments until the patient feels the stimulation at a comfortable level. Patients who are receiving replacement generators should also be titrated in the same manner to allow re-accommodation. See "Dosing Strategies" for more information.

 **Note: (Generators with AutoStim only)** A smaller output current step size of 0.125 mA is available (up to 2 mA) to allow for patient tolerability to device stimulation.

11.1.2.2 ***Long-term follow up***

The subsequent follow-up schedule and the nature of each examination should be determined by the physician on the basis of patient response to and tolerance of the implant. In all other respects, follow up should be performed in accordance with the standard medical practice for patients with epilepsy.

In the event intolerable adverse events are reported, physicians should always try reducing stimulation parameters as a means of eliminating or reducing the severity of an event. See "Tolerability Strategies" for parameter adjustment recommendations. Additionally, physicians should instruct patients or caregivers on the application of the magnet to turn the generator off (output current 0 mA) if an adverse event becomes intolerable.

11.1.2.3 ***Typical follow-up visit activities***

At each patient visit, the generator should be interrogated, using the appropriate version of the VNS Therapy programming software. Stimulation adjustments may also be performed depending on patient response and/or tolerability.

VNS Therapy System treatment should not be uncomfortable, nor should it cause bothersome side effects. Patients may be observed after the last stimulation adjustment to make certain that they are comfortable with all available programmed stimulation modes. Since each

patient may respond differently to the stimulation, the observation period may be at least 30 minutes, or as long as necessary as determined by the physician.

Diagnostic testing should also be performed at each visit to confirm proper functioning of the VNS Therapy System. Additionally, it is recommended that testing of the magnet output be performed while the patient is still in the physician's office to ensure tolerability of the Magnet Mode output.

For generators with AutoStim Mode, heartbeat detection performance should be evaluated at each visit.

After reprogramming and/or diagnostics testing, data should be printed out and filed. These data can be used for comparison with a patient's diary or own records to evaluate the VNS Therapy System, to confirm proper VNS Therapy System functioning, and to assess the need for reprogramming. At the end of the session, a final interrogation should be performed to confirm parameters are set to the intended dose before the patient leaves the office.

 **Note:** For instructions on printing out data, see the programming system physician's manuals.

11.2 Individualization of Treatment

11.2.1 Therapy Parameters Used in Clinical Trials

The average output current used during the clinical studies after 3 months of stimulation was about 1 mA.¹ Other standard Normal Mode treatment settings were 30 Hz, 500 µsec pulse width, 30 seconds ON time, and 5 minutes OFF time. There are no data to verify that these are optimal parameters.

 **Caution: (Generators with AutoStim only)** It is recommended that the output current for the AutoStim Mode not exceed the greater of output current for the Normal Mode or Magnet Mode, especially for patients who experience discomfort or adverse stimulation effects (e.g., during sleep).

 **Caution: (Model 106 Serial Numbers < 80000 only)** Magnet Mode output current should be set at least 0.125 mA higher than AutoStim output current, to prevent rare instances where a device safety feature disables stimulation due to repeated magnet applications.

Table 95 lists the range of stimulation parameters after 3 months of active treatment used in the randomized, blinded, active control trials.

¹ Heck C., Helmers SL, DeGiorgio CM. "Vagus nerve stimulation therapy, epilepsy, and device parameters: scientific basis and recommendations for use". *Neurology* 2002; 59 (6, Suppl 4): S31-7.

Table 95. High Stimulation Group Parameters

Stimulation Parameters	Normal Mode	Magnet Mode
Output current	0–3.5 mA	0–3.5 mA
Frequency	30 Hz	30 Hz
Pulse width	500 µsec	500 µsec
ON time	30 sec	30 sec
OFF time	5 min	N/A

There is no proven correlation at present between high output current (mAmps) and device effectiveness, nor is there a standard treatment level that needs to be achieved during treatment ramping. However, computational modeling of vagus nerve stimulation suggests an approximate target for nerve activation.¹

11.2.2 Dosing Strategies

In general, VNS Therapy should be set to a comfortable level for the patient and increased as tolerated to help achieve efficacy. Although LivaNova recommends adjusting output current as necessary, there are no controlled data at this time to indicate that higher current levels are associated with better efficacy. Patients whose seizures are well controlled at follow up should not have their settings changed unless they experience uncomfortable side effects.

Patients should be started on stimulation at a low current setting (0.25 mA), and the current should be increased gradually to allow accommodation to the stimulation. For patient comfort, the output current should be increased in 0.25 mA increments until a comfortable tolerance level is reached. Physicians should appreciate that some patients will accommodate to stimulation levels over time and should therefore allow further increases (in 0.25 mA steps) in output current, if needed.

 **Note:** The Guided Programming feature in select versions of the VNS programming software can help guide you through the initial titration process. See the programming system physician's manuals for details.

The magnet output should be programmed at each visit, if necessary, to a level that is perceptible to the patient. This is typically set 0.25 mA higher than the Normal Mode output current. Some patients have reported that it is easier to verify daily that stimulation is being delivered if the magnet output current is set to one step above normal stimulation settings. This slightly higher output current is intended to allow patients who have accommodated to normal stimulation to recognize or perceive the magnet stimulation, thereby confirming device function.

For generator models with AutoStim, the AutoStim output current should be set no greater than the Magnet Mode output current. You may choose to set AutoStim output current between the Normal and Magnet Mode output currents, or equal to Normal Mode for comfort or tolerability.

 **Caution: (Model 106 Serial Numbers < 80000 only)** Magnet Mode output current should be set at least 0.125 mA higher than AutoStim output current, to prevent rare instances where a device safety feature disables stimulation due to repeated magnet applications.

¹ Helmers SL, Begnaud J, Cowley A, et al. "Application of a computational model of vagus nerve stimulation". *Acta Neurol Scand.* 2012; 126 (5): 336-43.

Table 96 lists the suggested initial stimulation parameters to begin titration of VNS Therapy.

Table 96. Suggested Initial Stimulation Parameters (≥ 2 Weeks After Implant)

NORMAL	Output Current	0.25 mA
	Signal Frequency [†]	20 - 30 Hz
	Pulse Width [†]	250 - 500 μ sec
	Duty Cycle: 10%	
	Signal ON Time	30 sec
	Signal OFF Time	5 min
AUTOSTIM*	Output Current	0.25 - 0.375 mA
	ON Time	60 sec
	Pulse Width	250 - 500 μ sec
MAGNET	Current	0.5 mA
	ON Time	60 sec
	Pulse Width	250 - 500 μ sec

* Not available in all generator models.

† Some patients may find 20 Hz/250 μ sec more tolerable. For this reason, some physicians prefer to start at the lower settings, and increase as tolerable. Other physicians may prefer to start at the higher settings, and adjust downward if needed for tolerability (Heck, Helmers, and DeGiorgio. 2002).

11.2.3 Tolerability Strategies

After each output current increase, evaluate the patient for tolerability. If an increase in output current is not tolerable, other stimulation parameters may be adjusted as shown in Table 97 to help with patient tolerability.

Prior to each parameter adjustment, it is recommended to revert the output current to the last level that was tolerated by the patient.

Make the parameter adjustment and try the increase in output current again.

If the patient was already started at the lower recommended settings for pulse width and frequency, reductions in output current and further reductions in pulse width may be the only course of action. However, if the pulse width is reduced to 130 μ sec, the output current should be increased to minimize the impact to the overall amount of therapy delivered. Literature has

shown that a higher output current is needed to activate the vagus nerve when pulse widths below 250 μ sec are used.^{1 & 2}

Table 97. Parameter Adjustments for Tolerability³

Parameter	Adjustment
Pulse Width	500 $\rightarrow$ 250 μ sec
Signal Frequency	30 $\rightarrow$ 20 Hz*
Output Current	$\downarrow$ 0.125 mA [†] or $\downarrow$ 0.25 mA

* 25 Hz is also available

† Only available in certain generator models

Table 98 provides an example of how to titrate when adjusting for patient comfort. Each example includes what the starting signal frequency and/or pulse width might be.

Table 98. Example — Tolerability Adjustments During Titration

Programming Steps	Parameter	Adjustment	Purpose
1	Output Current	+0.25 mA	Titration attempt
If patient experiences discomfort:			
2	Output Current	-0.25 mA	Comfort adjustment
3	Pulse Width or Signal Frequency	500 → 250 μsec or 30 → 20 Hz	
If parameter reduction is tolerable, continue titration.			
4	Output Current	+0.25 mA	Titration attempt

If the output currents are reduced to address side effects, but the target level (i.e., adequate seizure control with minimal side effects) has not been reached, future attempts at increasing output current are recommended.

11.2.4 Example Dosing Approach

This section describes a 2-phase dosing approach.⁴ The goal of Phase 1 (0.5-3 months after implant) is to increase the output current to a target range. The goal of Phase 2 (3-18 months

¹ Koo B, Ham SD, Sood S, Tarver B. "Human vagus nerve electrophysiology: A guide to vagus nerve stimulation parameters". *J Clin Neurophysiol* 2001;18 (5): 429-33.

² Helmers SL, Begnaud J, Cowley A, et al. "Application of a computational model of vagus nerve stimulation". *Acta Neurol Scand*.2012; 126 (5):336-43.

³ Heck C, Helmers SL, DeGiorgio CM. "Vagus nerve stimulation therapy, epilepsy, and device parameters: Scientific basis and recommendations for use". *Neurology* 2002; 59 (6, Suppl 4):S31-7.

⁴ Heck C, Helmers SL, DeGiorgio CM. "Vagus nerve stimulation therapy, epilepsy, and device parameters: Scientific basis and recommendations for use". *Neurology* 2002; 59 (6, Suppl 4):S31-7.

after implant) is to increase the duty cycle. If the patient achieves desired outcomes at any point, further adjustments may be stopped.

11.2.4.1 Phase 1 (output current)

Note: The Guided Programming feature in select versions of VNS programming software can help guide you through Phase 1 of this dosing approach. See the programming system physician's manuals for details.

Two weeks following implantation surgery, apply the initial recommended settings as described in Table 96. You may choose to start the pulse width and frequency at 500 μ sec and 30 Hz respectively, and adjust down as needed for tolerability. Or, you may start at the lower range of the recommended settings, 250 μ sec and 20 Hz.

With a duty cycle of 10%, increase the output current upward in 0.25 mA steps over the next several weeks. The target for output current is 1.5-2.25 mA depending on pulse width selection:¹

- 1.5 mA if PW 500 μ sec
- 1.75 mA if PW 250 μ sec
- 2.25 mA if PW 130 μ sec

Multiple step (0.25 mA) increases can be made in output current during a single visit if tolerated by the patient. Frequent visits during this titration phase may allow for faster progress toward the target output current. Table 99 shows how all three stimulation modes can be adjusted:

Table 99. Output Current Adjustments

Mode	Step 1	Step 2	Steps 3, 4, 5 ...	Target*
Normal (mA)	0.25	0.50	+0.25	1.5- 2.25
AutoStim (mA) [†]	0.375	0.625	+0.25	1.625 - 2.25 [†]
Magnet (mA)	0.50	0.75	+0.25	1.75 - 2.5

* Target output current depends on pulse width selection. See combinations above.

† AutoStim Mode is not available for all generator models. Output currents for AutoStim Mode may be set in between Normal and Magnet mode selections (as shown), or equal to the Normal Mode for comfort or tolerability.

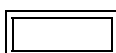
11.2.4.2 Phase 2 (duty cycle)

Once the output current has reached the target, duty cycle may be adjusted upward to assess better patient response. Allow adequate time between duty cycle adjustments for patient evaluation. Adjustments to duty cycle should be less frequent (approximately 3-6 months). Table 100 shows the recommended duty cycle increases.

¹ Helmers SL, Begnaud J, Cowley A, et al. "Application of a computational model of vagus nerve stimulation". *Acta Neurol Scand.* 2012; 126 (5):336-43.

Table 100. Duty Cycle Table of Adjustments

		OFF Time (min)								
		0.2	0.3	0.5	0.8	1.1	1.8	3	5	10
ON Time (sec)	7	58	←44	30	20	15	10	6	4	2
	14	69	56	↖41	29	23	15	9	6	3
	21	76	64	49	↖36	29	19	12	8	4
	30	81	71	57	44	↖35	←25	←16	←10	5
	60	89	82	71	59	51	38	27	18	10



For devices with AutoStim enabled, OFF times ≤ 0.8 minutes cannot be used.

Table 101 shows an example of Phase 1 and 2 adjustments over time.

Table 101. Example — Phase 1 and 2 Adjustments Over Time

		Months After Implant						
		0.5 -3	3-6	6-9	9-12	12-15	15-18	18+
Phase		Phase 1 Phase 2						
Duty Cycle*		10%	16%	25%	35%	36%	41%	

* Additional adjustments after 41% could include 44% and 58%. See Duty Cycle table for recommended ON-time / OFF-time combinations.

11.2.5 Optimize Generators Capable of AutoStim

11.2.5.1 Optimize the Heartbeat Detection setting

The AutoStim feature relies on accurate heartbeat detection in order to perform as intended. The device performs heartbeat detection by detecting the R-wave of the ECG morphology, which is known to vary based on the posture and body position of the patient. Therefore, a pre-operative assessment of R-wave amplitudes at different body positions is recommended in order to verify minimum detection requirements and to optimize Heartbeat Detection. Instructions for the assessment can be found in the Implantation Procedure chapter. Of the measurements recorded, use the average R-wave amplitude to choose an appropriate Heartbeat Detection setting based on the ranges listed in Table 102.

Table 102. Heartbeat Detection Mapping

Heartbeat Detection	Average Amplitude (mV) (across different postures)	
	Minimum	Maximum
5	0.40	0.50
4	0.51	0.70
3	0.71	0.85
2	0.86	1.25
1	1.26	--

If previous R-wave measurements are not available, either of the following options can be performed as an alternative:

- Repeat the measurements as instructed in the *Implantation Procedure* chapter to determine the average R-wave amplitude.
- Test each of the 5 Heartbeat Detection settings using the Verify Heartbeat Detection feature at each of the 2 body positions and choose the setting that accurately detects heartbeats in both positions.

 **Note:** The Verify Heartbeat Detection feature is described in the programming system physician's manuals.

At subsequent follow-up office visits, the physician should verify that the generator continues to accurately detect the heart rate, and adjust the heartbeat detection setting, if needed.

11.2.5.2 Optimize the AutoStim Threshold setting

The clinician can adjust the sensitivity of the underlying detection algorithm. Six AutoStim Threshold settings are available, 20% - 70% (in 10% increments), each of which correspond to the threshold that the heart rate must surpass in order to elicit a detection (only if detection is enabled) or a detection followed by the triggering of AutoStim (if detection is enabled and AutoStim output current is > 0 mA).

 **Note:** When Detection is "ON", the programming software prevents selection of Normal Mode OFF times less than 1.1 minutes to allow the device sufficient time to detect heart rate changes during each "OFF" cycle.

The objective to optimizing the AutoStim Threshold setting for an individual patient is to reduce the number of detections due to normal, autonomic heart rate changes while maintaining a sensitivity that will detect heart rate changes associated with many seizures.

Clinicians may use a variety of tools to establish a reasonable baseline (e.g., heart rate monitors, Holter monitors, etc.). To assess normal baseline heart rates, the clinician can measure heart rate while the patient is lying down, sitting, or standing (HR_{BL}). After a baseline is established, the clinician can assess a rise in heart rate (HR_{ACT}) during activity by monitoring the heart rate during normal day-to-day activities. The following equation calculates the percent rise from baseline to active ($\%HR_{NORM\ INCR}$).

$$(HR_{ACT} - HR_{BL}) / HR_{BL} \times 100 = \%HR_{NORM INCR}$$

To determine the heart rate rise during a seizure, the clinician may utilize the electrocardiogram (ECG) collected during the patient's epilepsy monitoring unit (EMU) stay.

i **Note:** See Figure 91 for an illustration of steps 1 and 2.

1. In the electroencephalography (EEG) recording, go to the beginning of a seizure. Scan up to 5 minutes before the electrographic or clinical onset of the seizure and pick a 10-second period of time to establish a baseline heart rate ($HR_{EEG BL}$). Within that 10-second window, count the number of R-R intervals and multiply by 6.

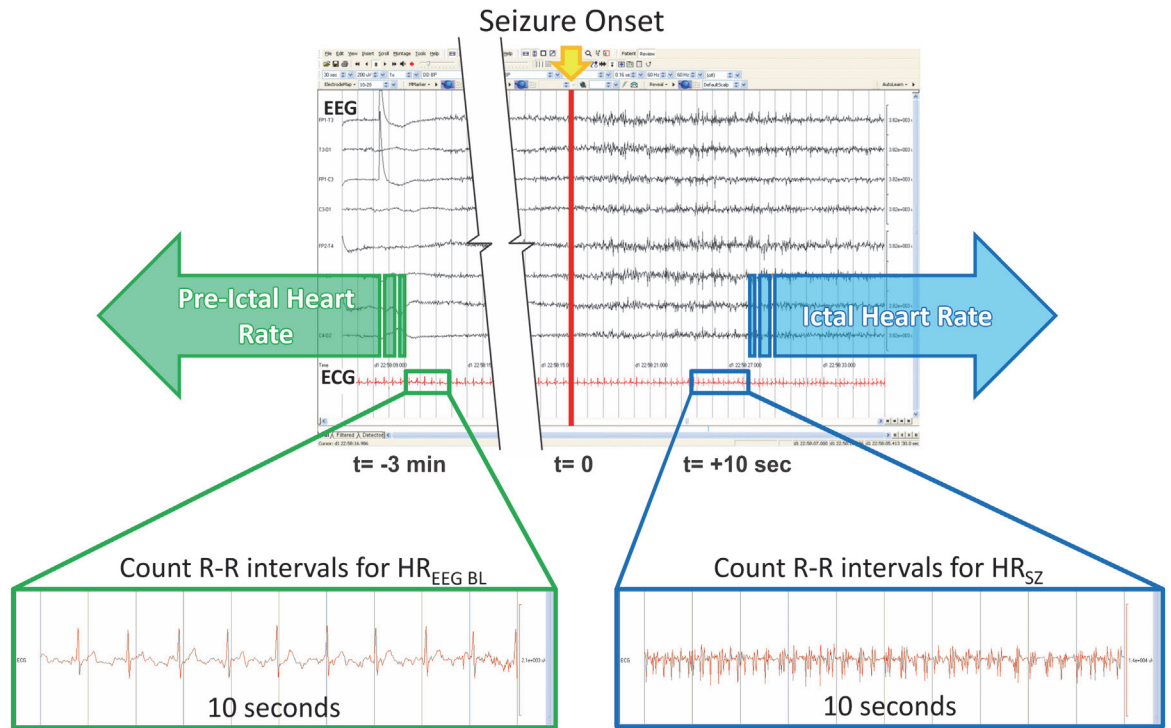
$$HR_{EEG BL} = (\# \text{ of R-R intervals}) \times 6$$

Alternatively, a different section of ECG recording may be used to calculate the pre-ictal heart rate:

- Obtain the simple average heart rate from at least two non-seizure epochs occurring at least 12 hours after or 1 hour prior to a seizure, with the patient in the same state as the start of the seizure.
 - Obtain the simple average heart rate from at least two clinical measurements of the patient's heart rate while sitting in the clinic, measured at least 5 minutes apart. These should occur at least 12 hours after or 1 hour prior to a seizure.
2. In the same recording, identify the beginning of the electrographic or clinical onset of the seizure. Scan the seizure and choose a 10-second period of time of maximum heart rate during the seizure (HR_{SZ}). Count the number of R-R intervals and multiply by 6.

$$HR_{SZ} = (\# \text{ of R-R intervals}) \times 6$$

Figure 91. Calculation of Baseline Heart Rate and Heart Rate During a Seizure



For this example (Figure 91), the baseline heart rate was assessed by scanning the ECG and finding a 10-second window of time approximately 3 minutes *prior to* seizure onset. Heart rate during the seizure was assessed by finding a 10-second window starting approximately 10 seconds *after* seizure onset.

3. Calculate the percent increase ($\%HR_{SZ\ INCR}$) from baseline:

$$(HR_{SZ} - HR_{EEG\ BL}) / HR_{EEG\ BL} \times 100 = \%HR_{SZ\ INCR}$$

If $\%HR_{SZ\ INCR} > \%HR_{NORM\ INCR}$ then choose an AutoStim Threshold setting that represents a threshold between the two values. For example, if $\%HR_{SZ\ INCR}$ is 51% and $\%HR_{NORM\ INCR}$ is 34%, then an AutoStim Threshold setting of 40% or 50% should be chosen. An AutoStim Threshold setting of 50% should be chosen if a lower potential false positive rate is desired or an AutoStim Threshold setting of 40% should be chosen if a higher sensitivity is desired.

If a patient's normal day-to-day heart rate increases are similar to or greater than their increases in heart rate during a seizure, then choose an AutoStim Threshold setting that represents a threshold lower than the $\%HR_{SZ\ INCR}$. For example, if $\%HR_{SZ\ INCR}$ is 62% and $\%HR_{NORM\ INCR}$ is 68%, then an AutoStim Threshold setting of 60% should be chosen. In this scenario the patient may expect to receive additional stimulations. If

bothersome, these stimulations can be inhibited by applying the magnet over the generator for at least 3 seconds.

11.3 Patient Counseling Information

Patients should be told to test their generator's operation daily by performing magnet stimulation and verifying that stimulation occurs. If stimulation does not occur, their physician should be contacted.

It should be noted that the magnet stimulation timing is not synchronized with the timing clock used for determining ON time and has a tolerance of $\pm 15\%$ or ± 7 seconds. Therefore, if the Magnet Mode ON time is programmed to 7 seconds and the generator is swiped at the end of the clock cycle, magnet stimulation may not be perceived by the patient. If the patient does not perceive the magnet stimulation, he or she should be instructed to swipe the generator a second time.

 **Note:** See the "Stimulation" section in the device-specific Technical Information chapter.

In the unlikely event of uncomfortable adverse events, continuous stimulation, or other malfunction, the patient must be advised to hold or tape the magnet directly over the implanted generator to prevent additional stimulation. If patients or caregivers find this procedure necessary, they should immediately notify the patient's physician.

12 Implantation Procedure

12.1 Physician Training / Information

All VNS Therapy System programming should be by or under the supervision of a physician familiar with the use and operation of the programming software.

Physicians implanting the VNS Therapy System should be thoroughly familiar with all associated training materials, including:

- Product labeling for the generator, lead, programming system, and accessories (magnet, tunneler, and accessory pack), and patient labeling
- *"Implant Guide for the VNS Therapy System"* training manual and other brochures
- Video on the proper implantation technique: "Implantation of the VNS Therapy System"
- Electrode practice fixture—a device used to practice placing the helices around the left vagus nerve

12.2 VNS Therapy Devices

12.2.1 Generator Sales Pack Contents

The generator sales pack contains the following:

- 1 VNS Therapy generator (single receptacle or dual receptacle)
 -  **Note:** Ensure that at least one back-up generator is available before starting the procedure.
- 1 hex screwdriver
- 1 resistor assembly
 -  **Caution:** The resistor assembly is not included in the Model 1000 generator sales pack. An accessory pack should be available during surgery to troubleshoot problems if they occur.
- Documentation

12.2.2 Lead Sales Pack Contents

The lead sales pack contains the following:

- 1 VNS Therapy lead
 -  **Note:** Ensure that at least one back-up lead is available before starting the procedure.
- 4 (or more) silicone tie-downs
- Documentation

12.2.3 Other LivaNova Products

- 1 tunneler (sterile)

- 1 accessory pack (sterile)
- 1 programming system (non-sterile)



Note: Remember to use proper technique for introducing non-sterile items into a sterile field.

12.2.4

Surgical Materials



Note: The materials and equipment listed below are not provided by LivaNova.

The following is a list of additional materials typically used during the VNS Therapy implantation procedure:

All Generators

- Sterile Laser Arm Bag or equivalent (required)
- Vessel loops and/or silicone sheet for manipulation of the vagus nerve (suggested but optional)

Generators with AutoStim feature

The following materials are required to identify acceptable implant locations:

- Commercial ECG monitor — The ECG monitor should have the capability to print out the ECG waveform/amplitudes on the lead I channel
- Standard, 10 mm Ag/AgCl skin electrodes
- Commercial ECG Instructions For Use



Note: See “Determine acceptable device implant locations (tachycardia detection only)” for details. This information is also summarized in the *Pre-Surgical Evaluation Tool* in the generator sales pack.

12.2.5

To Open the Sterile Pack



Caution: The sterile lead pack should only be opened after exposing the vagus nerve and selecting the VNS Therapy lead helical that best fits.



Caution: Do not open the sales pack if it has been exposed to extreme temperatures or if there is any indication of external damage or damage to the package seal. Instead, return it unopened to LivaNova.

Before the sterile pack is opened, it should be examined carefully for evidence of damage or compromised sterility. If the outer sales pack or inner sterile pack has been opened or damaged, LivaNova cannot guarantee sterility of the contents, and it should not be used. An opened or damaged product should be returned to LivaNova.

To open the sterile pack, do the following:

1. Grasp the tab, and peel back the outer cover.
2. Observing sterile technique, lift out the sterile inner tray.
3. Grasp the inner tray's tab, and carefully peel off the inner cover to expose the contents without dropping them.

12.3 Recommendations for Implantation

In general, implantation of the VNS Therapy System is similar to accepted practice for implantation of a cardiac pacemaker, with the exception of the placement of the helices and the subcutaneous routing of the lead body. The surgical approach and techniques will vary with the preference of the surgeon. To ensure correct lead placement, this chapter of the physician's manual provides recommendations for implantation, order of placement of the helical electrodes and anchor tether, and other essential steps.



Caution: To maximize system performance and minimize possible mechanical damage to the nerve or lead, **pay careful attention to helical placement and lead routing.**

For generators with the AutoStim feature, the physical location of the device critically affects its ability to properly sense heartbeats. Therefore, care must be taken to follow the implant location selection process outlined in "Determine acceptable device implant locations (tachycardia detection only)".



Note: The implant location selection procedure may be performed pre-operatively as part of the patient's surgical work-up.

Critical to the long-term success of the implant are proper techniques both for the attachment of the electrodes and the anchor tether to the left vagus nerve, and for the provision of adequate strain relief below and above the sternocleidomastoid muscle. For general placement of the generator and lead, see Figure 95.

It is recommended that the lead body be coiled and placed in the chest pocket to the side of the generator.

Adequate exposure of the vagus nerve (> 3 cm) facilitates placement of the helices on the nerve. Stretching the nerve or allowing it to dry during implantation may result in temporary swelling of the nerve. Constriction of the nerve or other nerve damage may result in vocal cord dysfunction.

LivaNova recommends that output of the generator and performance of the implanted system be tested at the time of implantation. Although an oscilloscope can be used for measurements, LivaNova recommends use of the appropriate version of the programming software and wand (placed in a sterile drape) for routine system verification.

After the electrode is placed on the nerve, test the electrode-nerve interface impedance by connecting the lead directly to the generator and performing a System Diagnostics (Lead Test). If required, a separate resistor assembly from the generator sales pack or accessory pack can be used while performing the optional Generator Diagnostics (Pre-Implant Test).



Note: See "Test the VNS Therapy System".

12.3.1 Before Surgery and Outside of the Sterile Field

12.3.1.1 Interrogate the device

To ensure proper device communication, interrogate the device while still in the sterile pack. See the Quick Start Guide or for a detailed explanation, see the programming system physician's manuals.



Caution: (For 103 and subsequent models only) If interrogating a generator that has been exposed to low temperatures within the last 24 hours, low battery status indicator(s) may be displayed. See “Troubleshooting” in the programming system physician’s manuals.

12.3.1.2 **Program patient data**

Program the patient identification and implant date into the generator. See the Quick Start Guide or for a detailed explanation, see the programming system physician’s manuals.

12.3.1.3 **Determine acceptable device implant locations (tachycardia detection only)**

The implant location of generators capable of tachycardia detection critically affects their ability to properly sense heartbeats. The following steps describe the recommended process in identifying acceptable implant locations for the generator and lead.



Note: The implant location selection process is also summarized in the *Pre-Surgical Evaluation Tool* in the generator sales pack.

12.3.1.3.1. **Equipment / Materials Required**

- Commercial ECG monitor
 - ◆ The ECG monitor should have the capability to print out the ECG waveform/amplitudes on the lead I channel.
 - ◆ The ECG monitor must be configurable to a lowpass filter setting up to 150 Hz.
- Standard, 10 mm Ag/AgCl skin electrodes
- Commercial ECG Instructions For Use

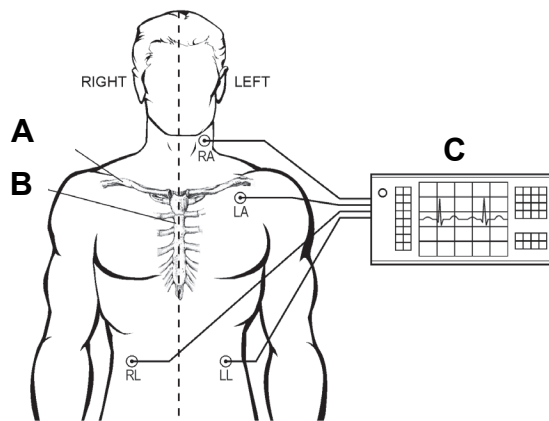


Note: Any commercial ECG system meeting the requirements in the section “Equipment / Materials Required” above is acceptable for use in the identification of potential implant locations procedure. Refer to the commercial ECG system Instructions For Use for proper operation or configuration.

12.3.1.3.2. **Procedure**

1. Verify that the ECG monitor printing scale is set to 10 mm/mV and the lowpass filter does not exceed 150 Hz.
2. Prepare the patient's skin in the left neck and chest area (e.g. remove excess body hair, perform alcohol wipe) to ensure proper contact with ECG skin electrodes.
3. Place ECG skin electrodes on the patient (a sample configuration is shown in Figure 92) as follows:
 - ◆ One electrode should be placed on the left neck, at the approximate intended implant location of the lead electrodes
 - ◆ One electrode should be placed on the chest, at the approximate intended implant location of the generator
 - ◆ One electrode should be placed on the right lower abdomen or leg
 - ◆ One electrode should be placed on the left lower abdomen or leg

Figure 92. Sample Electrode Configuration



A Collar Bone

B Sternum

C ECG Monitor

RA Intended implant position of the lead electrodes

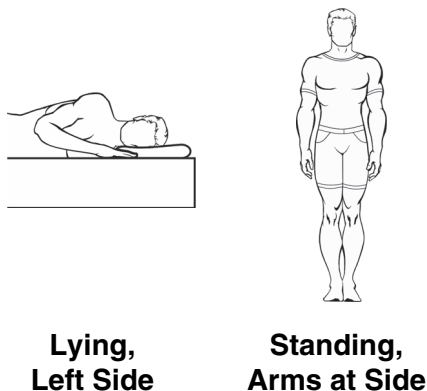
LA Intended implant position of the generator

4. Connect the ECG leads to the electrodes:
 - RA - neck
 - LA - chest
 - RL - right lower abdomen or right leg
 - LL - left lower abdomen or left leg
5. Verify that the lead I ECG waveform is showing on the ECG monitor, wait for the ECG signal to stabilize, and collect 10 seconds of ECG data with the patient positioned lying on the left side (first of the two positions in Figure 93).



Note: Refer to the commercial ECG Instructions For Use for proper operation or configuration.

Figure 93. Patient Positions



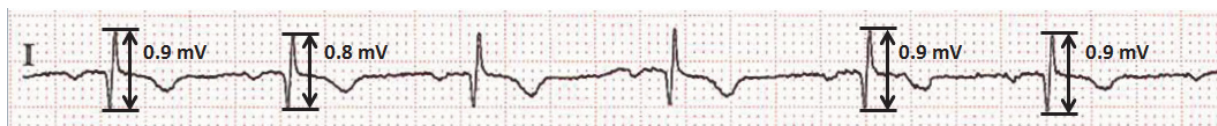
**Lying,
Left Side**

**Standing,
Arms at Side**

6. Print the ECG strip and label the patient position. On the ECG strip measure the peak-to-peak R-wave amplitude in the lead I channel (see Figure 94) by following the scaling in Step 1. Perform this for at least 4 representative R-waves in the 10 seconds of data and record the minimum amplitude value from the assessed R-waves. This value is

representative of the minimum peak-to-peak R-wave amplitude for the patient in the defined body position.

Figure 94. Sample ECG Trace with Peak-to-Peak R-wave Measurements



Note: 1 small division line = 0.1 mV, assuming a 10 mm/mV scale.

7. Verify that the minimum peak-to-peak R-wave amplitude measurement in Step 6 is 0.4 mV or greater. If this is the case, then repeat Steps 5-6 with the remaining body position defined in Figure 93 (e.g., standing with arms at side) until both body positions have been tested and the minimum peak-to-peak R-wave amplitude measurement for each body position is confirmed to be 0.4 mV or greater.

Note: Assuming a 10 mm/mV scale, the peak-to-peak R-wave amplitude measurements must span at least 4 lines on the ECG paper to meet the minimum requirement of 0.4 mV.

8. If the minimum peak-to-peak R-wave amplitude measurement for any one position is less than 0.4 mV, pick a new potential implant location for the generator which increases the distance between the neck electrode and the existing chest electrode, and/or is closer to the patient's heart. Place a new electrode on the new potential implant location (the old chest electrode may be removed if it is in the way), connect it to the LA lead, and repeat Steps 5-7 for both body positions until a location with adequate peak-to-peak R-wave amplitude can be identified.

Note: Try to place the generator at rib 4 or above, so the patient can have the maximum flexibility for MRI post-operatively. See *MRI with the VNS Therapy System* instructions for use for details.

9. When both body positions have been tested and the minimum peak-to-peak R-wave amplitude measurement for each body position is confirmed to be 0.4 mV or greater, the neck and chest electrode locations are acceptable selections for the implant. Mark the spots on the neck and chest where the electrodes are and use these locations as the intended implant location during surgery. The minimum peak-to-peak R-wave amplitude measurements from the different body positions are used to configure heartbeat detection and tachycardia detection (see "Heartbeat Detection and Tachycardia Detection Configuration") and post operatively to optimize the heartbeat detection setting (see "Optimize the Heartbeat Detection Setting" in the *Epilepsy Information—Patient Follow Up* chapter).

If all practical implant locations have been exhausted without identifying a location which yields a peak-to-peak R-wave amplitude of at least 0.4 mV at both body positions, the AutoStim Mode feature is not suitable for the patient.

12.3.2 Implantation Procedure Overview



Caution: This procedural overview is not a substitute for the complete implantation procedure. See detailed steps that follow.



Note: For generators capable of tachycardia detection, try to implant the lead and generator in the same approximate positions as determined in "Determine acceptable device implant locations (tachycardia detection only)".

The following overview summarizes the implantation procedure:

1. Expose the left carotid sheath and left vagus nerve.
2. Create a pocket in the chest for the generator.
3. Choose the correct size lead.
4. Tunnel the lead subcutaneously from the neck to the generator pocket in the chest.
5. Attach the electrodes and anchor tether to the left vagus nerve.
6. Secure the lead parallel to the nerve.
7. Form the strain relief bend and strain relief loop.
8. Connect the lead to the generator.
9. Verify that the connector pin is fully inserted, and tighten the setscrew.
10. Perform the System Diagnostics (Lead Test).
11. Place the generator in the chest pocket, with the extra coiled lead to the side of the generator, not behind it.
12. (For generators capable of tachycardia detection, configure detection settings and verify heartbeat.
13. Secure the generator to fascia; do not place sutures directly around or on the lead.
14. Perform the second System Diagnostics (Lead Test).
15. Interrogate the generator to verify current is 0 mA.
16. Irrigate the incision site with bacitracin or other solution.
17. Close the incisions.

12.3.3 Prepare for Surgery

The surgeon should ensure that the generator, lead, and tunneler are compatible.

LivaNova recommends that the patient be given antibiotics preoperatively and that both incision sites be irrigated frequently with generous amounts of bacitracin or equivalent solution prior to closure. (These incisions should be closed with cosmetic closure techniques to minimize scarring.) Also, antibiotics should be administered postoperatively at the discretion of the physician.



Caution: Infections related to any implanted device are difficult to treat, and explantation of the VNS Therapy System may be required.

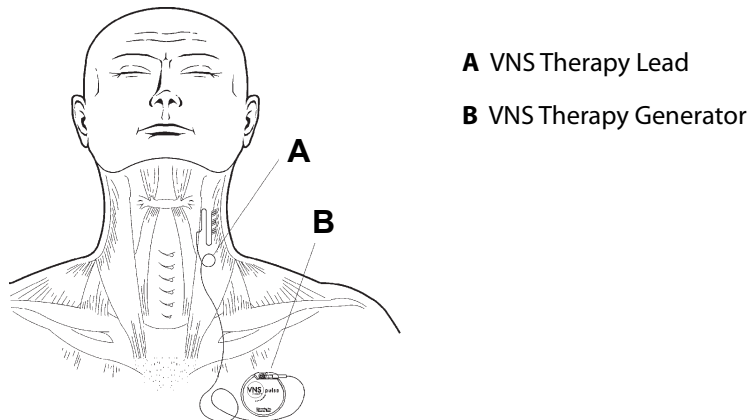
12.4 Lead and Pocket Location

The generator is usually implanted just below the clavicle in a subcutaneous pocket in the left upper chest. Try to place the generator at rib 4 or above, so the patient can have the maximum flexibility for MRI post-operatively. See *MRI with the VNS Therapy System* instructions for use for details.

Suggested placement for the lead is the area of the left vagus nerve half-way between the clavicle and the mastoid process, with the lead subcutaneously tunneled between the incision site in the neck and the pocket formed in the upper chest (see Figure 95). It is recommended that both the lead body and the generator be positioned on the left side of the body. The VNS Therapy tunneler is recommended for subcutaneous routing of the lead.

i Note: For placement of generators capable of tachycardia detection, see “Determine acceptable device implant locations (tachycardia detection only)”.

Figure 95. Placement of Generator and Lead



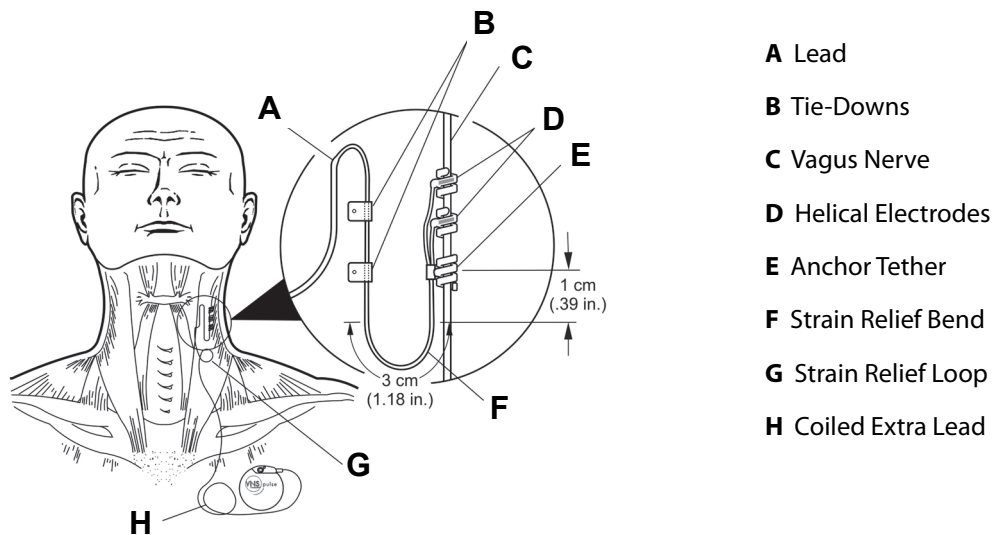
12.5 Begin the Procedure

While the specific surgical approach and techniques for implanting the lead will vary with the implant surgeon, the following detailed instructions are provided for guidance:

1. After administering appropriate anesthesia to the patient, expose the left carotid sheath as it extends along the anterior border of the sternocleidomastoid muscle.
2. Locate and expose *at least 3 centimeters (1.18 inches)* of the left vagus nerve. The recommended stimulation site is a 3-cm section of the vagus nerve, approximately half-way up between the clavicle and the mastoid process, where it is clear of branches (below where the superior and inferior cervical cardiac branches separate from the vagus nerve—see Figure 96 and Figure 98). The nerve usually lies in a posterior groove between the carotid artery and internal jugular vein.

i Caution: Avoid letting the vagus nerve become dry during surgery, because dehydration of the nerve can result in nerve damage and swelling.

Figure 96. Electrode Placement



3. Create a subcutaneous pocket in the chest below the clavicle for the generator. The pocket depth should not be deeper than 1 inch beneath the skin. It is not recommended to implant the generator below muscle. Doing so may contribute to communication difficulties once implanted.

Note: It is preferable to place the subcutaneous pocket along the axillary border, at or above rib 4.

12.6 Implant the Lead

To implant the lead, follow these steps:

12.6.1 Choose a Lead

1. Choose the appropriately sized lead (2.0 or 3.0 mm electrode inner diameter) carefully. It should fit snugly without constricting the nerve. The lead (2.0 mm/0.08 in) should accommodate most nerves.

Note: For lead size availability, see "Product Specifications" in the lead-specific Technical Information chapters.

Caution: The lead is available in multiple sizes. Since it is not possible to predict in patients what size lead will be needed, **It is recommended that at least one alternate lead size be available in the operating room.** In addition, backups for leads should be available in the event of compromised sterility or damage induced during surgery.

Caution: Do not expose the lead to dust or other similar particulates, because its silicone insulation can attract particulate matter.

Caution: Do not soak the lead in saline or similar solution before implanting it, because this may cause the insulated portions of the connector pin to swell and become difficult to insert into the generator.

12.6.2 Pass the Tunneler and Lead

The tunneler is used to tunnel the lead connector and lead body subcutaneously between the neck incision site and the generator in the chest pocket.



Note: A detailed description of the tunneling tool can be found in the *Tunnelers Directions for Use*.



Caution: To maximize system performance and minimize possible mechanical damage to the nerve or lead, pay careful attention to lead routing, lead stabilization, and electrode placement.



Caution: Never route the lead through muscle.



Caution: Never suture the lead or lead body to muscle tissue.



Caution: Always use the tie-downs.



Caution: Do not place sutures directly on the lead body. Doing so may result in insulation damage or wire failure, causing premature failure of the lead.

If necessary, the tunneler can be manually shaped to help direct it through the body.



Caution: Do not manually shape the tunneler **more than 25 degrees** because doing so may cause the sleeve to bend or kink.

To pass the tunneler, do the following:

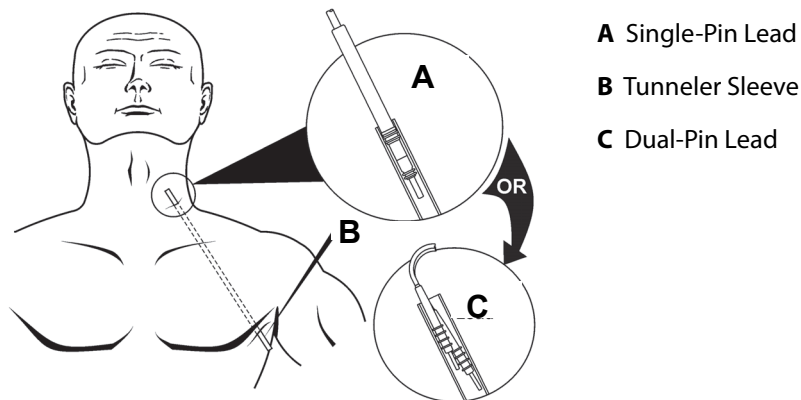
1. Place the bullet-tip end of the tunneler through the neck incision and tunnel subcutaneously toward the chest incision, exerting force on the handle end and directing the tunneler as necessary.

As an alternative, the lead connector and lead body can be tunneled subcutaneously from the neck incision site to the generator in the chest pocket *after placement of the electrodes and anchor tether on the nerve, and placement of strain relief with the tie-downs*. (See “Place the Electrodes” and “Provide Strain Relief”, respectively.)

2. After the bullet tip has passed from one incision site to the other, unscrew the bullet and withdraw the shaft from the sleeve, leaving the sleeve extended through both incisions (see Figure 97).

Figure 97. Position of Sleeve and Lead Connector(s)

Insert the lead into the sleeve at the neck incision until secure.



3. With the sleeve in place between the two incisions, carefully insert the lead connector(s) inside the end of the sleeve at the neck incision. For a dual-pin lead, the second connector will form a slight compression fit between the first lead connector tubing and the inside of the sleeve (see Figure 97).
4. Carefully pull the sleeve, along with the lead connector(s), from the chest incision end until the lead connector(s) completely exit(s) the chest incision.
5. Remove the lead connector(s) from the sleeve, leaving the electrode array at the neck incision site.
6. Discard the tunneler after use.

12.6.3 Place the Electrodes

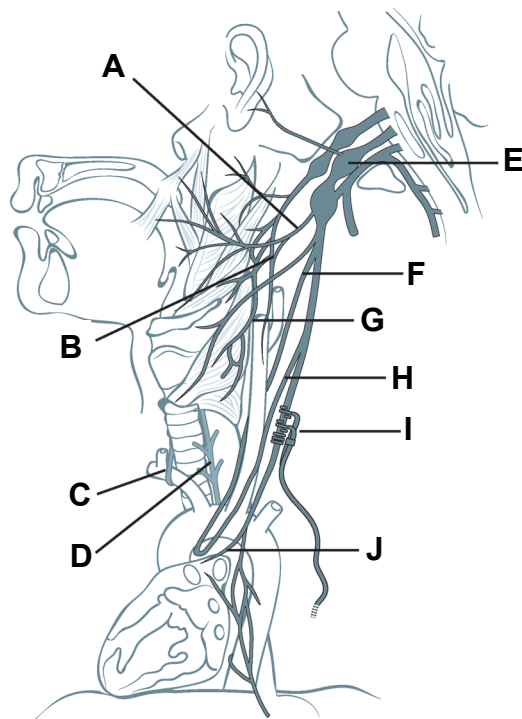
12.6.3.1 Anatomy

It is very important that the surgeon implanting the VNS Therapy System be familiar with vagus nerve anatomy, particularly the cardiac branches. The lead electrodes must not be placed on either the superior or the inferior cervical cardiac branches. **Place the lead below where the superior and inferior cardiac branches separate from the vagus nerve.** Stimulation of either of these two branches during the System Diagnostics (Lead Test) may cause **bradycardia and/or asystole**. Careful dissection laterally on the vagus nerve should aid the physician in determining proper electrode placement. In most but not all patients, the main vagus nerve is the largest of the three nerves. Figure 98 shows the correct anatomical placement of the helices.



Caution: Attachment of lead electrodes must not involve the superior cervical cardiac branch or the inferior cervical cardiac branch of the vagus nerve. Place the electrodes *below* where these two branches separate from the vagus nerve.

Figure 98. Vagus Nerve Anatomy and Placement of the Lead



- A** Pharyngeal Branch of Vagus Nerve
- B** Communicating Branch of Vagus Nerve to Carotid Sinus Branch of Glossopharyngeal Nerve
- C** Right Recurrent Laryngeal Nerve
- D** Left Recurrent Laryngeal Nerve
- E** Left Vagus Nerve
- F** Superior Cervical Cardiac Branch of Vagus Nerve
- G** Superior Laryngeal Nerve
- H** Inferior Cervical Cardiac Branch of Vagus Nerve
- I** Lead Electrode Location
- J** Thoracic Cardiac Branch of Vagus Nerve

Caution: Excessive manipulation of the vagus nerve during placement of the lead can result in noticeable post-operative hoarseness. Under most circumstances, this condition will resolve without additional medical intervention within three to four weeks, depending on the degree of stress applied to the nerve during surgery. LivaNova does not recommend that stimulation treatment be initiated until this condition has resolved, since it could aggravate the condition.

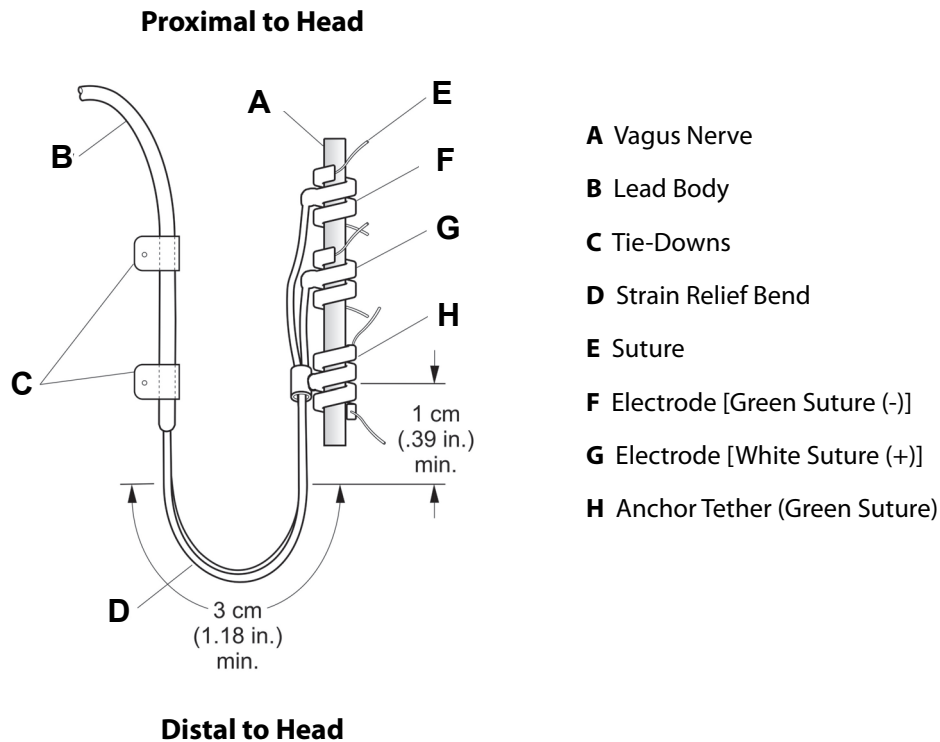
12.6.3.2 **Electrode Polarity**

The helical electrodes and anchor tether are coiled around the nerve, beginning with the electrode that is farthest from the lead bifurcation (with a green suture embedded in the helical material). This electrode should be nearest (proximal to) the patient's head.

Caution: The lead and helical electrodes are very delicate; be careful not to stretch, pinch, or crush them when using forceps, and not to over-straighten or stretch the helices when coiling them around the nerve, because doing so may damage the electrode or tether. Use soft rubber vessel loops to raise, or lift, the nerve, if necessary.

Depending on the surgeon's preference, the helices can alternately be placed by putting the anchor tether on first (distal to head), next placing the electrode closest to the lead bifurcation (with white suture), and then placing the electrode farthest from the lead bifurcation (with green suture). The polarity of stimulation does not change (see Figure 99).

Figure 99. Electrode Polarity



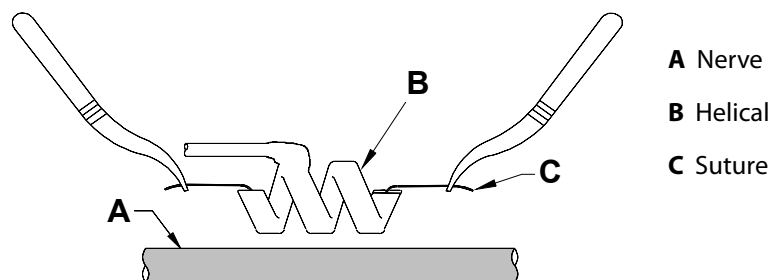
12.6.3.3 Place the helicals around the nerve

The helicals can be placed on the nerve as described below. As an alternative, each helical can be placed underneath the nerve before it is spread. A silicone sheet may be useful to separate the nerve from tissue during the procedure.

1. Locate the first helical (with green suture).
2. With forceps, gently pull each end of the helical, using the attached sutures to spread the helical (see Figure 100).

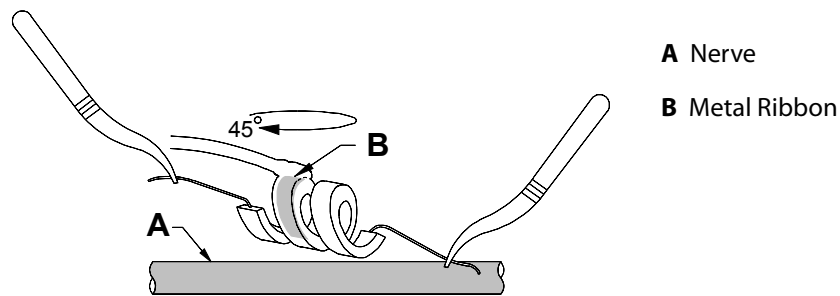
Caution: The suture may become dislodged from the helical if product labeling is not followed (i.e., grasping the elastomer and suture to manipulate the helical onto the nerve).

Figure 100. Spread the Helical



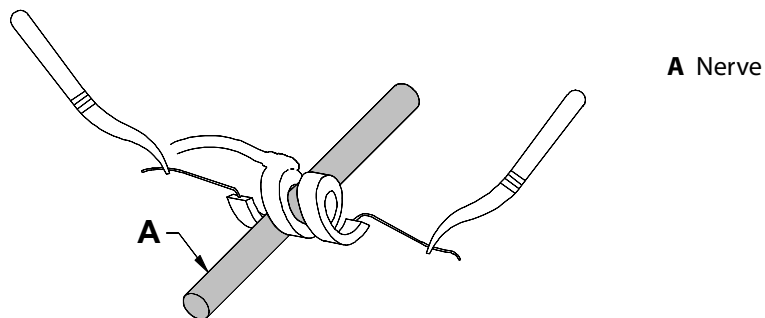
3. Spread the opened helical directly above and parallel to the exposed nerve and turn the helical clockwise at a 45 degree angle to the nerve (see Figure 101).

Figure 101. Turn the Helical



4. Place the turn of the helical where the lead wire connects to the helical (the section with the metal ribbon) onto the nerve (see Figure 102).

Figure 102. Placement of the Turn



5. Pass the *distal* suture portion of the helical under the nerve and back around so that it encircles the nerve (see Figure 103 and Figure 104).

Figure 103. Initial Placement of the Distal Portion of the Helical

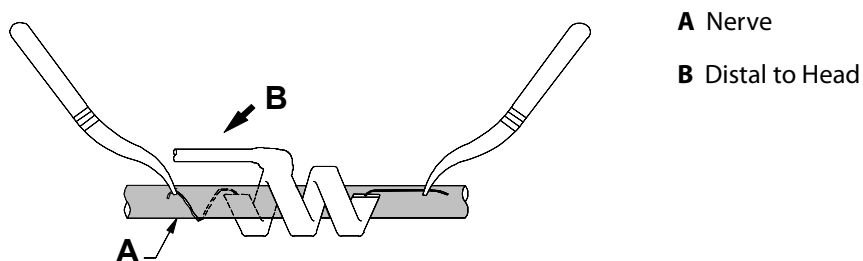
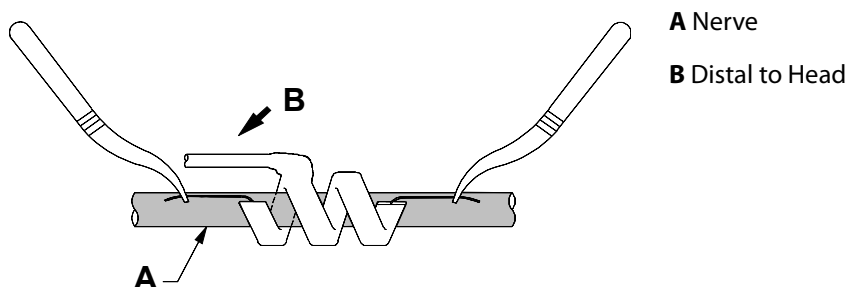
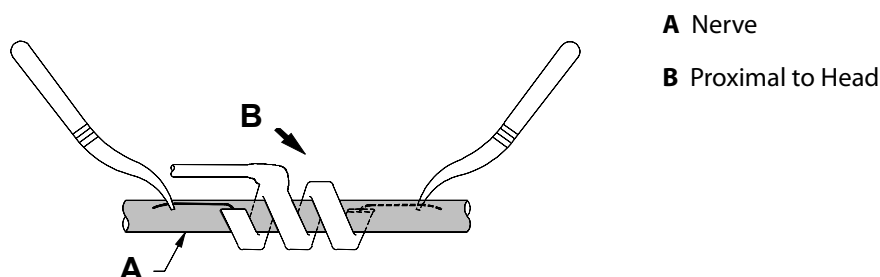


Figure 104. Helical Placement After Distal Portion Encircles the Nerve



6. Pass the *proximal* suture portion of the helical under the nerve and back around so that it encircles the nerve (see Figure 105).

Figure 105. Placement of the Proximal Portion of the Helical



7. Locate the middle helical (with white suture) and repeat steps 2 - 6.
8. Locate the third helical (with green suture) and repeat steps 2 - 6.
9. Verify all three helices have been coiled around the nerve, the lead body exits each helical in the same direction, and the two lead bodies are aligned parallel to each other and the nerve. The correct placement of the two helical electrodes and anchor tether is shown in Figure 106.

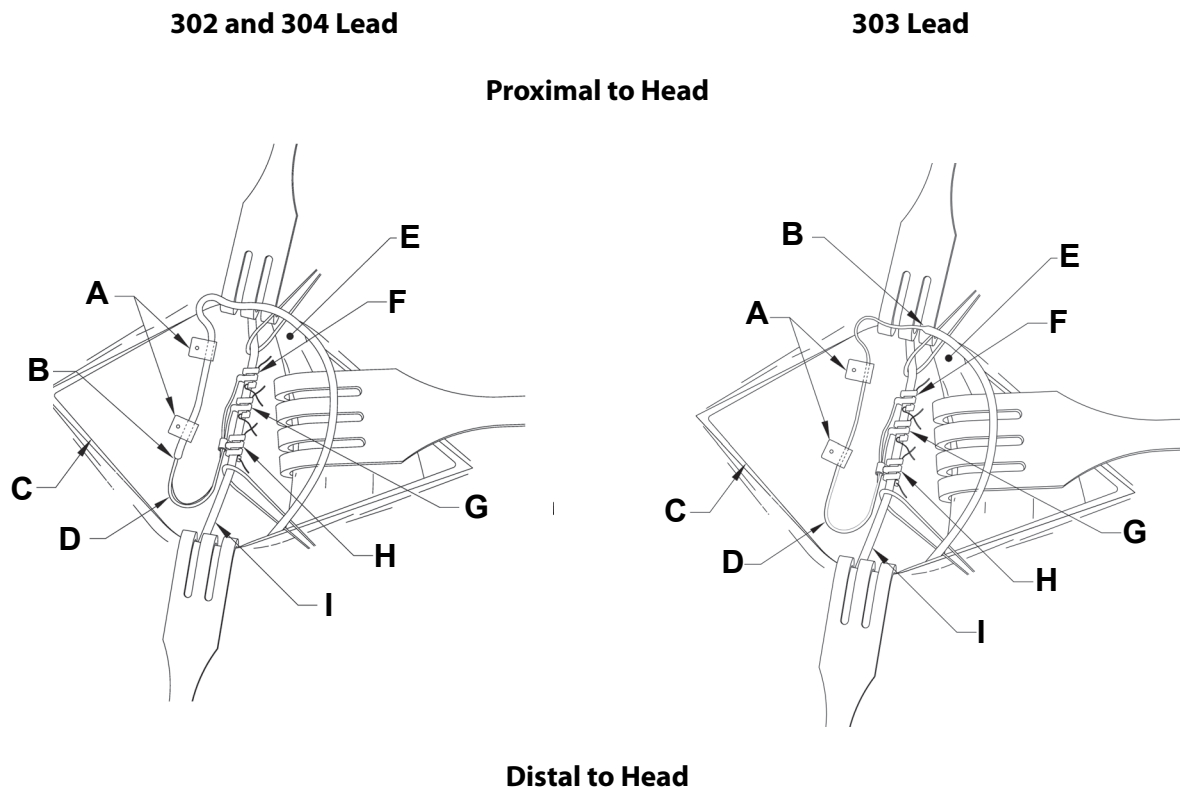


Caution: Sutures that are part of the lead (embedded in the helices of the electrodes and anchor tether) are meant to assist in helical placement around the vagus nerve. These sutures should not be tied to each other or around the nerve, since this may cause nerve damage.



Caution: Proper techniques for attaching the electrodes and the anchor tether to the left vagus nerve are critical to the long-term success of the implant.

Figure 106. Placement of Electrodes and Anchor Tether



A Sutured Tie-Downs

B Lead Transition

C Neck Incision

D Strain Relief Bend

E Sternocleidomastoid Muscle

F Electrode Farthest From Lead Transition [Green Suture (-)]

G Electrode [White Suture (+)]

H Anchor Tether (Green Suture)

I Vagus Nerve

12.6.4 Provide Strain Relief



Caution: Proper techniques for providing adequate strain relief below and above the sternocleidomastoid muscle are critical to the long-term success of the implant.



Caution: The lead wire has a potential for fracture if the recommended strain relief is not provided as described.

After attaching the two electrodes and the anchor tether, form a strain relief bend and a strain relief loop in the lead to provide adequate slack and allow for neck movement.

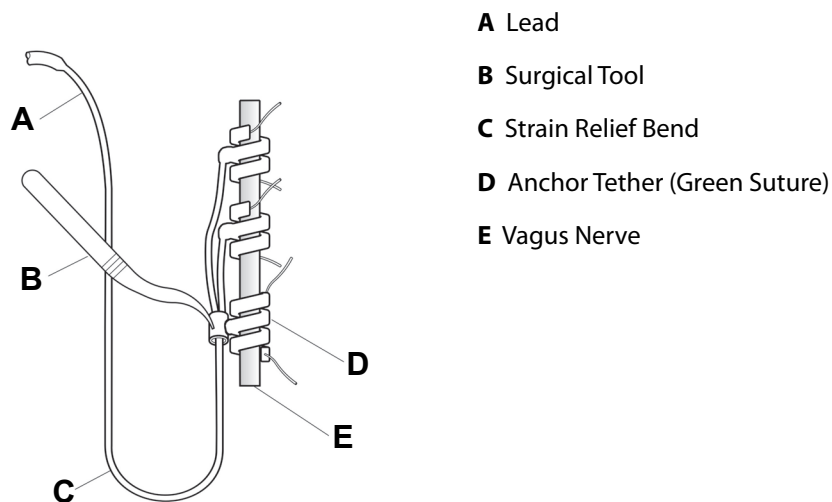
12.6.4.1 Form the strain relief bend

To form the *strain relief bend* (see Figure 96, Figure 107 [303 only], and Figure 108), do the following:

1. Form the lead body into a 3-cm (1.18 in) strain relief bend with at least 1 cm (0.39 in) of lead routed parallel to the nerve. (303 lead only—Pay careful attention to the

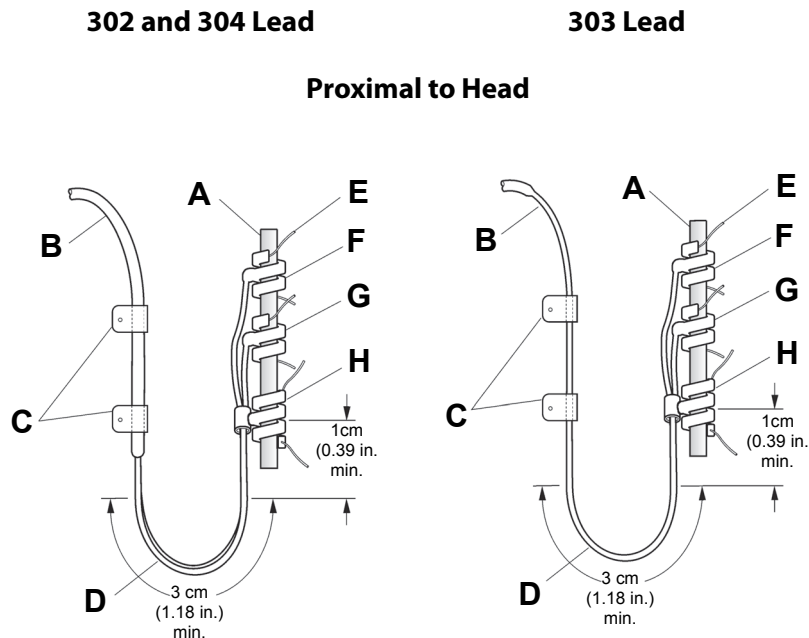
previously placed anchor tether and electrodes so they do not come unattached. Slight pressure may be placed against the anchor tether with a surgical instrument to ensure support to the anchor tether while the strain relief bend is being formed [see Figure 107]). The parallel portion can be placed in a pocket formed adjacent to the anchor tether.

Figure 107. (303 Lead only) Use of Surgical Tool (e.g., forceps) to Support Anchor Tether During Strain Relief Formation



2. Loosely attach the 3-cm strain relief bend to the adjacent fascia with tie-downs before routing the lead over the muscle. The first tie-down should be positioned laterally to the anchor tether (see Figure 108). Four (or more) tie-downs are provided in the lead sales pack.

Figure 108. Use of Tie-downs in Electrode Placement



Distal to Head

- | | |
|-----------------------------|---------------------------------------|
| A Vagus Nerve | E Suture |
| B Lead Body | F Electrode [Green Suture (-)] |
| C Tie Downs | G Electrode [White Suture (+)] |
| D Strain Relief Bend | H Anchor Tether (Green Suture) |

Caution: Sutures that are part of the lead coil are meant to assist in electrode placement around the left vagus nerve. These sutures should *not* be tied to each other since this may cause nerve damage (see Figure 108).

Caution: The lead and its electrodes are **very delicate**, and care should be taken not to over stretch or crush the helices.

12.6.4.2 Form the strain relief loop

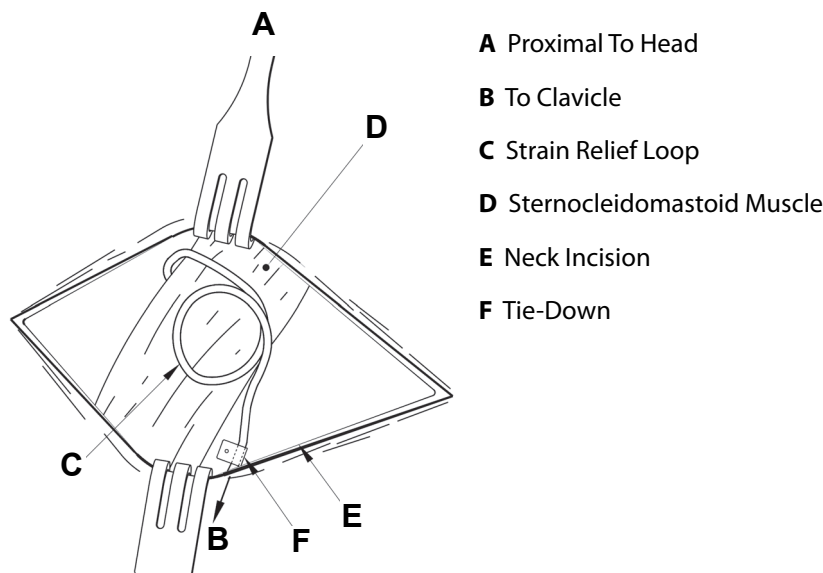
To form the *strain relief loop* (see Figure 109), do the following above the sternocleidomastoid muscle:

1. In the neck, form the lead into a large subcutaneous loop.
2. Loosely attach it to fascia with a tie-down before routing the lead over the clavicle. This strain relief loop should be large enough to provide several inches/centimeters of lead extension when the neck is turned to its maximum stretched positions.

Caution: Leave enough extra lead on both sides of the clavicle to prevent the tension over the clavicle from damaging the lead.

Caution: Placing the sutures directly on the lead body may result in insulation damage or wire failure, causing premature failure of the lead. Use only supplied tie-downs to secure the lead.

Figure 109. Strain Relief Loop



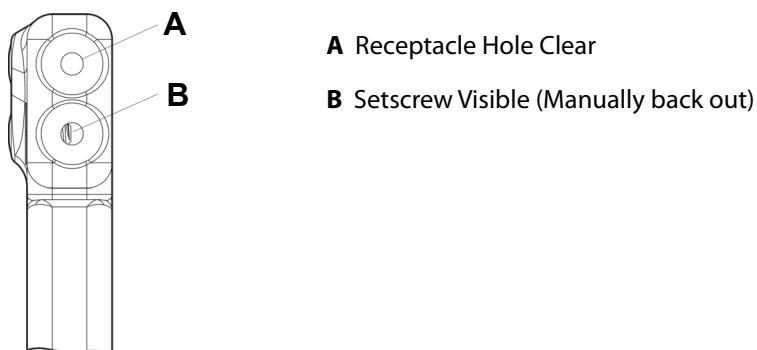
12.7 Connect the Lead to the Generator

Caution: Do not use electrosurgical equipment after the generator has been introduced to the sterile field. Exposure to this equipment may damage the generator.

To connect the lead directly to the generator:

1. Look inside the generator lead receptacle(s) to verify that no obstruction exists and that the setscrew(s) has been backed out adequately to allow full insertion of the connector pin(s). Avoid backing the setscrew(s) out further than needed for lead insertion (see Figure 110). The figure is intended to show the contrast between a blocked and a clear receptacle, and applies to single or dual pin headers.

Figure 110. Generator Receptacle and Setscrew



2. Keep the hex screwdriver perpendicular to the generator. Insert the hex screwdriver through the center of the setscrew plug(s) to vent back pressure accumulated during lead insertion.

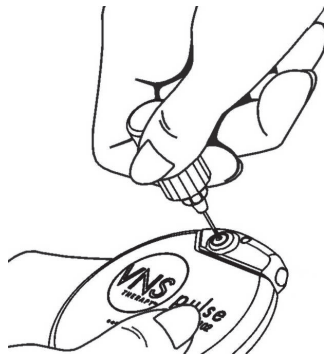


Caution: In the steps below, **always push down on the hex screwdriver while turning it clockwise until it clicks** (begins ratcheting) while ensuring that it is fully inserted in the setscrew. Also, the hex screwdriver must be inserted into the center of the silicone rubber setscrew plug and kept perpendicular to the generator to avoid stripping the setscrew and/or dislodging the setscrew plug.



Caution: When using the hex screwdriver, grasp it by the handle only, as shown in Figure 111. Do not grasp any other portion of the hex screwdriver during use, as this may affect its proper function. Touching the metal shaft while the hex screwdriver is engaged with the setscrew can conduct an electrostatic discharge into the device circuitry and may damage the generator.

Figure 111. Hex Screwdriver Position



3. When using a **single-receptacle** generator and VNS Therapy single-pin lead, insert the lead connector pin fully into the generator header. To allow escape of the back pressure created by insertion, leave the tip of the hex screwdriver in the slit in the setscrew plug.

When using a **dual-receptacle** generator and VNS Therapy dual-pin lead, insert the lead connector pins fully into the appropriate lead receptacles in the generator header. To allow escape of the back pressure created by insertion, leave the tip of the hex screwdriver in the slit in the setscrew plug of the connector being inserted. Insert the lead connector with the white marker band and with the embedded model number and serial number tag into the lead receptacle labeled “+” (see the Dual-Receptacle generator portion of Figure 112). The remaining lead connector is inserted into the remaining lead receptacle.

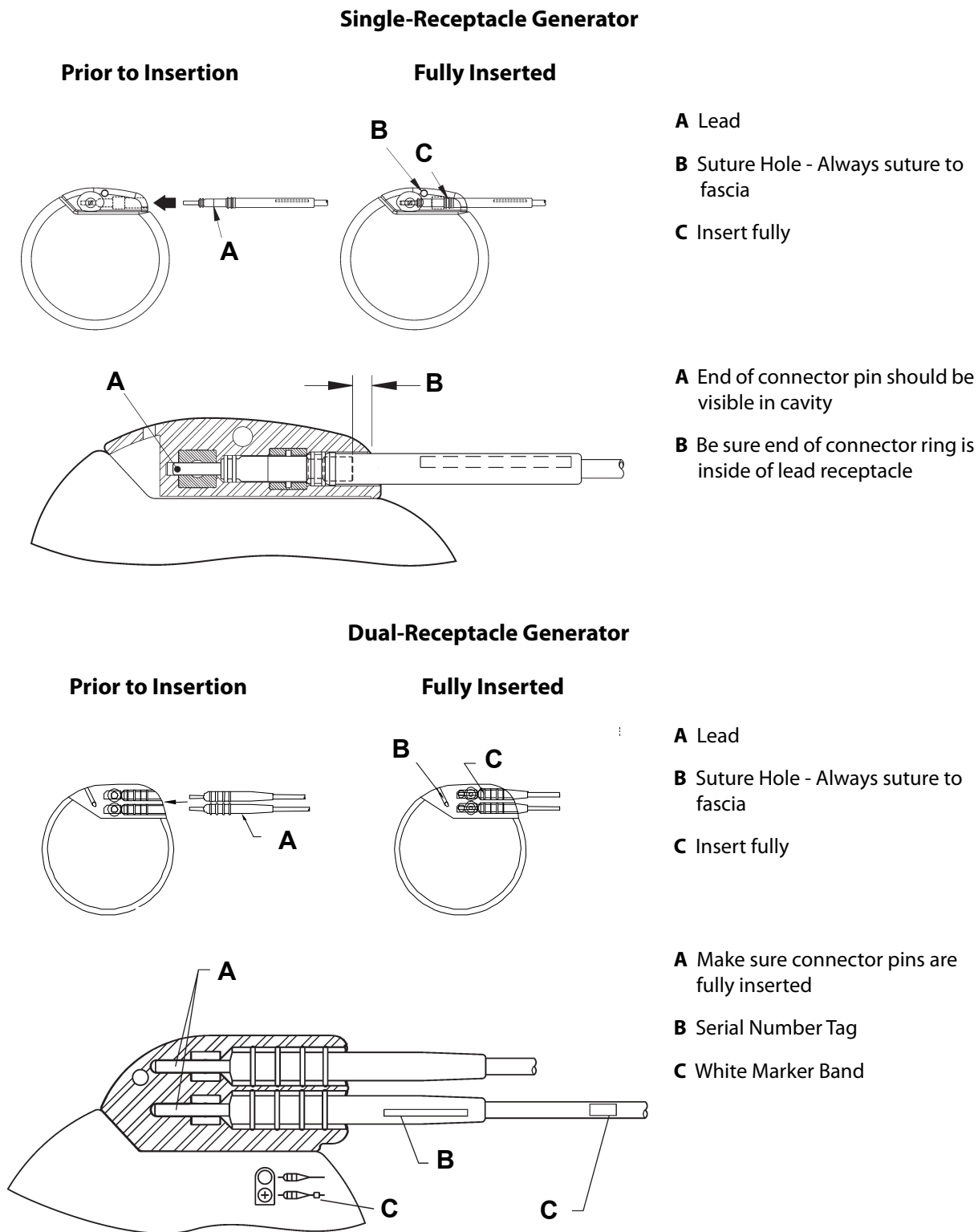


Caution: To avoid backing the setscrew out completely when loosening, during surgery, use no more than two counterclockwise turns.



Caution: Reversal of lead polarity has been associated with an increased chance of bradycardia in animal studies. It is important to make sure that the lead connector pins in the VNS Therapy dual-pin lead are correctly inserted (white marker band to + connection) into the generator dual receptacles.

Figure 112. Lead Connector(s) Prior to Insertion and Fully Inserted



4. With the hex screwdriver still inserted through the setscrew plug, verify that the connector pin is fully inserted. The pin should be visible in the area at the back end of the setscrew connector block. If it is not, remove the pin. To loosen the setscrew, engage the hex screwdriver into the setscrew, and turn it counterclockwise until the

connector pin can be fully inserted. Avoid backing the setscrew out further than needed for lead insertion. If using the dual-receptacle generator, repeat this procedure for each setscrew.

5. After verifying that the connector pin(s) has been fully inserted, tighten each setscrew by engaging the setscrew with the hex screwdriver and turning the hex screwdriver clockwise until it begins to click. Always push in on the hex screwdriver while turning it to ensure that the hex screwdriver is fully inserted in the setscrew.



Caution: It is important to do the following:

- ◆ Ensure that the lead receptacle(s) is clean and free of obstruction.
- ◆ Carefully insert the lead connector pin(s) into the lead receptacle(s) without bending the lead connector(s).
- ◆ Visually inspect that the connector pin(s) is clean and completely inserted.
- ◆ **Electrical connection to the generator is not established until the setscrew(s) is completely tightened with the hex screwdriver.** Failure to make a good connection can result in HIGH impedance during a System Diagnostics (Lead Test) or erratic stimulation at varying intensity due to rapid, unpredictable changes in lead impedance, which is expected to adversely affect device effectiveness and may have serious safety consequences. Additionally, for generators capable of tachycardia detection, heartbeat sensing may be compromised.
- ◆ Gently grasp and pull on lead connector boot(s) (the thick section of the lead) to verify the lead is properly secured inside the lead receptacle(s). Do not pull on lead body (thin section) or use excessive pull force, because doing so may cause lead damage.

12.8 Test the VNS Therapy System

The System Diagnostics (Lead Test), which should be conducted first, is performed with the lead and the generator connected. Thus, if the System Diagnostics (Lead Test) is successful, both components are working properly. However, if the System Diagnostics (Lead Test) fails, either of the two components could be defective, or there may not be a good electrical connection between the generator and the lead connector pin(s). If a defective component is suspected, disconnect the lead and perform the optional Generator Diagnostics (Pre-Implant Test), using the resistor assembly supplied with the generator or accessory pack.



Note: The programming wand should be placed into a sterile laser arm bag or equivalent (not provided by LivaNova) in order to introduce the programming wand into the sterile field. See the programming system physician's manuals for more information.



Caution: During the intraoperative System Diagnostics (Lead Test), infrequent incidents of bradycardia and/or asystole have occurred. If asystole, severe bradycardia (heart rate < 40 bpm), or a clinically significant change in heart rate is encountered during a System Diagnostics (Lead Test) or during initiation of stimulation, physicians should be prepared to follow guidelines consistent with Advanced Cardiac Life Support (ACLS).

Additionally, postoperative bradycardia can occur among patients with certain underlying cardiac arrhythmias. If a patient has experienced asystole, severe bradycardia (heart rate < 40 bpm), or a clinically significant change in heart rate during a System Diagnostics (Lead Test) at the time of initial device implantation, the patient should be placed on a cardiac monitor during initiation of stimulation.

The safety of this therapy has not been systematically established for patients experiencing bradycardia or asystole during VNS Therapy System implantation.

12.8.1 System Diagnostics (Lead Test)

The System Diagnostics is performed intraoperatively when the lead and the generator are connected. The test checks the connection between the lead, generator, and the nerve. Depending on the generator model and programmed Normal Mode output current, different test pulses may be conducted during the test (see Table 103).

Table 103. System Diagnostics (Lead Test)

Normal Mode Output Current	System Diagnostics Behavior		
	M102/102R	M103-106	M1000
0 mA	1 mA, 500 µsec for approximately 14 seconds	1 mA, 500 µsec for approximately 14 seconds	One brief pulse at 0.25 mA, 130 µsec, followed by delivery of programmed output for approximately 4 seconds.
>0 mA		One brief pulse at 0.25 mA, 130 µsec, followed by delivery of programmed output for the duration of the programmed ON time.	

To ensure proper system connection and functionality, perform the test and assess the following:

- Model 102/102R: Verify that the lead impedance status is “OK”.



Note: See the programming system physician’s manuals for lead impedance details.

- Model 103-1000: Verify that System Diagnostics is successful (output current and lead impedance are “OK”).

If lead impedance status is not “OK” for Model 102/102R, or if the System Diagnostics fails for Model 103-1000 (output current “LOW” or lead impedance “HIGH” or “LOW”) see “Troubleshooting” in the programming system physician’s manuals.



Caution: Electrical connection between the generator and the lead connector pin(s) may be at fault.

12.8.2 Generator Diagnostics (Pre-Implant Test)

The optional Generator Diagnostics is performed when the test resistor is attached to the generator in cases of troubleshooting. When the System Diagnostics fails (lead impedance “HIGH” or “LOW”), the Generator Diagnostics can be used to determine whether the lead or the generator is causing the problem. The Generator Diagnostics is performed with the test resistor that is included in the generator sales pack or accessory pack. This test will verify that the generator is functioning properly, independent of the lead.

To connect the test resistor to the generator, perform these steps:

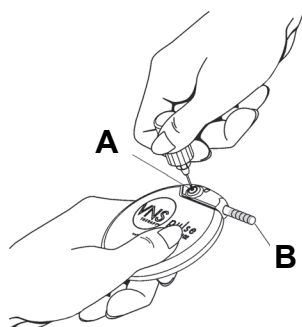
1. Remove the lead connector pin(s) from the lead receptacles by inserting the hex screwdriver through the center of the setscrew plug(s) and loosening the setscrew(s). Avoid backing out the setscrew(s) more than necessary to remove the lead. No more than a half-turn should be required to remove the lead.
2. Insert the connector pin(s) of the resistor assembly into the lead receptacle(s). Be careful while inserting the test resistor pin(s) into the lead receptacle(s). If binding or significant resistance is felt, remove the test resistor, inspect it, and clean it if necessary. Without the use of excessive force, reinsert the test resistor.

Note: Fully insert the hex screwdriver into the setscrew and push in on the hex screwdriver whenever the setscrew(s) is being tightened or loosened.

3. When the resistor assembly is in place, tighten the setscrew(s) until the hex screwdriver begins to click (see Figure 113). Again, always push in on the hex screwdriver while turning it to ensure that the hex screwdriver is fully inserted in the setscrew.

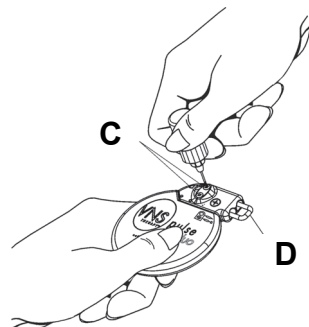
Figure 113. Connect the Resistor Assembly

Single-Receptacle Generator



- A** Setscrew Plug
B Resistor Assembly

Dual-Receptacle Generator



- C** Setscrew Plugs (2)
D Resistor Assembly

4. Perform the Generator Diagnostics (Pre-Implant Test).
 - If the Generator Diagnostics (Pre-Implant Test) is successful (lead impedance “OK”), the generator is working properly.
 - If the Generator Diagnostics fails (lead impedance “HIGH” or “LOW”), see “Troubleshooting” in the programming system physician’s manuals.

- If the component is damaged, contact LivaNova and return the item (following the disinfection procedure described in the “Precautions” section of the *Introduction to the VNS Therapy System* chapter), along with a completed Returned Product Form.

 **Note:** See the programming system physician’s manuals for details.

12.8.3 Optional Monitoring

Optional physiologic monitoring of VNS Therapy System operation may be done if surgery is performed under local anesthesia. Monitor the patient’s voice for signs of hoarseness while gradually increasing the generator output current. After performing the System Diagnostics and obtaining successful results, reset the current to 0 mA.

12.8.4 Heartbeat Detection and Tachycardia Detection Configuration

For generators capable of tachycardia detection, configure the heartbeat detection and tachycardia detection functions after the diagnostic testing is complete:

1. Place the generator in the chest pocket, coiling the remaining slack of the lead and placing it to the side of the generator. The generator can be placed with either side facing outward.
2. Use the programming software to turn on tachycardia detection and verify heartbeat detection.

 **Note:** Steps for configuring tachycardia and heartbeat detection are software specific. See the programming system physician’s manuals for details.

3. Use the following method to select a patient-specific **Heartbeat Detection** (sensitivity) value:
 - Average the two R-wave amplitude measurements obtained from the positional assessment. If this information is not available, go to step 6.
 - Map the average R-wave amplitude value to the appropriate Heartbeat Detection setting in Table 104 and select this value in the programming software.

 **Note:** To determine R-wave amplitude, see Step 6 in the “Determine acceptable device implant locations (tachycardia detection only)” section.

Table 104. Heartbeat Detection Mapping

Heartbeat Detection	Average Amplitude (mV) (across different postures)	
	Minimum	Maximum
5	0.40	0.50
4	0.51	0.70
3	0.71	0.85
2	0.86	1.25
1	1.26	--

4. During the heartbeat detection verification process, the programming software will display the heart rate detected by the generator for 2 minutes. The process will automatically stop after 2 minutes, or you may manually stop the process by pressing "Stop". The wand must stay over the generator during the entire process.
5. During the heartbeat detection verification process, use the ECG monitor to compare the heartbeat reported on the programming computer with that reported by the ECG monitor. If heartbeat detection is accurate, go to Step 8, Otherwise, go to Step 6.
6. If heartbeat detection is inaccurate in Step 5, or if the R-wave amplitude information from the "Determine acceptable device implant locations (tachycardia detection only)" section is not available, select a value of "1" from the Heartbeat Detection parameter list (1-5) and repeat steps 4 - 5.
7. Monitor and compare the heartbeat reported on the programming computer with that reported by the ECG monitor, and repeat Steps 4 - 5, as necessary, to test or configure other Heartbeat Detection settings (settings 2, 3, 4, 5) until the device accurately detects heartbeats. If more than one heartbeat detection setting results in accurate detection of heartbeats, select the lesser of these detection settings.
8. Select the **AutoStim Threshold** as appropriate (70%-least sensitive, 20%-most sensitive) and apply changes (i.e., program).
9. After configuration, proceed to "Complete the Implantation Procedure", step 2.

12.9 Complete the Implantation Procedure



Caution: Do not place the lead slack under the generator, because doing so could result in insulation failure and system malfunction.



Caution: This suturing is important to stabilize the generator and to prevent manipulation by the patient, which could damage the lead wires.



Caution: Do not place the sutures directly around the body of the lead; this could result in insulation failure and system malfunction, and possible lead breakage.

After the testing has been completed, finish the implantation procedure:

1. Place the generator in the chest pocket, coiling the remaining slack of the lead and placing it to the side of the generator. The generator can be placed with either side facing outward.
2. Secure the generator by placing a suture through the suture hole and attaching it to fascia (not to muscle).
3. Perform the second System Diagnostics and verify lead impedance status remains "OK."
4. Interrogate the generator to verify that Normal Mode, Magnet Mode, and AutoStim Mode (*106 and 1000 generators only*) output is 0 mA.
 - Output current: 0 mA
 - Magnet current: 0 mA
 - AutoStim current: 0 mA



Caution: Do not program the generator to an ON or periodic stimulation treatment for at least 14 days after the initial or replacement implantation. Failure to observe this precaution may result in patient discomfort or adverse events.

5. LivaNova recommends irrigation of both incision sites with generous amounts of bacitracin or equivalent solution before closure.
6. Close the surgical incisions. Use cosmetic closure techniques to minimize scarring.
7. Administer antibiotics postoperatively (at the discretion of the physician).

A neck brace can be used by the patient for the first week to help ensure proper lead stabilization.

12.10 Post Implant Patient Identification and Materials

12.10.1 Implant Warranty and Registration Form

Included with the generator is an Implant Warranty and Registration Form that *must* be completed and the top, white copy returned to LivaNova. Give a copy of this form to the patient or caregiver.

This information, as required by government agencies, becomes part of the LivaNova' registry of implantees and is used as a permanent record of implant recipient information.

12.10.2 Patient Magnet Kit

Give the patient a Patient Magnet kit, which contains magnets, accessories, and patient implant cards.

12.10.3 Patient Implant Card

Implant cards are provided in the following sales packs:

- Lead
- Generator
- Patient Magnet

The implant card contains information about the patient's VNS Therapy System. Give the card(s) to the patient and/or caregiver after the implant and tell them to carry it with them at all times.

13 Revision / Replacement / Removal Procedure

13.1 Introduction

Revision, replacement, or removal of the VNS Therapy System or any component of the system may be desired for several reasons:

- Replacement of the generator may be required due to pending End-of-Service (EOS) of the generator or if EOS has been reached and the generator cannot communicate or provide therapy.
- Revision/replacement of the lead may be necessary if a broken or damaged lead is suspected, based on diagnostic testing or x-ray evaluation.
- Removal of the VNS Therapy System may be required in cases of infection or for certain medical procedures (e.g., MRI) contraindicated by the labeling (see the *Introduction to the VNS Therapy System* chapter).

 **Note:** Return explanted or opened and unused component(s) of the VNS Therapy System to the company. A Return Product Kit is available from Clinical Technical Services (CTS) at 1 (866) 882-8804 (U.S. and Canada), +1 (281) 228-7330 (Worldwide), or +32 2 790 27 73 (Europe/EMMEA).

The following instructions are intended to be general guidelines. If you have questions about the procedures, contact Clinical Technical Services (CTS) at 1 (866) 882-8804 (U.S. and Canada), +1 (281) 228-7330 (Worldwide), or +32 2 790 27 73 (Europe/EMMEA).

13.2 VNS Therapy Components and Surgical Materials

The following materials should be available before performing a revision of any component of the VNS Therapy System.

13.2.1 Dual-Receptacle Generator Replacement

- Primary and backup dual-receptacle generators
- Two backup single-receptacle generators

13.2.2 Single-Receptacle Generator Replacement

- Primary and backup single-receptacle generators

13.2.3 Other Necessary VNS Therapy Components and Surgical Materials

- Primary and backup single-pin leads

 **Note:** Revision surgeries involving dual-pin leads require the availability of a new single-pin lead, and both single-receptacle and dual-receptacle generators.

- Tunneler
- Accessory pack
- Programming system
- Sterile laser arm bag or equivalent (not provided by LivaNova)

- Soft vessel loops or silicone sheet (not provided by LivaNova)

13.3 VNS Therapy System Revisions

For all revision surgeries, the patient should consent pre-operatively to receiving a new generator and new lead in case either is damaged during the revision surgery.

13.3.1 Procedure - Replacement of the Generator

13.3.1.1 Pre-operative steps

1. Use the programming system to interrogate the existing generator and perform System Diagnostics (Lead Test) before the patient enters the OR.

 **Note:** For detailed information about Systems Diagnostics, see “Test the VNS Therapy System” in the *Implantation Procedure* chapter.

2. It is recommended that the surgeon review an x-ray of the generator to determine the routing of the lead. This helps to avoid inadvertent damage to the lead during dissection to remove the generator.
3. If System Diagnostics results indicate “HIGH” or “LOW” lead impedance or the x-ray review shows a gross discontinuity in the lead [lead break or pin(s) disconnected], proceed to “Procedure – Replacement of the VNS Therapy Lead”.
4. If System Diagnostics results indicate “OK” lead impedance, use the programming system, outside the sterile field in the OR, to interrogate the replacement generator. This ensures clear communication.

 **Note:** If the replacement generator is capable of tachycardia detection, the existing generator pocket location may need to be revised.

5. If the replacement generator is capable of tachycardia detection (i.e., M106 or M1000), verify that the current generator implant location satisfies the requirements outlined in “Determine acceptable device implant locations (tachycardia detection only)” section of the *Implantation Procedure* chapter. If the current implant location does not satisfy the minimum R-wave amplitude requirements, use the same procedure to identify a suitable location close to the original implant site to place the new generator.

 **Note:** If possible, try to place the replacement generator at rib 4 or above, so the patient can have the maximum flexibility for MRI post-operatively. See *MRI with the VNS Therapy System* for details.

6. Program the patient data into the new generator.

13.3.1.2 Intra-operative steps

 **Caution:** Do not use electrosurgical equipment after the new generator has been introduced to the sterile field. Exposure to this equipment may damage the generator.

1. With the lead pin(s) still connected, remove the existing generator from the pocket.
2. Open the new generator sales pack. Use the hex screwdriver to disconnect the existing generator from the implanted lead. Remove the lead connector pin(s) from the lead

receptacles by inserting the hex screwdriver through the center of the setscrew plug(s) and loosening the setscrew(s). Avoid backing out the setscrew(s) more than necessary to remove the lead. No more than half a turn should be required to remove the lead.



Caution: When using the hex screwdriver, grasp it by the handle only. Touching the metal shaft while the hex screwdriver is engaged with the setscrew can conduct an electrostatic discharge into the device circuitry and may damage the generator.



Note: Extraneous pocket space left behind from the replacement of a larger generator with a smaller generator may increase the likelihood of certain adverse events (e.g., seroma, device manipulation, and device migration).



Note: Replacement of a smaller generator with a larger generator may require enlargement of the generator pocket during surgery. Physicians should assess the potential impact to post-surgical recovery time and likelihood of temporary patient discomfort due to surgical alteration of the generator pocket.

3. Connect the replacement generator to the lead following the steps in the “Connect the Lead to the Generator” section in the *Implantation Procedure* chapter and complete the remainder of the implantation procedure.

13.3.2 Procedure – Replacement of the VNS Therapy Lead



Note: Consult the prescribing physician before the surgery to determine parameter settings following placement of the new generator.

13.3.2.1 Pre-operative steps

1. Use the programming system to interrogate the existing generator and perform System Diagnostics (Lead Test) before the patient enters the OR. LivaNova recommends that the surgeon review x-rays to confirm the existence of a lead discontinuity [lead break or pin(s) disconnected], if possible.
2. If System Diagnostics results indicate “OK” lead impedance, there is no gross discontinuity in the lead from the x-ray review, and a short-circuit condition is not suspected, the implanted lead is functioning properly. Reassess proceeding with surgery or, if replacement of the generator is still desired, proceed to “Procedure - Replacement of the Generator”.
3. If System Diagnostic results indicate “HIGH” or “LOW” lead impedance or a gross lead discontinuity is observed, surgical intervention is required. Use the programming system, outside the sterile field in the OR, to interrogate all potential replacement generators. This ensures clear device communication.
4. If the replacement generator is capable of tachycardia detection (i.e., M106 or M1000), verify that the current generator implant location satisfies the requirements outlined in “Determine acceptable device implant locations (tachycardia detection only)” section of the *Implantation Procedure* chapter. If the current implant location does not satisfy the minimum R-wave amplitude requirements, use the same procedure to identify a suitable location close to the original implant site to place the new generator.
5. Proceed to “Intra-operative steps” below.

13.3.2.2 Intra-operative steps

 **Note:** For complete troubleshooting steps, see “Troubleshooting” in the programming system physician’s manuals.

13.3.2.2.1. “HIGH” lead impedance on System Diagnostics

If “HIGH” lead impedance is reported, perform the following steps:

1. With lead pin(s) still connected, remove the existing generator from the pocket.
2. Open the accessory pack and remove the hex screwdriver and test resistor.
3. Remove the lead connector pin(s) from the lead receptacle(s) by inserting the hex screwdriver through the center of the setscrew plug(s) and loosening the setscrew(s). Avoid backing out the setscrew(s) more than necessary to remove the lead. No more than a half turn should be required to remove the lead.
4. If foreign material (e.g., blood) is observed in the generator receptacle(s), flush the receptacle(s) with saline to remove the foreign material. Drain the excess fluid from the receptacle(s). Do not place any object (other than the connector pin) into the receptacle. Use saline to clean the lead connector pin(s), then wipe dry.
5. Re-insert the existing lead connector pin(s) into the existing generator following proper lead insertion techniques.

 **Note:** For proper lead insertion techniques, see “Connect the Lead to the Generator” in the *Implantation Procedure* chapter.



Caution: Visually inspect that the connector pin(s) is clean and completely inserted.

6. Introduce the programming system into the sterile field with a sterile laser arm bag (or equivalent) and perform an interrogation followed by System Diagnostics.
7. Record System Diagnostics results.
 - If the results indicate “OK” lead impedance, the previous “HIGH” lead impedance was resolved and the system appears to be functioning properly. Assess replacement of the generator.

If replacement of the generator is not desired, verify that all relevant steps outlined in “Test the VNS Therapy System” section of the *Implantation Procedure* chapter have been completed. Finish the procedure by following the steps in “Complete the Implantation Procedure” section in the *Implantation Procedure* chapter.

If replacement of the generator is desired, open a new compatible generator sales pack. Connect the replacement generator to the lead following the steps in the “Connect the Lead to the Generator” section in the *Implantation Procedure* chapter and complete the remainder of the implantation procedure. Ensure appropriate patient data has been programmed into the new generator.

 **Note:** The prescribing physician will program the stimulation parameters post-operatively based on the patient’s tolerance to the stimulation.

- If System Diagnostics results continue to report “HIGH” lead impedance, perform Generator Diagnostics (Pre-Implant Test) with the test resistor assembly from the

accessory pack to verify that the generator is functioning properly, independent of the lead. To perform Generator Diagnostics, follow the steps in "Generator Diagnostics (Pre-Implant Test)" below.

13.3.2.2.2. "LOW" lead impedance on System Diagnostics

Note: For complete troubleshooting steps see "Troubleshooting" in the programming system physician's manuals.

If System Diagnostics report "LOW" lead impedance, perform Generator Diagnostics (Pre-Implant Test) with the test resistor assembly from the accessory pack to verify that the generator is functioning properly, independent of the lead.

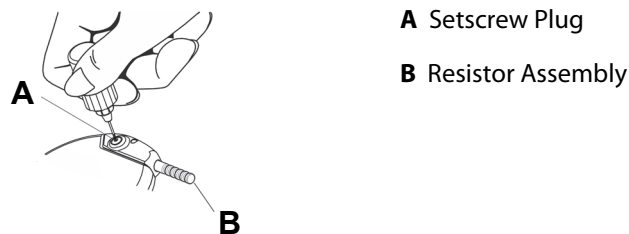
To perform Generator Diagnostics (Pre-Implant Test), follow the steps in "Generator Diagnostics (Pre-Implant Test)" below.

13.3.2.3 Generator Diagnostics (Pre-Implant Test)

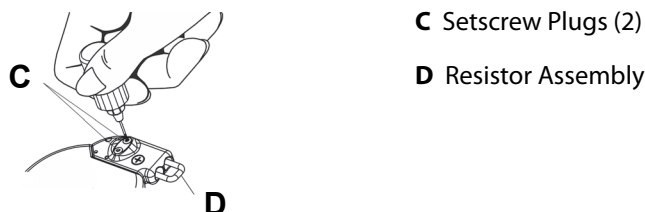
1. Insert the connector pin(s) of the resistor assembly into the lead receptacle(s). Be careful while inserting the test resistor pin(s) into the lead receptacle(s). If binding or significant resistance is felt, remove the test resistor, inspect it, and clean it if necessary. Without the use of excessive force, reinsert the test resistor.
2. When the resistor assembly is in place, tighten the setscrew(s) until the hex screwdriver begins to click (see Figure 114). Always push in on the hex screwdriver while turning it to ensure that the hex screwdriver is fully inserted in the setscrew.

Figure 114. Connect the Resistor Assembly

Models 102, 103, 105, 106 and 1000



Models 102R and 104



3. Perform Generator Diagnostics (Pre-Implant Test).

Note: For details, see the programming system physician's manuals.

- If Generator Diagnostics results indicate “HIGH” or “LOW” lead impedance, call CTS at 1 (866) 882-8804 (U.S. and Canada), +1 (281) 228-7330 (Worldwide), or +32 2 790 27 73 (Europe/EMMEA).
- If Generator Diagnostics results indicate “OK” lead impedance, the implanted lead should be replaced and Generator replacement assessed.

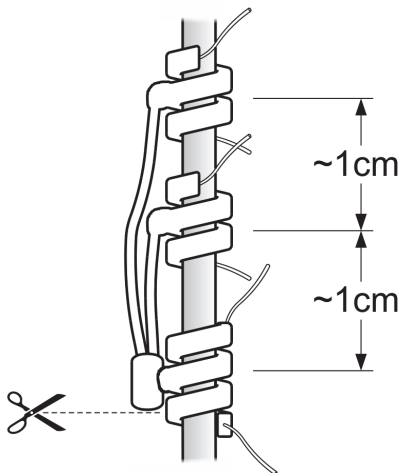
13.3.2.4 Remove existing helices and lead



Caution: Lead replacement or removal is a medical judgment that must be carefully weighed against the known and unknown risks of surgery. At present there are no known long-term hazards or risks associated with leaving the lead implanted, beyond those mentioned in this physician’s manual.

1. Open the neck incision and locate the vagus nerve/helices interface.
2. Assess the degree of fibrotic encapsulation to determine if the entire lead can be removed safely.
 - If removal of the existing helices can be accomplished, the new helices may be placed in the same location.
 - If complete removal of the helices from the nerve is not possible, transect as much of the lead as possible. With ≤ 2 cm of the lead remaining (see Figure 115) a full body MRI using the body coil to transmit RF is allowable. (See the *MRI with the VNS Therapy System* chapter for further details.).
- If it is not possible to leave ≤ 2 cm, then MRI can still be performed for brain or extremity imaging with the appropriate type of T/R coil. (See *MRI with the VNS Therapy System* chapter for further details.)

Figure 115. Transected Lead (≤ 2 cm)



3. The replacement helices can be placed above or below the existing helices if they must remain.

13.3.2.5 Complete the procedure

Complete the remainder of the implant procedure per the *Implantation Procedure* chapter, starting with the steps in the “Implant the lead” section. Pay particular attention to all cautions and warnings regarding the cardiac branches.

 **Note:** The prescribing physician will program the stimulation parameters post-operatively after the recommended 2-week recovery period to allow the nerve to heal.

13.4 Removal of the VNS Therapy System

If removal is medically necessary, LivaNova recommends removing as much of the VNS Therapy System as can be safely accomplished:

- Assess the degree of fibrotic in-growth in and around the helices.
- Remove the entire system, if possible.
- If fibrotic encapsulation hinders safe removal of the entire system, transect as much of the lead wire as possible (see Figure 115).
- Removal of the generator alone does not alter the hazards associated with certain MRI procedures.

 **Note:** For detailed information, see the *MRI with the VNS Therapy System* chapter.

- Diathermy procedures are contraindicated for patients with any portion of the VNS Therapy System remaining in the body.

 **Note:** For detailed information regarding the use of diathermy with VNS, see the *Introduction to the VNS Therapy System* chapter.

14 **Information and Support**

If there are questions regarding use of the VNS Therapy System or any of its accessories, contact LivaNova:



Manufacturer

LivaNova USA, Inc.

100 Cyberonics Boulevard

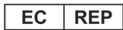
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Internet

www.livanova.com

15 Appendices

15.1 Appendix A — Model 103/104 Battery Longevity and Programmed Setting Choices

Parameters at 3kOhms (M103/104)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	μS	Years	Years	Years	Years	Years	Years	Years	Years	Years
0.5	10	130	>10	>10	>10	2.8	2.5	2.4	2.2	2.0	1.9
0.5	15	130	>10	>10	>10	2.7	2.2	1.9	2.1	1.7	1.5
0.5	20	130	>10	>10	>10	2.5	1.9	1.7	2.0	1.5	1.3
0.5	25	130	>10	>10	>10	2.4	1.7	1.4	1.9	1.4	1.2
0.5	30	130	>10	>10	9.5	2.3	1.6	1.3	1.8	1.3	1.0
0.5	10	250	>10	>10	>10	2.7	2.3	2.0	2.1	1.8	1.6
0.5	15	250	>10	>10	>10	2.5	1.9	1.6	2.0	1.5	1.3
0.5	20	250	>10	>10	>10	2.4	1.7	1.4	1.9	1.3	1.1
0.5	25	250	>10	>10	8.7	2.3	1.5	1.2	1.8	1.2	0.9
0.5	30	250	>10	9.8	7.6	2.1	1.3	1.0	1.7	1.0	0.8
0.5	10	500	>10	>10	>10	2.5	1.9	1.6	1.9	1.5	1.2
0.5	15	500	>10	>10	8.9	2.3	1.5	1.2	1.8	1.2	0.9
0.5	20	500	>10	9.3	7.2	2.1	1.2	1.0	1.6	1.0	0.8
0.5	25	500	>10	8.1	6.1	1.9	1.1	0.8	1.5	0.9	0.6
0.5	30	500	>10	7.1	5.2	1.8	0.9	0.7	1.4	0.8	0.6
0.5	10	750	>10	>10	9.4	2.3	1.6	1.3	1.8	1.2	1.0
0.5	15	750	>10	9.1	7.0	2.1	1.2	0.9	1.6	1.0	0.7
0.5	20	750	>10	7.5	5.6	1.9	1.0	0.7	1.5	0.8	0.6
0.5	25	750	>10	6.4	4.7	1.7	0.9	0.6	1.3	0.7	0.5
0.5	30	750	>10	5.5	4.0	1.5	0.7	0.5	1.2	0.6	0.4
0.5	10	1000	>10	>10	7.9	2.2	1.4	1.1	1.7	1.1	0.8
0.5	15	1000	>10	7.7	5.8	1.9	1.0	0.8	1.5	0.8	0.6
0.5	20	1000	>10	6.3	4.6	1.7	0.8	0.6	1.3	0.7	0.5
0.5	25	1000	>10	5.3	3.8	1.5	0.7	0.5	1.2	0.6	0.4
0.5	30	1000	>10	4.6	3.2	1.4	0.6	0.4	1.1	0.5	0.3
1	10	130	>10	>10	>10	2.6	2.1	1.9	2.0	1.5	1.3
1	15	130	>10	>10	>10	2.5	1.9	1.6	1.9	1.4	1.1
1	20	130	>10	>10	>10	2.4	1.6	1.3	1.8	1.2	0.9
1	25	130	>10	>10	9.3	2.2	1.5	1.2	1.7	1.1	0.8
1	30	130	>10	>10	8.2	2.1	1.3	1.0	1.6	1.0	0.8
1	10	250	>10	>10	>10	2.4	1.7	1.4	1.8	1.3	1.0
1	15	250	>10	>10	8.9	2.2	1.4	1.1	1.7	1.1	0.9
1	20	250	>10	9.4	7.2	2.1	1.2	0.9	1.6	0.9	0.7
1	25	250	>10	8.1	6.1	1.9	1.1	0.8	1.5	0.8	0.6

Parameters at 3kOhms (M103/104)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	μS	Years	Years	Years	Years	Years	Years	Years	Years	Years
1	30	250	>10	7.1	5.3	1.8	0.9	0.7	1.4	0.7	0.5
1	10	500	>10	>10	7.9	2.1	1.2	1.0	1.5	0.9	0.7
1	15	500	>10	7.8	5.8	1.8	1.0	0.7	1.4	0.7	0.5
1	20	500	>10	6.3	4.6	1.6	0.8	0.6	1.2	0.6	0.4
1	25	500	>10	5.3	3.8	1.5	0.7	0.5	1.1	0.5	0.4
1	30	500	>10	4.6	3.2	1.3	0.6	0.4	1.0	0.4	0.3
1	10	750	>10	8.0	6.0	1.8	1.0	0.7	1.3	0.7	0.5
1	15	750	>10	6.0	4.3	1.5	0.7	0.5	1.1	0.5	0.4
1	20	750	>10	4.7	3.4	1.3	0.6	0.4	1.0	0.4	0.3
1	25	750	9.3	3.9	2.8	1.2	0.5	0.3	0.9	0.4	0.3
1	30	750	8.3	3.4	2.3	1.1	0.4	0.3	0.8	0.3	0.2
1	10	1000	>10	6.6	4.9	1.6	0.8	0.6	1.2	0.5	0.4
1	15	1000	>10	4.8	3.4	1.3	0.6	0.4	1.0	0.4	0.3
1	20	1000	9.0	3.8	2.7	1.1	0.5	0.3	0.8	0.3	0.2
1	25	1000	7.8	3.1	2.2	1.0	0.4	0.3	0.7	0.3	0.2
1	30	1000	6.9	2.7	1.8	0.9	0.3	0.2	0.6	0.2	0.2
1.5	10	130	>10	>10	8.8	2.2	1.4	1.1	1.6	1.0	0.8
1.5	15	130	>10	>10	7.9	2.1	1.3	1.0	1.6	0.9	0.7
1.5	20	130	>10	9.3	7.1	2.0	1.1	0.9	1.5	0.8	0.6
1.5	25	130	>10	8.3	6.3	1.9	1.0	0.8	1.4	0.7	0.5
1.5	30	130	>10	7.6	5.7	1.8	0.9	0.7	1.3	0.6	0.5
1.5	10	250	>10	>10	8.8	2.1	1.3	1.0	1.5	0.8	0.6
1.5	15	250	>10	8.9	6.8	1.9	1.0	0.8	1.3	0.7	0.5
1.5	20	250	>10	7.5	5.6	1.7	0.9	0.6	1.2	0.6	0.4
1.5	25	250	>10	6.4	4.7	1.6	0.8	0.5	1.1	0.5	0.4
1.5	30	250	>10	5.6	4.0	1.4	0.7	0.5	1.0	0.5	0.3
1.5	10	500	>10	7.3	5.4	1.7	0.8	0.6	1.2	0.6	0.4
1.5	15	500	>10	5.7	4.1	1.4	0.7	0.5	1.0	0.4	0.3
1.5	20	500	>10	4.7	3.3	1.2	0.5	0.4	0.9	0.4	0.2
1.5	25	500	9.2	3.9	2.7	1.1	0.4	0.3	0.8	0.3	0.2
1.5	30	500	8.2	3.3	2.3	1.0	0.4	0.3	0.7	0.3	0.2
1.5	10	750	>10	5.3	3.8	1.4	0.6	0.4	0.9	0.4	0.3
1.5	15	750	9.5	4.1	2.9	1.1	0.5	0.3	0.8	0.3	0.2
1.5	20	750	8.1	3.3	2.3	1.0	0.4	0.3	0.6	0.2	0.2
1.5	25	750	7.0	2.7	1.9	0.8	0.3	0.2	0.6	0.2	0.1
1.5	30	750	6.2	2.3	1.6	0.7	0.3	0.2	0.5	0.2	0.1
1.5	10	1000	9.7	4.2	3.0	1.1	0.5	0.3	0.8	0.3	0.2
1.5	15	1000	7.8	3.1	2.2	0.9	0.4	0.2	0.6	0.2	0.2
1.5	20	1000	6.5	2.5	1.7	0.8	0.3	0.2	0.5	0.2	0.1

Parameters at 3kOhms (M103/104)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	μS	Years	Years	Years	Years	Years	Years	Years	Years	Years
1.5	25	1000	5.6	2.1	1.4	0.7	0.2	0.2	0.4	0.2	0.1
1.5	30	1000	4.9	1.8	1.2	0.6	0.2	0.1	0.4	0.1	0.1
2	10	130	>10	8.7	6.6	1.9	1.1	0.8	1.4	0.7	0.5
2	15	130	>10	7.2	5.3	1.7	0.9	0.6	1.2	0.6	0.4
2	20	130	>10	6.2	4.5	1.6	0.8	0.5	1.1	0.5	0.4
2	25	130	>10	5.5	4.0	1.4	0.7	0.5	1.0	0.5	0.3
2	30	130	>10	5.0	3.5	1.3	0.6	0.4	1.0	0.4	0.3
2	10	250	>10	6.4	4.7	1.6	0.8	0.6	1.2	0.5	0.4
2	15	250	>10	5.2	3.8	1.4	0.6	0.4	1.0	0.4	0.3
2	20	250	>10	4.4	3.1	1.2	0.5	0.4	0.9	0.4	0.3
2	25	250	9.1	3.8	2.7	1.1	0.5	0.3	0.8	0.3	0.2
2	30	250	8.3	3.4	2.3	1.0	0.4	0.3	0.7	0.3	0.2
2	10	500	9.5	4.1	2.9	1.2	0.5	0.3	0.8	0.3	0.2
2	15	500	7.8	3.1	2.2	1.0	0.4	0.3	0.7	0.3	0.2
2	20	500	6.7	2.6	1.8	0.8	0.3	0.2	0.6	0.2	0.1
2	25	500	5.9	2.2	1.5	0.7	0.3	0.2	0.5	0.2	0.1
2	30	500	5.2	1.9	1.3	0.6	0.2	0.1	0.4	0.1	0.1
2	10	750	7.5	2.9	2.0	0.9	0.3	0.2	0.6	0.2	0.2
2	15	750	5.9	2.2	1.5	0.7	0.3	0.2	0.5	0.2	0.1
2	20	750	5.0	1.8	1.2	0.6	0.2	0.1	0.4	0.1	0.1
2	25	750	4.3	1.5	1.0	0.5	0.2	0.1	0.3	0.1	0.1
2	30	750	3.7	1.3	0.9	0.4	0.1	0.1	0.3	0.1	0.1
2	10	1000	6.1	2.3	1.6	0.7	0.3	0.2	0.5	0.2	0.1
2	15	1000	4.7	1.7	1.1	0.6	0.2	0.1	0.4	0.1	0.1
2	20	1000	3.9	1.3	0.9	0.5	0.2	0.1	0.3	0.1	0.1
2	25	1000	3.3	1.1	0.8	0.4	0.1	0.1	0.3	0.1	0.1
2	30	1000	2.9	1.0	0.6	0.3	0.1	0.1	0.2	0.1	0.0
2.5	10	130	>10	7.2	5.3	1.7	0.9	0.6	1.3	0.6	0.5
2.5	15	130	>10	6.0	4.4	1.5	0.7	0.5	1.1	0.5	0.4
2.5	20	130	>10	5.1	3.7	1.4	0.6	0.4	1.0	0.4	0.3
2.5	25	130	>10	4.5	3.2	1.2	0.5	0.4	0.9	0.4	0.3
2.5	30	130	9.3	4.0	2.8	1.1	0.5	0.3	0.8	0.3	0.2
2.5	10	250	>10	5.4	3.9	1.4	0.6	0.5	1.0	0.4	0.3
2.5	15	250	9.6	4.1	2.9	1.2	0.5	0.3	0.8	0.3	0.2
2.5	20	250	8.4	3.4	2.4	1.0	0.4	0.3	0.7	0.3	0.2
2.5	25	250	7.4	2.9	2.0	0.9	0.3	0.2	0.6	0.2	0.2
2.5	30	250	6.7	2.5	1.7	0.8	0.3	0.2	0.5	0.2	0.1
2.5	10	500	8.0	3.2	2.2	1.0	0.4	0.3	0.7	0.3	0.2
2.5	15	500	6.3	2.4	1.6	0.8	0.3	0.2	0.5	0.2	0.1

Parameters at 3kOhms (M103/104)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	μS	Years	Years	Years	Years	Years	Years	Years	Years	Years
2.5	20	500	5.3	1.9	1.3	0.6	0.2	0.2	0.4	0.2	0.1
2.5	25	500	4.6	1.6	1.1	0.5	0.2	0.1	0.4	0.1	0.1
2.5	30	500	4.0	1.4	0.9	0.5	0.2	0.1	0.3	0.1	0.1
2.5	10	750	6.1	2.3	1.6	0.7	0.3	0.2	0.5	0.2	0.1
2.5	15	750	4.7	1.7	1.1	0.6	0.2	0.1	0.4	0.1	0.1
2.5	20	750	3.9	1.3	0.9	0.5	0.2	0.1	0.3	0.1	0.1
2.5	25	750	3.3	1.1	0.7	0.4	0.1	0.1	0.3	0.1	0.1
2.5	30	750	2.8	0.9	0.6	0.3	0.1	0.1	0.2	0.1	0.0
2.5	10	1000	5.0	1.8	1.2	0.6	0.2	0.1	0.4	0.1	0.1
2.5	15	1000	3.7	1.3	0.9	0.4	0.1	0.1	0.3	0.1	0.1
2.5	20	1000	3.0	1.0	0.7	0.4	0.1	0.1	0.2	0.1	0.1
2.5	25	1000	2.5	0.8	0.6	0.3	0.1	0.1	0.2	0.1	0.0
2.5	30	1000	2.2	0.7	0.5	0.3	0.1	0.1	0.2	0.1	0.0
3	10	130	>10	6.3	4.6	1.6	0.7	0.5	1.1	0.5	0.4
3	15	130	>10	5.0	3.6	1.3	0.6	0.4	1.0	0.4	0.3
3	20	130	9.6	4.2	2.9	1.2	0.5	0.3	0.8	0.3	0.2
3	25	130	8.6	3.6	2.5	1.0	0.4	0.3	0.7	0.3	0.2
3	30	130	7.8	3.1	2.2	0.9	0.4	0.3	0.7	0.2	0.2
3	10	250	>10	4.4	3.1	1.2	0.5	0.4	0.8	0.3	0.2
3	15	250	8.1	3.3	2.3	1.0	0.4	0.3	0.7	0.3	0.2
3	20	250	6.8	2.6	1.8	0.8	0.3	0.2	0.6	0.2	0.1
3	25	250	5.9	2.2	1.5	0.7	0.3	0.2	0.5	0.2	0.1
3	30	250	5.3	1.9	1.3	0.6	0.2	0.2	0.4	0.1	0.1
3	10	500	6.6	2.5	1.7	0.8	0.3	0.2	0.5	0.2	0.1
3	15	500	5.0	1.8	1.2	0.6	0.2	0.1	0.4	0.1	0.1
3	20	500	4.0	1.4	0.9	0.5	0.2	0.1	0.3	0.1	0.1
3	25	500	3.4	1.2	0.8	0.4	0.1	0.1	0.3	0.1	0.1
3	30	500	3.0	1.0	0.7	0.3	0.1	0.1	0.2	0.1	0.1
3	10	750	4.9	1.7	1.2	0.6	0.2	0.1	0.4	0.1	0.1
3	15	750	3.6	1.2	0.8	0.4	0.1	0.1	0.3	0.1	0.1
3	20	750	2.8	0.9	0.6	0.3	0.1	0.1	0.2	0.1	0.0
3	25	750	2.4	0.8	0.5	0.3	0.1	0.1	0.2	0.1	0.0
3	30	750	2.0	0.7	0.4	0.2	0.1	0.1	0.2	0.1	0.0
3	10	1000	3.8	1.3	0.9	0.4	0.2	0.1	0.3	0.1	0.1
3	15	1000	2.8	0.9	0.6	0.3	0.1	0.1	0.2	0.1	0.0
3	20	1000	2.2	0.7	0.5	0.3	0.1	0.1	0.2	0.1	0.0
3	25	1000	1.8	0.6	0.4	0.2	0.1	0.0	0.1	0.0	0.0
3	30	1000	1.5	0.5	0.3	0.2	0.1	0.0	0.1	0.0	0.0
3.5	10	130	>10	4.7	3.4	1.3	0.6	0.4	0.9	0.4	0.3

Parameters at 3kOhms (M103/104)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	μS	Years	Years	Years	Years	Years	Years	Years	Years	Years
3.5	15	130	9.0	3.8	2.6	1.1	0.4	0.3	0.8	0.3	0.2
3.5	20	130	7.7	3.1	2.1	0.9	0.4	0.3	0.6	0.2	0.2
3.5	25	130	6.8	2.6	1.8	0.8	0.3	0.2	0.6	0.2	0.1
3.5	30	130	6.1	2.3	1.6	0.7	0.3	0.2	0.5	0.2	0.1
3.5	10	250	8.0	3.2	2.2	1.0	0.4	0.3	0.7	0.3	0.2
3.5	15	250	6.4	2.4	1.7	0.8	0.3	0.2	0.5	0.2	0.1
3.5	20	250	5.3	1.9	1.3	0.6	0.2	0.2	0.4	0.2	0.1
3.5	25	250	4.6	1.6	1.1	0.5	0.2	0.1	0.4	0.1	0.1
3.5	30	250	4.0	1.4	0.9	0.5	0.2	0.1	0.3	0.1	0.1
3.5	10	500	4.7	1.7	1.1	0.6	0.2	0.1	0.4	0.1	0.1
3.5	15	500	3.6	1.2	0.8	0.4	0.1	0.1	0.3	0.1	0.1
3.5	20	500	3.0	1.0	0.7	0.3	0.1	0.1	0.2	0.1	0.1
3.5	25	500	2.5	0.8	0.5	0.3	0.1	0.1	0.2	0.1	0.0
3.5	30	500	2.1	0.7	0.5	0.2	0.1	0.1	0.2	0.1	0.0
3.5	10	750	3.2	1.1	0.7	0.4	0.1	0.1	0.3	0.1	0.1
3.5	15	750	2.4	0.8	0.5	0.3	0.1	0.1	0.2	0.1	0.0
3.5	20	750	2.0	0.6	0.4	0.2	0.1	0.0	0.2	0.1	0.0
3.5	25	750	1.7	0.5	0.4	0.2	0.1	0.0	0.1	0.0	0.0
3.5	30	750	1.4	0.5	0.3	0.2	0.1	0.0	0.1	0.0	0.0
3.5	10	1000	2.4	0.8	0.5	0.3	0.1	0.1	0.2	0.1	0.0
3.5	15	1000	1.9	0.6	0.4	0.2	0.1	0.0	0.2	0.0	0.0
3.5	20	1000	1.5	0.5	0.3	0.2	0.1	0.0	0.1	0.0	0.0
3.5	25	1000	1.3	0.4	0.3	0.1	0.0	0.0	0.1	0.0	0.0
3.5	30	1000	1.1	0.3	0.2	0.1	0.0	0.0	0.1	0.0	0.0

15.2 Appendix B — Model 105 Battery Longevity and Programmed Setting Choices

Parameters at 3kOhms (M105)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	µS	Years	Years	Years	Years	Years	Years	Years	Years	Years
0.5	10	130	>10	>10	>10	2.5	1.6	1.3	1.9	1.2	0.9
0.5	15	130	>10	>10	>10	2.5	1.5	1.2	1.8	1.1	0.9
0.5	20	130	>10	>10	>10	2.4	1.4	1.1	1.7	1.0	0.8
0.5	25	130	>10	>10	>10	2.2	1.3	1.0	1.7	1.0	0.7
0.5	30	130	>10	>10	>10	2.2	1.2	0.9	1.6	0.9	0.7
0.5	10	250	>10	>10	>10	2.5	1.6	1.3	1.9	1.2	1.0
0.5	15	250	>10	>10	>10	2.4	1.4	1.1	1.8	1.1	0.8
0.5	20	250	>10	>10	>10	2.3	1.3	1.0	1.7	1.0	0.7
0.5	25	250	>10	>10	>10	2.2	1.2	0.9	1.6	0.9	0.7
0.5	30	250	>10	>10	>10	2.1	1.1	0.8	1.5	0.8	0.6
0.5	10	500	>10	>10	>10	2.4	1.4	1.1	1.7	1.0	0.8
0.5	15	500	>10	>10	>10	2.2	1.2	0.9	1.6	0.9	0.7
0.5	20	500	>10	>10	>10	2.0	1.1	0.8	1.5	0.8	0.6
0.5	25	500	>10	>10	9.0	1.9	0.9	0.7	1.4	0.7	0.5
0.5	30	500	>10	>10	8.6	1.8	0.9	0.6	1.3	0.7	0.5
0.5	10	750	>10	>10	>10	2.2	1.3	1.0	1.7	0.9	0.7
0.5	15	750	>10	>10	>10	2.0	1.1	0.8	1.5	0.8	0.6
0.5	20	750	>10	>10	8.9	1.9	0.9	0.7	1.4	0.7	0.5
0.5	25	750	>10	>10	7.7	1.7	0.8	0.6	1.3	0.6	0.4
0.5	30	750	>10	9.6	6.8	1.6	0.7	0.5	1.2	0.5	0.4
0.5	10	1000	>10	>10	>10	2.1	1.2	0.9	1.6	0.9	0.6
0.5	15	1000	>10	>10	8.9	1.9	0.9	0.7	1.4	0.7	0.5
0.5	20	1000	>10	>10	7.3	1.7	0.8	0.6	1.2	0.6	0.4
0.5	25	1000	>10	9.2	6.5	1.5	0.7	0.5	1.1	0.5	0.4
0.5	30	1000	>10	8.0	5.7	1.4	0.6	0.4	1.0	0.4	0.3
1	10	130	>10	>10	>10	2.4	1.4	1.1	1.7	0.9	0.7
1	15	130	>10	>10	>10	2.3	1.4	1.0	1.6	0.9	0.7
1	20	130	>10	>10	>10	2.3	1.3	1.0	1.6	0.9	0.7
1	25	130	>10	>10	>10	2.2	1.2	0.9	1.5	0.8	0.6
1	30	130	>10	>10	>10	2.1	1.1	0.8	1.5	0.8	0.6
1	10	250	>10	>10	>10	2.2	1.2	0.9	1.5	0.8	0.6
1	15	250	>10	>10	>10	2.1	1.1	0.8	1.5	0.8	0.6
1	20	250	>10	>10	>10	2.0	1.0	0.7	1.4	0.7	0.5
1	25	250	>10	>10	9.7	1.9	0.9	0.7	1.3	0.6	0.5
1	30	250	>10	>10	8.9	1.8	0.8	0.6	1.2	0.6	0.4

Parameters at 3kOhms (M105)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	μS	Years	Years	Years	Years	Years	Years	Years	Years	Years
1	10	500	>10	>10	>10	2.0	1.0	0.7	1.3	0.6	0.5
1	15	500	>10	>10	9.6	1.8	0.8	0.6	1.2	0.5	0.4
1	20	500	>10	>10	7.8	1.6	0.7	0.5	1.1	0.5	0.3
1	25	500	>10	9.3	6.6	1.4	0.6	0.4	1.0	0.4	0.3
1	30	500	>10	8.4	5.9	1.3	0.6	0.4	0.9	0.4	0.3
1	10	750	>10	>10	9.7	1.7	0.8	0.6	1.2	0.5	0.4
1	15	750	>10	>10	7.4	1.5	0.7	0.5	1.0	0.4	0.3
1	20	750	>10	8.6	6.0	1.3	0.6	0.4	0.9	0.4	0.3
1	25	750	>10	7.3	5.1	1.2	0.5	0.3	0.8	0.3	0.2
1	30	750	>10	6.4	4.4	1.1	0.4	0.3	0.7	0.3	0.2
1	10	1000	>10	>10	8.0	1.5	0.7	0.5	1.0	0.4	0.3
1	15	1000	>10	8.8	6.2	1.3	0.5	0.4	0.9	0.4	0.2
1	20	1000	>10	7.1	4.9	1.1	0.5	0.3	0.8	0.3	0.2
1	25	1000	>10	6.0	4.1	1.0	0.4	0.3	0.7	0.3	0.2
1	30	1000	>10	5.1	3.5	0.9	0.3	0.2	0.6	0.2	0.2
1.5	10	130	>10	>10	>10	2.0	1.1	0.8	1.5	0.7	0.6
1.5	15	130	>10	>10	>10	1.9	1.0	0.7	1.4	0.7	0.5
1.5	20	130	>10	>10	9.4	1.8	0.9	0.7	1.3	0.6	0.5
1.5	25	130	>10	>10	8.8	1.8	0.8	0.6	1.3	0.6	0.4
1.5	30	130	>10	>10	7.8	1.7	0.8	0.6	1.2	0.6	0.4
1.5	10	250	>10	>10	9.3	1.8	0.9	0.6	1.3	0.6	0.4
1.5	15	250	>10	>10	7.9	1.6	0.7	0.5	1.1	0.5	0.4
1.5	20	250	>10	>10	7.6	1.6	0.7	0.5	1.1	0.5	0.3
1.5	25	250	>10	9.1	6.5	1.4	0.6	0.4	1.0	0.4	0.3
1.5	30	250	>10	8.5	6.0	1.3	0.6	0.4	0.9	0.4	0.3
1.5	10	500	>10	9.4	6.6	1.4	0.6	0.4	1.0	0.4	0.3
1.5	15	500	>10	7.4	5.2	1.2	0.5	0.3	0.8	0.3	0.2
1.5	20	500	>10	6.5	4.5	1.1	0.4	0.3	0.7	0.3	0.2
1.5	25	500	>10	5.7	4.0	1.0	0.4	0.3	0.7	0.2	0.2
1.5	30	500	>10	5.1	3.5	0.9	0.3	0.2	0.6	0.2	0.1
1.5	10	750	>10	7.2	5.0	1.2	0.5	0.3	0.8	0.3	0.2
1.5	15	750	>10	5.5	3.8	1.0	0.4	0.2	0.7	0.2	0.2
1.5	20	750	>10	4.7	3.2	0.8	0.3	0.2	0.6	0.2	0.1
1.5	25	750	>10	4.0	2.7	0.7	0.3	0.2	0.5	0.2	0.1
1.5	30	750	10.0	3.6	2.4	0.7	0.2	0.2	0.4	0.2	0.1
1.5	10	1000	>10	5.7	4.0	1.0	0.4	0.3	0.7	0.2	0.2
1.5	15	1000	>10	4.3	2.9	0.8	0.3	0.2	0.5	0.2	0.1
1.5	20	1000	9.9	3.5	2.4	0.7	0.2	0.2	0.4	0.2	0.1

Parameters at 3kOhms (M105)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	µS	Years	Years	Years	Years	Years	Years	Years	Years	Years
1.5	25	1000	8.7	3.1	2.1	0.6	0.2	0.1	0.4	0.1	0.1
1.5	30	1000	7.8	2.7	1.8	0.5	0.2	0.1	0.3	0.1	0.1
2	10	130	>10	>10	9.4	1.8	0.9	0.6	1.3	0.6	0.4
2	15	130	>10	>10	8.0	1.7	0.8	0.5	1.2	0.5	0.4
2	20	130	>10	9.8	7.0	1.5	0.7	0.5	1.1	0.5	0.3
2	25	130	>10	8.8	6.2	1.4	0.6	0.4	1.0	0.4	0.3
2	30	130	>10	8.1	5.7	1.3	0.6	0.4	0.9	0.4	0.3
2	10	250	>10	9.7	6.9	1.5	0.7	0.5	1.0	0.4	0.3
2	15	250	>10	8.2	5.7	1.3	0.5	0.4	0.9	0.4	0.3
2	20	250	>10	6.8	4.7	1.1	0.5	0.3	0.8	0.3	0.2
2	25	250	>10	5.9	4.1	1.0	0.4	0.3	0.7	0.3	0.2
2	30	250	>10	5.2	3.6	0.9	0.4	0.2	0.6	0.2	0.2
2	10	500	>10	6.5	4.5	1.1	0.4	0.3	0.7	0.3	0.2
2	15	500	>10	5.0	3.4	0.9	0.3	0.2	0.6	0.2	0.1
2	20	500	>10	4.0	2.7	0.7	0.3	0.2	0.5	0.2	0.1
2	25	500	9.6	3.4	2.3	0.7	0.2	0.2	0.4	0.2	0.1
2	30	500	8.7	3.0	2.1	0.6	0.2	0.1	0.4	0.1	0.1
2	10	750	>10	4.8	3.3	0.9	0.3	0.2	0.6	0.2	0.1
2	15	750	>10	3.6	2.4	0.7	0.2	0.2	0.4	0.2	0.1
2	20	750	8.1	2.8	1.9	0.5	0.2	0.1	0.4	0.1	0.1
2	25	750	7.0	2.4	1.6	0.5	0.2	0.1	0.3	0.1	0.1
2	30	750	6.2	2.1	1.4	0.4	0.1	0.1	0.3	0.1	0.1
2	10	1000	>10	3.8	2.6	0.7	0.2	0.2	0.5	0.2	0.1
2	15	1000	8.0	2.8	1.9	0.5	0.2	0.1	0.4	0.1	0.1
2	20	1000	6.5	2.2	1.5	0.4	0.1	0.1	0.3	0.1	0.1
2	25	1000	5.5	1.8	1.2	0.4	0.1	0.1	0.2	0.1	0.1
2	30	1000	4.8	1.6	1.1	0.3	0.1	0.1	0.2	0.1	0.0
2.5	10	130	>10	>10	8.3	1.7	0.8	0.6	1.2	0.5	0.4
2.5	15	130	>10	9.6	6.8	1.5	0.6	0.5	1.0	0.4	0.3
2.5	20	130	>10	8.5	6.0	1.4	0.6	0.4	0.9	0.4	0.3
2.5	25	130	>10	7.4	5.2	1.2	0.5	0.4	0.9	0.3	0.2
2.5	30	130	>10	6.7	4.7	1.1	0.5	0.3	0.8	0.3	0.2
2.5	10	250	>10	8.3	5.9	1.3	0.6	0.4	0.9	0.4	0.3
2.5	15	250	>10	6.5	4.5	1.1	0.4	0.3	0.8	0.3	0.2
2.5	20	250	>10	5.5	3.8	1.0	0.4	0.3	0.7	0.2	0.2
2.5	25	250	>10	4.6	3.2	0.8	0.3	0.2	0.6	0.2	0.1
2.5	30	250	>10	4.1	2.8	0.8	0.3	0.2	0.5	0.2	0.1
2.5	10	500	>10	5.4	3.7	0.9	0.4	0.2	0.6	0.2	0.2

Parameters at 3kOhms (M105)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	µS	Years	Years	Years	Years	Years	Years	Years	Years	Years
2.5	15	500	>10	4.0	2.7	0.7	0.3	0.2	0.5	0.2	0.1
2.5	20	500	9.0	3.2	2.1	0.6	0.2	0.1	0.4	0.1	0.1
2.5	25	500	7.8	2.7	1.8	0.5	0.2	0.1	0.3	0.1	0.1
2.5	30	500	6.8	2.3	1.5	0.5	0.2	0.1	0.3	0.1	0.1
2.5	10	750	>10	3.9	2.7	0.7	0.3	0.2	0.5	0.2	0.1
2.5	15	750	8.2	2.9	1.9	0.5	0.2	0.1	0.4	0.1	0.1
2.5	20	750	6.6	2.2	1.5	0.4	0.1	0.1	0.3	0.1	0.1
2.5	25	750	5.5	1.8	1.2	0.4	0.1	0.1	0.2	0.1	0.1
2.5	30	750	4.9	1.6	1.1	0.3	0.1	0.1	0.2	0.1	0.0
2.5	10	1000	8.8	3.1	2.1	0.6	0.2	0.1	0.4	0.1	0.1
2.5	15	1000	6.5	2.2	1.5	0.4	0.1	0.1	0.3	0.1	0.1
2.5	20	1000	5.2	1.7	1.1	0.3	0.1	0.1	0.2	0.1	0.0
2.5	25	1000	4.3	1.4	0.9	0.3	0.1	0.1	0.2	0.1	0.0
2.5	30	1000	3.7	1.2	0.8	0.2	0.1	0.1	0.2	0.1	0.0
3	10	130	>10	>10	7.3	1.5	0.7	0.5	1.1	0.5	0.3
3	15	130	>10	8.5	6.0	1.3	0.6	0.4	0.9	0.4	0.3
3	20	130	>10	7.4	5.1	1.2	0.5	0.3	0.8	0.3	0.2
3	25	130	>10	6.2	4.3	1.1	0.4	0.3	0.7	0.3	0.2
3	30	130	>10	5.5	3.8	1.0	0.4	0.3	0.7	0.2	0.2
3	10	250	>10	6.9	4.8	1.2	0.5	0.3	0.8	0.3	0.2
3	15	250	>10	5.3	3.7	0.9	0.4	0.2	0.6	0.2	0.2
3	20	250	>10	4.4	3.0	0.8	0.3	0.2	0.5	0.2	0.1
3	25	250	>10	3.7	2.5	0.7	0.2	0.2	0.5	0.2	0.1
3	30	250	9.2	3.2	2.2	0.6	0.2	0.1	0.4	0.1	0.1
3	10	500	>10	4.1	2.8	0.8	0.3	0.2	0.5	0.2	0.1
3	15	500	8.7	3.1	2.1	0.6	0.2	0.1	0.4	0.1	0.1
3	20	500	7.1	2.4	1.6	0.5	0.2	0.1	0.3	0.1	0.1
3	25	500	6.1	2.0	1.4	0.4	0.1	0.1	0.3	0.1	0.1
3	30	500	5.3	1.7	1.2	0.3	0.1	0.1	0.2	0.1	0.0
3	10	750	8.4	2.9	2.0	0.6	0.2	0.1	0.4	0.1	0.1
3	15	750	6.3	2.1	1.4	0.4	0.1	0.1	0.3	0.1	0.1
3	20	750	5.1	1.7	1.1	0.3	0.1	0.1	0.2	0.1	0.0
3	25	750	4.2	1.4	0.9	0.3	0.1	0.1	0.2	0.1	0.0
3	30	750	3.6	1.2	0.8	0.2	0.1	0.1	0.2	0.0	0.0
3	10	1000	6.6	2.2	1.5	0.4	0.1	0.1	0.3	0.1	0.1
3	15	1000	4.9	1.6	1.1	0.3	0.1	0.1	0.2	0.1	0.0
3	20	1000	3.9	1.3	0.8	0.3	0.1	0.1	0.2	0.1	0.0
3	25	1000	3.2	1.0	0.7	0.2	0.1	0.0	0.1	0.0	0.0

Parameters at 3kOhms (M105)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	μS	Years	Years	Years	Years	Years	Years	Years	Years	Years
3	30	1000	2.7	0.9	0.6	0.2	0.1	0.0	0.1	0.0	0.0
3.5	10	130	>10	6.7	4.7	1.2	0.5	0.3	0.9	0.4	0.2
3.5	15	130	>10	6.0	4.1	1.1	0.4	0.3	0.8	0.3	0.2
3.5	20	130	>10	5.0	3.4	0.9	0.4	0.2	0.7	0.2	0.2
3.5	25	130	>10	4.6	3.1	0.8	0.3	0.2	0.6	0.2	0.1
3.5	30	130	>10	4.1	2.8	0.8	0.3	0.2	0.5	0.2	0.1
3.5	10	250	>10	4.6	3.1	0.9	0.3	0.2	0.6	0.2	0.1
3.5	15	250	>10	3.6	2.5	0.7	0.2	0.2	0.5	0.2	0.1
3.5	20	250	8.7	3.0	2.1	0.6	0.2	0.1	0.4	0.1	0.1
3.5	25	250	7.5	2.6	1.7	0.5	0.2	0.1	0.3	0.1	0.1
3.5	30	250	6.7	2.3	1.5	0.5	0.2	0.1	0.3	0.1	0.1
3.5	10	500	7.2	2.4	1.6	0.5	0.2	0.1	0.4	0.1	0.1
3.5	15	500	5.9	2.0	1.3	0.4	0.1	0.1	0.3	0.1	0.1
3.5	20	500	5.0	1.6	1.1	0.3	0.1	0.1	0.2	0.1	0.0
3.5	25	500	4.3	1.4	0.9	0.3	0.1	0.1	0.2	0.1	0.0
3.5	30	500	3.7	1.2	0.8	0.2	0.1	0.1	0.2	0.1	0.0
3.5	10	750	5.2	1.7	1.1	0.4	0.1	0.1	0.3	0.1	0.1
3.5	15	750	4.1	1.3	0.9	0.3	0.1	0.1	0.2	0.1	0.0
3.5	20	750	3.4	1.1	0.7	0.2	0.1	0.0	0.2	0.0	0.0
3.5	25	750	3.0	0.9	0.6	0.2	0.1	0.0	0.1	0.0	0.0
3.5	30	750	2.6	0.8	0.5	0.2	0.1	0.0	0.1	0.0	0.0
3.5	10	1000	4.4	1.4	1.0	0.3	0.1	0.1	0.2	0.1	0.0
3.5	15	1000	3.4	1.1	0.7	0.2	0.1	0.0	0.2	0.0	0.0
3.5	20	1000	2.8	0.9	0.6	0.2	0.1	0.0	0.1	0.0	0.0
3.5	25	1000	2.3	0.7	0.5	0.2	0.0	0.0	0.1	0.0	0.0
3.5	30	1000	2.0	0.6	0.4	0.1	0.0	0.0	0.1	0.0	0.0

15.3 Appendix C — Model 106 Battery Longevity and Programmed Setting Choices (With AutoStim Feature Disabled)

Parameters at 3kOhms (M106)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	µS	Years	Years	Years	Years	Years	Years	Years	Years	Years
0.5	10	130	>10	>10	>10	3.0	2.5	2.2	2.2	1.8	1.6
0.5	10	250	>10	>10	>10	2.9	2.3	2.0	2.2	1.7	1.5
0.5	10	500	>10	>10	>10	2.7	1.9	1.6	2.0	1.4	1.2
0.5	10	750	>10	>10	>10	2.6	1.7	1.3	1.9	1.2	1.0
0.5	10	1000	>10	>10	>10	2.4	1.5	1.1	1.8	1.1	0.8
0.5	15	130	>10	>10	>10	2.9	2.2	1.9	2.1	1.6	1.4
0.5	15	250	>10	>10	>10	2.8	2.0	1.7	2.1	1.5	1.2
0.5	15	500	>10	>10	>10	2.5	1.6	1.3	1.9	1.2	0.9
0.5	15	750	>10	>10	>10	2.3	1.4	1.0	1.7	1.0	0.8
0.5	15	1000	>10	>10	>10	2.1	1.2	0.9	1.6	0.9	0.6
0.5	20	130	>10	>10	>10	2.8	2.0	1.7	2.1	1.5	1.2
0.5	20	250	>10	>10	>10	2.7	1.8	1.5	2.0	1.3	1.1
0.5	20	500	>10	>10	>10	2.4	1.4	1.1	1.7	1.0	0.8
0.5	20	750	>10	>10	>10	2.1	1.1	0.9	1.6	0.8	0.6
0.5	20	1000	>10	>10	9.3	1.9	1.0	0.7	1.4	0.7	0.5
0.5	25	130	>10	>10	>10	2.7	1.8	1.5	2.0	1.4	1.1
0.5	25	250	>10	>10	>10	2.5	1.6	1.3	1.9	1.2	1.0
0.5	25	500	>10	>10	>10	2.2	1.2	0.9	1.6	0.9	0.7
0.5	25	750	>10	>10	9.6	1.9	1.0	0.7	1.4	0.7	0.5
0.5	25	1000	>10	>10	7.8	1.7	0.8	0.6	1.3	0.6	0.4
0.5	30	130	>10	>10	>10	2.6	1.7	1.3	1.9	1.3	1.0
0.5	30	250	>10	>10	>10	2.4	1.5	1.2	1.8	1.1	0.9
0.5	30	500	>10	>10	>10	2.1	1.1	0.8	1.5	0.8	0.6
0.5	30	750	>10	>10	8.3	1.8	0.9	0.6	1.3	0.6	0.5
0.5	30	1000	>10	9.5	6.7	1.6	0.7	0.5	1.2	0.5	0.4
1	10	130	>10	>10	>10	2.7	1.8	1.5	1.9	1.2	1.0
1	10	250	>10	>10	>10	2.5	1.6	1.2	1.7	1.0	0.8
1	10	500	>10	>10	>10	2.2	1.2	0.9	1.5	0.8	0.6
1	10	750	>10	>10	>10	2.0	1.0	0.7	1.3	0.6	0.4
1	10	1000	>10	>10	9.7	1.8	0.8	0.6	1.1	0.5	0.4
1	15	130	>10	>10	>10	2.6	1.7	1.4	1.8	1.2	0.9
1	15	250	>10	>10	>10	2.4	1.4	1.1	1.6	0.9	0.7
1	15	500	>10	>10	>10	2.0	1.1	0.8	1.3	0.7	0.5
1	15	750	>10	>10	8.7	1.7	0.8	0.6	1.1	0.5	0.4
1	15	1000	>10	9.8	7.0	1.5	0.7	0.5	1.0	0.4	0.3

Parameters at 3kOhms (M106)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	µS	Years	Years	Years	Years	Years	Years	Years	Years	Years
1	20	130	>10	>10	>10	2.5	1.6	1.3	1.8	1.1	0.9
1	20	250	>10	>10	>10	2.3	1.3	1.0	1.6	0.8	0.6
1	20	500	>10	>10	9.3	1.8	0.9	0.7	1.2	0.6	0.4
1	20	750	>10	9.7	6.9	1.5	0.7	0.5	1.0	0.4	0.3
1	20	1000	>10	7.8	5.5	1.3	0.5	0.4	0.8	0.3	0.2
1	25	130	>10	>10	>10	2.4	1.5	1.2	1.7	1.0	0.8
1	25	250	>10	>10	>10	2.1	1.2	0.9	1.5	0.8	0.6
1	25	500	>10	>10	7.8	1.7	0.8	0.6	1.1	0.5	0.3
1	25	750	>10	8.2	5.7	1.4	0.6	0.4	0.9	0.4	0.2
1	25	1000	>10	6.5	4.5	1.2	0.5	0.3	0.7	0.3	0.2
1	30	130	>10	>10	>10	2.4	1.4	1.1	1.7	1.0	0.7
1	30	250	>10	>10	>10	2.0	1.0	0.8	1.4	0.7	0.5
1	30	500	>10	9.5	6.7	1.5	0.7	0.5	1.0	0.4	0.3
1	30	750	>10	7.0	4.9	1.2	0.5	0.3	0.8	0.3	0.2
1	30	1000	>10	5.6	3.8	1.0	0.4	0.3	0.7	0.2	0.2
1.5	10	130	>10	>10	>10	2.3	1.3	1.0	1.6	0.9	0.7
1.5	10	250	>10	>10	>10	2.0	1.0	0.8	1.4	0.7	0.5
1.5	10	500	>10	>10	7.9	1.6	0.7	0.5	1.1	0.5	0.3
1.5	10	750	>10	8.1	5.7	1.2	0.5	0.4	0.8	0.3	0.2
1.5	10	1000	>10	6.4	4.4	1.0	0.4	0.3	0.7	0.3	0.2
1.5	15	130	>10	>10	>10	2.2	1.2	0.9	1.5	0.8	0.6
1.5	15	250	>10	>10	9.7	1.8	0.9	0.6	1.3	0.6	0.4
1.5	15	500	>10	8.5	6.0	1.3	0.5	0.4	0.9	0.4	0.3
1.5	15	750	>10	6.1	4.2	1.0	0.4	0.3	0.7	0.3	0.2
1.5	15	1000	>10	4.7	3.2	0.8	0.3	0.2	0.6	0.2	0.1
1.5	20	130	>10	>10	>10	2.0	1.1	0.8	1.5	0.7	0.6
1.5	20	250	>10	>10	8.5	1.7	0.8	0.5	1.1	0.5	0.4
1.5	20	500	>10	7.2	5.0	1.2	0.5	0.3	0.8	0.3	0.2
1.5	20	750	>10	5.0	3.4	0.9	0.3	0.2	0.6	0.2	0.1
1.5	20	1000	>10	3.8	2.6	0.7	0.2	0.2	0.5	0.2	0.1
1.5	25	130	>10	>10	>10	1.9	1.0	0.7	1.4	0.7	0.5
1.5	25	250	>10	>10	7.5	1.5	0.7	0.5	1.1	0.5	0.3
1.5	25	500	>10	6.3	4.4	1.0	0.4	0.3	0.7	0.3	0.2
1.5	25	750	>10	4.3	2.9	0.8	0.3	0.2	0.5	0.2	0.1
1.5	25	1000	9.2	3.3	2.2	0.6	0.2	0.1	0.4	0.1	0.1
1.5	30	130	>10	>10	9.8	1.8	0.9	0.7	1.3	0.6	0.4
1.5	30	250	>10	9.5	6.8	1.4	0.6	0.4	1.0	0.4	0.3
1.5	30	500	>10	5.5	3.8	0.9	0.4	0.2	0.6	0.2	0.2

Parameters at 3kOhms (M106)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	μS	Years	Years	Years	Years	Years	Years	Years	Years	Years
1.5	30	750	>10	3.8	2.6	0.7	0.2	0.2	0.5	0.2	0.1
1.5	30	1000	8.2	2.8	1.9	0.5	0.2	0.1	0.4	0.1	0.1
2	10	130	>10	>10	>10	2.0	1.0	0.8	1.4	0.7	0.5
2	10	250	>10	>10	8.2	1.6	0.7	0.5	1.1	0.5	0.3
2	10	500	>10	7.2	5.0	1.2	0.5	0.3	0.8	0.3	0.2
2	10	750	>10	5.2	3.6	0.9	0.3	0.2	0.6	0.2	0.1
2	10	1000	>10	4.0	2.8	0.7	0.3	0.2	0.5	0.2	0.1
2	15	130	>10	>10	9.5	1.8	0.9	0.6	1.3	0.6	0.4
2	15	250	>10	8.9	6.3	1.4	0.6	0.4	0.9	0.4	0.3
2	15	500	>10	5.3	3.7	0.9	0.3	0.2	0.6	0.2	0.2
2	15	750	>10	3.8	2.6	0.7	0.2	0.2	0.5	0.2	0.1
2	15	1000	8.3	2.9	2.0	0.5	0.2	0.1	0.4	0.1	0.1
2	20	130	>10	>10	8.1	1.6	0.8	0.5	1.1	0.5	0.4
2	20	250	>10	7.3	5.1	1.2	0.5	0.3	0.8	0.3	0.2
2	20	500	>10	4.2	2.9	0.8	0.3	0.2	0.5	0.2	0.1
2	20	750	8.4	2.9	2.0	0.6	0.2	0.1	0.4	0.1	0.1
2	20	1000	6.6	2.2	1.5	0.4	0.1	0.1	0.3	0.1	0.1
2	25	130	>10	>10	7.2	1.5	0.7	0.5	1.1	0.5	0.3
2	25	250	>10	6.4	4.4	1.1	0.4	0.3	0.7	0.3	0.2
2	25	500	>10	3.6	2.4	0.7	0.2	0.2	0.5	0.2	0.1
2	25	750	7.2	2.5	1.7	0.5	0.2	0.1	0.3	0.1	0.1
2	25	1000	5.6	1.9	1.2	0.4	0.1	0.1	0.2	0.1	0.1
2	30	130	>10	9.0	6.4	1.4	0.6	0.4	1.0	0.4	0.3
2	30	250	>10	5.6	3.9	1.0	0.4	0.3	0.7	0.2	0.2
2	30	500	8.9	3.1	2.1	0.6	0.2	0.1	0.4	0.1	0.1
2	30	750	6.4	2.1	1.4	0.4	0.1	0.1	0.3	0.1	0.1
2	30	1000	4.9	1.6	1.1	0.3	0.1	0.1	0.2	0.1	0.0
2.5	10	130	>10	>10	9.9	1.8	0.9	0.7	1.3	0.6	0.4
2.5	10	250	>10	9.3	6.6	1.4	0.6	0.4	1.0	0.4	0.3
2.5	10	500	>10	5.8	4.0	1.0	0.4	0.3	0.6	0.2	0.2
2.5	10	750	>10	4.1	2.8	0.7	0.3	0.2	0.5	0.2	0.1
2.5	10	1000	9.1	3.2	2.2	0.6	0.2	0.1	0.4	0.1	0.1
2.5	15	130	>10	>10	8.0	1.6	0.7	0.5	1.1	0.5	0.3
2.5	15	250	>10	7.2	5.0	1.2	0.5	0.3	0.8	0.3	0.2
2.5	15	500	>10	4.2	2.9	0.8	0.3	0.2	0.5	0.2	0.1
2.5	15	750	8.5	2.9	2.0	0.5	0.2	0.1	0.4	0.1	0.1
2.5	15	1000	6.7	2.3	1.5	0.4	0.1	0.1	0.3	0.1	0.1
2.5	20	130	>10	9.3	6.6	1.4	0.6	0.4	1.0	0.4	0.3

Parameters at 3kOhms (M106)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	µS	Years	Years	Years	Years	Years	Years	Years	Years	Years
2.5	20	250	>10	5.8	4.0	1.0	0.4	0.3	0.7	0.3	0.2
2.5	20	500	9.3	3.3	2.2	0.6	0.2	0.1	0.4	0.1	0.1
2.5	20	750	6.8	2.3	1.5	0.4	0.1	0.1	0.3	0.1	0.1
2.5	20	1000	5.3	1.7	1.2	0.3	0.1	0.1	0.2	0.1	0.0
2.5	25	130	>10	8.1	5.7	1.3	0.5	0.4	0.9	0.4	0.3
2.5	25	250	>10	4.9	3.4	0.9	0.3	0.2	0.6	0.2	0.1
2.5	25	500	7.9	2.7	1.8	0.5	0.2	0.1	0.4	0.1	0.1
2.5	25	750	5.7	1.9	1.3	0.4	0.1	0.1	0.3	0.1	0.1
2.5	25	1000	4.4	1.4	1.0	0.3	0.1	0.1	0.2	0.1	0.0
2.5	30	130	>10	7.2	5.1	1.2	0.5	0.3	0.8	0.3	0.2
2.5	30	250	>10	4.3	2.9	0.8	0.3	0.2	0.5	0.2	0.1
2.5	30	500	7.0	2.4	1.6	0.5	0.2	0.1	0.3	0.1	0.1
2.5	30	750	4.9	1.6	1.1	0.3	0.1	0.1	0.2	0.1	0.0
2.5	30	1000	3.8	1.2	0.8	0.2	0.1	0.1	0.2	0.1	0.0
3	10	130	>10	>10	8.4	1.7	0.8	0.6	1.1	0.5	0.4
3	10	250	>10	7.5	5.3	1.2	0.5	0.3	0.8	0.3	0.2
3	10	500	>10	4.4	3.0	0.8	0.3	0.2	0.5	0.2	0.1
3	10	750	8.6	3.0	2.0	0.6	0.2	0.1	0.4	0.1	0.1
3	10	1000	6.8	2.3	1.5	0.4	0.1	0.1	0.3	0.1	0.1
3	15	130	>10	9.3	6.6	1.4	0.6	0.4	1.0	0.4	0.3
3	15	250	>10	5.7	3.9	1.0	0.4	0.3	0.7	0.2	0.2
3	15	500	9.0	3.2	2.1	0.6	0.2	0.1	0.4	0.1	0.1
3	15	750	6.4	2.2	1.5	0.4	0.1	0.1	0.3	0.1	0.1
3	15	1000	5.0	1.6	1.1	0.3	0.1	0.1	0.2	0.1	0.0
3	20	130	>10	7.7	5.4	1.2	0.5	0.4	0.8	0.3	0.2
3	20	250	>10	4.6	3.1	0.8	0.3	0.2	0.5	0.2	0.1
3	20	500	7.3	2.5	1.7	0.5	0.2	0.1	0.3	0.1	0.1
3	20	750	5.1	1.7	1.1	0.3	0.1	0.1	0.2	0.1	0.0
3	20	1000	3.9	1.3	0.8	0.3	0.1	0.1	0.2	0.1	0.0
3	25	130	>10	6.6	4.6	1.1	0.4	0.3	0.8	0.3	0.2
3	25	250	>10	3.9	2.6	0.7	0.2	0.2	0.5	0.2	0.1
3	25	500	6.1	2.1	1.4	0.4	0.1	0.1	0.3	0.1	0.1
3	25	750	4.3	1.4	0.9	0.3	0.1	0.1	0.2	0.1	0.0
3	25	1000	3.3	1.0	0.7	0.2	0.1	0.0	0.1	0.0	0.0
3	30	130	>10	5.8	4.0	1.0	0.4	0.3	0.7	0.3	0.2
3	30	250	9.3	3.3	2.2	0.6	0.2	0.1	0.4	0.1	0.1
3	30	500	5.3	1.7	1.2	0.3	0.1	0.1	0.2	0.1	0.0
3	30	750	3.7	1.2	0.8	0.2	0.1	0.0	0.2	0.0	0.0

Parameters at 3kOhms (M106)			Time from BOL to IFI			Time from IFI to N EOS			Time from N EOS to EOS		
			10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle	10% Duty Cycle	33% Duty Cycle	50% Duty Cycle
mA	Hz	µS	Years	Years	Years	Years	Years	Years	Years	Years	Years
3	30	1000	2.8	0.9	0.6	0.2	0.1	0.0	0.1	0.0	0.0
3.5	10	130	>10	7.2	5.1	1.3	0.5	0.4	0.9	0.4	0.3
3.5	10	250	>10	4.7	3.2	0.9	0.3	0.2	0.6	0.2	0.2
3.5	10	500	7.3	2.5	1.7	0.5	0.2	0.1	0.4	0.1	0.1
3.5	10	750	5.3	1.7	1.2	0.4	0.1	0.1	0.2	0.1	0.1
3.5	10	1000	4.5	1.4	1.0	0.3	0.1	0.1	0.2	0.1	0.0
3.5	15	130	>10	6.1	4.2	1.1	0.4	0.3	0.8	0.3	0.2
3.5	15	250	>10	3.7	2.5	0.7	0.3	0.2	0.5	0.2	0.1
3.5	15	500	5.9	2.0	1.3	0.4	0.1	0.1	0.3	0.1	0.1
3.5	15	750	4.2	1.4	0.9	0.3	0.1	0.1	0.2	0.1	0.0
3.5	15	1000	3.4	1.1	0.7	0.2	0.1	0.0	0.2	0.0	0.0
3.5	20	130	>10	5.2	3.6	1.0	0.4	0.2	0.7	0.3	0.2
3.5	20	250	8.9	3.1	2.1	0.6	0.2	0.1	0.4	0.1	0.1
3.5	20	500	5.0	1.6	1.1	0.3	0.1	0.1	0.2	0.1	0.0
3.5	20	750	3.5	1.1	0.7	0.2	0.1	0.0	0.2	0.0	0.0
3.5	20	1000	2.8	0.9	0.6	0.2	0.1	0.0	0.1	0.0	0.0
3.5	25	130	>10	4.6	3.1	0.9	0.3	0.2	0.6	0.2	0.1
3.5	25	250	7.7	2.7	1.8	0.5	0.2	0.1	0.3	0.1	0.1
3.5	25	500	4.3	1.4	0.9	0.3	0.1	0.1	0.2	0.1	0.0
3.5	25	750	2.9	0.9	0.6	0.2	0.1	0.0	0.1	0.0	0.0
3.5	25	1000	2.4	0.7	0.5	0.2	0.0	0.0	0.1	0.0	0.0
3.5	30	130	>10	4.1	2.8	0.8	0.3	0.2	0.5	0.2	0.1
3.5	30	250	6.8	2.3	1.6	0.5	0.2	0.1	0.3	0.1	0.1
3.5	30	500	3.7	1.2	0.8	0.2	0.1	0.1	0.2	0.1	0.0
3.5	30	750	2.6	0.8	0.5	0.2	0.1	0.0	0.1	0.0	0.0
3.5	30	1000	2.0	0.6	0.4	0.1	0.0	0.0	0.1	0.0	0.0

15.4 Appendix D — Model 1000 Battery Longevity and Programmed Setting Choices

15.4.1 AutoStim Feature Disabled

Parameters at 3kOhms			AutoStim Feature Disabled Normal Mode Duty Cycle								
			10% (30s ON/5 min OFF)			35% (30s ON/1.1 min OFF)			51% (60s ON/1.1 min OFF)		
			BOL to IFI	IFI to N EOS	N EOS to EOS	BOL to IFI	IFI to N EOS	N EOS to EOS	BOL to IFI	IFI to N EOS	N EOS to EOS
mA	Hz	µS	Years	Years	Years	Years	Years	Years	Years	Years	Years
0.5	20	250	11.9	1.2	1.2	6.1	0.6	0.6	4.6	0.5	0.5
0.5	20	500	11.8	1.2	1.2	6.0	0.6	0.6	4.5	0.5	0.5
0.5	30	250	10.2	1.0	1.0	4.7	0.5	0.5	3.5	0.4	0.4
0.5	30	500	10.1	1.0	1.0	4.6	0.5	0.5	3.4	0.3	0.3
1	20	250	11.7	1.2	1.2	5.9	0.6	0.6	4.5	0.5	0.4
1	20	500	11.6	1.2	1.1	5.8	0.6	0.5	4.4	0.4	0.4
1	30	250	10.0	1.0	1.0	4.5	0.5	0.5	3.3	0.3	0.3
1	30	500	9.9	1.0	1.0	4.4	0.4	0.4	3.2	0.3	0.3
1.5	20	250	11.4	1.1	1.1	5.7	0.6	0.5	4.2	0.4	0.4
1.5	20	500	9.4	0.9	0.8	4.1	0.4	0.3	3.0	0.3	0.2
1.5	30	250	9.8	1.0	0.9	4.4	0.4	0.4	3.2	0.3	0.3
1.5	30	500	7.7	0.7	0.7	3.1	0.3	0.2	2.2	0.2	0.2
2	20	250	9.7	0.9	0.8	4.3	0.4	0.3	3.2	0.3	0.2
2	20	500	7.2	0.7	0.6	2.8	0.3	0.2	2.0	0.2	0.2
2	30	250	8.2	0.8	0.7	3.3	0.3	0.3	2.4	0.2	0.2
2	30	500	5.6	0.5	0.5	2.0	0.2	0.2	1.4	0.1	0.1
2.5	20	250	7.9	0.7	0.7	3.2	0.3	0.2	2.3	0.2	0.2
2.5	20	500	5.5	0.5	0.4	1.9	0.2	0.1	1.4	0.1	0.1
2.5	30	250	6.5	0.6	0.5	2.4	0.2	0.2	1.7	0.2	0.1
2.5	30	500	4.2	0.4	0.3	1.4	0.1	0.1	1.0	0.1	0.1
3	20	250	6.4	0.6	0.5	2.4	0.2	0.2	1.7	0.2	0.1
3	20	500	4.2	0.4	0.3	1.4	0.1	0.1	1.0	0.1	0.1
3	30	250	5.1	0.5	0.4	1.8	0.2	0.1	1.2	0.1	0.1
3	30	500	3.1	0.3	0.2	1.0	0.1	0.1	0.7	0.1	0.1
3.5	20	250	5.2	0.5	0.4	1.8	0.2	0.1	1.3	0.1	0.1
3.5	20	500	3.2	0.3	0.2	1.0	0.1	0.1	0.7	0.1	0.1
3.5	30	250	4.0	0.4	0.3	1.3	0.1	0.1	0.9	0.1	0.1
3.5	30	500	2.3	0.2	0.2	0.7	0.1	0.1	0.5	0.0	0.0

15.4.2 AutoStim Feature Enabled (7 AutoStims/Hour and AutoStim ON time 60 seconds)

Parameters at 3kOhms			AutoStim Feature Enabled (7 AutoStims/Hour and AutoStim ON time 60s)* Normal Mode Duty Cycle					
			10% (30s ON/5 min OFF)			35% (30s ON/1.1 min OFF)		
			BOL to IFI	IFI to N EOS	N EOS to EOS	BOL to IFI	IFI to N EOS	N EOS to EOS
mA	Hz	µS	Years	Years	Years	Years	Years	Years
0.5	20	250	6.3	0.6	0.7	4.6	0.5	0.5
0.5	20	250	6.3	0.6	0.6	4.5	0.5	0.5
0.5	20	500	5.5	0.6	0.6	3.6	0.4	0.4
0.5	30	250	5.4	0.5	0.6	3.5	0.4	0.4
1	20	250	6.3	0.6	0.6	4.5	0.5	0.5
1	20	500	6.2	0.6	0.6	4.4	0.4	0.4
1	30	250	5.4	0.5	0.5	3.5	0.4	0.4
1	30	500	5.3	0.5	0.5	3.4	0.3	0.3
1.5	20	250	6.1	0.6	0.6	4.3	0.4	0.4
1.5	20	500	5.0	0.5	0.4	3.2	0.3	0.3
1.5	30	250	5.2	0.5	0.5	3.4	0.3	0.3
1.5	30	500	4.1	0.4	0.4	2.4	0.2	0.2
2	20	250	5.2	0.5	0.4	3.4	0.3	0.3
2	20	500	3.8	0.4	0.3	2.2	0.2	0.2
2	30	250	4.4	0.4	0.4	2.6	0.2	0.2
2	30	500	3.0	0.3	0.2	1.6	0.1	0.1
2.5	20	250	4.2	0.4	0.4	2.5	0.2	0.2
2.5	20	500	2.9	0.3	0.2	1.6	0.1	0.1
2.5	30	250	3.5	0.3	0.3	1.9	0.2	0.2
2.5	30	500	2.2	0.2	0.2	1.1	0.1	0.1
3	20	250	3.4	0.3	0.3	1.9	0.2	0.1
3	20	500	2.2	0.2	0.2	1.2	0.1	0.1
3	30	250	2.7	0.3	0.2	1.5	0.1	0.1
3	30	500	1.7	0.1	0.1	0.8	0.1	0.1
3.5	20	250	2.8	0.3	0.2	1.5	0.1	0.1
3.5	20	250	1.7	0.2	0.1	0.8	0.1	0.1
3.5	20	500	2.1	0.2	0.2	1.1	0.1	0.1
3.5	30	250	1.2	0.1	0.1	0.6	0.1	0.0

*Longevity values with Prone Detection ON vary by no more than 4%.

15.4.3 AutoStim Feature Enabled (7 AutoStims/Hour, AutoStim ON time 30 seconds)

Parameters at 3kOhms			AutoStim Feature Enabled (7 AutoStims/Hour and AutoStim ON time 30s)* Normal Mode Duty Cycle					
			10% (30s ON/5 min OFF)			35% (30s ON/1.1 min OFF)		
			BOL to IFI	IFI to N EOS	N EOS to EOS	BOL to IFI	IFI to N EOS	N EOS to EOS
			Years	Years	Years	Years	Years	Years
0.5	20	250	7.0	0.71	0.72	4.8	0.49	0.49
0.5	20	250	7.0	0.71	0.71	4.7	0.48	0.49
0.5	20	500	6.2	0.64	0.64	3.9	0.39	0.40
0.5	30	250	6.2	0.63	0.63	3.8	0.38	0.39
1	20	250	6.9	0.71	0.71	4.7	0.48	0.48
1	20	500	6.9	0.70	0.68	4.6	0.46	0.44
1	30	250	6.1	0.62	0.63	3.7	0.38	0.38
1	30	500	6.1	0.62	0.60	3.7	0.37	0.35
1.5	20	250	6.8	0.69	0.66	4.5	0.45	0.41
1.5	20	500	5.8	0.56	0.52	3.4	0.32	0.28
1.5	30	250	6.0	0.61	0.58	3.6	0.37	0.33
1.5	30	500	5.0	0.48	0.44	2.6	0.25	0.21
2	20	250	6.0	0.58	0.53	3.6	0.34	0.29
2	20	500	4.7	0.44	0.40	2.4	0.22	0.19
2	30	250	5.2	0.50	0.46	2.8	0.26	0.23
2	30	500	3.8	0.36	0.32	1.8	0.16	0.14
2.5	20	250	5.1	0.48	0.44	2.7	0.25	0.21
2.5	20	500	3.7	0.34	0.30	1.7	0.16	0.13
2.5	30	250	4.3	0.40	0.36	2.1	0.19	0.17
2.5	30	500	2.9	0.27	0.23	1.3	0.11	0.10
3	20	250	4.2	0.40	0.35	2.1	0.19	0.16
3	20	500	2.9	0.27	0.23	1.3	0.11	0.10
3	30	250	3.5	0.32	0.28	1.6	0.14	0.12
3	30	500	2.2	0.20	0.17	0.9	0.08	0.07
3.5	20	250	3.5	0.33	0.29	1.6	0.15	0.12
3.5	20	250	2.3	0.21	0.18	0.9	0.08	0.07
3.5	20	500	2.8	0.26	0.22	1.2	0.11	0.09
3.5	30	250	1.7	0.15	0.13	0.7	0.06	0.05

*Longevity values with Prone Detection ON vary by no more than 5%.

15.5 Appendix E — Mean Sensitivity and Potential False Positive Rates per Threshold for AutoStim Setting

The following table is applicable for generators with the Autostim feature only and supplements the “Receiver Operating Characteristic (ROC) Curve for Tachycardia Detection Associated with Seizures” figure within the *model specific Technical Information* chapter.

This table lists the mean values and 95% confidence intervals (CI) from the E36 and E37 clinical study performance data.

Threshold for AutoStim	Mean Sensitivity (%) (95% CI) ^a			Potential False Positives per Hour (95% CI) ^a
	Ictal Tachycardia Seizures Only (—◆—) n=11 pts, 28 sz	Seizures ≥ 20% Heart Rate Change (-●-) n=25 pts, 82 sz	All Patients (pts), All Seizures (sz) (—■—) n=34 pts, 170 sz	Applies to All Categories n=50 pts, 4516 hrs
70% Threshold	60.7 (40.0, 81.8)	26.8 (14.2, 42.9)	18.8 (10.5, 34.4)	0.4 (0.3, 0.5)
60% Threshold	67.9 (46.9, 88.0)	39.0 (23.8, 53.9)	27.1 (12.9, 41.0)	0.6 (0.5, 0.8)
50% Threshold	85.7 (70.4, 96.0)	56.1 (38.1, 73.0)	41.2 (20.9, 50.9)	1.0 (0.8, 1.3)
40% Threshold	96.4 (86.2, 100)	70.7 (52.5, 84.4)	53.5 (28.9, 61.3)	1.9 (1.5, 2.3)
30% Threshold	100 ^b	91.5 (78.6, 97.5)	67.1 (39.0, 71.2)	3.7 (3.2, 4.5)
20% Threshold	100 ^b	98.8 (94.4, 100)	80.0 (56.0, 82.1)	7.6 (6.6, 8.8)

a. 95% confidence intervals constructed using 3000 bootstrap samples

b. Confidence Intervals cannot be calculated when mean sensitivity equals 100%

16 **Limited Replacement Warranty**

LivaNova USA, Inc. warrants the VNS Therapy generator and lead against any defects due to faulty material or workmanship for a period of two (2) years from the date of implantation. This warranty applies only to the original purchaser of the VNS Therapy generator and lead and the patient implanted with it. This Limited Replacement Warranty also applies only when the product is used in accordance with the product's physician's manual and excludes damage due to improper handling, defacing, accident (including dropping), or misuse. This product is not warranted when used or implanted by a person(s) not trained in or familiar with the VNS Therapy System and Programming System physician's manuals. This Limited Replacement Warranty is not a representation that any one VNS Therapy generator or lead will last the entire time of the Limited Replacement Warranty.

In no event shall LivaNova USA, Inc. be liable for any special, incidental, indirect, or consequential damages based on the failure of the device to function within normal tolerances, or resulting from damage to the device by external forces, whether the claim is based on warranty, contract, tort, or otherwise, or in connection with the purchase, use, or surgical implantation of this device or associated components or costs over and above the original purchase price from LivaNova USA, Inc.

To qualify for the Limited Replacement Warranty, the following conditions must be met:

1. A properly completed Implant and Warranty Registration form for both the VNS Therapy generator and the VNS Therapy lead must be returned to LivaNova USA, Inc. within sixty (60) days of device implantation;
2. The battery in the VNS Therapy generator cannot have been depleted as a result of programming to unusually high output currents, pulse widths, or duty cycles, which will cause a high energy/current drain;
3. The VNS Therapy lead cannot have been cut or damaged due to excessive handling or abuse during surgical implantation;
4. The product must have been used and prescribed in accordance with the VNS Therapy System and Programming System physician's manuals;
5. The VNS Therapy generator or lead must have been implanted prior to its "Expiration Date;"
6. The defective VNS Therapy generator or lead must be returned to LivaNova USA, Inc. with an accompanying Authorization number, available from Technical Support at 1 (866) 882-8804 (U.S. and Canada) or +1 (281) 228-7330 (Worldwide), and confirmed defective by the Quality Assurance Department; and
7. All returned VNS Therapy generators and leads shall become the property of LivaNova USA, Inc.



Caution: Return explanted generators and leads to LivaNova USA, Inc. for examination and proper disposal, along with a completed Returned Product Report form. Before returning the lead, disinfect the device components with Betadine[®], Cidex[®] soak, or another similar disinfectant, and double-seal them in a pouch or other container properly labeled with a biohazard warning.

If the VNS Therapy generator or lead becomes defective within the warranty period, contact LivaNova USA, Inc. Customer Service for a no-cost replacement. LivaNova USA, Inc. reserves the right to replace a defective product with the most comparable product currently available.

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Returned biohazardous product should be clearly identified as such on the outside surface of the package.

No implied warranty, including, but not limited to, any implied warranty of merchantability or fitness for a particular purpose, shall extend beyond the period specified above. This replacement warranty shall be the exclusive remedy available to any person. No person has any authority to bind LivaNova USA, Inc. to any representation, condition, or warranty except this Limited Replacement Warranty.

While this warranty gives you specific legal rights, you may also have other rights that vary from state to state or that encroach upon the above.